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Anesthesia and Postoperative Care

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SECTION I: ANESTHESIA FOR ADULT CARDIOVASCULAR SURGERY

Cardiac anesthesiologists are important contributors to the perioperative management of patients undergoing cardiovascular surgery. More complex critically ill patients are undergoing cardiac surgery with the increasing use of percutaneous procedures for ischemic cardiovascular disease, advanced surgical approaches for heart failure management, and increasing patient age and comorbidities. The evolution of dedicated cardiothoracic anesthesiologists is an important aspect of the team approach. Advanced hemodynamic monitoring, management of the complex coagulopathies associated with cardiac surgery, and evolution of evidence-based medical practices are important perspectives for their consultative role that we will focus on in this chapter.^{1,2}

Preoperative Preparation and Evaluation

Consultation with a cardiothoracic anesthesiologist is an important preoperative consideration. A thorough history, physical examination, and understanding of the presenting cardiac pathology and proposed surgical procedures are critical and include concomitant patient comorbidity, current medications, especially anticoagulants, challenges to airway management and invasive monitoring, and current clinical status.

Management of Preoperative Medications

Cardiovascular Medications. Patients undergoing cardiac surgery often receive multiple pharmacologic agents preoperatively. Most antihypertensive and antianginal cardiac medications should be continued preoperatively. Whether to continue or when to withhold preoperative antihypertensive therapy necessitates patient risk/benefit. Acute withdrawal of antihypertensive medications such as β -blockers or clonidine may result in potential perioperative hemodynamic instability, and these medications should be continued unless there are specific hemodynamic or other concerns.^{3,4}

However, decisions to continue angiotensin-converting enzyme (ACE) inhibitors and angiotensin II receptor subtype I antagonists (ARA) are less clear. Patients in whom ACE therapy was maintained until the morning of surgery had an increased

probability of hypotension at anesthesia induction.⁵ ARA-treated patients had more postinduction hypotension refractory to conventional vasopressors than those on other antihypertensive therapy (β -blockers, calcium channel blockers, or ACE inhibitors).⁶

Most recommendations suggest discontinuing ARA before surgery.^{7,8} Chronic ACE inhibitor use or ARA therapy less than 10 hours prior to general anesthesia was reported to increase risk of developing moderate hypotension but was not associated with increased postoperative complications. Further, intraoperative hemodynamic instability related to ACE or ARA therapy is not consistent in all studies. Patients treated long-term with ACE inhibitors with normal left ventricular function did not have altered hemodynamic instability during cardiac surgery.⁹ Although continuing therapy has potential beneficial effects,^{8,10} continued use of these agents on the day of surgery may necessitate episodic vasopressor support. Although the continuation of preoperative cardiovascular medications is recommended, holding ACE inhibitors or ARA therapy is likely warranted.

Statins. Beyond lipid-lowering properties, statins possess pleiotropic effects that have been reported to reduce postoperative morbidity and mortality.¹¹⁻¹³ Among patients undergoing percutaneous coronary interventions, Pascual reported that statin pretreatment was associated with reduced early ischemic events, primarily in those with high levels of inflammatory markers.¹³ Pretreatment with statins in patients undergoing coronary artery bypass grafting (CABG) reduced P-selectin levels, an adhesion molecule that plays a role in the pathogenesis of arteriosclerosis, below those of patients given a placebo.¹⁴ Berkan reported less use of inotropic agents among patients treated with statins, speculating that myocardial injury caused by cardiopulmonary bypass (CPB)-induced inflammatory changes was reduced.¹⁴ Others have reported a protective effect of statin pretreatment in reducing myocardial damage after CABG.^{13,15} Based on these beneficial pleiotropic effects, statin therapy should be continued routinely in the perioperative period.

Medications Affecting Hemostasis. In patients with cardiovascular disease, the use of oral anticoagulants is common and includes both antithrombotic agents and antiplatelet agents. Oral antithrombotic agents include warfarin and other vitamin K antagonists (VKAs), as well as the non-vitamin K direct oral anticoagulants (DOACs).¹⁶ Warfarin is the vitamin K oral anticoagulant most often used in North America, and the DOACs include the direct

thrombin inhibitor, dabigatran, and the factor Xa (FXa) inhibitors, apixaban, edoxaban, and rivaroxaban.¹⁶ The oral antiplatelet agents include aspirin and the P2Y₁₂ receptor antagonists clopidogrel, prasugrel, and ticagrelor. Clopidogrel and prasugrel are prodrugs, of which clopidogrel is the most commonly used P2Y₁₂ receptor antagonist. It is a prodrug that undergoes two metabolic steps to be activated and has a potential resistance reported to be around 30%.¹⁷ Ticagrelor is a direct-acting agent with no reported resistance. These agents will be considered separately. There are multiple guidelines currently reported regarding management strategies for surgery, including those from the American College of Chest Physicians (ACCP) in 2022 to inform best practices for the perioperative management of patients who are receiving a VKA and may require heparin bridging for the perioperative management of patients who are receiving a DOAC and for patients who are receiving one or more antiplatelet drugs and will be included. Considerations in managing the different anticoagulants are as follows.

Warfarin. Warfarin is the VKA used in the United States. The mechanism of action is inhibition of the post-translational modification of factors II, VII, IX, and X, as well as proteins C and S; the drug functions as an anticoagulant by lowering circulating levels of these factors. An international normalized ratio (INR) of 2 corresponds to an approximate level of about 45% circulating factors, while an INR of 3 is approximately 25%.¹⁸ Current ACCP guidelines recommend against heparin bridging for patients receiving VKA therapy for a mechanical heart valve who require VKA interruption for an elective surgery/procedure (conditional recommendation, very low certainty of evidence). A similar recommendation against heparin bridging was made for patients receiving VKA therapy for atrial fibrillation (AF) who require VKA interruption for an elective surgery/procedure (strong recommendation, moderate certainty of evidence).¹⁹ If there is a need for urgent reversal for patients who require emergency surgery, ACCP guidelines recommend the use of intravenous (IV) vitamin K along with a four-component prothrombin complex concentrate.

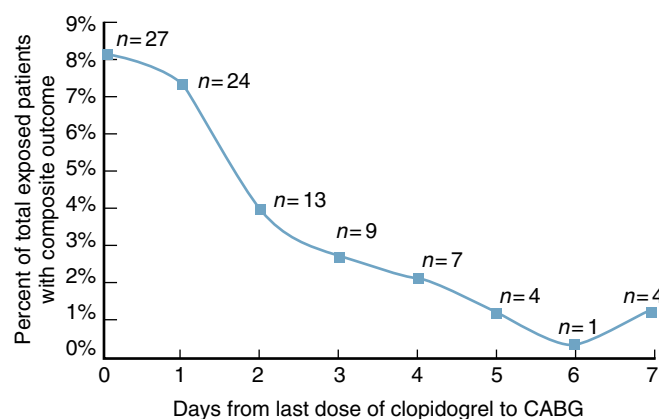
Dabigatran (Direct Thrombin Inhibitor). Dabigatran, a non-vitamin K DOAC, is an oral direct thrombin inhibitor that was the first DOAC approved in the United States. Unlike the other DOACs, renal elimination is the primary route for removing the drug from the body. In patients with acute illness, dabigatran levels can accumulate. But also unlike the other agents there is a specific antidote called idarucizumab, a monoclonal antibody that reverses its effect to facilitate postsurgical and procedural interventions, which is currently approved for urgent surgical and procedural interventions in patients receiving dabigatran.²⁰ Although dabigatran currently constitutes a small percentage of DOAC use, when it becomes available as a generic product, it may be increasingly prescribed. One of the benefits of this being a direct thrombin inhibitor is that standard coagulation tests will detect its effect, including partial thromboplastin times (PTT), although thrombin times and diluted thrombin times are more sensitive assays. For elective surgery, guidelines suggest the drug should be stopped 3 to 5 days before surgery, depending upon renal function.¹⁹

Apixaban and Rivaroxaban (Factor Xa Inhibitors). Apixaban and rivaroxaban are the two DOACs used by most patients in the United States. Unlike dabigatran, these agents are less dependent on renal elimination for metabolism. Recommendations are to stop these drugs at least 2 days before cardiac surgery, although current American College of Cardiology (ACC) guidelines suggest stopping apixaban and rivaroxaban for 1 or 2 days preoperatively.¹⁹ Edoxaban is not commonly used, but it is recommended that the drug be stopped 3 days before cardiac surgery. Unlike

dabigatran, which has a specific antidote approved for and studied in cardiac and other surgical patients, the specific antidote for apixaban and rivaroxaban is a recombinant modified FXa protein; this is indicated for reversal of life-threatening or uncontrolled bleeding and not surgical patients. Further, andexanet alfa reverses heparin, causes heparin resistance, and should not be used to reverse FXa inhibitors in cardiac surgical patients. Standard coagulation tests (prothrombin time [PT], PTT) do not detect the effects of factor Xa inhibitors; rather, specific calibrated anti-Xa assays (specific chromogenic anti-FXa assays) are required.²⁰

Aspirin. Older guidelines from The Society of Thoracic Surgeons suggested that recent acetylsalicylic acid (ASA) use was associated with perioperative bleeding.²¹ The Society guideline recommended (class IIa) discontinuing ASA 3 to 5 days before elective CABG to reduce bleeding risk but continuation of ASA in urgent and emergency CABG, recognizing that the benefits of ASA outweigh the small bleeding risk. Current ACCP guidelines suggest that patients receiving ASA and undergoing CABG surgery continue ASA without interruption.

P2Y₁₂ inhibitors: Clopidogrel, Prasugrel, Ticagrelor. The P2Y₁₂ inhibitors clopidogrel, prasugrel, and ticagrelor are inhibitors of adenosine diphosphate (ADP)-mediated platelet activation. Although patients undergoing elective CABG within 3 to 5 days of clopidogrel treatment are at increased risk for bleeding, transfusion, and reoperation, there may be a protective effect in patients undergoing emergency surgery (Fig. 4.1).²²⁻²⁵ However, variability of bleeding, especially with clopidogrel, is likely because of the potential for clopidogrel resistance that can occur in at least 30% of patients as clopidogrel is a prodrug and undergoes metabolic transformation. This is not the case with ticagrelor, as it is not a prodrug and does not require metabolic transformation to be active. Patients receiving clopidogrel within ~3 days of surgery compared to 5 days or more had higher transfusion requirements (95% vs. 52%).²⁶ However, others have suggested this does not occur.²⁷ Ebrahimi and colleagues examined the effect of clopidogrel in patients with non-ST-segment elevation acute coronary syndromes (NSTE-ACS) requiring CABG in the Acute



• **Figure 4.1** Prevalence of composite outcome (reoperation or major bleeding) by number of days clopidogrel was stopped before surgery. Denominator for points on curve is the total number of patients who were exposed to clopidogrel ≤ 7 days before surgery ($n = 329$); 89 patients experienced the composite outcome. Red line = clopidogrel exposure. (From Berger JS, Frye CB, Harshaw Q, Edwards FH, Steinhilb SR, Becker RC. Impact of clopidogrel in patients with acute coronary syndromes requiring coronary artery bypass surgery: a multicenter analysis. *J Am Coll Cardiol*. 2008;52:1693-1701.)

Catheterization and Urgent Intervention Triage strategy (ACUITY) trial.²⁸ The trial enrolled 13,819 patients with NSTEMI/ACS undergoing early invasive management; 11.1% underwent CABG before discharge. Clopidogrel-exposed patients had a longer median duration of hospitalization than nonexposed patients but experienced fewer ischemic events within 30 days and had a similar occurrence of non-CABG-related major bleeding and post-CABG major bleeding. Multivariable analysis demonstrated that clopidogrel use before CABG was a predictor of reduced 30-day composite ischemia. Current ACCP guidelines suggest that for patients receiving a P2Y₁₂ inhibitor drug, clopidogrel should be stopped 5 days before, ticagrelor 3 to 5 days before, and prasugrel 7 days before the surgery (conditional recommendation, low certainty of evidence).¹⁹

For patients with coronary stents placed within the prior 3 to 12 months who are receiving dual therapy with a P2Y₁₂ inhibitor and aspirin, the P2Y₁₂ inhibitor should be held before elective surgery, and routine bridging therapy with a glycoprotein (GP) IIb/IIIa inhibitor, cangrelor, or low-molecular-weight heparin is not advised.

Combination Antiplatelet Therapy. Cannon reported that dual antiplatelet therapy within 5 days of CABG was associated with a moderate increase in bleeding; however, combined therapy conferred no appreciable risk for bleeding if discontinued more than 5 days before CABG.²⁹ ASA and clopidogrel taken together until 2 days before operation is associated with higher postoperative blood loss but not increased occurrence of reoperation for bleeding.³⁰ Others report increased blood product requirement with preoperative dual antiplatelet therapy.³¹ For patients undergoing off-pump CABG (OPCAB), Shim and colleagues reported that preoperative ASA and clopidogrel exposure, even within 2 days of operation, did not increase perioperative blood loss and blood transfusion requirements.³² Of note, the authors used strict transfusion guidelines and intraoperative blood salvage techniques. Current ACCP guidelines suggest that patients receiving ASA and a P2Y₁₂ inhibitor who had coronary stents placed within the last 3 to 12 months and are undergoing an elective surgery/procedure should stop the P2Y₁₂ inhibitor prior to surgery.

Glycoprotein IIb/IIIa Inhibitors. Both short- and long-acting GP IIb/IIIa inhibitors cause profound platelet dysfunction. In patients requiring an emergency operation, they are associated with an increased risk of bleeding. Recommendations for discontinuation prior to surgery vary, depending on whether the agent used is a short- or long-acting inhibitor (4 to 6 hours for short-acting; 12 to 24 hours for long-acting) (Table 4.1).²¹ Lee and colleagues recommend delaying surgery for the appropriate time interval and transfusing platelets as needed rather than prophylactically.³³ De Carlo and colleagues state that emergency surgery can be performed safely in patients treated with all GP IIb/IIIa inhibitors and similarly note that platelet transfusion should be reserved for clinically relevant bleeding.³⁴ Lincoff and colleagues report that urgent CABG can be performed for abciximab-treated patients without excess mortality or important morbidity.³⁵

Herbal Supplements. Herbal supplements are taken by one-third of Americans,^{36,37} frequently underreported, and often taken with conventional drugs that may have effects on their own or complex drug interactions.³⁸⁻⁴¹

Major adverse effects include increased bleeding.³⁷ Garlic, ginkgo biloba, and ginger possess antiplatelet activity that may increase bleeding risk, particularly in patients receiving other antiplatelet agents.^{37,38,40,42} Garlic and ginkgo inhibit platelet

TABLE 4.1 Platelet Inhibitors, Mechanism, and Half-Life

Platelet Inhibitor	Mechanism	Half-Life	Minimum Wait
Clopidogrel	ADP-platelet aggregation Irreversible	8 hours	5-7 days
Abciximab	GPIIb/IIIa Noncompetitive	30 minutes	12 hours
Tirofiban	GPIIb/IIIa Reversible	2.2 hours	4 hours
Eptifibatide	GPIIb/IIIa Reversible	2.5 hours	4 hours

ADP, Adenosine 5'-diphosphate; GP, glycoprotein.

Modified from Lee LY, DeBois W, Krieger KH, et al. The effects of platelet inhibitors on blood use in cardiac surgery. *Perfusion*. 2002;17:33-37.

activation and/or aggregation, and ginkgo should be discontinued at least 36 hours preoperatively.⁴⁰ Certain medications also contain warfarin derivatives (coumarin), including chamomile, horse chestnut, motherwort, and tamarind, that may enhance bleeding risk.^{38,43}

Prolongation of sedative effects may result from kava, valerian, and St. John's wort. Kava, an anxiolytic and sedative, may prolong benzodiazepine sedation secondary to its ability to potentiate central nervous system depressants.^{37,40,42} Concomitant use of opioids with valerian and kava may increase central nervous system depression. Valerian produces dose-dependent sedation, which appears to be mediated through modulation of γ -aminobutyric acid (GABA) neurotransmission.⁴⁰

Other effects of herbal medicines in the operative setting include hypertension from ephedra ("ma huang") and hypoglycemia from ginseng.⁴⁰ Ephedra contains alkaloids, including ephedrine and pseudoephedrine, that increase blood pressure and heart rate.⁴⁰ Ginseng may lower postprandial glucose, resulting in hypoglycemia, particularly in fasting patients.⁴⁰ Immunosuppressant properties of echinacea theoretically increase the risk for poor wound healing and infection.^{37,44,45} Short-term use of echinacea has immunostimulatory effects that may diminish the effectiveness of immunosuppressive medications in the perioperative period.⁴⁰

In general, current recommendations are for patients to discontinue herbal medicines at least 2 weeks before surgery (Tables 4.2 and 4.3).^{37,40,46}

Monitoring

Cannulae

In addition to standard American Society of Anesthesiologists monitoring, large-bore IV and brachial or radial arterial cannulae are placed prior to induction of anesthesia. Central venous access is commonly obtained following induction of anesthesia. Decisions about whether to use a central venous triple-lumen catheter versus a pulmonary artery flotation catheter are case and surgeon/anesthesiologist-specific, and current practice is the use of ultrasound guidance for placement.

TABLE 4.2 Commonly Used Herbal Supplements, Relevant Pharmacologic Effects, and Perioperative Considerations

Herbal Supplement	Relevant Pharmacologic Effects	Perioperative Considerations
Echinacea	Activation of cell-mediated immunity	Allergic reactions; decreased effectiveness of immunosuppressants; potential for immunosuppression with long-term use
Ephedra ("ma huang")	Increase in heart rate and blood pressure through direct and indirect sympathomimetic effects	Risk of myocardial infarction and stroke from tachycardia and hypertension; ventricular arrhythmias with halothane; long-term use depletes endogenous catecholamines and may cause intraoperative hemodynamic instability
Garlic	Inhibition of platelet aggregation (may be irreversible); increased fibrinolysis; equivocal antihypertensive activity	Potential to increase risk of bleeding, especially when combined with other medications that inhibit platelet aggregation
Ginkgo	Inhibition of platelet activating factor	Potential to increase risk for bleeding, especially when combined with other medications that inhibit platelet aggregation
Ginseng	Lowers blood glucose; inhibition of platelet aggregation (may be irreversible); increased PT and PTT in animals	Hypoglycemia; potential to increase risk of bleeding
Kava	Sedation; anxiolysis	Potential to increase sedative effect of anesthetics; potential for addiction, tolerance, and withdrawal after abstinence unstudied
St. John's wort	Inhibition of neurotransmitter reuptake	Induction of cytochrome P450 enzymes, affecting cyclosporine, warfarin, steroids, protease inhibitors, possibly benzodiazepines; decreased serum digoxin levels
Valerian	Sedation	Potential to increase sedative effect of anesthetics; benzodiazepine-like acute withdrawal; potential to increase anesthetic requirements with long-term use

PT, Prothrombin time; PTT, partial thromboplastin time.

Modified from Ang-Lee MK, Moss J, Yuan CS. Herbal medicines and perioperative care. *JAMA*. 2001;286:208-216.

TABLE 4.3 Herbal Supplements and Interactions with Cardiovascular Medicines

Conventional Medicine	Herbal Supplement	Result of Interaction	Possible Mechanism of Interaction
Interaction with: Digoxin	Guar gum	Decreased plasma digoxin levels	Reduced absorption; guar gum reduces gastric emptying, which results in transient delayed digoxin absorption
	St. John's wort	Decreased plasma digoxin concentration	Induction of P-glycoprotein; digoxin is a substrate of P-glycoprotein, which is induced by St. John's wort
	Siberian ginseng	Increased plasma digoxin levels	Some component of Siberian ginseng might impair digoxin elimination or interfere with digoxin assay
	Wheat bran	Decreased plasma digoxin levels	Reduced absorption; bran contains fibers that can trap digoxin
Interactions with: Antihypertensive drugs	Ginkgo	Increased blood pressure	Unknown
	Licorice	Hypokalemia	Additive effect on potassium excretion; licorice has mineralocorticoid effects that may cause potassium excretion
Interactions with: Antiplatelet drugs	Ginkgo	Spontaneous hyphema	Additive inhibition of platelet aggregation; ginkgolides have antiplatelet activities and are platelet activating factor receptor antagonists
Interactions with: Anticoagulants	Boldo/fenugreek	Increased anticoagulant effect	Additive effect on coagulation mechanism; boldo and fenugreek contain anticoagulant coumarin
	Devil's claw	Increased anticoagulant effect, purpura	Unknown
	Garlic	Increased anticoagulant effect; increased clotting time	Additive effect on coagulation mechanisms; garlic has antiplatelet activity

TABLE 4.3 Herbal Supplements and Interactions with Cardiovascular Medicines—cont'd

Conventional Medicine	Herbal Supplement	Result of Interaction	Possible Mechanism of Interaction
Interactions with: Antilipidemic drugs	Ginkgo	Reports of intracerebral hemorrhage	Additive effect on coagulation mechanism; ginkgolides from ginkgo have antiplatelet activity and are platelet activating factor receptor antagonists
	Green tea	Decreased anticoagulant effect	Pharmacologic antagonism; warfarin produces anticoagulation by inhibiting production of vitamin K–dependent clotting factors; green tea contains vitamin K and thus antagonizes effect of warfarin
	St. John's wort	Decreased anticoagulant effect	Hepatic enzyme induction; warfarin is metabolized by CYP1A2 in the liver, which is induced by St. John's wort
	Simvastatin	Decreased plasma levels simvastatin concentration	Hepatic enzyme induction; simvastatin is extensively metabolized by CYP3A4 in the intestinal wall and liver, which is induced by St. John's wort
	Lovastatin	Decreased lovastatin absorption	Decreased absorption of lovastatin resulted in an increase in LDL levels that led to abortion of trial. Lovastatin pharmacokinetics and LDL returned to normal after bran discontinuation

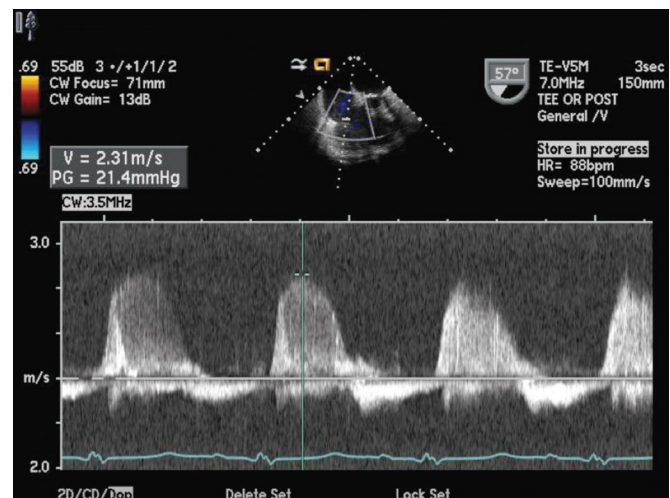
Modified from Izzo AA, Di Carlo G, Borrelli F, Ernst E. Cardiovascular pharmacotherapy and herbal medicines: the risk of drug interaction. *Int J Cardiol*. 2005;98:1-14.

Transesophageal Echocardiography

As discussed in Chapter 6, transesophageal echocardiography (TEE) has become a mainstay of monitoring cardiovascular surgical patients to evaluate cardiac pathology, facilitate adequacy of surgical repair, assist with separation from CPB, and rapidly evaluate acute cardiac dysfunction.⁴⁷⁻⁵² In 12,566 consecutive cardiac surgical patients, Eltzschig found that TEE influenced surgical decision-making in 7% of patients pre-CPB and 2.2% post-CPB.⁴⁷ Minhaj reported that TEE demonstrated new cardiac pathology in one-third of patients and influenced decisions regarding the use of CPB.⁵³ Qaddoura evaluated TEE in primary CABG and noted it provided new findings pre- and post-CPB in 13% of patients.⁵⁴ Leung in early intraoperative TEE noted an association between new and persistent post-CPB wall motion abnormalities and increased risk for postoperative mortality and myocardial infarction (MI).⁵⁵

Doppler-derived hemodynamic indices by TEE can provide noninvasive quantitative information on intracardiac velocities, pressure gradients, and valve area.^{50,56} Estimates of forward flow (stroke volume and cardiac output) provide useful information, particularly in cases where a pulmonary artery catheter is not used (Fig. 4.2). Pulmonary artery systolic pressure can be estimated from tricuspid valve regurgitant velocity using the *modified Bernoulli equation* (Box 4.1), which converts instantaneous velocities to pressure gradients.⁵⁰ Stroke volume can be calculated as the product of the cross-sectional area and time velocity integral using two-dimensional and Doppler measurements.⁵⁰ Fig. 4.3 and Box 4.2 illustrate TEE-derived noninvasive calculation of the aortic valve area.

Intraoperative TEE is particularly important for patients undergoing mitral valve repair to examine residual or new valvular regurgitation. Whether to intervene for residual mitral regurgitation following repair is controversial. Gillham suggested not returning to CPB for a second attempt at mitral valve repair for mild mitral regurgitation based on TEE findings alone.⁵⁷ They reported that 61% of patients with mild mitral regurgitation



• **Figure 4.2** Transesophageal echocardiographic image depicting peak tricuspid regurgitant velocity in $\text{m} \cdot \text{s}^{-1}$. CW, Continuous wave; HR, heart rate; PG, pressure gradient; V, velocity.

• BOX 4.1 Modified Bernoulli Equation for Calculating Intracardiac Pressure, and an Example

$$\Delta P = 4V^2$$

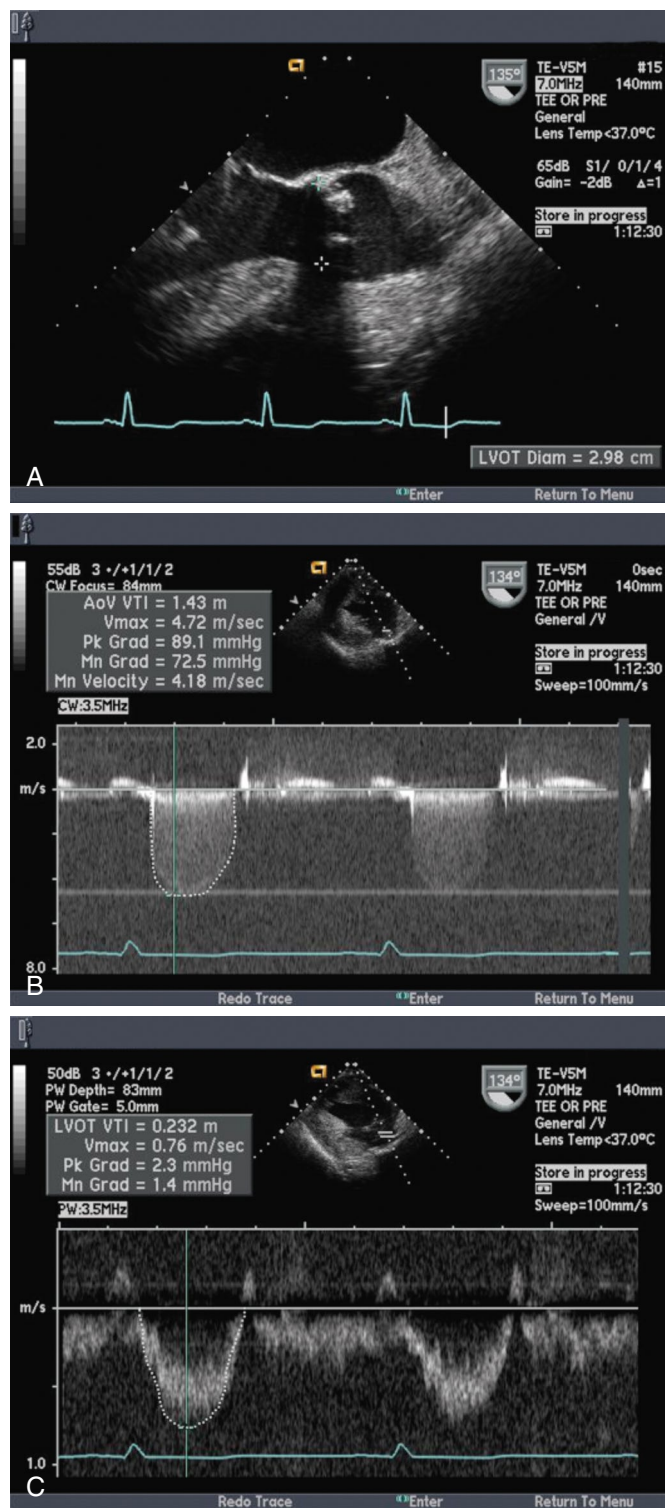
$$\text{PPA (systolic) pressure} = 4V_{\text{TR}}^2 + \text{PRA}$$

Example:

$$P = 4(2.3 \text{ m} \cdot \text{s}^{-1})^2 + 10 \text{ mmHg}$$

$$\text{PPA (systolic) pressure} = 31 \text{ mmHg}$$

ΔP , Change in pressure; P , pressure; PPA, pulmonary artery pressure; PRA, right atrial pressure, estimated = 10 mmHg; TR, tricuspid regurgitation; V, velocity.



• **Figure 4.3** Use of TEE for noninvasive calculation of aortic valve area. **A**, Midesophageal imaging plane for left ventricular outflow tract diameter measurement (2.98 cm). **B**, Continuous-wave Doppler velocity time integral (VTI) of aortic valve (1.43 m). **C**, Pulsed-wave Doppler of left ventricular outflow tract (0.232 m).

identified by intraoperative TEE had zero to trace regurgitation at follow-up transthoracic echocardiography. However, other reports note a trend toward an increased need for reoperation when TEE identifies residual mitral regurgitation following mitral valvuloplasty.⁵⁸ TEE is critical in assessing and detecting patients at risk

• **BOX 4.2** Use of Transesophageal Echocardiography for Noninvasive Calculation of Aortic Valve Area, and an Example

Continuity Equation = $CSA_{(LVOT)} \times TVI_{(LVOT)}$
= $CSA_{(aorticvalve)} \times TVI_{(aorticvalve)}$

$CSA_{(aorticvalve)} = CSA_{(LVOT)} \times TVI_{(LVOT)} / TVI_{(aorticvalve)}$

Example:

= $0.785 (2.98)^2 \times (23 \text{ cm}) / (143 \text{ cm})$
= $7.06 \times 23 / 143$
Aorticvalve area = 1.13 cm^2

LVOT, left ventricular outflow tract; TVI, time velocity integral.
Modified from Oh JS, Seward JB, Tajik AJ. Hemodynamic assessment. In: *The Echo Manual*. Philadelphia: Lippincott William & Wilkins; 1999:59-71.

TABLE 4.4 Echocardiographic Factors Related to Poor Outcome after Mitral Valve Repair for Degenerative Mitral Regurgitation

Pre-CPB	Post-CPB
Posterior leaflet height (>15 mm)	Residual mitral regurgitation > mild
Anterior leaflet height (>45 mm)	Persistent prolapse
Anterior leaflet to posterior leaflet > 1.5	Increased mean MV pressure gradient
Coaptation point of MV leaflets to septum (C-septal) distance < 15 mm	
Bileaflet prolapse	

MV, mitral valve.
Modified from Iglesias I. Intraoperative TEE assessment during mitral valve repair for degenerative and ischemic mitral valve regurgitation. *Semin Cardiothorac Vasc Anesth*. 2007;11:301-305.

for left ventricular outflow obstruction secondary to systolic anterior motion of the anterior mitral valve following mitral valve repair (Table 4.4).⁶³ Advances in ultrasound technology with real-time three-dimensional TEE increase the ability to image and add to the utility of intraoperative echocardiography in cardiac surgery.⁵⁹⁻⁶²

Epiaortic Scanning

Epiaortic scanning to detect atherosclerotic lesions in the ascending aorta can be an important aid in surgical decision-making, particularly in regard to arterial cannulation.^{64,65} Djaiani and colleagues modified their surgical management in one-third of patients based on its results in CABG.⁶⁶ Similarly, Rosenberger and colleagues reported that it changed surgical decision-making in 4.1% of patients, including using cardiac arrest, performing aortic atherectomy or replacement, using off-pump support, avoiding aortic clamping, using ventricular fibrillatory arrest, changing arterial cannulation site, and avoiding aortic cannulation.⁶⁴

Cerebral Oximetry

The use of noninvasive measures of regional cerebral oxygen saturation in adult cardiac surgical patients is controversial because there are conflicting data regarding the ability of cerebral oximetry to predict outcomes following cardiac surgery. Hong and colleagues reported cerebral oximetry was not predictive of cognitive decline following heart surgery; however, patients who exhibited intraoperative desaturation required longer postoperative hospitalization.⁶⁷ Similarly, Reents reported that cerebral oximetry was not predictive of postoperative cognitive performance.⁶⁸ In contrast, Slater reported that intraoperative cerebral oxygen desaturation was associated with an increased risk for cognitive decline and prolonged hospital stay after CABG.⁶⁹ More data are needed to demonstrate whether complications associated with modifications in patient care are reduced by noninvasive monitoring of regional cerebral oxygenation.

Medications

Premedication

With most patients being admitted the same day as their surgical procedure, preoperative medications for anxiolysis are commonly administered in the preoperative holding area or operating room (OR) once the IV cannula is placed. A short-acting sedative such as midazolam is administered in doses of 1 to 2 mg and is preferred over longer-acting agents.

Induction Agents

Propofol and etomidate are commonly used induction agents in combination with a low-dose opioid and muscle relaxant, with the goal of facilitated recovery (Table 4.5). Propofol has several properties that make it an advantageous induction agent, particularly for procedures of short duration. It enables more rapid awakening compared with other induction drugs, and its antiemetic effects minimize postoperative nausea.⁷⁰ Etomidate possesses minimal cardiovascular side effects, making it an ideal agent for induction in hemodynamically unstable patients and those with impaired ventricular function. However, it has several side effects, including pain on injection and transient adrenocortical suppression through inhibition of 11- β -hydroxylase.⁷⁰ Use of etomidate, particularly in the critical care setting, is controversial

because of its adrenal suppression side effects.⁷¹ Ketamine (a phencyclidine derivative) is less commonly used as an induction agent, primarily because of cardiovascular side effects, including increases in heart rate and blood pressure, myocardial depression, and episodic emergence delirium.⁷⁰ Thiopental is no longer available in the United States.

Maintenance of Anesthesia

The goal of maintenance anesthesia is to maintain stable hemodynamics while facilitating recovery in most patients. Maintenance of general anesthesia typically consists of a balanced anesthetic technique employing low-dose opioid in combination with a volatile inhalational anesthetic agent or IV agent. The choice of opioid, inhalational agent, and muscle relaxant depends on the surgical procedure and patient hemodynamics.

No specific anesthetic maintenance regimen is more advantageous than another in patient outcomes, but evidence supports the potential role of inhalational anesthetic agents in myocardial preconditioning.⁷² In a double-blind, randomized controlled trial of patients undergoing CABG, Meco reported beneficial preconditioning effects of desflurane on myocardial injury (lower troponin I) and myocardial functional recovery following surgery.⁷²

Regional anesthesia, with or without a general anesthetic, has produced mixed patient outcomes. There have been reports of regional anesthetic techniques using epidural blockade in off- and on-pump cardiac surgical patients⁷³⁻⁷⁵; however, regional anesthetic techniques have not been generally adopted.

Opioids

Semisynthetic opioids—fentanyl, sufentanil, and remifentanyl—are differentiated by potency, onset, and duration of action. All have demonstrated safety and effectiveness for use in cardiac surgery (Table 4.6).^{76,77} Cheng conducted a multicenter randomized controlled trial on the efficacy and resource utilization of remifentanyl and fentanyl in fast-track recovery from cardiac surgery.⁷⁸ Both anesthetic techniques permitted early and similar times to tracheal extubation, less intense monitoring, and reduced resource utilization after CABG. Similarly, Howie compared remifentanyl to fentanyl combined with isoflurane/propofol for early extubation following CABG.⁷⁹ Both allowed for fast-track cardiac anesthesia. In a randomized clinical trial (RCT), Mollhoff demonstrated the efficacy and safety of remifentanyl and fentanyl for

TABLE 4.5 Induction Agents

Drug	Dosage	Mechanism of Action	Systolic Blood Pressure Response	Heart Rate Response
Propofol	1.5-2.5 mg · kg ⁻¹ IV	Interaction with GABA	Decreased	Decreased
Etomidate	0.2-0.4 mg · kg ⁻¹ IV	Interaction with GABA	No change to decreased	No change
Ketamine	1-2 mg · kg ⁻¹ IV	Interaction with NMDA, opioid, monoaminergic, muscarinic receptors and voltage-sensitive calcium channels	Increased	Increased
Thiopental	3-5 mg · kg ⁻¹ IV	Interaction with GABA	Decreased	Increased

GABA, γ -Aminobutyric acid; IV, intravenous; NMDA, N-methyl-D-aspartate.

Modified from Stoelting R. *Pharmacology and Physiology in Anesthetic Practice*. Philadelphia: Lippincott Williams & Wilkins; 1999, p. 141.

TABLE 4.6 Commonly Used Opioids

Opioid	Elimination Half-Time (hours)	Effect-Site (Blood-Brain Equilibration [minutes])	Analgesic Potency
Fentanyl	3.1-6.6	6.8	75-125 times more potent than morphine
Sufentanil	2.2-4.6	6.2	5-10 times more potent than fentanyl
Alfentanil	1.4-1.5	1.4	1/10-1/5 as potent as fentanyl
Remifentanyl	0.17-0.33	1.1	Similar potency to fentanyl

Modified from Stoelting R. *Pharmacology and Physiology in Anesthetic Practice*. Philadelphia: Lippincott Williams & Wilkins; 1999, p. 83.

fast-track CABG.⁸⁰ Time to extubation was longer, and the occurrences of shivering and hypertension were higher in the remifentanyl group. However, the groups had similar intensive care unit (ICU) and hospital lengths of stay. Engoren compared three opioids used for fast-track cardiac anesthesia: fentanyl, sufentanil, and remifentanyl.⁸¹ Extubation times and costs were equivalent. Shorter duration of action of remifentanyl allowed for faster recovery, but it is more expensive than fentanyl, and tracheal extubation times were similar.⁸²

Antifibrinolytic Drugs

Antifibrinolytic therapy is a critical pharmacologic management strategy for cardiac surgical patients. Tissue injury and blood interfacing with extracorporeal life support (ECLS) structures, especially during CPB, are major contributors to bleeding and coagulopathy.^{83,84} Tranexamic acid is the most commonly used antifibrinolytic drug worldwide. It is a lysine analog, similar to epsilon aminocaproic acid. These drugs inhibit fibrinolysis by binding to plasminogen at the lysine binding site to displace the binding of plasminogen to fibrin clot and prevent binding to fibrinogen.⁸⁴ Aprotinin, another fibrinolytic inhibitor, is a broad-spectrum protease inhibitor isolated from cow lung that directly inhibits plasmin and other serine proteases but is available only in certain countries.⁸⁴

Antifibrinolytics are clot stabilizers; however, despite the extensive use of epsilon aminocaproic acid in the United States, there is no evidence the drug consistently reduces bleeding in cardiac surgical patients.⁸⁴ The safety and efficacy of tranexamic acid for reducing bleeding in cardiac surgery have been demonstrated in multiple clinical trials, including a recent large series from Australia.⁸⁵ Although effective in reducing bleeding, there is a small increase in the risk of seizures with tranexamic acid, especially at higher doses.⁸⁶ Tranexamic acid, a molecule similar to GABA, is thought to potentially inhibit GABA receptors in the brain, although other factors are likely involved that are specific to cardiac surgery, including open chamber procedures with risk for air emboli.^{83,84}

Heparin Management

Monitoring. Several automated devices are available for anticoagulation management following heparin administration; the

activated clotting time (ACT) monitor is the most common. Unfractionated heparin (UFH) is commonly administered on a weight-based protocol (300-500 units per kilogram) with the goal of achieving an ACT of 480 or more seconds prior to initiation of CPB.^{87,88} In general, if an ACT of 480 or greater has not been obtained with doses of heparin (up to 500-600 units · kg⁻¹), one should suspect heparin resistance.⁸⁹

Heparin Resistance. Heparin resistance is generally suspected when excessive doses of heparin are required to produce the desired anticoagulation. Reports suggest that heparin resistance in cardiac surgical patients requiring CPB occurs in 4% to 22% of patients.^{90,91} Preoperative risk factors include preprocedural heparin administration, thrombocytosis, and hyperfibrinogenemia.

One of the issues regarding heparin resistance is the actual definition. An early important report in cardiac surgery suggested that a need for more than 500 units/kg was consistent with heparin resistance. Defining heparin resistance is further complicated by the lack of knowledge defining an ideal ACT required for CPB. The number of 480 seconds has been repeated in many textbooks and guidance reports and has become a standard threshold value, although earlier studies suggest 350 seconds may provide satisfactory anticoagulation for CPB. Of note is that a target level of 2 to 4 units per milliliter of heparin is another potential goal to consider for coagulation monitoring.⁹²

Treatment options for heparin resistance include antithrombin (AT) III therapy or, potentially, the use of fresh frozen plasma (FFP). In most clinical situations with time constraints of preparation for CPB, use of FFP is impractical.¹⁷ Several clinical trials that reported AT III is effective in restoring heparin responsiveness for most patients exhibiting heparin resistance prior to CPB defined heparin resistance as an ACT < 480 seconds after 400 units · kg⁻¹ of heparin.¹⁷

Heparin-Induced Thrombocytopenia. Heparin-induced thrombocytopenia (HIT) is an immune-mediated prothrombotic response caused by IgG antibodies that develop to platelet factor 4 (PF4), a polypeptide stored in platelet alpha granules.⁹³ Following heparin exposure, the IgG antibodies bind to platelet epitopes, causing aggregation and further activation that release highly prothrombotic microparticles. Following cardiac surgery and CPB, up to 50% of patients form HIT antibodies.⁹³ Due to this unique immunologic response, IgG antibodies that produce HIT last approximately 3 months.⁹³ Thus, patients with a previous history of HIT may not be at increased risk for subsequent antibody formation as this immunologic response is not a true amnestic response but rather a transient event due to platelet activation.

Despite the high incidence of antibody formation, clinically apparent HIT occurs in 1% to 5% following UFH.⁹³ The initial assay to determine whether a patient has HIT is the enzyme-linked immunoassay (ELISA). However, the assay has a high sensitivity but less specificity. Results return as optical density. The definitive test is a functional assay, usually a serotonin release assay.⁹³ Warkentin noted that patients with higher optical densities were at greater risk for true HIT, as documented by definitive testing with serotonin release assays.⁹⁴

HIT usually presents with a 50% decrease in platelet count or a thrombotic event after heparin exposure; however, heparin is a ubiquitous drug in cardiovascular medicine, and prior recent exposure may not be known.⁹³ ELISA should be used for clinical decision-making for intraoperative anticoagulation. If cardiac surgery cannot be delayed until HIT assays are negative, other techniques are currently used. Alternative

TABLE 4.7 Alternative Anticoagulants in Patients with Heparin-Induced Thrombocytopenia Requiring Anticoagulation for Cardiopulmonary Bypass

Drug	Half-Life	Reversal	Metabolism	Monitoring	Dosing
Bivalirudin	25 minutes	None	Metabolic > renal	ACT, ECT	1.5 mg · kg ⁻¹ , 50 mg in pump, 2.5 mg · kg ⁻¹ · h ⁻¹ infusion
Lepirudin	80 minutes	None	Renal	PTT, ECT	0.25 mg · kg ⁻¹ , 0.2 mg · kg ⁻¹ in pump prime, 0.5 mg · min ⁻¹ infusion
Argatroban	30 minutes	None	Hepatic > renal	PTT, ACT	0.1 mg · kg ⁻¹ bolus, 5–10 µg · kg ⁻¹ · min ⁻¹ infusion
Danaparoid	20 hours	None	Renal	Factor Xa levels	125 units · kg ⁻¹ , 3 units · kg ⁻¹ in pump prime, 7 units · kg ⁻¹ · h ⁻¹ infusion

ECT, ecarin clotting time.
Data from Despotis.⁹⁷

anticoagulation includes bivalirudin, a direct thrombin inhibitor, which is the agent most extensively studied for on-and-off pump cardiac surgery. Alternatively, plasmapheresis to decrease the IgG titer is another strategy using heparin for cardiac surgery and then postoperatively switching to a direct thrombin inhibitor such as bivalirudin or argatroban (Table 4.7).^{95,96}

Heparin Reversal. Protamine is the primary agent for heparin reversal. Protamine, a highly basic polypeptide, forms a nonspecific acid-base, polyionic-polycationic interaction with heparin that neutralizes its anticoagulant effect. Indeed, mixing the two drugs *ex vivo* forms a precipitate, and the protamine heparin interaction may be associated with hypersensitivity reactions, including anaphylaxis.⁹⁸ Many cardiac surgical centers reverse on an empirical basis, administering typically 1 mg of protamine for every 100 units of heparin. However, this routinely administered far too much protamine, and excess protamine can contribute to coagulopathy post-bypass. The optimal management for protamine dosing for reversal is using a point-of-care titration to determine the exact level required. Because the half-life of heparin is approximately 1 hour, protamine dosing, administering 0.5 mg to 0.7 mg of protamine for every 100 units of the initial heparin dose for CPB, is another potential strategy.^{97,99}

Weaning from Cardiopulmonary Bypass (see Chapter 2)

A number of factors require attention before CPB separation: achieving normothermia; proper electrolyte balance (potassium, glucose, ionized calcium); adequate hematocrit; the anticipation of inotropic and vasopressor support; reestablishing ventilatory support; heart rate and rhythm; and need for pacing. TEE plays an integral role in weaning from CPB, particularly for patients who have undergone valve repair or compromised ventricular function. TEE permits rapid recognition in the partial bypass period of circumstances that could complicate separation from CPB, such as persistent valvar regurgitation, intracardiac air, or regional wall motion abnormalities related to graft failure.

Inotropic and Vasopressor Support

Factors related to the need for inotropic support following CPB include preoperative ventricular dysfunction, advanced New York

Heart Association functional class, and prolonged aortic clamp time.¹⁰⁰ TEE examination and understanding clinical risk factors can predict inotropic requirements.¹⁰¹ Ahmed and colleagues evaluated laboratory and hemodynamic factors and identified low cardiac index (CI), left ventricular end-diastolic pressure 20 mmHg or higher, left ventricular ejection fraction 40% or less, and chronic kidney disease stage 3 to 5 as factors associated with the need for inotropic support at the time of separation from CPB.¹⁰²

Table 4.8 lists the mechanism of action for commonly used sympathomimetic and vasopressor agents. Milrinone, an inodilator, inhibits phosphodiesterase III to increase intracellular cyclic adenosine monophosphate (cAMP) levels in both the myocardium to increase contractility and in the vasculature to produce vasodilation in the pulmonary and systemic circulation. Levosimendan, a calcium sensitizer inodilator, induces a calcium-dependent conformational change of troponin C and enhances both rate and extent of cardiac contraction.^{103–105} Levosimendan has demonstrated utility in facilitating weaning from CPB.¹⁰³

Specific Management Issues

Fast-Track Anesthesia

Early work by Cheng demonstrated the safety and feasibility of facilitated recovery from cardiac surgery¹⁰⁶ evaluating extubation within 1 to 6 hours versus conventional tracheal extubation within 12 to 22 hours after CABG in an RCT. Compared to patients receiving conventional care, patients having early extubation had improved postextubation intrapulmonary shunt fraction and shorter ICU and hospital lengths of stay without additional morbidity.¹⁰⁶ Others also noted that fast-track and ultra-fast-track (extubated on ICU arrival) anesthesia were not associated with increased patient morbidity and mortality following CABG.¹⁰⁷

Perioperative Glucose Control

Poor perioperative glucose control, defined as four consecutive blood glucose concentrations > 200 mg · dL⁻¹ despite insulin therapy, is associated with a sevenfold increase in morbidity,¹⁰⁸ and perioperative hyperglycemia is a risk factor for morbidity.¹⁰⁹ Although intensive insulin therapy to maintain blood glucose of 110 mg · dL⁻¹ or lower appears to reduce morbidity and mortality in surgical ICU patients,¹¹⁰ there are conflicting results about

TABLE 4.8 Sympathomimetics

Sympathomimetics	α -Receptor	β_1 -Receptor	β_2 -Receptor	Mechanism of Action	Cardiac Output	Heart Rate	Arrhythmias	Pao	Peripheral Vascular Resistance	Renal Blood Flow
Natural Catecholamine					++					
Epinephrine	+	++	++	Direct	++	++	+++	+	+/-	-
Norepinephrine	+++	++	0	Direct	-	-	+	+++	+++	-
Dopamine	++	++	+	Direct	+++	+	+	+	+	+++
Synthetic Catecholamine										
Isoproterenol	0	+++	+++		+++	+++	+++	+/-	- -	-
Dobutamine	0	+++	0		+++	+	+/-	+	NC	++
Synthetic Noncatecholamine										
Indirect acting: Ephedrine	++	+	+	Indirect, some direct	++	++	++	++	+	- -
Direct acting: Phenylephrine	+++	0	0	Direct	-	-	NC	+++	+++	- - -

NC, No change; Pao, mean arterial pressure.
 Modified from Stoelting R. *Pharmacology and Physiology in Anesthetic Practice*. Philadelphia: Lippincott Williams & Wilkins; 1999, p. 260.

whether intensive insulin therapy to normalize glucose perioperatively improves outcomes. One randomized study of the impact of intensive intraoperative insulin therapy found that lowering glucose concentrations to near normal levels intraoperatively by IV insulin infusion did not reduce short-term death, morbidity, or length of stay (LOS) in the ICU or hospital. Rather, the investigators found an increased incidence of death and stroke in the intensive treatment group, raising concern about the routine implementation of this intervention.¹¹¹ At the Cleveland Clinic, the target intraoperative glucose range is 70 to 150 mg · dL⁻¹, and the intraoperative insulin management protocol and adjustment schedule are described in [Appendix Table 4A-1](#).

Blood Management

Patient blood management (PBM) protocols reduce allogeneic blood exposure through strategies based on specific transfusion thresholds and algorithms. However, transfusion practices vary considerably despite published guidelines.^{85,112,113} However, most PBM efforts focus on red blood cell (RBC) reductions using restrictive targeted hemoglobin.¹¹⁴ The TRICS II and TRICS III studies of RBC transfusions in higher-risk cardiac surgical patients randomized 5243 to hemoglobin levels of <7.5 g/dL versus <9.5 g/dL using a composite primary outcome of adverse events that included mortality, MI, stroke, or renal failure up to 28 days or discharge.^{115,116} The study had CABG and/or valve surgery with ~12% reoperations and ~91% of patients receiving tranexamic acid. In this noninferiority study, there was no difference in the composite primary outcome, which occurred in 11.4% of patients in the restrictive group versus 12.5% in the higher threshold. RBCs were transfused in 52.3% compared with 72.6% of the two groups. The follow-up study of

these patients (TRICS-III) 6 months postoperatively reported no difference in the composite outcome in the two groups, 17.4% versus 17.1%, or mortality of 6.2% versus 6.4%. However, patients undergoing heart transplantation or mechanical circulatory support/ventricular assist device (VAD) insertion were excluded.¹¹³

Unfortunately, these studies cannot readily measure specific oxygen delivery or oxygen debt and, as a result, be used to treat specific hemoglobin values, especially in the critically ill patient population. Further, postoperatively, patients are also lying in bed, physically inactive, and remain relatively immobile. Clinical trials focusing on RBC transfusions increase hemoglobin levels, but oxygen delivery to tissues may not be increased due to multiple factors.¹¹ As noted from an NHLBI Transfusion Medicine Symposium, decisions to transfuse RBCs should be optimally based on how anemia contributes to tissue O₂ delivery as other physiologic thresholds are needed, especially in critically ill patients, including those not studied in these clinical trials.¹¹³

As previously mentioned, most PBMs focus on RBCs, not other important hemostatic factors. For example, platelet transfusions are commonly administered empirically. Established algorithms are generally based on consensus as high-quality evidence to determine the appropriateness of platelet transfusions in bleeding patients is not known.¹¹³ Most algorithms to guide therapeutic approaches for bleeding in cardiac surgical patients use platelet counts as the laboratory determinant for platelet transfusions. Viscoelastic testing using thromboelastography (TEG), rotational thromboelastometry (ROTEM) and newer techniques provides insight into whole blood clotting.

Although transfusions are reported to be associated with excess risk of multiorgan complications, prolonged ventilatory support, and mortality, most supportive studies are retrospective and

observational.^{117,118} Sicker, higher-risk patients often require more blood transfusion and may have infectious complications such as sternal wound infections.¹¹⁸ Although immunomodulatory effects known as *transfusion-associated immunomodulation* (TRIM) are reported, again, this is often in more critically ill patients to start.¹¹⁹

Structural and functional changes occur during RBC storage, and it has been suggested that older blood contributes to complications associated with transfusion.¹²⁰ Many of these changes are time-dependent and include decreases in pH, 2,3-diphosphoglycerate, and adenosine triphosphate (ATP), and increases in free hemoglobin, potassium, and lactate.¹²¹ Reduced deformability is also time-dependent and begins at 2 weeks and progresses throughout the storage period.¹²² In laboratory studies, fresh blood transfusion is more effective in relieving the effects of microcirculatory hypoxia compared to banked blood, and in particular, 2-week stored blood was found to have a limited capacity for improving tissue oxygenation.¹²³ In another laboratory investigation, Sweeney and colleagues found increased thrombin generation for RBC products with longer storage duration.¹²⁴ In the centrifuged supernatant of stored blood, some RBC microvesicles express phosphatidylserine that is capable of facilitating thrombin generation. This may be a mechanism for adverse thrombotic effects of RBC with increased storage duration that has been observed in clinical investigations.

Despite the concerns regarding complications related to the length of RBC storage, a blinded randomized study of the effects of red-cell storage duration on patients undergoing cardiac surgery reported that the duration of red-cell storage was not associated with significant differences in the change in organ dysfunction.¹²⁵ Further, evaluating multiple biomarkers, the investigators also noted the transfusion of fresh versus aged RBCs does not result in substantial changes in hemostasis, immune, or nitric oxide parameters.¹²⁶

Although transfusion is beneficial in subsets of patients, there is uncertainty about whom to transfuse and when. Many clinicians have called for closer scrutiny of unrestricted liberal use of RBC transfusion. Hebert and colleagues, in the Transfusion Requirements in Critical Care trial, reported that restrictive RBC transfusion strategies were as effective as liberal ones in a general ICU population.¹²⁷ Blood conservation methods during and after cardiac surgery should be more widely applied and instituted, including optimization of preoperative hematocrit, use of intraoperative cell salvage, lower hemoglobin thresholds for transfusion, and use of antifibrinolytic agents. Evidence suggests that centers that have implemented evidence-based transfusion guidelines reduce use of RBC products without increasing patient morbidity, and specific transfusion protocols with proposed order sets for cardiac surgical patients are available.¹²⁸

Reoperation

Principles of anesthetic induction, maintenance, and facilitated recovery are similar for patients undergoing reoperation (see Chapter 5). Because of an increased risk for complications on sternal reentry, patients typically have additional IV access and RBC immediately available prior to sternotomy. Communication with the surgical team is essential before sternal reentry, particularly if alternative cannulation techniques (e.g., femoral or axillary artery cannulation) will be used. Surgical dissection can be challenging and lengthy, with a risk for excess bleeding, ischemia resulting from manipulation of prior grafts, and arrhythmias caused by the positioning of the heart.

Off-Pump Coronary Artery Bypass Grafting

Revascularization without CPB has several reported benefits over on-pump surgery: fewer transfusions, less early neurocognitive dysfunction, and less renal insufficiency (see Chapter 9).¹²⁹ Goals for anesthetic management are similar to those for on-pump surgery; however, several special considerations pertain to case management for off-pump coronary artery bypass (OPCAB). Invasive monitoring is similar, as is facilitated recovery with extubation in the OR or soon after in the ICU. Heparin management is surgeon- and institution-specific, ranging from partial-to full-dose heparinization. TEE is particularly useful for detecting regional wall motion abnormalities, accessing volume status, and determining the effect of lifting and retracting the heart and stabilizer devices on hemodynamics. OPCAB can be particularly challenging in managing acute hemodynamic changes that occur with necessary positioning and stabilization of the heart to perform coronary artery anastomosis.

TEE monitoring for OPCAB typically identifies transient regional wall-motion abnormalities during vessel occlusion.^{130,131} Temporizing measures to maintain hemodynamic stability include the Trendelenburg position, vasopressors, and IV fluids. Couture and colleagues examined mechanisms of hemodynamic changes during OPCAB, noting that mobilization and stabilization of the heart or myocardial ischemia can produce important changes in hemodynamics.¹³² They report that suction- and compression-type stabilizers produce hemodynamic changes via different mechanisms. For suction-type stabilizers, hemodynamic changes are due to heart dislocation (90-degree anterior displacement) and right ventricular (RV) compression. Compression-type stabilizers compress the left ventricular outflow tract and produce abnormal diastolic expansion secondary to direct deformation of the left ventricular geometry. Depending on collateral flow, coronary occlusion during the anastomosis can cause left ventricular dysfunction.¹³² Bainbridge and Cheng highlighted anesthetic considerations for less invasive direct CABG and OPCAB procedures, noting goals similar to those for conventional CABG but with some key differences, including the use of regional techniques for postoperative pain (unilateral paravertebral blocks or intercostal blocks).¹³³

Heart Transplantation

The approach to anesthetic management for heart transplant recipients follows principles similar to those for heart failure patients, recognizing, however, that transplant recipients often have ingested food or drink within hours of surgery. Potential heart transplant recipients typically have IV and arterial access prior to rapid-sequence or modified rapid-sequence anesthetic induction. Although alternative induction agents may be used successfully, etomidate and succinylcholine with small doses of narcotic are effective agents for this, recognizing that circulation time is longer and important hemodynamic instability secondary to reduced ejection fraction may occur. Central venous access is obtained following anesthetic induction. Maintenance of anesthesia is similar to that for other cardiovascular procedures, using a balanced technique of opioids, muscle relaxation, and low-dose inhalation agents. Similar to other procedures, anesthetic goals are to maintain hemodynamic stability and facilitate recovery following surgery.

The newly implanted heart is denervated without vagal tone. In general, postimplantation, agents with direct-acting catecholamines are preferred; agents with indirect activity (e.g., ephedrine)

may have a diminished effect. Similarly, medications such as atropine and glycopyrrolate will not provide typical heart rate responses.^{134,135}

TEE has particular utility in heart transplantation as a guide for separating the patient from CPB, detecting the presence of intracardiac air, and evaluating ventricular function. It is particularly useful for detecting RV dysfunction, especially in patients with pulmonary hypertension.

Lung Transplantation

Similar to heart transplant recipients, lung transplant recipients typically have eaten within hours of surgery. A special consideration for anesthetic induction of lung transplant recipients is avoiding prolonged positive pressure with mask ventilation, which may lead to important hypotension. Central venous access is obtained following anesthetic induction. Some centers use a pulmonary artery flotation catheter pretransplant and pull it back into the pulmonary trunk before clamping the pulmonary artery. Lung isolation is commonly achieved with a left-sided double-lumen endotracheal tube (ETT). However, using a single-lumen ETT with a bronchial blocker for lung isolation is also an option. The ability to tolerate one-lung ventilation is assessed early because failure to do so necessitates using CPB. TEE may guide in assessing ventricular function during transplantation, RV function upon pulmonary artery clamping and pulmonary vein stenosis, and also in detecting intracardiac air.

Feltracco and colleagues report that lung transplantation for severe pulmonary hypertension has distinct challenges compared with other etiologies.¹³⁶ Common echocardiographic features of patients with severe pulmonary hypertension include RV enlargement, tricuspid valve regurgitation, and diastolic dysfunction. The authors recommend ventilation strategies that avoid excessively increased intrathoracic pressure, moderate hypercapnia, and optimal positive end-expiratory pressure (PEEP) to prevent increases in pulmonary vascular resistance (PVR). They recommend CPB support in the following circumstances: intractable hypoxemia, greater than 30% reduction in cardiac output during the trial of pulmonary artery clamping, doubling of PVR, an increase in systolic pulmonary artery pressure to greater than 80% of systemic systolic pressure, surgical manipulation that severely compromises cardiac function, and severe ventricular wall motion abnormalities.¹³⁶

Baez and colleagues similarly noted that hemodynamic instability may ensue from myocardial depressant effects of induction agents, with excessive positive pressure ventilation on anesthetic induction, and with surgical retraction to gain exposure while removing the lungs.¹³⁷ They recommend reducing lung hyperinflation by decreasing tidal volumes, lowering the respiratory rate to maximize expiratory time, and permissive hypercapnia. TEE can be used to assess the presence and degree of RV dysfunction, which may necessitate the use of pulmonary vasodilators. Those available vary in selectivity for pulmonary vasculature and cost. Agents used include milrinone, inhaled nitric oxide, preferred because of greater selectivity for the pulmonary vasculature versus systemic vasculature, and inhaled epoprostenol, which is similarly effective and less costly.¹³⁷

If the chance of needing CPB is low, some centers place epidural catheters prior to anesthetic induction.¹³⁸ Goals of ventilation management include prolonging expiratory time to allow for more complete emptying of the lungs, along with permissive hypercapnia, with the thought that hypoventilation reduces hemodynamic effects of dynamic hyperinflation and auto-PEEP.

Miranda and colleagues have listed methods to decrease PVR, such as the use of vasodilators and induction with a 100% fraction of inspired oxygen to reverse hypoxic pulmonary vasoconstriction.¹³⁸

Descending Thoracic Artery Aneurysm (see Chapter 23)

Open repair of descending thoracic aorta aneurysm requires considerable anesthetic preparation. Monitoring typically necessitates large-bore IV access, right-sided brachial and femoral arterial catheter placement, central access with a large French introducer and pulmonary artery flotation catheter, a cerebrospinal fluid (CSF) drainage catheter, and TEE. Depending on the extent of the disease, repair may require CPB and circulatory arrest. A double-lumen ETT or single-lumen ETT with an endobronchial blocker is necessary for lung isolation. Additional venous access, if needed, is obtained via the right femoral vein because the surgeon may choose to use a left atrial-to-left femoral artery bypass. If partial bypass is used, the perfusionist can adjust flow to maintain upper-extremity mean blood pressure typically greater than 80 mmHg and lower mean blood pressure greater than 70 mmHg. Intrathecal preservative-free papaverine may be requested prior to clamping as a spinal cord protective measure. Additional neuroprotective measures include maintaining CSF pressure at less than 10 mmHg and passive cooling. Estrera recommends early management using free CSF drainage to maintain CSF pressure at less than 10 mmHg but later limiting cerebrospinal drainage unless a neurologic deficit occurs.¹³⁹

In addition to hypothermia and CSF drainage for spinal protective measures, monitoring of somatosensory (SSEP) or motor-evoked potentials (MEP) has been reported to provide additional protection.¹⁴⁰ Keyhani and colleagues note that MEPs and SSEPs are highly correlated only when intraoperative changes are irreversible, and these irreversible changes are associated with immediate neurologic deficits.¹⁴¹ Normal SSEP and MEP findings have strong negative predictive value, indicating that patients without signal loss were likely to be without a neurologic deficit.¹⁴¹ Some centers have used epidural cooling with epidural catheters placed at thoracic (T) T-12 to lumbar (L) L-1 and a 4F intrathecal thermistor catheter placed at L-3 to L-4.¹⁴²

Jacobs and colleagues reported that MEP is a highly reliable technique to assess spinal cord ischemia and is useful in reducing paraplegia during thoracoabdominal aneurysm repair.¹⁴³ Their protocol includes CSF drainage, moderate hypothermia, and left heart bypass with selective organ perfusion. MEP was used to monitor spinal cord function, and when important decreases occurred, hemodynamic (raising distal aortic and mean arterial pressure [MAP]) and surgical (reattachment of visible intercostal arteries) strategies were employed.¹⁴³

Endovascular stenting of thoracic aortic aneurysms is a less invasive approach to repair. Anesthetic management may be regional (spinal or epidural) or general. Disadvantages of regional anesthesia include patient movement, inability to use TEE as a monitoring tool, potential for hypotension with sympathectomy, and difficulty establishing an airway should complications occur during the procedure.¹⁴⁴

Transcatheter Aortic Valve Replacement

Transcatheter aortic valve replacement (TAVR) is currently used for patients with aortic valve stenosis who have important comorbidity

and are not surgical candidates because of high operative risk (see Chapter 12). Cheung and Ree summarized four key steps to the procedure: surgical access (via femoral vein or artery or left ventricular apex), native aortic valvuloplasty (predilation by balloon valvuloplasty), positioning and deployment of the prosthesis, and surgical closure.¹⁴⁵ TEE can assist with proper sizing by providing measurement of aortic annular dimension, guidewire advancement, and valve prosthesis positioning. TEE has particular utility in identifying complications with placement, such as device embolization, tamponade, perivalvar regurgitation, and coronary ostial obstruction with resultant regional wall-motion abnormalities.¹⁴⁵ Pharmacologic agents (adenosine or β -blockers) or, more commonly, rapid ventricular pacing are used during device deployment to attenuate left ventricular ejection. The procedure is performed under general anesthesia, which is beneficial for patient immobility, tolerance of rapid ventricular pacing, and better management of complications.¹⁴⁵

Ventricular Assist Devices

Anesthetic management for placing left ventricular assist devices (LVAD) focuses on considerations similar to those for severe heart failure, recognizing that these patients are critically ill with limited cardiac reserve and may have considerable hemodynamic instability on anesthetic induction.¹⁴⁶ TEE is a critical monitoring tool (Table 4.9). Identifying intracardiac shunts, such as a patent foramen ovale (PFO), has implications postimplantation. An unrecognized PFO may lead to hypoxemia because unloading of the

left ventricle (LV) leads to decreased left atrial pressure, which may result in substantial right-to-left shunting of blood. TEE detection of aortic valve regurgitation is also critical because this will reduce forward flow from the LVAD. In addition, TEE is useful for identifying RV dysfunction, the position of the inflow cannula, and de-airing.¹⁴⁷

Bleeding is not uncommon following LVAD insertion and is often multifactorial, with hepatic dysfunction, preoperative anticoagulation, and excessive fibrinolysis.¹⁴⁸ Patients with severe RV dysfunction may need an RV assist device, the use of pulmonary vasodilators (inhaled prostaglandins or nitric oxide), or both.¹⁴⁹

SECTION II: ANESTHESIA FOR NEONATES AND CHILDREN

Introduction

Approximately 4 million children are born each year in the United States, and nearly 40,000 of these have some form of congenital heart disease (CHD). It is estimated that 50% of these children present for a therapeutic or palliative intervention within the first year of life, almost all of them requiring general anesthesia. The anesthetic mortality in neonates and children with CHD is low (<5%), but perioperative management of pediatric patients with CHD requires training and understanding of the pathophysiologic principles of each lesion and surgical procedure planned, as well as a broad knowledge of pediatric medicine and pediatric cardiology.¹⁵⁰ Substantial improvements have been made in the anesthetic management of children with CHD. This is particularly true for neonates and patients undergoing staged repair of single-ventricle malformations.

For the acutely ill child who requires ventilatory and inotropic support preoperatively, anesthetic management must be carefully constructed to optimize cardiac output, and, in those with shunt-dependent physiology, to balance systemic (Qs) and pulmonary blood flow (Qp). Increasingly, children outside the newborn period arrive for cardiac operations on the same day as admission. Thus, the anesthesiologist must have access to the history and all cardiac diagnostic information before meeting with the child and the family. In many centers, a formal multidisciplinary conference occurs weekly to review historical, radiographic, echocardiographic, and cardiac catheterization data, and discuss surgical planning. Associated problems (e.g., reactive airway disease, airway anomalies, renal dysfunction, congenital syndromes) are also discussed often in the presence of members of other specialties.

Preoperative optimization of the patients requiring elective cardiac surgery is key. As an example, there is a growing interest in perioperative identification and treatment of iron deficiency and anemia.¹⁵¹ This approach is derived from the increasing evidence of the negative impact of iron deficiency anemia on perioperative outcomes in adult cardiac surgical patients and patients with heart failure.¹⁵² Another frequent problem to assess in children requiring cardiac surgery (or any surgery) is the presence of active or recent upper respiratory tract infections (URIs). This issue has evolved during the COVID-19 pandemic, where preoperative identification of symptoms and testing have played a crucial role in the operative planning of cardiac surgical procedures.¹⁵³ Respiratory infections are known to increase the perioperative risk of airway-related adverse events because of airway hyperreactivity that can persist up to 6 to 8 weeks following a symptomatic

TABLE 4.9 Utility of Transesophageal Echocardiography for Left Ventricular Assist Device Placement

Pre-CPB	On CPB	Post-CPB
Optimize left ventricular filling	Appropriate inlet cannula orientation (oriented to mitral valve)	Monitor cannula position
Exclude patent foramen ovale, aortic regurgitation, mitral stenosis	Verify device is functioning	Right ventricular function and tricuspid regurgitation
Monitor right ventricular function and assess tricuspid regurgitation	Exclude right-to-left shunting	Decompression of left ventricle and left atrium
	Monitor decompression of left ventricle and left atrium	Possible air entrainment if left ventricle collapses and subatmospheric intradevice pressures occur
	Exclude aortic insufficiency	Doppler-determined LVAD flows
	De-airing	

Modified from Mets B. Anesthesia for left ventricular assist device placement. *J Cardiothorac Vasc Anesth.* 2000;14:316-326.

infection. In a prospective study of children with heart disease undergoing elective surgery, those with a recent URI had increased postoperative respiratory complications (29% vs. 17%), bacterial infections (fivefold increased risk), marginally prolonged length of ICU stay, and multiple complications compared with those not having a URI.¹⁵⁴ The decision to proceed with elective surgery must be carefully weighed against the inconvenience of postponing surgery or the increased risk from postponement.

This section summarizes important principles and considerations of anesthesia for clinicians involved in the perioperative care of pediatric patients undergoing cardiac surgery with and without CPB.

Laboratory Evaluation

Laboratory evaluation should include analysis of hemoglobin, hematocrit, creatinine, blood urea nitrogen, and serum electrolytes in patients receiving diuretics or those with renal insufficiency. Liver function tests can be useful for patients at risk of liver dysfunction, such as occurs in patients with Fontan circulation. An elevated hematocrit in a normovolemic child reflects the magnitude and chronicity of cyanosis in the absence of iron deficiency. Hematocrit levels above 60% may predispose the patient to capillary sludging and stroke, and anemia may contribute to left-to-right shunting by decreasing the relative PVR. Because of liberalized feeding guidelines that allow the administration of clear liquids up to 1 to 3 hours before induction,¹⁵⁵ admission for preoperative IV hydration is seldom required for most patients. However, it is important to achieve adequate hydration in cyanotic infants.

Premedication

Premedication is used to achieve adequate sedation and maintain respiratory and hemodynamic stability. In children with complex CHD, premedication is directed toward decreasing oxygen consumption, improving systemic oxygen saturation, and promoting satisfactory anesthetic induction. Oral administration is effective and widely accepted. Dosage is adjusted based on age and clinical condition, but most centers use 0.5 mg/kg of midazolam orally 10 to 20 minutes before anesthetic induction, up to a maximum dose of 20 mg. Dosages of 0.7 mg/kg may be used in hemodynamically stable younger children, with lower doses of 0.3 mg/kg in children with reduced myocardial reserve. Generally, children younger than 6 months do not require premedication. In uncooperative children or those with autism, a 3- to 5-mg/kg intramuscular dose of ketamine is safe and effective. In the presence of IV access, an IV dose of midazolam (0.1 mg/kg) can be administered to decrease separation anxiety. In recent years, dexmedetomidine has been increasingly used as an alternative to midazolam. Intranasal dexmedetomidine (4 µg/kg) is sometimes used in children who do not accept oral medications.¹⁵⁶ Concentrated preparations should be preferred to limit the volume of drug administered nasally.

Physiologic Monitoring

Physiologic monitoring includes routine noninvasive monitoring (e.g., pulse oximeter, 5-lead electrocardiography (ECG), noninvasive blood pressure), as well as an arterial catheter, central venous catheter, and temperature probes. In term newborns, a 22-gauge radial arterial catheter is preferred, although some anesthesiologists prefer to exchange it for a 2.5F 2.5-cm catheter that can be better

secured. In small babies or premature infants, a 24-gauge catheter can be used, keeping in mind that the quality of the curve and the ability to draw blood samples from it can be suboptimal, especially when hypothermia is needed. Posterior tibial and dorsalis pedis arterial catheters should be avoided because they tend to function poorly after CPB. Femoral artery catheters may be used, but because of the future need for cardiac catheterizations, some surgeons prefer not to use them. In extremely low-birth-weight babies, a 22-gauge axillary arterial line can be placed. Use of an umbilical artery catheter for up to 7 days is appropriate for newborns.

In some centers, a combination of upper extremity and femoral arterial accesses is used to measure gradients across an aortic coarctation or a hypoplastic arch. A right radial arterial line is preferred in the case of arch repair, although a contralateral line would be preferred when a Blalock-Taussig-Thomas shunt (BTTS) is inserted. If selective cerebral perfusion (SCP) is used, an ipsilateral arterial line will be used to measure the perfusion pressure during SCP. Preoperative identification of an aberrant subclavian artery is important as well. In a recent report by Gleich and colleagues, the overall incidence of major complications of arterial cannulation for monitoring purposes in children was low (0.2%).¹⁵⁷ All complications occurred in femoral arterial lines in children younger than 5 years of age, with the greatest complication rates in infants and neonates. The standard practice is now to place the arterial line under ultrasound guidance.¹⁵⁸

Many centers employ percutaneous central venous catheters as a standard monitoring tool. These catheters must be placed under ultrasound guidance to increase safety and decrease the risk of complications.¹⁵⁹ Placing percutaneous central catheters in the superior vena cava (SVC) should be carefully considered in infants with single-ventricle physiology because thrombosis of upper-extremity vessels could preclude or complicate a future bidirectional Glenn procedure. The size and length of central venous catheters placed percutaneously should be based on the age and weight of the patient.^{160,161} Others rely on directly placed transthoracic catheters placed before or after repair of the malformation to obtain information for separation from CPB.¹⁶² However, these do not allow central venous pressure (CVP) monitoring in the prebypass period or effective monitoring of SVC pressure during CPB. Although not universally employed, direct measurement of left atrial, right atrial, and pulmonary artery pressures via small indwelling catheters provides a more accurate assessment of central pressures than other methods used to guide treatment in the postbypass and postoperative periods.

Temperature

Temperature probes are placed for measuring rectal (core) and either nasopharyngeal or tympanic membrane temperatures. Nasopharyngeal and tympanic membrane temperatures provide a reasonable estimate of brain temperature. Large gradients between rectal and nasopharyngeal or tympanic membrane temperature may reflect inadequate total-body cooling and may predispose the patient to unanticipated warming during periods of circulatory arrest or low-flow CPB.

Intraoperative Echocardiography

Intraoperative TEE is important for monitoring myocardial function and detecting air emboli, in addition to providing a morphologic map for surgical repair. In the postinduction period, TEE provides an opportunity to assess the anatomy and revise the

operative plan if necessary. It permits assessment of systolic and diastolic function, identification of valvar dysfunction, and estimation of pulmonary artery pressure. These observations may lead to modifying the anesthetic plan. After CPB, previously unidentified malformations and residual defects can be identified and corrected in the same operative setting, which may reduce morbidity and mortality.¹⁶³ In patients weighing less than 2.5 kg, the TEE probe should be placed with caution because of the risk of esophageal injury and airway obstruction. In such instances, the use of an epicardial echocardiography probe should be considered.

Neurologic Monitoring

Near-infrared spectroscopy (NIRS) monitoring is now the standard of care, either alone or with transcranial Doppler (TCD) and some form of electroencephalography (EEG) monitoring.^{164,165} It is our practice to routinely monitor NIRS during heart surgery, on or off CPB. TCD is valuable in detecting emboli and during antegrade cerebral perfusion to determine optimal flow. Positioning the TCD probe in infants is challenging. TEE is also an invaluable monitor of emboli, a use that must not be overlooked. Of the modalities, EEG is least useful in the operative setting because it is susceptible to changes in intraoperative temperature and the type of anesthetic agents used.

Near-Infrared Spectroscopy. Light in the near-infrared (700–1300 nm) range has three important physical properties that make it useful for diagnostic assessment:

- It penetrates tissue.
- It is non-ionizing.
- It is absorbed differentially by relevant chromophores depending on their oxygen-binding state.

When near-infrared light is emitted across a tissue (e.g., brain) and detected at its exit, absorption of the light can be used to calculate chromophore concentration using variants of the Beer-Lambert equation. All optical spectrometers consist of the same basic components: a light source of known intensity and wavelength, a light detector to measure the intensity of the light exiting the tissue, and a computer to translate the changes in light intensity into clinically useful information such as the concentrations of HbO₂, hemoglobin, or oxidized cytochrome aa3.¹⁶⁴

When photons impinge on biological materials, their transmission depends on a combination of reflectance, scattering, and absorption effects. A light source (light-emitting diode or laser source) emits near-infrared light that passes through a “banana-shaped” reflectance path in the frontal cerebral cortex to two to

three detectors placed 3 to 5 cm from the emitter. Absorption occurs at specific wavelengths, determined by the molecular properties of the material in the light path. Optical path length for reflected light is linearly related to spacing between transmission and receiving sites, so many NIRS measurement instruments place the transmitting diode and light detector several centimeters apart on the head. Although this spacing results in a measurable signal intensity, it affects the amount and depth of tissue monitored. On a practical level, the available instruments space their transmitting and receiving sites differently, thus measuring different quantities and depths of tissue, which makes comparisons between instruments difficult (Table 4.10).

Shallow arcs of light travel across skin and skull but do not penetrate the cerebral tissue. Deep arcs of light cross skin, skull, dura, and cortex. Subtracting the absorbance measured in the narrow arc from that measured in the deep arc leaves absorbance that is due to intracerebral chromophores. This is one of the distinguishing characteristics of cerebral oximeters compared with pulse oximeters. Cerebral oximeters use spatial resolution techniques to differentiate cortical from extracranial blood, whereas pulse oximeters differentiate pulsatile (arterial) from nonpulsatile (venous/capillary) blood. Cerebral oximetry measures predominantly venous saturation (75:25 or 85:15, depending on age and model used).¹⁶⁶ NIRS could be a surrogate for jugular venous oxygen saturation (Sjvo₂) monitoring without being invasive. It does not depend on pulse, blood pressure, or body temperature. This makes the technique ideally suited for monitoring oxygenation during CPB, hypothermic circulatory arrest, shock, or cardiovascular collapse.

Clinical Applications of NIRS. To use the device, one or two cerebral oximeter probes are placed on the forehead below the hairline. Andropoulos and colleagues reported that during antegrade cerebral perfusion, left-sided cerebral saturation was substantially lower than the right (92%–94% right and 60%–65% left).¹⁶⁷ However, the left-sided values were well within normal limits, and the right-sided values suggest luxury perfusion.

The landmark study by Austin and colleagues using multimodality neurologic monitoring spurred interest in NIRS for pediatric open-heart surgery.¹⁶⁸ These authors reported a 26% incidence of adverse postoperative neurologic outcomes when intraoperative desaturations were not treated versus only 6% when the changes were treated. Central venous oxygen saturation (ScO₂) is a balance between oxygen delivery and utilization. If the latter remains unchanged, then any decrements in cerebral saturation must be caused by decreased cerebral oxygen delivery. This could be due to a decrease in arterial saturation, hemoglobin, or cerebral blood

TABLE 4.10 Currently Available Near-Infrared Spectrometry Devices

Device/Manufacturer	Light Source	Wavelengths Used	Readout	FDA Status
INVOS Covidien (United States)	LED	2	rSO ₂ i%	Approved for pediatric use
Niro Hamamatsu (Japan)	LED	3	c-TOI	Not FDA approved
Foresight Casmed (United States) ^a	Laser	4	ScO ₂	Approved for pediatric use
Equanox 7600 ^a Nonin Medical, Inc. (United States)	LED	3	rSO ₂ %	Approved for pediatric use >40 kg

^aMeasures “absolute” cerebral saturation. All other devices are marketed as trend monitors.

c-TOI, Cerebral tissue oxygenation index; FDA, US Food and Drug Administration; LED, light-emitting diode; rSO₂i%, regional cerebral oxygen saturation index; rSO₂%, regional oxygen saturation; ScO₂, cerebral tissue oxygen saturation.

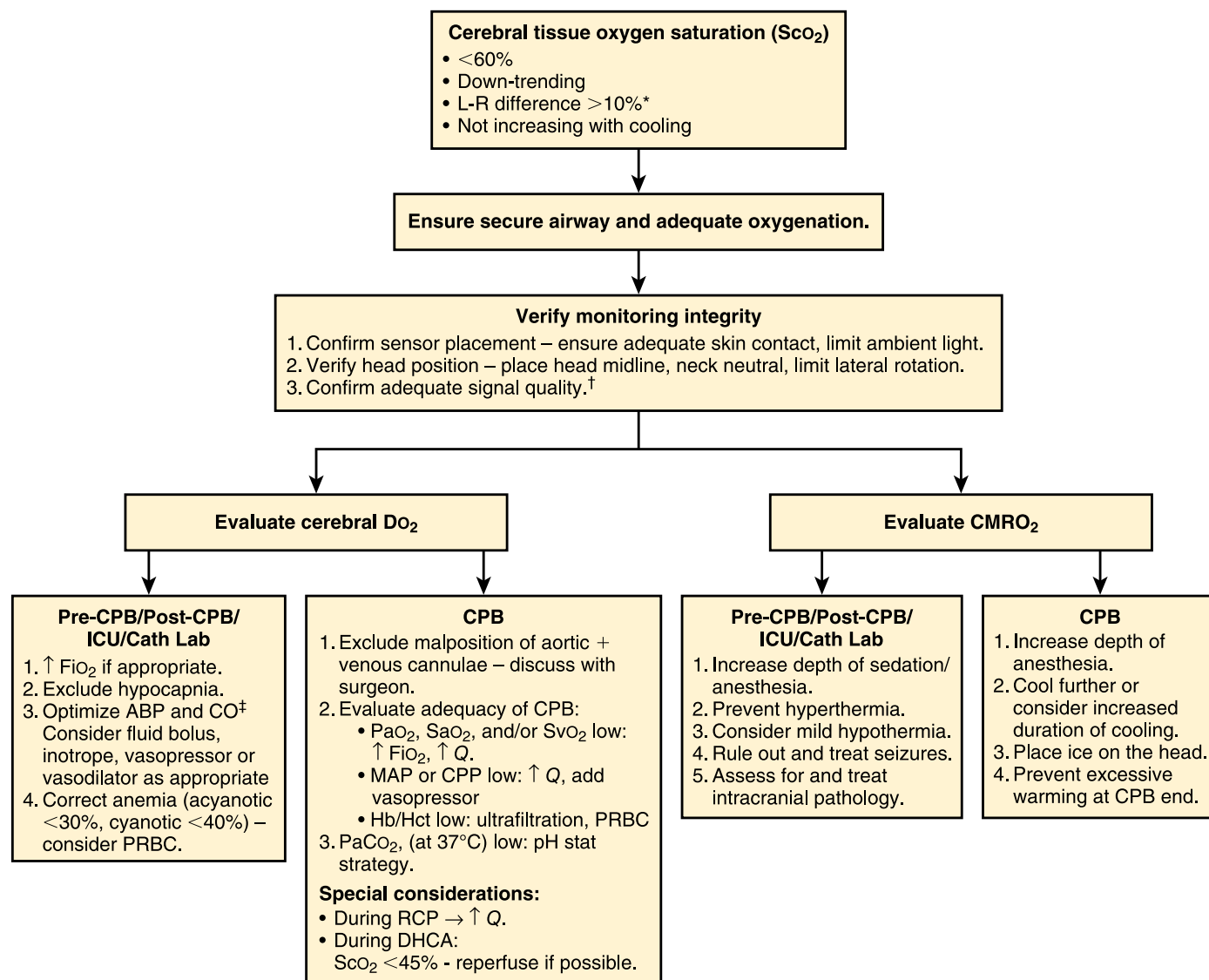
flow. Thus, if ScO_2 decreases in the face of normal pulse oximetry (SpO_2), it is important to decide why cerebral oxygen delivery has changed. Conditions of increased utilization include hyperthermia, seizures, and change in level of arousal; these must be treated. During the initiation of CPB, one of the common causes for decreased regional oxygen saturation (rSO_2) is arterial cannula malposition or occlusion to venous drainage, which decreases cerebral perfusion pressure and reduces cerebral blood flow. Moreover, cerebral oximetry could be a guide to hypothermic circulatory arrest and intermittent perfusion, making circulatory arrest safer.

Outcomes after Heart Surgery and NIRS Monitoring. The Austin study was a retrospective cohort study; no randomized trial of NIRS has been conducted in children. Dent and colleagues¹⁶⁹ showed a correlation between prolonged low regional oxygen saturation (rSO_2) ($<45\%$ rSO_2 for ≥ 180 min) and new magnetic resonance imaging (MRI) abnormalities in a group of neonates who underwent the Norwood procedure with antegrade cerebral perfusion (ACP). In a prospective study of neonates with either single- or two-ventricle physiology undergoing surgery with CPB and ACP, the authors were unable to show an association between

new white matter injury on postoperative MRI and prolonged low perioperative $\text{rSO}_2\%$. However, it is important to note that in this study, 50% of patients in the single-ventricle group had prolonged low rSO_2 ($<45\%$ for 240 min), one third of whom had new white matter injury in the postoperative period. No patients in the two-ventricle group had prolonged $\text{rSO}_2\%$ below 45% for 120 minutes. Although stroke and chorea are obvious neurologic abnormalities, subtle neurocognitive changes are difficult to establish. Thus, despite the lack of level 1A evidence that NIRS improves neurocognitive outcomes, we recommend intraoperative cerebral oximetry as a tool to optimize anesthesia (ventilation, oxygenation); perfusion (alpha-stat, pH-stat, hemoglobin, flow, temperature); and surgical techniques (cannulation). Fig. 4.4 shows the protocol recently published by Zaleski and Kussman from Boston Children's Hospital.¹⁶⁴

Anesthetic Agents

The variations of severity and pathophysiology of CHD mandate individualized anesthetic management based on the



• **Figure 4.4** Intraoperative cerebral oximetry protocol. (From Zaleski KL, Kussman BD. Near-Infrared Spectroscopy in Pediatric Congenital Heart Disease. *J Cardiothorac Vasc Anesth.* 2020;34(2):489-500)

pharmacologic properties of anesthetics and other drugs used during general anesthesia. A wide variety of anesthetic drugs have been used successfully and safely, including inhalation agents such as sevoflurane, and IV agents such as propofol, fentanyl, midazolam, thiopental, and ketamine (IV or intramuscular). For critically ill neonates, opioid drugs with or without benzodiazepine are generally preferred. Fentanyl is most often used, titrated in 2- to 5- $\mu\text{g}/\text{kg}$ increments with or without midazolam (0.1 mg/kg per increment) until the patient is no longer responsive. A nondepolarizing muscle relaxant is then administered (e.g., vecuronium or rocuronium). Alternatively, combined infusions of opioids and benzodiazepines may be used. Ketamine in doses of 1 to 2 mg/kg is an IV agent, has minimal effects on hemodynamics, and allows the concentration of inhalation agent to be reduced or turned off altogether. Regardless of the anesthetic used, 80% of children with poor myocardial function experience hypotension requiring treatment.¹⁷⁰ Presence of intracardiac shunts affects anesthetic induction. The presence of a right-to-left shunt leads to rapid IV induction but can slow inhalational induction with a volatile anesthetic because of decreased pulmonary blood flow.¹⁷¹ Left-to-right shunts generally do not affect the speed of induction. These physiologic effects must be considered when choosing an anesthetic. This also makes de-bubbling all medications and IV fluid mandatory.

Intubation and Ventilation

Endotracheal intubation can be performed orally or nasally; the preference is based less on science and more on institutional choice. Cuffed ETTs were traditionally avoided in children younger than 8 years of age; however, the use of microcuffed tubes has been found not to increase the risk of tracheal stenosis or complications.¹⁷² In a multicenter randomized trial in patients younger than age 5, including neonates, Weiss and colleagues found that cuffed ETTs were reliable in infants and children.¹⁷³ In fact, the number of attempts at tube changes to place the correct-sized tube was reduced, and cuffed tubes did not increase the risk of postextubation stridor. It is important to measure ETT cuff pressure and maintain it below 20 cm H₂O to minimize the risk of tracheal mucosal ischemia. In neonates and extremely

low-birth-weight infants, it is prudent to obtain an intraoperative chest radiograph to confirm ETT position.

High fractional concentration of inspired oxygen (FiO_2) is avoided in children with shunt physiology or a nonrestrictive ventricular septal defect (VSD). Oxygen is a potent pulmonary vasodilator, and the use of high concentrations can reduce Q_s by diverting more of the cardiac output through the shunt into the Q_p . Similarly, a low partial pressure of arterial carbon dioxide (PaCO_2) can reduce PVR, increase Q_p , and reduce Q_s . Increased PaCO_2 increases PVR and thus may also be hazardous in patients with intracardiac or extracardiac shunts. Therefore, induction with reduced FiO_2 and a normal or slightly elevated PaCO_2 is helpful in balancing blood flow between the systemic and pulmonary circulations.

Maintenance of Anesthesia

A combination of an opioid, usually fentanyl or sufentanil, and an inhalation anesthetic, usually isoflurane or sevoflurane, is used for anesthesia maintenance. In general, children with limited cardiac reserve are maintained primarily on an opioid anesthetic, with low concentrations of inhalation agents as a supplement when tolerated. Historically, use of high-dose opioids has been advocated to blunt the stress response in neonates and infants. Lower doses of opioids have proved to be equally effective, with less release of inflammatory mediators and a lesser degree of endothelial injury.¹⁷⁴

In patients in whom early extubation is planned, anesthesia in the postbypass period is maintained with an inhalation anesthetic, and use of fentanyl is limited. Remifentanyl is a synthetic ultra short-acting narcotic metabolized by plasma cholinesterase with a half-life of 3 to 5 minutes. It has been used for fast-track anesthesia in children.¹⁷⁵

Dexmedetomidine is a novel sedative/hypnotic agent that acts at central nervous system α_2 -adrenergic receptor binding sites as a highly selective agonist. Dexmedetomidine is increasingly used intraoperatively as part of a balanced anesthetic technique in pediatric patients undergoing cardiac surgery CPB. The pharmacokinetics of dexmedetomidine have been investigated in a few pediatric studies.^{176,177} Based on those studies, the dose recommendations for pediatric patients are summarized in Table 4.11.¹⁷⁶ The use of

TABLE 4.11 Dosing Recommendations for Dexmedetomidine Steady-State Concentrations

Age group (days)	Target C_{ss} ($\mu\text{g mL}^{-1}$)	Initial loading dose ($\mu\text{g kg}^{-1}$)	Infusion 1: pre-CPB, first 60 min of CPB ($\mu\text{g kg}^{-1}\text{h}^{-1}$)	Loading dose to CPB prime volume ($\mu\text{g mL}^{-1}$)	Infusion 2: after 60 min of CPB until end of CPB ($\mu\text{g kg}^{-1}\text{h}^{-1}$)	Infusion 3: 60 min after CPB ($\mu\text{g kg}^{-1}\text{h}^{-1}$)
Neonatal (0–21)	200	0.24	0.22	0.004	0.04	0.14
Neonatal (0–21)	500	0.6	0.55	0.01	0.1	0.35
Neonatal (0–21)	700	0.84	0.77	0.014	0.14	0.49
Neonatal (0–21)	1000	1.2	1.1	0.02	0.2	0.7
Infant (22–180)	200	0.29	0.26	0.005	0.05	0.17
Infant (22–180)	500	0.72	0.66	0.012	0.12	0.42
Infant (22–180)	700	1.01	0.92	0.017	0.17	0.59
Infant (22–180)	1000	1.44	1.32	0.024	0.24	0.84

C_{ss}, steady-state concentration.

dexmedetomidine has increased in patient populations where intraoperative extubation and fast-track protocol have been implemented.¹⁷⁸ In a meta-analysis, Ghimire and colleagues found that dexmedetomidine significantly reduced the incidence of junctional ectopic tachycardia (JET) after pediatric CHD surgery.¹⁷⁹ Recently, dexmedetomidine has been reported to protect cardiomyocytes from ischemia-reperfusion injury in cellular models and adult rodent models.¹⁸⁰ However, whether and how dexmedetomidine may protect human cardiomyocytes in young children remains largely unknown.¹⁸¹

Cardiopulmonary Bypass

With initiation of CPB, cooling is started. In neonates and infants undergoing hypothermia, the pump prime may be maintained at a temperature of 18°C to 22°C (cold), 30°C (moderate), or 37°C (warm). When deep hypothermic circulatory arrest (DHCA) is needed, the patient is cooled to between 15°C and 20°C nasopharyngeal or tympanic membrane temperature and rectal temperature. The goal is to achieve optimal uniform cooling through a combination of core cooling using CPB and surface cooling using a cooling blanket beneath the patient. Room temperature is lowered after arterial and venous catheter placement. When circulatory arrest is planned, ice packs are placed around the child's head after initiating CPB.

The duration of cooling before reducing flow to low levels or initiating circulatory arrest is generally 20 to 25 minutes. Cooling should proceed at a controlled rate so that temperature does not fall more than 1°C per minute (see Chapter 2). A reduced rate of head or rectal cooling may indicate suboptimal tissue perfusion or a malpositioned temperature probe. If pump flow is inadequate, vasodilators (e.g., phentolamine) can be added directly to the CPB circuit. During cooling and before circulatory arrest, arterial blood gases and hematocrit are measured, and necessary adjustments made.

In children, the appropriate arterial blood gas management strategy during hypothermia is the pH-stat technique.¹⁸² During cooling and rewarming periods, pH-stat provides better cerebral blood flow, particularly in patients with aortopulmonary collaterals, and is also useful during antegrade cerebral perfusion because the increased cerebral blood flow allows for rapid cooling. Data are unclear but suggest that overall, pH-stat is useful in most pediatric patients; however, crossover strategies may be another option (see Chapter 2).

Anticoagulation during CPB is necessary to prevent thrombin formation and subsequent clotting as blood comes into contact with the non-endothelial CPB surfaces. UFH is the primary anticoagulant used to inhibit not only the formation of thrombin but also the activity of circulating thrombin through its binding to AT.¹⁸³ UFH is usually administered at a dose of 300 to 400 IU/kg. The administration of 400 IU/kg of heparin has been shown to produce adequate prolongation of ACT > 480 seconds. Individualized heparin doses based on heparin dose-response analysis have been used in some studies.¹⁸⁴ However, in a recent large cohort of children who received UFH for CPB, Nakamura and colleagues found that heparin responsiveness before CPB was not reliably predicted by either in vitro heparin dose-response slopes estimated by the HMS Plus System (Medtronic, Minneapolis, MN) or regression models using commonly available preoperative clinical and laboratory data.¹⁸⁵ The strength of the correlation between predicted and observed heparin responsiveness was found to be, at best, moderate for young children <5 years.

Different techniques have been used to monitor anticoagulation intraoperatively. Standard whole blood clotting times are still routinely used to monitor anticoagulation in children undergoing cardiac surgery. However, some centers have recommended monitoring heparin concentration along with the ACT as a more accurate guide for the administration of heparin to infants during CPB.¹⁸⁶ In a study by Manliot and colleagues, low circulating AT activity was associated with lower heparin efficacy, which ultimately leads to a decreased ability to suppress thrombin generation during CPB.¹⁸⁷ Determination of risk factors for heparin resistance, and potentially AT supplementation, may individualize and improve anticoagulation treatment.¹⁸⁸ In the presence of AT deficiency, FFP is usually administered because of its frequent availability in the OR. Even though AT concentrate has sometimes been used in children supported with extracorporeal membrane oxygenation (ECMO), there is a lack of studies assessing the efficacy and safety of AT supplementation in neonates and children undergoing cardiac surgery. AT levels are sometimes included as part of the preoperative laboratory testing. Because of the low prevalence of congenital AT deficiency, routine measurement of AT levels is not recommended.¹⁸⁹ However, preoperative measurement of AT level should be considered in the presence of risk factors (e.g., neonates exposed to preoperative heparin infusion). After the patient is weaned off CPB, protamine is administered to neutralize the effect of heparin. The protamine dose is usually based on the total amount of heparin administered during CPB. However, both residual heparin and protamine overdose are associated with increased postoperative bleeding. Most centers use a protamine-to-heparin ratio of 1:1. Based on the pharmacokinetic of heparin, a 1:1 ratio could lead to protamine overdose and bleeding; it is often recommended to use a protamine-to-heparin ratio of 1:2 ratio. Protamine dose can also be estimated based on heparin concentration monitoring.^{184,190}

Although most centers still use RBC and FFP in the CPB prime for neonates, there is little high-grade evidence to support its use in older patient populations.¹⁹¹ Some institutions use albumin to maintain oncotic pressure and reduce fluid accumulation, although proponents of FFP argue that in addition to maintaining oncotic pressure, FFP increases the level of coagulation factors and reduces intraoperative transfusion requirements in complex neonatal and cyanotic patients. In a study of 60 patients weighing 7 kg to 15 kg randomized to FFP or crystalloid prime, Dieu and colleagues did not identify any significant differences in median postoperative blood loss or transfusion requirements.¹⁹² Other studies have demonstrated that patients with FFP prime have higher fibrinogen concentrations and improved thromboelastography values at the end of CPB, but this was not associated with a decrease in postoperative transfusions. Unfortunately, the composition of the prime is often a matter of opinion and institutional practice and differs widely. The only consensus regarding CPB prime is minimizing transfusion requirements and hemodilution. Additional pediatric-specific trials are needed to determine the optimal CPB prime composition.

The optimal hemoglobin values for congenital cardiac surgery are unknown and likely vary depending on age, degree of cyanosis, and complexity of the surgical procedure. Multiple studies have demonstrated that a restrictive transfusion practice does not increase morbidity or mortality in pediatric ICU patients, congenital cardiac patients, and preterm neonates.¹⁹³⁻¹⁹⁶ However, many of the studies exclude neonates or patients with single ventricle physiology. One of the few studies examining the impact of a restrictive (hemoglobin 11.1 g/dL) versus liberal (13.9 g/dL)

RBC transfusion strategy in children with single-ventricle physiology after cavopulmonary anastomosis found no difference between the two study groups.¹⁹³ Although guidelines have recommended the adoption of restrictive transfusion thresholds even in the cyanotic population,¹⁹⁷ this remains a topic of intense debate.¹⁹⁸ Clearly, further studies are needed to evaluate the effect of RBC transfusion on oxygen delivery (DO_2) and oxygen consumption (VO_2) at the level of the microcirculation. Whereas RBC transfusion remains the treatment of choice to treat acute anemia, a single transfusion threshold is still not a viable approach.¹⁹⁹

Separation from Cardiopulmonary Bypass

The patient is rewarmed to a core temperature of 35°C to 36°C, the heart is filled and allowed to eject, arterial blood gases are obtained to ensure adequate acid-base balance, and calcium level is corrected to normal values for neonates and infants. Pacing wires are applied to the heart and tested, and the heart rate is maintained at an age-appropriate level using atrial or atrioventricular (AV) sequential pacing if needed. If high doses of inotropic agents are required despite adequate preload and ventilatory support, the presence of a residual anatomic defect or poor adaptation to new loading conditions resulting from the operative repair may be contributing factors. TEE is helpful for determining the cause of the low output state.

Rationale for Specific Therapies

Right Ventricular (Pulmonary Ventricle) Dysfunction. Primary RV dysfunction may occur after intracardiac surgery in neonates, infants, and children. Diagnosis of RV dysfunction is suggested by high right-sided filling pressures, liver distension, hypotension, tachycardia, reduced cardiac output, and systemic venous desaturation (low mixed-venous saturation). Treatment is directed toward improving oxygen delivery by increasing preload, augmenting contractility directly or indirectly, enhancing coronary perfusion, and reducing afterload.

The right ventricle is generally less responsive to inotropic support than the LV and, therefore, may require higher doses of inotropic agents. Epinephrine enhances RV contractility. By improving systemic arterial pressure, epinephrine can augment RV coronary blood flow. Maintaining a normal to slightly elevated systolic arterial pressure will maximize coronary perfusion and augment RV contractility. Milrinone, a phosphodiesterase-3 inhibitor, is a useful inotrope with pulmonary vasodilatory properties. In a randomized controlled trial of infants and children after heart surgery, a bolus of 75 $\mu\text{g}/\text{kg}$ followed by a 0.75 $\mu\text{g}/\text{kg}/\text{min}$ infusion reduced the occurrence of low cardiac output syndrome by 55%.²⁰⁰ The use of milrinone varies depending on institutional preferences; the loading dose usually ranges between 25 to 50 $\mu\text{g}/\text{kg}$ during rewarming, followed by an infusion of 0.375 to 0.5 $\mu\text{g}/\text{kg}/\text{min}$. Communication with the surgeon is important prior to administering the bolus, even on CPB, because some surgeons are concerned with the hypotension that could result.

RV afterload can be decreased by mechanical ventilation with or without inhaled nitric oxide (iNO). Mechanical ventilation should be adjusted to optimize preload and decrease afterload. The right ventricle is extremely sensitive to alterations in intrathoracic pressure; therefore, ventilation that enables the lowest possible mean airway pressure should be the goal. Increased mean intrathoracic pressure increases RV afterload by direct compression of alveolar and extra-alveolar pulmonary vessels. iNO is an endothelium-derived smooth

muscle relaxant used in neonates with persistent pulmonary hypertension and in pulmonary hypertension related to CHD. Nitric oxide (NO) decreases R_p and reduces intrapulmonary shunting, which may improve oxygenation. However, results of its use in the postoperative period are conflicting. In a randomized controlled trial of more than 100 infants at high risk for pulmonary hypertension, 20 ppm not only failed to show any benefit, it did not prevent pulmonary hypertensive crisis in children after congenital heart surgery.²⁰¹ However, in a similar population, others have shown it to be effective even at lower doses.²⁰² If inhaled nitric oxide (iNO) is used, NO_2 levels should be monitored, and iNO should not be abruptly withdrawn because rebound pulmonary hypertension can occur.

If these measures are unsuccessful, ECMO should be implemented. EMCO unloads the right ventricle and favorably shifts the oxygen supply-demand ratio, often allowing the injured myocardium to recover.

Left Ventricular (Systemic Ventricle) Dysfunction. After separation from CPB, the contractile state of the systemic ventricle may be depressed. Contributing factors include the preoperative condition of the myocardium (myocardial hypertrophy, elevated end-diastolic pressure, systolic dysfunction), response of the myocardium to the new loading conditions imposed by the operative repair, effects of hypothermia on myocardial compliance, suboptimal myocardial protection, and residual anatomic problems.

Systemic ventricular dysfunction is managed by optimizing preload, afterload, and heart rate. Tachycardia (>180 – 190 beats/min) may impair ventricular function in newborns and infants and should be treated with β -adrenergic blocking agents and, if necessary, vasopressors. When the heart rate is less than 120 to 130 beats/min, atrial or AV sequential pacing is appropriate.

Management of Hypoplastic Left Heart Physiology. In the preinduction period, ductal patency must be maintained with prostaglandins to ensure systemic cardiac output. Management depends on optimizing systemic oxygen delivery (by increasing cardiac output) and restricting Q_p .

Before discontinuing CPB, ionized calcium levels and hematocrit must be optimized to ensure adequate oxygen-carrying capacity. Myocardial function is supported by the judicious use of inotropes. Tidal volume is increased to account for a reduction in lung compliance, and minute ventilation is adjusted to maintain normocarbia. After separation from CPB, FiO_2 is adjusted to maintain SaO_2 between 75% and 85% and an arterial Pao_2 of 40 to 50 mmHg. Modified ultrafiltration has been shown to improve myocardial function, decrease lung water, and remove inflammatory mediators in patients with hypoplastic left heart physiology, as well as other complex malformations (see Chapter 2). Excessive Q_p is less common in the immediate postbypass period. After modified ultrafiltration and before chest closure, the anesthesiologist should estimate $\text{Q}_\text{p}/\text{Q}_\text{s}$ and attempt to adjust FiO_2 and minute ventilation accordingly.

Chest closure can markedly reduce lung compliance and worsen hemodynamics. Leaving the chest open may improve Q_p and heart filling by reducing mean airway pressure.

Rationale for Managing Fontan, Hemi-Fontan, and Bidirectional Glenn Procedures

Patients undergoing a bidirectional Glenn or hemi-Fontan procedure usually have had either a pulmonary trunk band or a systemic-to-pulmonary artery shunt in the neonatal period. Cardiac performance may be impaired by either a small noncompliant ventricle or a large dilated ventricle, the latter resulting from

excessive aortopulmonary shunt flow. Inotropic support may be necessary in the prebypass period as well as postoperatively.

After the bidirectional Glenn or hemi-Fontan procedure, cardiac output is generally well maintained because inferior vena cava flow mixes with pulmonary venous blood in the physiologic left atrium. Low systemic arterial saturation and reduced $\dot{Q}_P$, however, are problems in the postoperative period. A marked discrepancy between end-tidal carbon dioxide (P_{etCO_2}) and P_{aCO_2} is an early sign of reduced $\dot{Q}_P$. If $\dot{Q}_P$ is reduced with no residual cardiac abnormalities, cardiac output should be optimized and interventions to lower R_p should be employed. Children undergoing bidirectional Glenn or Fontan operations should be considered for fast-track anesthesia to optimize cardiopulmonary interactions.

Infants and Children Not Undergoing Cardiopulmonary Bypass

The most common procedures that do not involve CPB are *palliative* (systemic–pulmonary artery shunting and pulmonary trunk banding) or *corrective* (ligation of patent ductus arteriosus [PDA] or repair of coarctation of the aorta).

Palliative Procedures

Palliative procedures are performed under general anesthesia with monitoring of systemic arterial pressure. Measurement of arterial pressure, arterial oxyhemoglobin saturation (SaO_2), and end-tidal carbon dioxide pressure (P_{etCO_2}) is necessary to assess the effectiveness of the procedure. An important reduction in P_{etCO_2} after pulmonary trunk banding indicates that $\dot{Q}_P$ may be excessively reduced. This is followed by a precipitous drop in SaO_2 . If the banding procedure is optimal, systemic arterial blood pressure should increase by approximately 10 to 15 mmHg. The gradient between P_{etCO_2} and P_{aCO_2} should be about 6 to 10 mmHg. SaO_2 should be no lower than 75% to 80%, and pulmonary arterial pressure should decrease to about 50% of systemic pressure.

Closure of Patent Ductus Arteriosus

Over the last decade, management of PDA in preterms and infants has shifted from the OR to the cardiac catheterization laboratory.²⁰³ Although most centers are still in the learning curve for the percutaneous approach, PDA device closure is now the preferred option in most large heart centers. The perioperative management of PDA device closure has been described and differs from the surgical approach.²⁰⁴ That leaves patients with PDA not suitable for percutaneous occlusion as the only ones requiring surgical ligation. Anesthetic considerations for PDA ligation depend on the ductus size and clinical condition and age of the patient. Babies with a large PDA and low R_p generally present with excessive $\dot{Q}_P$ and heart failure. Neonates and premature infants may also have left ventricular dysfunction from coronary ischemia caused by substantial diastolic runoff to the pulmonary circuit. Thus, patients range from relatively healthy young children to sick, ventilator-dependent premature infants on inotropic agents. Healthy children can tolerate a variety of anesthetic techniques with extubation in the OR and use of regional anesthesia analgesia. Symptomatic neonates and premature infants require a carefully controlled anesthetic and fluid management plan.

Most preterm infants who fail medical management consisting of indomethacin, diuretics, and fluid restriction require admission to a neonatal ICU. A common finding is sepsis, so it is important

to ascertain a history of medical treatment and verify negative blood cultures before surgical intervention. Premature neonates with ductal patency are often operated on in the neonatal ICU, thereby avoiding transport hazards such as hypothermia, multiple transfers to and from infant incubators, accidental extubation, and venous access disruption.

In the neonatal ICU, the patient is positioned on a warmer, and access to the patient must be shared among the anesthesiologist, surgeon, surgical assistant, and scrub nurse. Careful positioning of an IV catheter and rapid access to a manual resuscitator or equivalent should be established before the baby is draped. Anesthesia is induced with fentanyl (usually in 1- to 5- μ g aliquots) to maintain appropriate arterial pressure and perfusion. Muscle relaxation is obtained with vecuronium or rocuronium to prevent a reduction in heart rate and preserve cardiac output. Hypotension following anesthesia induction should be anticipated because these neonates are often on large doses of diuretics to manage their ventilation. The patient may temporarily require ventilation with 100% oxygen if SpO_2 drops below 90% or is associated with changes in heart rate and blood pressure. Oxygen is weaned once both lungs are allowed to expand after ductal closure. Manual ventilation is often necessary during retraction of the lung in small neonates or in those with preexisting increased oxygen requirements.

Complications include ligation of a pulmonary artery or the aorta. If SaO_2 remains low and P_{etCO_2} decreases, this alerts the perioperative team to possible pulmonary artery ligation. Similarly, placing a pulse oximeter on the foot ensures that aortic blood flow below the duct is maintained and can signal a possible aortic ligation.

Coarctation of the Aorta

Coarctation of the aorta is a common cardiac defect and, in infants, is often associated with anomalies of the mitral valve and left ventricular outflow tract and with malformation of the great arteries. As with PDA, neonatal repairs are performed in critically ill patients. Coarctation in the newborn is typically associated with left ventricular dysfunction, and these patients may be receiving prostaglandins to ensure ductal patency. Usually, they are also receiving mechanical ventilation and inotropic agents. Central IV and arterial and peripheral IV catheters are normally placed for operation. Optimal placement of an arterial pressure catheter is in the right radial artery so that pressure can be monitored during aortic clamping. If right-sided arterial access is not achieved, a blood pressure cuff is placed on the right arm, and the arterial catheter in the lower extremity. Some centers always require upper and lower extremity arterial lines to monitor the gradient, but that is not mandatory. Anesthesia is administered with a combination of fentanyl and an inhalation agent. During aortic clamping, proximal systemic arterial pressure is allowed to rise by 20% to 25% over baseline to optimize spinal cord perfusion. Intravascular volume loading with 10 to 20 mL/kg of crystalloid is given just before removal of the aortic clamp. The anesthetic concentration is decreased, and additional fluid is administered until arterial pressure rises.

Avoidance of hyperthermia along with mild cooling is appropriate for patients undergoing aortic coarctation repair. Intraoperative hyperthermia has been associated with a risk of spinal cord ischemia and paraplegia.²⁰⁵ A target core temperature of approximately 35°C is appropriate. In older subjects, postrepair rebound hypertension caused by heightened baroreceptor reactivity often occurs and requires therapy. After aortic clamp release, systemic hypertension is most effectively lowered by institution of β -adrenergic blockade using esmolol or combined α and β blockade with labetalol.

Sodium nitroprusside may be a necessary adjunct to control refractory hypertension; however, it increases calculated ventricular wall stress in the absence of β -adrenergic blockade by accelerating dP/dt . An effective alternative to nitroprusside is the calcium channel blocker nicardipine. Again, neonates are unlikely to require or tolerate lung isolation.

One-Lung Ventilation in Children. Lung isolation is helpful during thoroscopic procedures in older children and adolescents. Lung isolation in infants and toddlers can be achieved with endobronchial intubation or bronchial blocker. The latter can be placed under fiberoptic or fluoroscopic guidance. Bronchial blocker or endobronchial intubation can be used for lung isolation in children.²⁰⁶ The former allows expansion of the surgical side, if necessary, by deflating the blocker to improve oxygenation. Both techniques are associated with complications and should be undertaken judiciously. Lung isolation in cyanotic pediatric patients requires close communication between the surgeon and anesthesiologist, as further reductions in oxygen saturation may be associated with ischemic changes on the ECG.

Regional Anesthesia and Fast-Track

Given the increased risk of opioid dependency after in-hospital exposure to opioids and the increased risk of morbidity, regional techniques offer an adjunct for perioperative analgesia. Although regional anesthesia can provide potent analgesia, its use for cardiac anesthesia is not without risk. Neuraxial techniques, epidural and spinal, have been used in patients not requiring anticoagulation. In patients requiring full anticoagulation, anesthesiologists and surgeons have been reluctant to use neuraxial techniques because of the increased risk of hematoma. Other regional anesthesia techniques, such as paravertebral and interfascial plane blocks, may be associated with less risk for spinal hematoma. However, bilateral blocks are typically needed for sternotomy incisions. Regarding superficial blocks, there are limited data regarding adverse events, but scientific societies and experts recommend consideration of site vascularity, compressibility, and consequences of bleeding. Beyond the risk of placing blocks in anticoagulated patients, other side effects, including decreased peripheral vascular resistance from sympathectomy and local anesthetic cardiac toxicity, are concerning to cardiac anesthesiologists attempting to maintain stable physiology.

The erector spinae plane block (ESPB) is a fascial plane block performed by injecting local anesthetic into the fascial plane deep to the erector spinae muscle overlying the transverse processes of the thoracic vertebrae.²⁰⁷ Though not yet fully elucidated, its proposed mechanism of action is thought to be via blockade of the dorsal and ventral rami of the thoracic spinal nerves and sympathetic fibers, with spread both cranially and caudally to cover multiple dermatomal levels.²⁰⁸ Unlike neuraxial techniques and paravertebral blocks, ESPBs are relatively superficial and easy to place because of their clear ultrasonographic landmarks. They have a lower risk of major complications such as dural puncture and pneumothorax as there are no important structures near the injection site. Bilateral thoracic ESPB catheters have been described as effective regional anesthesia during cardiac surgery.²⁰⁹ No large studies have examined ESPBs in pediatric patients undergoing cardiac surgery. Unlike paravertebral blocks (PVBs) and ESPBs, which require significant patient positioning to access the back, chest wall blocks have gained popularity because of their ease of access and

potential opioid-sparing properties.²¹⁰ Further studies are needed to assess the efficacy and safety of those different regional anesthesia techniques in children. Table 4.12 summarize the different blocks currently used, as well as advantages and disadvantages of each technique.

In recent years, several institutions have designed multidisciplinary enhanced recovery after cardiac surgery (ERAS) protocols to reduce hospital LOS and complications.²¹¹ As an example, a team from Boston Children's Hospital recently published its initial experience with an enhanced recovery program for congenital cardiac surgery that clearly highlighted the need for the implementation of a multidisciplinary and multimodal strategy that includes regional anesthesia techniques and early extubation.²¹² Although the topic of early extubation remains a subject of intense debate,^{213,214} prolonged mechanical ventilation (>6 h) in "reasonably stable" pediatric patients postcardiac surgery no longer can be considered good clinical practice.

Coagulation Management and Transfusion Practices

Pediatric cardiac surgery is associated with a substantial risk of bleeding, frequently requiring the administration of allogeneic blood products and coagulation factor concentrates. It is important for pediatric cardiac anesthesiologists to understand the coagulation pathway, its maturation, and alterations associated with the pathophysiology of congenital heart defects.²¹⁵ Like most organ systems in the newborn, the hematologic system is not fully developed at birth but matures over the first year of life. In the late 1980s, Dr. Maureen Andrew termed the age-related changes of the hemostatic system *developmental hemostasis*.²¹⁶ This parallel maturation of the coagulation and fibrinolytic systems maintains the delicate hemostatic balance between thrombosis and bleeding in healthy neonates and infants. An understanding of these changes in the coagulation system is critical for practitioners involved in caring for pediatric cardiac patients.

Implementation of blood conservation techniques are integral parts of pediatric cardiac anesthesiology and surgery.¹⁹¹ Maintenance of adequate anticoagulation during CPB, rapid assessment of coagulopathy, and the implementation of transfusion institutional algorithms based on monitoring of coagulopathy and the use of targeted therapy are keys (Table 4.13).²¹⁷

Despite the progress made in the last decades, further clinical studies are urgently needed to assess the safety and efficacy of the most recent diagnostic and therapeutic options in children.

Effects of Anesthetic Medications on the Developing Brain

In the last decade, interest has focused on the role anesthesia and sedation might play in affecting the developing brain and has been the subject of considerable debate and review.²¹⁸ Although studies in animal pups exposed to prolonged volatile anesthetic agents resulted in learning disabilities and associated histopathologic changes in the brain, those results were never confirmed in retrospective studies of children exposed to single anesthetics. Although from the published study one cannot answer whether anesthesia caused learning disabilities, the repeated need for anesthesia could be a marker for them. Prospective multicenter studies are currently underway to answer whether anesthesia affects the developing brain. It is, however, very unlikely that the findings of prospective studies could affect timing of complex neonatal heart surgery due

TABLE 4.12 Common Regional Anesthesia Techniques for Cardiac Surgery

Regional Technique	Advantages	Disadvantages	Physiologic Effects	Catheter
Thoracic epidural anesthesia (TEA)	- Well studied - Proven efficacy - Reduced risk of myocardial infarction, respiratory depression, atrial arrhythmia - Reduced mortality	- Risk of hematoma, potentially exacerbated by cardiopulmonary bypass - Risk of dural puncture	Somatic and sympathetic blockade	Well suited
Spinal anesthesia	- Well studied - Proven efficacy - Decreased stress response to cardiac surgery	Risk of hematoma	Somatic and sympathetic blockade	Well suited
Caudal anesthesia	- May facilitate early extubation in pediatric patients - Decreased stress response to cardiac surgery	- Risk of hematoma - Risk of dural puncture - Typically only used in pediatric patients - Effectiveness not well studied in cardiac patients	Somatic and sympathetic blockade	Well suited
PVB	- Equivalent analgesia to neuraxial anesthesia - Lower incidence of minor complications compared with neuraxial techniques - Improved safety profile with ultrasound guidance	- Risk of hematoma - Risk of pneumothorax - Effectiveness not well studied in cardiac patients	Somatic and sympathetic blockade	Well suited
ESPB	- Potentially opioid-sparing - Less invasive compared with neuraxial and PVB - Theoretically low risk of hematoma and other major complications	- Effectiveness not well studied - Large studies are lacking	Somatic blockade, possible sympathetic blockade	Well suited
ICNB	Potentially opioid-sparing	- Effectiveness not well studied - Risk of pneumothorax - Large doses of anesthetic needed for analgesic effect	Somatic blockade only	Poorly suited
PECs block	- Opioid-sparing - Theoretically low risk of hematoma and other major complications	Effectiveness not well studied	Somatic blockade only	Poorly suited
SAPB	- Potentially opioid-sparing - Theoretically low risk of hematoma and other major complications	Effectiveness not well studied	Somatic blockade only	Poorly suited
TTMPB	Theoretically low risk of hematoma and other major complications	Effectiveness not well studied	Somatic blockade only	Poorly suited
Parasternal nerve block	- Opioid-sparing - May facilitate early extubation - Theoretically low risk of hematoma and other major complications	- Requires 10 injection points - Effectiveness not well studied	Somatic blockade only	Poorly suited

ESPB, erector spinae plane block; ICNB, intercostal nerve block; PECs, pectoral interfascial; PVB, paravertebral block; SAPB, serratus anterior plane block; TEA, thoracic epidural anesthesia; TTMPB, transversus thoracic muscle plane block.

From Caruso TJ, Lawrence K, Tsui BCH. Regional anesthesia for cardiac surgery. *Curr Opin Anaesthesiol*. 2019;32(5):674-682.

to the high risk associated with delayed treatment of cardiac diseases that outweigh the risk of exposure to general anesthesia. It is, however, a conversation that anesthesiologists will frequently have with parents prior to congenital cardiac surgery.

Enhanced Recovery After Surgery

Enhanced recovery programs have been developed to facilitate return to normal function after surgery with proven effectiveness in multiple surgical populations, especially in adult cardiac surgery. More recently, efforts have been made by several

programs to develop enhanced recovery protocols (ERPs) in pediatric cardiac surgical populations. The goals of ERAS are to optimize fluid balance, nutrition, use multimodal pain management, improve lung function, and accelerate the return of normal gastrointestinal function. These interventions have led to shortened and improved recovery, reduced morbidity and LOS, improved patient experience, and optimization of resource use around surgical care.²¹⁹ ERAS protocols have been developed specifically for the pediatric congenital heart surgical population, and summarizes the program developed at Boston Children's Hospital. In a study by Roy and colleagues, the authors

TABLE 4.13 Consensus Recommendations for Blood Conservation Interventions in Children Having Cardiac Operations

Intervention	Type of Evidence	Expected Outcome
Preoperative interventions Treat preoperative anemia	Indirect evidence only	Anemic children undergoing noncardiac operations have increased in-hospital mortality
Treat iron deficiency	Indirect evidence from adults only	- Intravenous iron preparations are preferred for quicker results - rEPO for short preoperative course may be effective adjunct to increase red cell mass
Search for congenital bleeding disorders (von Willebrand is most common undiagnosed congenital defect) Can have an acquired vWD in patients on LVAD Intraoperative interventions	Anecdotal at best	May be important in ventricular assist patients
Prophylactic transfusion to avoid excessive hemodilution while on CPB	Multivariate adjusted study comparing prophylactic transfusion for Hct <20%	Prophylactic transfusion to keep Hct >20% was associated with reduced composite morbidity and mortality compared to therapeutic transfusion after CPB
Intraoperative cell salvage	Observational data only	Benefit is not clear, but extrapolation from adults supports subgroup use in children
Miniaturized CPB circuits Topical hemostatic agents Low-dose antifibrinolytic therapy especially with tranexamic acid Point-of-care tests of clotting (TEG and ROTEM)	Observation data only Only observational data in adults In vitro evidence coupled with observational study in children In children only observational studies	Decreases intraoperative blood utilization Low risk, effective therapy Plasma concentration of TXA of ≤ 20 $\mu\text{g/mL}$ provides effective antifibrinolytic effect Reduced transfusion with use of point-of-care technology
Postoperative interventions		
Limit blood sampling and minimize blood wastage in the PICU	Observational studies only, in children	Reduced blood loss, especially in neonates
Postoperative cell saver	Single randomized trial	- Reduced transfusions - Supported by single randomized trial
Restrictive transfusion practice	Three randomized trials and 1 subgroup analysis of a randomized trial	- Trial data showed tolerance of restrictive transfusion Hb threshold in all trials - Varied Hb thresholds - Varied outcome measures - Some key exclusions in certain trials (neonates, bleeding, hemodynamically unstable)

CPB, cardiopulmonary bypass; Hb, hemoglobin; Hct, hematocrit; LVAD, left ventricular assist device; PICU, pediatric intensive care; rEPO, recombinant erythropoietin alpha; TEG, thromboelastography; TXA, tranexamic acid; vWD, von Willebrand disease.

From Cholette JM, Faraoni D, Goobie SM, Ferraris V, Hassan N. Patient blood management in pediatric cardiac surgery: a review. *Anesth Analg*. 2018;127(4):1002-1016.

reported their initial experience with ERAS in their population.²¹² From October 1, 2018, to February 28, 2019, 155 of 448 patients were eligible for the ERAS program. Key metrics included early extubation (<8 hours), achieved in 54% of patients, and multimodal pain regimen used in all patients (100%) postoperatively but in only 57% intraoperatively. The authors also performed a matched analysis highlighting that median mechanical ventilation time was 7.6 hours (3.8-12.2) in ERAS versus 8.2 (4.0-17.0) in pre-ERAS era. Raw median ICU LOS was shorter with ERAS: 1.12 days (0.93-2.01) versus 1.28 days (0.96-2.09) pre-ERAS, but there was no difference in hospital LOS. There was no increase in complications, readmissions, and reinterventions. The authors published a follow-up study in

2022 where they looked at adherence to guidelines over a 12-month period postimplementation (implementation phase of 5 months).²¹¹ Adherence to many aspects of guidelines improved from between the implementation phase and the follow-up. The following improvements were notable: OR extubation, 27% versus 16% and a decrease in median ventilation time from 7.6 hours (3.8-12.3) to 6.0 hours (0-9.2). In addition, there was a reduction in opioids, reported as oral morphine equivalents, which was most significant for intraoperative oral morphine equivalents. There was no difference in overall ICU and postoperative lengths of stay, except in lower-risk surgical procedures. Surgical outcomes were similar in the two periods. Although the implementation of ERAS protocols remains rare in pediatric

cardiac surgical programs, the recent publication of the American Association for Thoracic Surgery Congenital Cardiac Surgery Working Group 2021 consensus document on a comprehensive perioperative approach to enhanced recovery after pediatric cardiac surgery will certainly increase awareness and hopefully promote the implementation in the years to come.²²⁰

SECTION III: POSTOPERATIVE CARE

Introduction

Whether or not a cardiac operation is successful is determined not only by events in the OR but also by the level of preoperative and postoperative care. When the perioperative team is well prepared and able to provide comprehensive intensive care, even patients who are seriously ill can survive their operations and achieve good long-term results.

Although the cardiac surgery population is growing older, more frail, and more vulnerable, patients should reasonably expect a successful recovery from their procedure.^{221,222} To this end, the contemporary perioperative team will need to identify patient risk and the appropriateness and/or timing of surgery preoperatively, take steps to minimize the risk for intraoperative and postoperative hemodynamic perturbations, and comprehensively plan for successful hospital discharge.

Recovery after cardiac surgery will be affected by the patient's altered physiology and the degree of preoperative morbidity from both circulatory derangements and comorbid subsystem abnormalities. For instance, care early after open intracardiac surgery is complicated by the whole-body inflammatory response to CPB. Failure to realize that, for a time, patients who undergo CPB will be in a special biological situation to which the knowledge and rules applicable to other humans may or may not apply can lead to conceptual and management errors as well as unnecessary tests and interventions.

In the early postoperative phase after cardiac surgery, patient status can be broadly categorized as either optimal or suboptimal/critically ill. Each category carries therapeutic implications:

- **Optimal:** Patients in this category require routine care; although no change or important modification is currently necessary or foreseeable, ongoing vigilance and effective team communication are required to ensure best outcomes.^{223,224} This category represents *normal* recovery.
- **Suboptimal/critically ill:** Patients in this category range from those who are hemodynamically stable with appropriate support (e.g., catecholamine support for low cardiac output or amiodarone infusion for AF with rapid ventricular rate) to those who are hemodynamically unstable or deteriorating (e.g., low cardiac output syndrome) and require rapid assessment and intervention (e.g., need to return to the OR for bleeding).

The patient recovering normally and without complications after cardiac surgery will usually have adequate function of all subsystems according to standard criteria. Although pain is always present and will vary in intensity from patient to patient, no restlessness, agitation, or anxiety appears. Eyes and skin appear normal, and the pulse is full (and may be rapid). Breathing is neither labored nor excessively rapid. The patient is oriented and lucid and exhibits generally appropriate behavior, whether a neonate, infant, or adult. So long as this pattern of normal convalescence

continues, testing and intervention can be safely minimized. Expedient discharge from the ICU can be accomplished, and the subsequent hospital stay is likely to be short.

Any deviation from the pattern of an uncomplicated recovery is an indication that closer observation and possibly more intensive testing and treatment are needed.²²⁵ Bedside assessment and interventions must be intensive and, at times, rapid, and they may be complex. Care must be well organized and follow specific patient-management and communication protocols that allow all members of the intensive care team to be clear about management details.

Protocol use can be facilitated by considering that the patient is a complex, integrated system comprising various separate but interrelated subsystems (i.e., cardiovascular, pulmonary, renal, nervous, gastrointestinal). One effective approach for patient care, then, is "subsystems analysis."²²⁶ For the postoperative team, this analysis begins with a detailed examination of the patient's medical history and preoperative testing, integrating information from both the surgeon and anesthesia team in the OR and from the ICU care team in the early and late postoperative periods.

Cardiothoracic surgery critical care may one day become a subspecialty of its own.^{227,228} Around the world, numerous institutions with extensive surgical experience in both congenital and acquired heart disease have developed their protocols and specific management systems. These include "fast-track" protocols,^{229,230} critical pathways, and, more recently, ERPs that integrate the goals of all healthcare providers, the patient, family and caregivers, and other key stakeholders.²³¹⁻²³³ Yet even with clear protocols in place, optimal care after cardiac surgery requires overall direction by a knowledgeable and experienced perioperative team with cardiac-focused skillsets and access to dedicated cardiovascular ICUs and specialized methods for securing information (such as machine learning algorithms and artificial intelligence).²³⁴⁻²³⁷ That said, the optimal intensive care staffing pattern is still being elucidated.²³⁸

Within the context of these developments, this chapter discusses general principles and specific details that may be helpful for those desiring to adapt existing protocols or develop their own.

Early Recovery after Cardiac Surgery

Cardiovascular Subsystem

Cardiac Reserve. Cardiac reserve is the capacity to increase (or at least maintain) cardiac output in response to a variety of stressful sudden developments. It is determined by the integrative ability of cross-system mechanisms, including increased neurohumoral total (i.e., central and peripheral) body VO_2 , increased ventricular afterload, and decreased ventricular preload.^{239,240} That capacity is provided by all cardiac and extracardiac mechanisms for maintaining and increasing the force of ventricular contraction and cardiac output, primarily myocardial contractility and coronary blood flow.¹ After cardiac surgery, the adequacy of cardiac performance alone is insufficient to ensure normal recovery and survival. Cardiac reserve also must be adequate.

¹The familiar exercise stress test is a test for cardiac reserve.

Cardiac reserve inadequacy is most common in the first 6 to 12 hours after cardiac surgery, particularly during periods of increased VO_2 (e.g., from agitation, tachypnea, or hyperthermia); suddenly increased ventricular afterload (e.g., from paroxysmal pulmonary arterial hypertension in a neonate); or acute reduction in ventricular preload (e.g., from sudden blood loss). Such cardiac reserve inadequacies can be categorized as “pre-pump” issues (i.e., lack of preload), pump issues (i.e., myocardial ischemia), or “post-pump” issues (i.e., vasoplegia), which can produce hypotensive emergencies or cardiac arrest early after cardiac surgery.²⁴¹

Cardiac reserve is also highly dependent on the patient's preoperative condition. When disease causes reserves to be nearly maximally used to maintain adequate cardiac performance preoperatively, that which remains may be insufficient to meet the stresses of the intraoperative and postoperative period successfully. Reserves probably cannot be increased before the operation unless they have been acutely impaired by a reduced myocardial energy charge.²⁴² Energy charge may be increased by the cardioplegic technique used in the OR. Limited cardiac reserves are also specifically compensated for by many features of early postoperative care.

Adequate Cardiac Function

In the early postoperative period, the perioperative team should focus on preload assessment and optimization of DO_2 .²⁴³ The arteries and veins are infrequently the primary limiting factors, so the emphasis is on the ability of the heart itself to provide adequate blood flow to the body.

Adequacy of DO_2 is determined by the patient's cardiac output (CO; or CI if indexed to the body surface area) and arterial oxygen content (CaO_2) by using the following formula:

$$\text{DO}_2 = \text{CO (L/minute)} \times \text{CaO}_2 (\text{mL O}_2/\text{dL})$$

CO is determined by using heart rate multiplied by stroke volume, whereas CaO_2 is determined by using hemoglobin concentration, SaO_2 , and arterial oxygen tension (PaO_2), according to the following formula:

$$\text{CaO}_2 = (1.34 \times \text{Hgb} \times \text{SaO}_2) + (0.0031 \times \text{PaO}_2)$$

The value of 1.34 is the milliliters of oxygen carried by the hemoglobin (Hgb) molecule at 100% saturation. The small amount of dissolved oxygen ($0.0031 \times \text{PaO}_2$) is generally ignored clinically, but it does provide meaningful information about oxygen content and delivery. At the bedside, one can approximate CaO_2 by using the most recently measured hemoglobin concentration and SpO_2 level monitored continuously with a pulse oximeter. This will be further detailed in the following section.

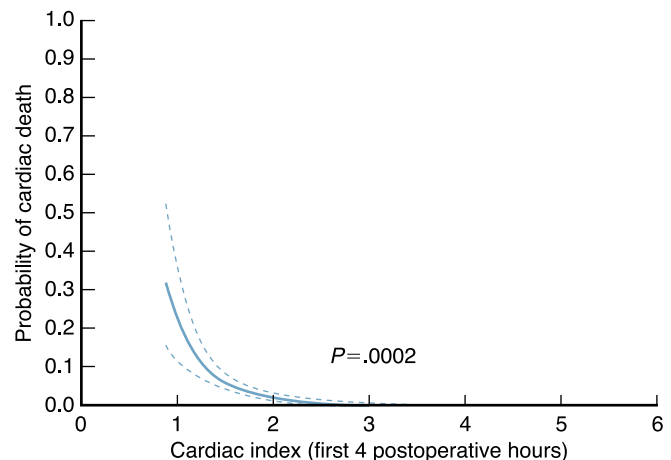
Given this fundamental equation, oxygen delivery can be improved at the bedside by²⁴⁴:

- Modifying stroke volume, by administering fluids (preload) or using inotropic or vasoactive drugs (afterload);
- Improving the SaO_2 and PaO_2 , by administering additional O_2 therapy (which may necessitate mechanical ventilation); and/or
- Increasing the concentration of hemoglobin by transfusing red cells (when appropriate²⁴⁵).

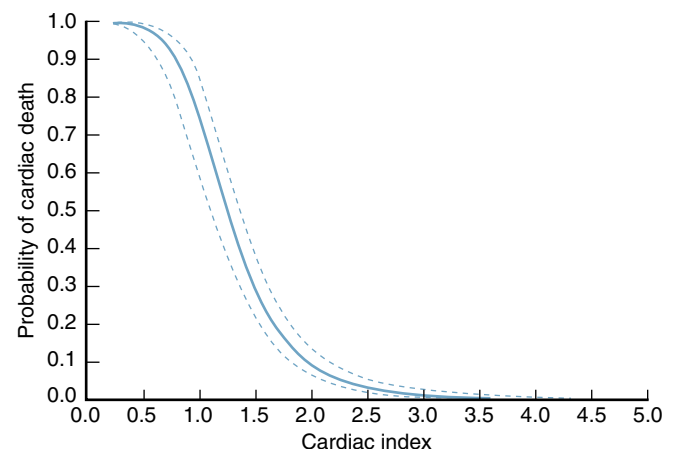
Cardiac Index. CI (cardiac output expressed as $\text{L}/\text{min}/\text{m}^2$) is one measure of adequacy of the cardiovascular subsystem, as evidenced by the oft-noted relationship between CI and survival

(described by Dietzman and colleagues in 1969²⁴⁶). In adults, a CI of at least $2.2 \text{ L}/\text{min}/\text{m}^2$ or higher is generally adequate oxygen delivery in the early postoperative period (Fig. 4.5). This is at the lower end of the range of normal ($2.2\text{--}4.4 \text{ L}/\text{min}/\text{m}^2$).²⁴⁷ In general, infants and small children appear to require a somewhat higher CI for normal recovery (Fig. 4.6). In young patients, CI tends to be lower for about 4 hours after surgery than it was soon after discontinuing CPB; it begins to rise after 9 to 12 hours.^{248,249}

Cardiac indices lower than these values are usually inadequate for maintaining normal recovery^{250,251}; this can be formalized in the inverse relation between CI early after surgery and the probability of hospital death. This relation can be refined by considering not only cardiac output but also mixed venous oxygen levels,



• **Figure 4.5** Relationship of cardiac index in the early hours after mitral valve replacement to the probability of cardiac death (UAB, 1975-1979). Solid line is the point estimate, and dashed lines are the 70% confidence limits. (From Conti VR, Wideman F, Blackstone EH, Kirklin JW. Unpublished study; 1979.)



• **Figure 4.6** Relationship of early postoperative cardiac index (average of all early postoperative values) to probability of cardiac death in infants and small children. This graph suggests that convalescence cannot be considered normal in infants and small children unless cardiac index is about 2.0 to $2.2 \text{ L} \cdot \text{min}^{-1} \cdot \text{m}^{-2}$, somewhat higher than the value for adults. (From Parr GV, Blackstone EH, Kirklin JW, Pacifico AD, Lauridsen P. Cardiac performance early after interatrial transposition of venous return in infants and small children. *Circulation*. 1974;50(2 suppl):II2-II8.)

with lower levels indicating a worsening prognosis at any given value of cardiac output.²

Arterial Blood Pressure. Arterial blood pressure is an insensitive method of estimating adequacy of cardiac output early after surgery, primarily because systemic vascular resistance is usually altered from baseline.²⁵⁴ This may be related to increased levels of circulating catecholamines,²⁵⁵ plasma renin,²⁵⁶ or angiotensin II, or to other mechanisms. Systemic vascular resistance may result in normal, high, or low arterial blood pressure even when cardiac output is low.

Some patients tend to have low systemic vascular resistance and arterial blood pressure early after surgery, even when cardiac performance is good. This may occur more frequently in children with cyanotic heart disease, adults with diabetes, and patients with sepsis or drug interactions (especially preoperative use of ACE inhibitors).^{257,258} Arterial hypotension is an indication for thoughtful evaluation.

Children are not convalescing normally when mean arterial blood pressure is >10% below normal for the patient's age (Table 4.14). For adults, particularly those older than 65 years, arterial blood pressure may mandate maintenance at or above commonly accepted normal values to ensure adequate perfusion of various organs, such as the brain, viscera, and kidneys.

Pedal Pulses. Simple observation of pedal pulses is common in the postoperative period as a useful, but not infallible, method of estimating cardiac output adequacy in children, young adults, and those without peripheral vascular disease. Normal (grade 4) pedal pulses early after surgery are highly but not perfectly correlated

with adequate cardiac output and a high probability of survival.^{226,259} In adults aged ≥ 65 years, using the amplitude of pedal pulses to estimate perfusion adequacy is often confounded by the presence of peripheral arterial occlusive disease.

Whole-Body Oxygen Consumption. Vo_2 is the quantity (in mL) of oxygen consumed by the tissue per minute. Whole-body Vo_2 is infrequently calculated, but knowledge of it is useful; in some circumstances, it provides a better basis for prognostic and therapeutic inferences than cardiac output or mixed venous oxygen levels. Whole-body Vo_2 can be calculated by using a rearranged Fick equation.³

The normal value for Vo_2 at 37°C is 155 mL/min/m² (range 110–160 mL/min/m²); however, this value may increase several-fold during periods of stress.²⁴⁴ The value for whole-body Vo_2 in the patient recovering from cardiac surgery must be interpreted in light of body temperature: Residual hypothermia contributes to low Vo_2 , usually present within the first few hours after open heart surgery. This reduced Vo_2 is due in part to reduced capillary density (reduced area of capillary flow) and increased heterogeneity of capillary flow through the muscle mass and other body tissues in the early hours after CPB.²⁶⁰ Normally recovering patients operated on with mild hypothermic CPB generally require 2 to 8 hours for this effect to disappear and for peripheral perfusion to return to normal.²⁶¹

When Vo_2 is appreciably lower than the normal level for the existing body temperature, a hazardous condition exists; indeed, one useful definition of shock is “a condition characterized by an acute reduction in Vo_2 .” Abnormally low Vo_2 may result from reduction in or extreme heterogeneity of capillary flow (of which “no reflow” is an extreme example) in one or more organs of the body (sometimes termed a *reduction in capillary density*), lengthening of the diffusion path between capillaries and cells, or intracellular metabolic derangement. One or all of these may exist in patients early after cardiac surgery. When important Vo_2 reduction relative to temperature persists for more than a few hours, the probability of death increases.

A further consideration is oxygen extraction (O_2ER), which is the fraction of DO_2 per min (expressed as $\text{O}_2\text{ER} = \text{Vo}_2/\text{DO}_2$).²⁴⁴ Normal O_2ER is approximately 25% (or a ratio 0.25) but can

TABLE 4.14 Normal Values for Blood Pressure According to Age

AGE		Systolic Pressure/ Diastolic Pressure (mmHg)	Mean ^a (mmHg)	10% > Mean Normal Value (mmHg)	10% < Mean Normal Value (mmHg)
≤Years	<Years				
	0.5	80/46	57	63	51 ^b
0.5	1.0	89/60	70	77	63
1.0	2.0	99/64	76	84	68
2.0	4.0	100/65	77	85	69
4.0	12.0	105/65	78	86	70
12.0	15.0	118/68	85	94	74
15.0		120/70	87	96	78

^aMean arterial blood pressure has been calculated as the diastolic pressure plus one third of the pulse pressure.

^b40 mmHg in infants <1 month of age.

Data from Nadas AS, Fyler DC. *Pediatric Cardiology*. Philadelphia: WB Saunders; 1972.

²Although dye dilution or thermodilution is the standard for measuring cardiac output, caution is required in interpreting measurement values. In small individuals, confidence intervals around the measurement may be large; low core temperature may diminish accuracy; and tricuspid regurgitation, absence of an adequate mixing chamber, lack of steady state, and intracardiac shunts (as originally articulated by Stewart and Hamilton^{252,253}) invalidate the measurement.

³The Fick equation²⁴⁴ states that:

Vo_2 (mL/min) = cardiac output (CO) $\times$ [arterial O_2 content (CaO_2) – venous O_2 content (CvO_2)]

Vo_2 (mL/min) = $(\text{HR} \times \text{SV}) \times [((1.34 \times \text{Hgb} \times \text{SaO}_2) + (0.0031 \times \text{PaO}_2)) - ((1.34 \times \text{Hgb} \times \text{SvO}_2) + (0.0031 \times \text{PvO}_2))]$

- HR is the heart rate.
- SV is the stroke volume.
- 1.34 is the mL of oxygen carried by hemoglobin at 100% saturation.
- Hgb is the amount of hemoglobin in g/dL.
- SaO_2 is the O_2 saturation in arterial blood.
- 0.0031 is the solubility coefficient of oxygen.
- PaO_2 is the partial O_2 pressure of arterial blood.
- SvO_2 is the O_2 saturation in venous blood.
- PvO_2 is the partial O_2 pressure of venous blood.

The equation can be rewritten for cardiac output in the following manner:

$$\text{CO} = \frac{\text{Vo}_2}{\text{CaO}_2 - \text{CvO}_2} (\text{L/min}) \text{ and then}$$

$$\text{CO} = \frac{\text{Vo}_2}{((1.34 \times \text{Hgb} \times \text{SaO}_2) + (0.0031 \times \text{PaO}_2)) - ((1.34 \times \text{Hgb} \times \text{SvO}_2) + (0.0031 \times \text{PvO}_2))} (\text{L/min})$$

increase to meet the metabolic demands of the body's tissues (i.e., maintain VO_2).

Mixed Venous Oxygen Level. Mixed venous oxygen level, generally expressed as oxygen tension (PvO_2) or saturation (SvO_2) within the venous system, can be a useful index of circulatory adequacy because it somewhat reflects mean tissue oxygen levels.²²⁶ When PvO_2 is <30 mmHg, oxygen delivery is impaired, probably secondary to inadequate cardiac output; when it is <23 mmHg, the inadequacy is likely to be severe (Fig. 4.7). However, normal or near-normal venous oxygen levels are not reassuring as to the adequacy of cardiac output unless VO_2 is approximately normal for the existing body temperature; that is, normal mixed venous oxygen level does not necessarily denote normal DO_2 , whereas an abnormal level usually portends poor oxygen delivery.

After CPB, changes in venous oxygen level may be useful in detecting low cardiac output, one of many causes of decreased oxygen delivery. Identifying decreases may be of particular value in patients coming to surgery with a high severity-of-illness index; however, the method of measurement in real time is imperfect. In a nonsurgical but critically ill ICU population, Jain and colleagues found a weak relationship between CI and SvO_2 , as measured from a Swan-Ganz catheter.²⁶² Other studies of indwelling fiberoptic reflectance oximetry have found no or only a very weak relationship between SvO_2 or PvO_2 and measured CI.^{263,264} In contrast, more recent studies of the utility of monitoring SvO_2 value as a response to inotropic therapy²⁶⁵ have observed that persistently low SvO_2 values (below 60%) after cardiac surgery were associated with higher 30-day and 1-year mortality.²⁶⁶

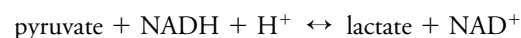
As postoperative critical care has evolved, intraoperative echocardiography and the use of the Swan-Ganz catheter have become standard routines.^{267,268} In lower-risk patients, some centers have begun to use a central-line catheter that can obtain

a venous oxygen sample from the SVC to be used interchangeably with SvO_2 ²⁶⁹ or as a trend correlate.²⁷⁰ However, caution is warranted because of concerns about accuracy across the spectrum of contemporary cardiac surgery patients.^{271,272}

Metabolic Acidosis and Hyperlactatemia. The acid-base status of blood is a frequently used, but somewhat nonspecific and insensitive, indicator of cardiac output adequacy. Metabolic acidosis during and after cardiac surgery is most commonly a result of lactic acidemia. Lactate production is a byproduct of anaerobic metabolism, which most often occurs under conditions in which oxygen delivery is suboptimal (typically secondary to low cardiac output), as is oxygen consumption. Our understanding of lactate metabolism has grown considerably over the last 30 years. Lactate measurement remains an important tool for clinicians in the ICU, but lactate elevation should be thought of as a sign of metabolic stress and not assumed to represent an oxygen debt. The presence of hyperlactatemia portends worse patient outcomes and should trigger an evaluation for the underlying cause.

Lactate has been shown to be an important metabolic fuel for both the heart and brain during periods of stress. Under normal conditions, the heart derives approximately 60% to 100% of its energy from free fatty acids and the remainder from glucose and lactate (0%–20% from each).²⁷³ However, myocardial uptake and utilization of lactate has been shown to increase during times of metabolic stress, such as during exercise, shock, and β -adrenergic stimulation, and can account for up to 60% of the cardiac oxidative substrate.^{274–276} The human brain also becomes a lactate consumer during periods of increased metabolic demand.²⁷⁷ Lactate has been shown to account for approximately 7% of cerebral energy requirements under basal conditions and up to 25% during exercise.²⁷⁸ Thus, lactate appears to be an important mobile fuel source for aerobic metabolism during times of stress, as it can be rapidly created and exchanged between tissues.

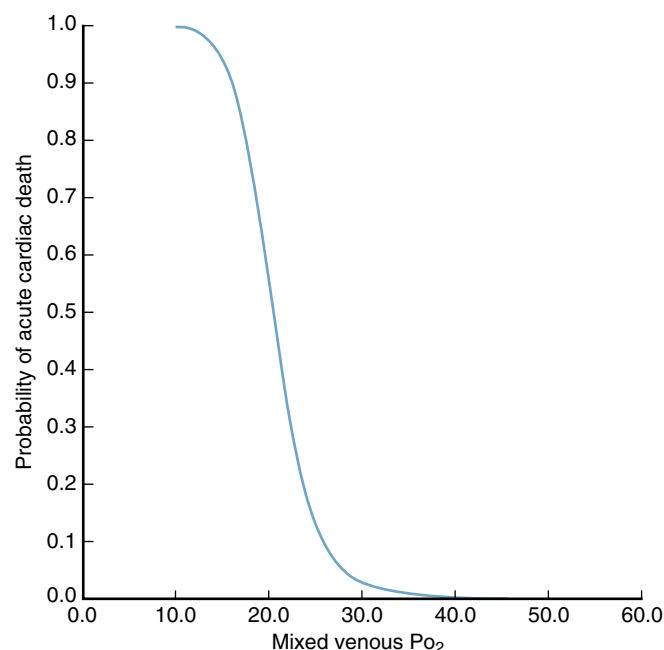
Lactate is formed from pyruvate in the cytosol of the cell following this equation:



During glycolysis, glucose is broken down through several chemical reactions to form 2 molecules of pyruvate as well as 2 molecules of ATP. Cytoplasmic pyruvate can then undergo either (1) oxidative phosphorylation via the citric acid cycle in the mitochondria or (2) enzymatic conversion to lactate, a process that does not require oxygen. The conversion of lactate to pyruvate is held in equilibrium by the enzyme lactate dehydrogenase, which favors the production of lactate with a lactate:pyruvate ratio of approximately 10:1 under normal conditions.²⁷⁹ Therefore, any condition that increases pyruvate production will also increase lactate formation.

The average lactate turnover at a physiologic steady state is approximately 20 mmol/kg per day, with 70% to 75% of this being cleared by the liver.^{280,281} The two major mechanisms of lactate clearance are (1) oxidation via pyruvate and the citric acid cycle and (2) gluconeogenesis in the liver and kidneys via the Cori cycle.^{282,283} At rest, approximately half of available lactate is metabolized through oxidation and 75% to 80% during exercise.²⁷⁸

Hyperlactatemia can occur in any clinical setting where lactate production exceeds lactate clearance. It is a common occurrence in patients undergoing cardiac surgery, with a reported incidence as high as 80% during the perioperative period, and it has been associated with worse patient outcomes.^{284,285} A lactate level of 3 to 5 mmol/L is typically used to define hyperlactatemia (0–2 mmol/L is considered normal).



• **Figure 4.7** Relationship of mixed venous Po_2 to probability of acute cardiac death in infants and young children. Convalescence cannot be considered normal if the value is less than about 28 mmHg. (From Parr GV, Blackstone EH, Kirklin JW. Cardiac performance and mortality early after intracardiac surgery in infants and young children. *Circulation*. 1975;51(5):867-874.)

Lactic acidosis can be classified into two categories (type A and B) based on the presence or absence of clinical signs of tissue hypoxia. This distinction can provide a useful framework for identifying causes and appropriate management in patients with hyperlactatemia.

- **Type A lactic acidosis** is defined by lactate accumulation in the setting of either global or regional tissue hypoxia. In these scenarios, lactate is overproduced and underutilized as a result of impaired mitochondrial oxidation.²⁷⁹
- **Type B lactic acidosis** is defined by lactate elevation in the absence of cellular hypoxia. It is further subdivided into type B1, which occurs in specific disease processes; type B2, which is caused by a drug or toxin; and type B3, which is caused by inborn errors of metabolism.

An important and likely underappreciated mechanism of type B lactic acidosis is adrenergic-driven accelerated aerobic glycolysis.²⁸⁶ When carbohydrate metabolism exceeds the oxidative capacity of the mitochondria, intracellular pyruvate concentration increases, which in turn increases lactate production by mass effect.²⁸⁷ Under stressful conditions, epinephrine-dependent stimulation of the β_2 -adrenoceptor augments the glycolytic flux, both directly and through enhancement of sarcolemmal Na^+K^+ -ATPase.²⁸⁸ Disorders associated with elevated endogenous epinephrine levels, such as severe asthma, sepsis, severe trauma, cardiogenic or hemorrhagic shock, and pheochromocytoma, can cause hyperlactatemia through this mechanism. Aerobic glycolysis and tissue hypoxia are not mutually exclusive and, in many situations, may both contribute to hyperlactatemia.^{279,288}

The diagnostic utility of lactate is diverse. It functions as a marker of resuscitation, it can identify patients with occult hypoperfusion, and it can provide important prognostic information. In the clinical setting, hyperlactatemia is commonly used as a marker of hypoperfusion and tissue ischemia.^{289,290} Postoperatively, its presence is associated with a complicated recovery and should trigger the clinician to evaluate the patient for a new shock state or evidence of tissue hypoxia. Resuscitative efforts to normalize serum lactate are often designed to improve oxygen delivery to reduce tissue hypoxia. These interventions traditionally include initiation of volume resuscitation, addition of inotropes and vasopressors, and transfusion of packed RBCs.²⁸⁹

A patient with elevated lactate should be evaluated for evidence of inadequate perfusion. Capillary refill, urine output, and mental status are easily assessed at the bedside, and abnormal findings should raise concern about inadequate perfusion. Central venous oxygen saturation $<70\%$ or a venoarterial CO_2 difference >6 mmHg may signify inadequate oxygen delivery.²⁹¹

Patients with evidence of global hypoperfusion should be thoroughly evaluated for the underlying cause of shock. Information from a pulmonary artery catheter, such as CI, filling pressures, and mixed venous oxygen saturation, along with an echocardiogram to evaluate ventricular and valvular function, can be invaluable for diagnosing cardiogenic shock and tamponade physiology. Chest tube outputs should be evaluated, and if evidence of significant postoperative bleeding is found, the patient should be evaluated for coagulopathy. Any coagulopathy should be treated, and the patient should be resuscitated with blood products. If bleeding persists, the patient may require re-exploration.

In the absence of global hypoperfusion, the patient should be evaluated for regional ischemia, such as mesenteric ischemia, limb ischemia, or compartment syndrome, and surgical remediation should be initiated when warranted. Serial lactate testing can guide ongoing resuscitation efforts, as lactate clearance has been

associated with improved outcomes; failure to clear lactate is a strong predictor of death.^{292,293}

If there is no clinical evidence of hypoperfusion, interventions aiming at increasing DO_2 are not likely to be of any benefit and may potentially cause harm. In this situation, non-hypoxic causes of lactic acidosis should be considered. The patient's medical history and medications should be reviewed for potential causes of increased lactate production or impaired clearance.

Summary Points

- Lactate measurement is an important tool in the cardiac surgery ICU.
- Lactate elevation should be thought of as a sign of metabolic stress and not assumed solely to represent an oxygen debt requiring intervention to improve oxygen delivery.
- The presence of hyperlactatemia portends worse patient outcomes and should trigger an evaluation for both hypoxic and nonhypoxic causes, with rapid intervention if required.

Determinants of Cardiac Output

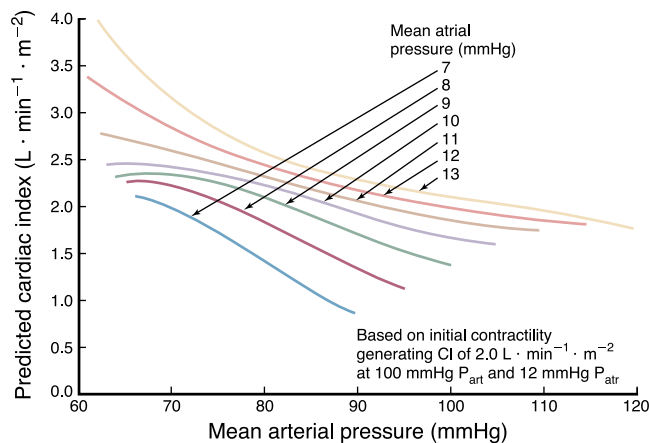
In adults recovering normally after cardiac surgery performed with modern methods of myocardial management, the CI is often 2.2 to 3.5 L/min/m². For most patients, some degree of postoperative myocardial dysfunction and decreased ventricular compliance (a pump issue) should be anticipated in the early postoperative period.^{294,295} Hemodynamic perturbation may be further exacerbated by intravascular hypovolemia (a pre-pump issue) or vasodilation (a post-pump issue). It is incumbent on the perioperative team to rapidly assess and appropriately manage preload, afterload, and inotropic support. However, as wisely stated by Stephens and Whitman, "Equanimity and vigilance are required to avoid overreacting to disquieting but self-limited hemodynamic swings while appropriately intervening on concerning trends or sudden deterioration."²⁹⁴

Risk factors for low cardiac output seem primarily to be those that affect cardiac output in the OR, which in turn correlates strongly with cardiac output 4 to 6 hours later and the next day.²⁹⁶ Cardiac output after CPB is usually correlated with the patient's age (older patients have lower output), cardiac condition, and functional state just before surgery (the higher the New York Heart Association [NYHA] class, the lower the output), as well as with the duration of CPB and the duration of global myocardial ischemia. During the early postoperative period, the heart rate within usual ranges correlates directly with cardiac output (the higher the atrial pressure, the higher the cardiac output [Fig. 4.8]) but inversely with arterial blood pressure.²⁹⁷

Heart rate and stroke volume are determinants of cardiac output; stroke volume is affected by ventricular preload, afterload, and myocardial contractility. Many patients with normal ventricular function and stable hemodynamics do not require special measures to adjust these fundamental determinants. In contrast, patients with impaired or inadequate cardiac performance require at least adjustment of preload and afterload, and at times, adjustment of heart rate and/or pharmacologic or interventional augmentation of contractility. In many cardiac surgery patients, it is specifically the ventricles (one or both) that limit cardiac output (see "Relative Performance of Left and Right Ventricles" later).

Ventricular Preload

Ventricular preload, which correlates directly with the force of contraction, is equated with sarcomere length at end-diastole and,



• **Figure 4.8** Relationship among mean arterial blood pressure, mean atrial pressure, and cardiac output after cardiac surgical procedures in infants. Nomogram depicts specific solutions of the multivariable regression equation developed by Appelbaum and colleagues.²⁹⁷ All patients in the study were in good clinical condition. Mean arterial pressure is depicted as a continuous variable along the horizontal axis, and mean atrial pressure (the higher of the two) is represented by isobars. Note that in general, the lower the arterial pressure, the higher the cardiac index; and the higher the atrial pressure, the higher the cardiac index. *CI*, Cardiac index; *P_{art}*, mean arterial blood pressure; *P_{atr}*, mean atrial pressure. (Data from Appelbaum A, Blackstone EH, Kouchoukos NT, Kirklin JW. Afterload reduction and cardiac output in infants early after intracardiac surgery. *Am J Cardiol*. 1977;39(3):445-451.)

thus, with a change in ventricular volume between end-systole and end-diastole. This volume change is determined by transmural pressure during diastole, compliance and thickness of the ventricular wall, and curvature of the wall (the La Place effect). Transmural geometric arrangement of fibers also plays a role but changes little during the postoperative period.

Intraventricular and intrapericardial pressures determine transmural pressure. Intraventricular pressure at end-diastole (which is a determinant of the force of contraction) is related to phasic changes in atrial pressure, and these are affected by blood volume and systemic venous capacitance. The latter decreases early after CPB.²⁹⁸ As transmural pressure is affected by intrapericardial pressure, it is also affected by closure of the pericardium and sternum, which increases intrapericardial pressure and decreases transmural pressure. Daughters and colleagues²⁹⁹ have shown that pericardial closure during cardiac surgery, independent of sternal closure, increases intrapericardial pressure, decreases transmural pressure, and unfavorably affects cardiac performance. Changes in myocardial compliance during and after cardiac operations are due primarily to changes in myocardial water content.

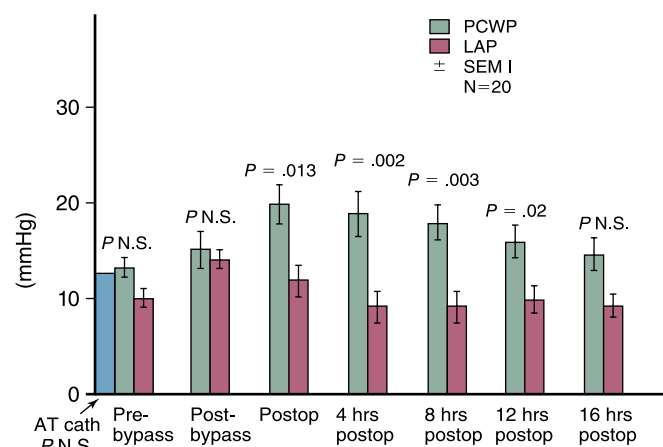
In patients with normal AV valves, most acute changes in pre-load after cardiac surgery are equated with acute changes in mean left atrial pressure (in the case of the LV) or right atrial pressure (in the case of the RV). In this setting, when the atria are functioning normally as reservoirs,³⁰⁰ ventricular end-diastolic pressure is similar to the mean pressure in the corresponding atrium (assuming normal ventricular compliance, which often is not the case in older adults). Therefore, mean atrial pressure is measured in cardiac surgery patients to determine ventricular end-diastolic pressure. Right atrial pressure is usually measured by using a fine polyvinyl catheter (a central line) introduced through the internal jugular or subclavian vein. Left atrial pressure can be measured through a fine catheter introduced through the right superior

pulmonary vein (or in neonates and young infants, the left atrial appendage), but this is rarely done in the current postoperative clinical setting. In the absence of pulmonary vascular disease and important pulmonary congestion or edema, left atrial or ventricular compliance abnormalities, AV valvular pathology, pulmonary artery diastolic pressure (PADP) can be used as a proxy for left atrial pressure.

In older adult patients undergoing cardiac surgery, values obtained from invasive venous or pulmonary artery monitoring should be considered in combination with trends from baseline or over time rather than as absolute values to guide therapy. In addition, pulmonary capillary wedge pressure exceeds left atrial pressure after CPB, and this discrepancy increases through the first 12 postoperative hours. It is thought that this difference is due to accumulation of interstitial lung water after CPB and with volume resuscitation and alteration in myocardial performance and compliance (Fig. 4.9).³⁰¹

Ventricular Afterload. In the intact ventricle, afterload is defined as systolic wall stress. This is the analog of the load that resists shortening in the isolated papillary muscle. Other things being equal, increased afterload results in decreased stroke volume. In the intact ventricle, afterload is related to (1) ventricular transmural pressure during systole, (2) ventricular wall curvature, as determined by ventricular volume (La Place effect), (3) ventricular wall thickness, and (4) the shape of the ventricle.

Ventricular wall determinants of afterload are not likely to change during and early after surgery. Instead, acute changes in LV and RV afterload are usually produced by changes in intraventricular pressures during systole. These changes are equated with changes in proximal aortic and pulmonary arterial systolic pressures. During and early after surgery, proximal pulmonary arterial pressures can be monitored directly (by using a pulmonary artery catheter), but proximal aortic pressures are not. They must be inferred from measured radial (or brachial or femoral) artery pressures. As many factors contribute to the magnitude of systolic



• **Figure 4.9** Pulmonary capillary wedge pressure (PCWP) compared with left atrial pressure (LAP) expressed as mean $\pm$ standard error (SEM). In 20 consecutive patients, PCWP exceeded LAP in the early postbypass period and was most significantly increased at 4, 8, and 12 hours after operation. These data suggest the LAP more accurately reflects left ventricular filling and is more accurate than PCWP to monitor hemodynamics postoperatively. (From Mammanna RB, Hiro S, Levitsky S, Thomas PA, Plachetka J. Inaccuracy of pulmonary capillary wedge pressure when compared to left atrial pressure in the early postsurgical period. *J Thorac Cardiovasc Surg*. 1982;84(3):420-425.)

amplification, systolic blood pressure is usually higher at the radial artery than in the ascending aorta, except in the presence of peripheral vasoconstriction secondary to low cardiac output or high-dose α -adrenergic agents. In most instances, systolic pressure variability between the aorta and peripheral arteries is not clinically important, but an awareness of it is advantageous in some situations. Mean pressures are similar in the two areas.

A tendency toward arterial hypertension is present in as many as 90% of adult patients soon after surgery, related to increased systemic arteriolar resistance.^{302,303} This complication increases (1) ventricular afterload and thereby decreases stroke volume, (2) aortic wall tension and thereby increases the likelihood of tearing aortic purse-string sutures and/or aortotomy suture lines, and (3) LV metabolic demands, which can exacerbate any latent myocardial ischemia.³⁰⁴ Nonetheless, there is a surprising lack of robust data to guide blood pressure management after cardiac surgery. Although mean arterial blood pressure (but typically not systolic pressure) is monitored due to the interrelations mentioned earlier between peripheral and central arterial pressures, contemporary early postoperative blood pressure management seeks to target a MAP of 65 to 90 mmHg and a systolic blood pressure <140 mmHg.^{294,295} However, the patient's preoperative blood pressure must be considered, and to avoid cerebral complications, markedly hypertensive patients must not be rendered hypotensive through overtreatment. For example, a higher target MAP may be indicated in patients with baseline hypertension or abnormal cerebral autoregulation.^{305,306} Conversely, a lower MAP may be desirable in patients with poor ventricular function that is sensitive to afterload or to protect a fragile aorta.²⁹⁴

Negative intrathoracic pressure can increase LV load-resisting shortening by increasing LV transmural pressure. Positive-pressure ventilation negates this effect, but labored spontaneous ventilation may augment afterload, which in turn may decrease cardiac output.^{307,308} Sodium nitroprusside has traditionally been used for pharmacologic management of postoperative hypertension in a cardiac surgery ICU (see Appendix), but nitroglycerin may be preferred when myocardial ischemia is present, as it can decrease coronary resistance. Other agents include low-dose labetalol (5–10 mg) if an adequate heart rate and functional epicardial pacing wires are present or hydralazine (5–20 mg) when rapid normalization of blood pressure is required. Here again, it is important that the perioperative specialist judiciously avoids overshoot and iatrogenic hemodynamic swings. A reasonable approach is to set an initial MAP target of 10% above the normal value.

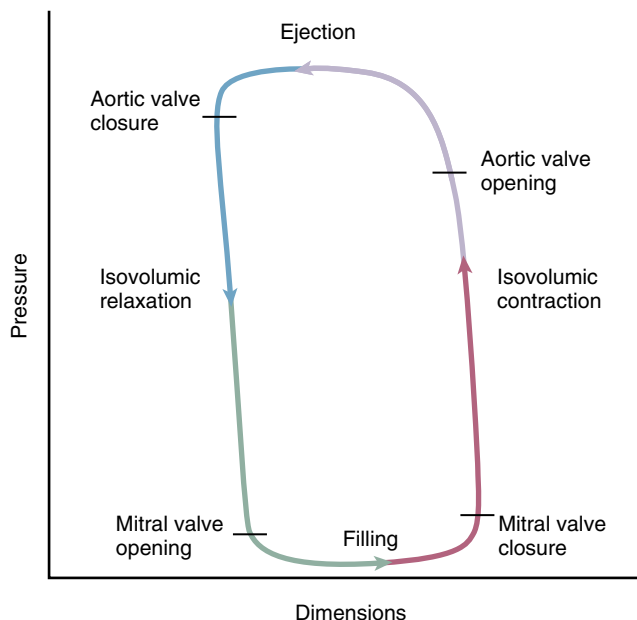
A newer agent being used in the postoperative cardiac surgery patient is clevidipine. Clevidipine quickly reduces blood pressure by selectively acting on the L-type Ca^{2+} channels on arteriolar smooth muscle.³⁰⁹ Clevidipine has a potential advantage over other agents (including nicardipine) due to its ultrashort action and rapid hydrolysis by blood and extravascular tissue esterases, and it does not depend on hepatorenal metabolism and excretion.^{309,310} In addition to its rapid action and metabolism, clevidipine may be more cost-effective than nitroprusside in patients undergoing cardiac surgery and in those with an aortic dissection; however, this requires additional study.^{311,312}

Myocardial Contractility. When a change in stroke volume cannot be explained by a change in end-diastolic fiber length (a preload or pre-pump issue) or load-resisting shortening (an afterload or post-pump issue), it is considered to result from a change in the contractile state (a pump issue). Contractility in a given ventricle can be acutely depressed or increased.

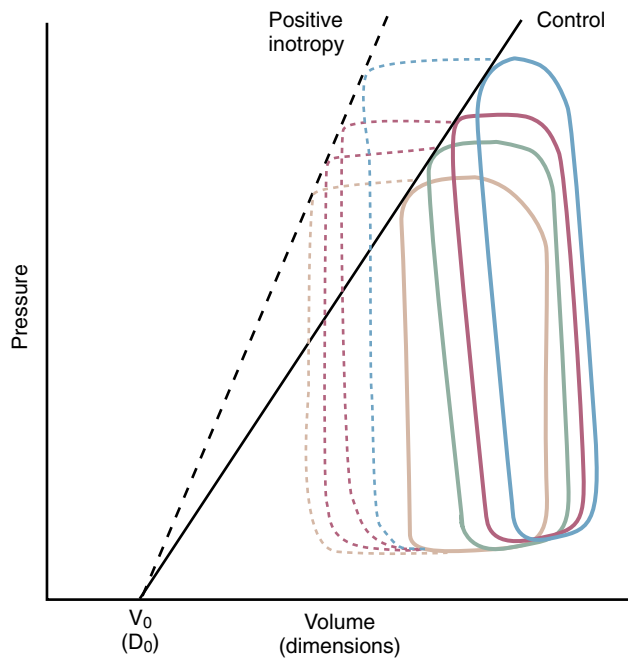
Bedside assessment of myocardial contractility and the resultant quantification of ventricular pump function are desirable goals postoperatively. The simplest way to represent the heart's capacity as a pump is to determine any of several modifications of the Frank-Starling mechanism. For instance, the measured change in cardiac output (or stroke volume) with a fluid or blood-product challenge serves as a surrogate for contractile function. In truth, this is not reflective of intrinsic contractile properties of the myocardium because the pressure-volume relationship is affected not only by preload but also by load-resisting shortening, myocardial compliance, and intact vagal and sympathetic reflex activity. Changes in instantaneous ventricular or aortic pressure over time may reflect myocardial contractility, but this value is exquisitely sensitive to afterload and preload.

The relationship between ventricular pressure and volume (the pressure-volume loop) is currently the nearest approximation to an in vivo assessment of contractility.³¹³ Additionally, the area within the loop represents stroke work. The end-systolic pressure and the pressure at end-diastole of several different loops represent contractility and stiffness, respectively. The loops are composed of four segments: isovolumic contraction, ejection, isovolumic relaxation, and filling (Fig. 4.10). When ventricular volume or resistance is altered, a group of points at end-systole fall along a line, the slope of which Suga and colleagues³¹³ called *E_{max}*. *E_{max}* is an index of contractility at zero volume (Fig. 4.11); changes in the slope in a steeper direction reflect increased inotropy and a shift rightward represents negative inotropy. With the use of catheters for LV pressure measurement and TEE for instantaneous border detection, pressure-volume loops, and *E_{max}* can be interpreted online in the ICU.

Relative Performance of Left and Right Ventricles. During and early after cardiac surgery, one or both ventricles are typically what limits cardiac performance. It is usually advantageous to keep this concept clearly in mind when considering and treating



• **Figure 4.10** Diagrammatic representation of a pressure-dimension relationship of the left ventricle, on which events of the cardiac cycle have been indicated. (From Foex P, Leone BJ. Pressure-volume loops: a dynamic approach to the assessment of ventricular function. *J Cardiothorac Vasc Anesth*. 1994;8:84.)



• **Figure 4.11** When resistance to ejection is altered, pressure-dimension loops at end-systole extend to a straight line termed the *end-systolic pressure-dimension line*. Slope of this line is an index of contractility. Increase in inotropy causes an increase in the slope of the line. It can also be seen that an increase in inotropy causes widening of the loop as ejection shortening is increased. Extrapolation of end-systolic pressure-dimension line to zero pressure defines V_0 (or D_0), the dimension the ventricle would attain if intracavitary pressure became zero. (From Foex P, Leone BJ. Pressure-volume loops: a dynamic approach to the assessment of ventricular function. *J Cardiothorac Vasc Anesth*. 1994;8:84.)

patients. When the AV valves are normal, the most important indicator is the relationship between left and right atrial pressures because this represents the closest available approximation of ventricular end-diastolic pressure and, by implication, sarcomere length.³¹⁴ When the cardiac valves are normal, the ventricle with the highest corresponding atrial pressure is more commonly the one limiting cardiac performance. Echocardiography can often provide useful, supportive information. In an analysis of more than 1.2 million patients undergoing cardiac surgery, the use of echocardiography (particularly in higher-risk patients) was associated with lower mortality and identification of important valvular pathologies needing intervention.³¹⁵

Heart Rate. The optimal target for the postoperative patient is sinus rhythm at a normal heart rate (see Table 4.15). The normal compensatory response to increased O_2 demand is increased heart rate. However, in older adults and patients with diseased myocardium, this response is often absent. In this situation, it may be prudent to manipulate heart rate in otherwise normally recovering patients with slow sinus (or junctional) rhythm to approximate AV electrical synchrony (even if not the same as normal sinus rhythm), with the goal of improving cardiac output. For this, atrial pacing via two temporary atrial leads placed during surgery is used (see DO_2 equation earlier). Atrial pacing is also helpful in suppressing premature atrial and ventricular beats and may limit the onset of an established arrhythmia.

Cardiac Rhythm. Disturbances of cardiac rhythm also may contribute to low cardiac output. Junctional (AV nodal) rhythm reduces cardiac output by 10% to 15%. Atrial contraction may

TABLE 4.15 Ranges of Heart Rate during Sinus Rhythm in Normally Convalescing Patients

AGE		
≤Years	<Years	Heart Rate (beats · min ⁻¹)
	1/12	120-190
1/12	6/12	110-180
6/12	12/12	100-170
1	3	90-160
3	6	80-150
6	15	80-140
15		70-130

From Kirklin JW, Archie JP. The cardiovascular subsystem in surgical patients. *Surg Gynecol Obstet*. 1974;139(1):17-23.

contribute more in patients who have abnormal ventricular compliance and who are, therefore, more reliant on atrial contraction for adequate preload. Nonetheless, a junctional rhythm is less efficient than sinus rhythm because the atrial contribution to ventricular filling is absent with junctional rhythm. A junctional rhythm is usually transient, and its effects are easily overcome by atrial pacing (unless the rate is rapid); its presence does not connote an immediate risk.³¹⁶

Bradyarrhythmias caused by edema or damage to the AV node or His bundle, hypoxemia, or drugs can lower cardiac output. Tachyarrhythmias in the form of AF or flutter, or paroxysmal atrial tachycardia may also result in hypotension. Risk for tachyarrhythmia increases during infusion of catecholamines. A complete discussion of postoperative rhythm disturbances and their treatment is found later under “Cardiac Arrhythmias.”^{317,318}

Causes of Acute Dysfunction (Low Cardiac Output) after Cardiac Surgery

Inadequate Operation. The surgeon’s responsibility for performing an adequate operation demands that he or she, with the perioperative team, continue to search for evidence of this postoperatively, particularly when the patient has signs and symptoms of a low cardiac output state. Using the methods described in this and other chapters, the postoperative care team should search for residual intracardiac or extracardiac shunting, pathway obstructions, valvular regurgitation, graft or conduit dysfunction, or cardiac compression. If the operation is found to be inadequate in any of these regards, prompt reoperation is usually indicated.

Myocardial Dysfunction. Myocardial dysfunction was once thought to explain low cardiac output after cardiac surgery when atrial pressures were elevated above the usual postoperative values—in the absence of any other explanation. The availability of two-dimensional echocardiography in the OR and ICU, particularly TEE, enables direct visualization of ventricular wall motion and assessment of end-diastolic and end-systolic volumes.^{315,319} These kinds of studies support the inference that low cardiac output is caused by myocardial injury or stunning or by impaired cardiac reserve in the presence of increased stress. This inference can be supported by the finding of increased creatine kinase-myocardial

band isoenzyme or troponin in the serum (see also the Fourth Universal Definition of Myocardial Infarction³²⁰).

Reduced Preload

Hypovolemia. The most common cause of reduced preload is hypovolemia, which may be a relative intravascular loss secondary to vasodilation or bleeding into either drained or undrained cavities (pleural spaces, retroperitoneum, or free peritoneal space). The most obvious cause of hypovolemia is bleeding associated with cardiectomy or CPB, reflected by excessive chest tube output. Low cardiac output or low arterial blood pressure associated with low filling pressures (left or right atrial pressure, CVP, or pulmonary capillary wedge pressure) is a precondition for hypovolemia. Echocardiography showing hyperdynamic wall motion and small chamber size is simply confirmatory.

Infrequently, excessive diuresis leads to relative hypovolemia and lowering of cardiac output. In this instance, the picture may be complicated by hypokalemia leading to arrhythmias. Occasionally, a sympathetic response supports blood pressure, but often this is blunted early after anesthesia. Even if reflex tachycardia develops, cardiac output ultimately suffers.

Diastolic Dysfunction. In the presence of left ventricular hypertrophy, fibrosis, or myocardial edema, measured filling pressures do not accurately reflect ventricular volume. In this situation, ventricular compliance is diminished. The root problem is inadequate resting (diastolic) sarcomere length. In these situations, echocardiography is especially useful for visualizing typical manifestations such as a small ventricular chamber in the presence of high filling pressure, tachycardia, small stroke volume, low arterial blood pressure, and low cardiac output.^{321,322} Appropriate interventions should aim to improve ventricular filling to improve heart rate; beta-blockade and subsequent volume infusion should be considered, depending on the patient's hemodynamic state.

Acute Cardiac Tamponade. Acute pericardial tamponade (with its resultant acute decrease in ventricular preload in the presence of elevated atrial pressures) must always be considered when low cardiac output exists soon after surgery. Acute cardiac tamponade can be caused by undrained intrapericardial bleeding, marked myocardial edema, and chamber dilation inside the closed chest; the pericardium can constrict under these circumstances even when it has not been re-sutured. Acute dilation of the right ventricle during an acute pulmonary hypertensive crisis can produce acute atypical tamponade in neonates and infants.³²³ For these reasons, it can be advantageous to leave the sternum open and cover the mediastinum with an impermeable sheet sutured to the skin edges or to open the chest in the ICU when this form of cardiac tamponade is limiting cardiac output.³²⁴⁻³²⁶

In patients with early adequate, stable cardiac output, rapid deterioration that cannot be easily explained otherwise is likely to be caused by cardiac tamponade. It is usually associated with rapidly rising right and left atrial pressures that often, but not always, equalize. Drainage from chest tubes tends to be initially brisk and then ceases (due to blood clots in the lumen of the chest tube), and serial chest radiographs show progressive widening of the cardiac and superior mediastinal shadows. Arterial pressure falls, and a narrow pulse pressure may replace a paradoxical pulse. Characteristically, arterial pressure shows minimal response to an inotrope bolus injection. Echocardiographic examination is indicated as soon as cardiac tamponade from retained intrapericardial blood is suspected, and it is often diagnostic.

Cardiac tamponade can also manifest in multiple atypical presentations that must always be considered when acute low cardiac output develops. For example, right and left atrial pressures may

differ widely in the setting of a localized, impacted clot adjacent to the right atrium. Neither TEE nor transthoracic echocardiography is reliable for detecting impacted clots, creating the potential for misdiagnosing tamponade as acute RV failure secondary to other causes, such as pulmonary hypertension. Therefore, when the diagnosis of cardiac tamponade is considered as a possible etiology of low cardiac output that does not promptly respond to nonsurgical intervention, emergency reoperation or reopening the sternum at the bedside is advisable.²⁴¹

Summary Points

- Postoperative cardiac tamponade is a clinical diagnosis.
- No one test or investigation can absolutely rule in or rule out a diagnosis of postoperative cardiac tamponade.
- The perioperative team must maintain a high index of suspicion of this syndrome, using the convergence of clinical examination and adjunct investigative information in unstable patients with signs and symptoms of low cardiac output.

Increased Ventricular Afterload. Increased RV afterload may appear quickly as a result of a sudden rise in pulmonary artery pressure and vascular resistance. Consequently, during an episode of paroxysmal pulmonary hypertension (e.g., intratracheal suctioning, coughing, or ventilatory dyssynchrony), cardiac output may fall rapidly, and “sudden” death may occur, particularly in neonates and infants. These outcomes are probably not purely the result of increased RV afterload because they reflect impaired RV reserve as well.

Increased left ventricular afterload may result from a sudden elevation of systemic arterial pressure (hypertension) during suctioning, restlessness, or hypoxia. Hypertension can result in increased bleeding and strain vascular anastomoses, and cause a sudden increase in left ventricular afterload that, combined with impaired left ventricular reserves, can lower cardiac output soon after surgery. Sustained increase in systemic vascular resistance and left ventricular afterload is present early after cardiac operations in at least half of adults who undergo surgery for acquired heart disease.^{303,304}

Often, disturbances in afterload (afterload mismatch) and preload (preload reserve)³²⁷ are neither independent nor isolated events. Increased RV afterload leads to decreased left ventricular preload and interventricular dependence. Similarly, increased left ventricular afterload leads to decreased RV preload. A common situation in which a corrective operation leads to increased left ventricular afterload is restoration of mitral valve competence or closure of a VSD. In a physics analogy, mitral regurgitation or left-to-right flow through a VSD represents a pair of resistors in a parallel circuit in which $R_T = 1/r_1 + 1/r_2$, where r_1 and r_2 represent resistances in the two outflow streams and R_T is total resistance. Closure of 1 outflow (r) increases downstream resistance to ventricular shortening; by inference, wall tension and myocardial oxygen consumption increase.

Risk Factors for Low Cardiac Output

Various circumstances increase the probability of low cardiac output after cardiac surgery. These have been determined largely through multivariable analyses of outcomes after surgery for specific conditions.

Patient-Specific Factors. Chronic impairment of ventricular preload, afterload, and/or contractility by any mechanism (ventricular hypertrophy, stiffness, chronic heart failure) increases the risk for low cardiac output after a cardiac surgical procedure. These are considered immutable risk factors for low cardiac output because they do not change quickly after the operation. However, when an ambulatory patient preoperatively has inadequate

cardiac output postoperatively, the patient's previous hemostasis has been disrupted by factors impairing myocardial performance that occurred during the operation, in the ICU, or both. Conversely, surgical correction of significant valvular regurgitation or closure of a defect with a large left-to-right shunt has an immediately favorable impact on cardiac output.

Intraoperative maneuvers can sometimes ameliorate acute preoperative reduction in ventricular contractility. For the most part, these maneuvers are directed toward increasing an acutely reduced energy charge.

Procedural Factors. The most important intraoperative risk factor for low cardiac output soon after surgery, other than an incomplete operation, is a discrepancy between the duration of any global myocardial ischemia and the efficacy of the measures used for myocardial management (see complete discussion in Chapter 3). This is often reflected in the finding of long global myocardial ischemic time as a risk factor for low cardiac output and for death postoperatively. Coronary air embolization during CPB is said to affect cardiac performance after the operation adversely, but most air entering the coronary arteries while the heart is not supporting the circulation passes quickly into the coronary sinus and may have little deleterious effect.

The extent of the whole-body inflammatory response to CPB affects the heart as well and relates to the probability of low cardiac output after surgery. This is one aspect of the correlation between CPB duration and postoperative death; another is that long CPB duration is sometimes the result, rather than the cause, of poor cardiac performance.

Low cardiac output due to surgical factors can result from (1) an incomplete or inadequate operation; (2) acute myocardial injury, with or without necrosis, from impaired coronary blood flow resulting from, for example, failure of CABG; (3) incomplete relief of ventricular inflow or outflow obstruction; (4) important residual or created AV valve or semilunar valve regurgitation that may increase the stroke volume requirements of the ventricle; and (5) a residual VSD or large left-to-right shunt that may similarly increase left ventricular stroke volume requirements.

Fine polyvinyl catheters are sometimes placed temporarily in the left atrium, right atrium, occasionally the pulmonary trunk via the right ventricle, and uncommonly, the LV to investigate whether an inadequate or incomplete operation is causing the low cardiac output; temporary epicardial atrial and ventricular wires are placed for control of heart rate. More commonly, TEE with color-flow Doppler imaging has been used to detect an incomplete or inadequate operation.

The possibility that residual left-to-right shunting is contributing to low cardiac output must always be considered after repair of congenital heart lesions and acquired or iatrogenic septal defects. Left-to-right shunting can be estimated in terms of the pulmonary (Q_p) to systemic (Q_s) flow ratio by simultaneously removing samples from the radial artery, right atrium, and pulmonary artery. Given the materials now used as patches for VSD repair, an appreciable left-to-right shunt ($Q_p/Q_s > 1.5$) early postoperatively must be assumed to represent an incomplete repair or an overlooked defect. The simplified shunt equation is:

$$\frac{Q_p}{Q_s} = \frac{SaO_2 - SRAO_2}{SaO_2 - SAPO_2}$$

where SaO_2 is the oxygen saturation percentage of arterial blood, $SRAO_2$ is the volume of blood withdrawn from the right

atrium, and $SAPO_2$ is the volume of blood withdrawn from the pulmonary artery.

TEE with color-flow Doppler imaging can often identify a left-to-right shunt, but quantitation is not precise. Intraoperative TEE has become standard of care during adult cardiac surgery procedures and for guiding intracardiac repair of CHD for patients ≥ 3 kg in size.³²⁸⁻³³⁰

Course of Low Cardiac Output

The heart is especially vulnerable early postoperatively because its impaired functioning adversely affects coronary blood flow; this, in turn, may further worsen cardiac function. Therefore, vigilant attention by the perioperative team, even in lower-risk cases, is required to identify and manage low cardiac output early after cardiac operations.

With treatment, most patients with low cardiac output early after surgery recover, and unless the low output was produced by a large area of myocardial necrosis or compromised organ perfusion (e.g., acute kidney injury), most patients have no persistent, demonstrable ill effects.

Treating Low Cardiac Output

Managing low cardiac output starts with prevention in the OR.²⁹⁴ Goal-directed perfusion can be described as a protocol for optimizing CPB flow rates, hemoglobin levels, and vasopressor titration, all with the goal of achieving a targeted DO_2 threshold. Recent studies have indicated that maintaining DO_2 at ≥ 280 mL/min/m² during CPB was associated with a reduction in mortality and organ dysfunction after cardiac surgery.³³¹⁻³³³

The extension of this process in the ICU is goal-directed hemodynamic and fluid therapy, which similarly denotes an algorithmic approach to maintaining adequate organ perfusion and oxygen delivery. This approach includes optimizing blood pressure and indices of preload (cardiac filling pressures) and assessing cardiac function and output.^{294,334} Although a universal consensus is lacking as to what is required and for how long,³³⁵⁻³³⁷ the integration of clinical assessment, laboratory data, and hemodynamic monitoring information is needed to ensure that the low cardiac output state identified by such an algorithm is reversible.

Investigating whether cardiac tamponade or compression is the cause is one of the first steps. If tamponade caused by retained blood in the pericardium is present, emergency reoperation is indicated (see "Postoperative Cardiac Arrest" later). In patients at an increased risk for low cardiac output, this complication can be prevented by leaving the sternum open after the operation and not closing it until 24 to 48 hours later; notwithstanding, attention to appropriate antimicrobial prophylaxis with an individual institution's microbiogram is necessary.^{325,338,339}

When cardiac constriction is believed not to be present, treatment is directed at increasing cardiac output by manipulating preload, afterload, contractile state, and heart rate and improving tissue oxygen delivery levels. When these measures fail, use of devices to support the circulation must be considered (see "Extracorporeal Life Support" later). All such devices have their risks, and the decision to use them must always be made with an awareness of the possibility the patient could survive but be left seriously disabled and with knowledge of the costs of such interventions.³⁴⁰⁻³⁴⁵

Noninvasive Methods. When postoperative cardiac output is low, preload-appropriate fluid administration is the most common

hemodynamic intervention. Consideration of what type of fluid and how much to infuse is typically driven by the presence of one or more of the following factors: existing or ongoing blood loss, alteration in vascular capacitance due to vasodilation with rewarming, secondary to an inflammatory response from CPB, and reduced ventricular compliance due to myocardial stunning or ischemia.^{294,346} Conversely, excessive fluid administration can result in heart failure, pulmonary edema, acute kidney injury, and other organ dysfunction.^{347,348}

Given the patient-specific and dynamic variations in physiology after surgery, a target pressure within 15 mmHg of the higher of the two atrial pressures is reasonable. If a patient requires more than 3 L of crystalloid, this should prompt the perioperative team to consider alternative diagnoses. As an example, for a patient whose left ventricular wall is unusually thick or has lost its contractility or compliance, it may be helpful to raise mean left atrial pressure to 20 mmHg.²⁹⁵ However, the tendency toward pulmonary edema is increased when left atrial pressure is elevated to this level. When the RV is the limiting factor in cardiac performance, right atrial pressure usually can be raised advantageously only to about 18 mmHg. Above this, a descending limb on the Starling curve usually becomes apparent, and cardiac output falls. Also, the tendency toward whole-body fluid retention, pleural effusion, and ascites is increased by high right atrial pressure. When left ventricular performance is the limiting factor and systemic arterial blood pressure is more than 10% above normal (see Table 4.14), vasodilating agents should be used to reduce left ventricular afterload to a level between normal and 10% above normal.

RV dysfunction refers to impaired RV filling or ejection, but it is not associated with overt signs or symptoms of heart failure.³⁴⁹ RV failure describes reduced forward flow through the pulmonary circulation, resulting in a low cardiac output state, as evidenced by hypotension, hypoxemia, cool extremities, neurologic manifestations, and signs of systemic congestion (jugular venous distension, hepatojugular reflux, peripheral edema, pericardial effusion, oliguria, congestive hepatopathy/splenomegaly, ascites, anasarca).³⁵⁰⁻³⁵² Acute RV failure is a "...progressive, often rapid, syndrome characterized by systemic congestion secondary to impaired RV filling and/or reduced RV flow output."³⁴⁹ Both RV dysfunction and RV failure limit cardiac performance and are associated with significant morbidity and mortality in postoperative cardiac surgery patients.^{352,353}

Contemporary management of RV dysfunction or RV failure includes optimizing heart rate (sinus rhythm, ideally) and volume status (right arterial pressure of 5–12 mmHg), avoiding hypotension (MAP >65), and using inotropes, inodilators, and/or vasodilators to reduce RV afterload to improve cardiac stroke volume.^{349,354}

There is insufficient information to determine the superiority of one agent over another. Traditionally, nitroprusside (0.5–3.0 µg/kg/min), nitroglycerin (0.5–3.0 µg/kg/min), or phentolamine (1.5–2.0 µg/kg/min) have been used; however, these agents may not be advisable in the hypotensive patient. Inodilators (agents with inotropic and vasodilatory properties) such as dobutamine and milrinone can improve hemodynamics in RV failure and have the additional benefit of reducing PVR, although an additional vasopressor (vasopressin or norepinephrine) will probably be needed to support systemic blood pressure.^{355,356}

In infants, maintaining near-anesthesia for 24 to 48 hours with fentanyl or another intravenously administered agent may minimize paroxysms related to pulmonary artery hypertension and the consequent increase in RV afterload. Alternatively, neonates and

infants could be given the long-acting α -receptor-blocking agent phenoxymethamine. It is administered first at the commencement of CPB. An additional dose is usually given about 12 hours after the patient returns to the ICU.

Heart rate is adjusted to optimal levels when necessary by using atrial pacing, ventricular pacing when AF is present, or AV sequential pacing when AV dyssynchrony or dissociation is present. When tachyarrhythmias are present, pharmacologic means of control may be used.

If these relatively simple measures do not quickly bring cardiac performance to an adequate level, inotropic agents are begun (see Appendix 4B), despite well-known disadvantages. There is no ideal inotropic agent, nor are there specific indications for specific agents. In their review, Doyle and colleagues³⁵⁷ classified inotropic drugs according to their effect on intracellular cAMP: cAMP-independent drugs include calcium, digoxin, and α -adrenergic agonists; cAMP-dependent agents include epinephrine, dobutamine, and isoproterenol. These are β -adrenergic agonists that, coupled with dopaminergic drugs (dopamine), have variable effects on peripheral resistance. Phosphodiesterase inhibitors (e.g., milrinone) and calcium sensitizers (levosimendan)^{358,359} may enhance contractility while producing myocardial relaxation (lusitropism) and relaxing vascular smooth muscle. They are not susceptible to receptor downregulation.

Dopamine, although used less commonly in contemporary practice, may initially be infused at 2.5 µg/kg/min. This dose can be increased to 15 to 20 µg/kg/min if needed; however, if a favorable response is not obtained at 10 µg/kg/min, it is not likely to be obtained at higher doses. A previous notion held that the advantage of dopamine was that it increases renal blood flow and cardiac contractility,³⁶⁰ but more recent data and expert consensus indicate that it does not deter organ dysfunction and may be associated with increased rates of AF.³⁵⁷⁻³⁵⁹ Dopamine is typically initiated at low doses (2–4 µg/kg/min) to maintain systemic peripheral vascular resistance, whereas higher doses (>6 µg/kg/min) increase peripheral resistance. Dobutamine is gradually added in similar doses and appears to augment myocardial blood flow and afterload reduction to a greater degree and blood pressure to a lesser degree.³⁶¹

Occasionally, hypotension exists in the presence of normal and adequate cardiac output. Under that special circumstance, norepinephrine can be administered through a central venous catheter; a low initial dose (0.01–0.05 µg/kg/min) is often sufficient. In cases of progressive shock, larger doses can be used, but potentially at the expense of digital ischemia, mesenteric ischemia, and other complications. Vasopressin can be administered alone as a first-line agent or in combination (0.01–0.06 U/min) with other agents to achieve desired blood pressure in patients who develop vasoplegic shock after cardiac surgery. Although vasopressin may improve clinical outcomes, its superiority over norepinephrine is debatable.³⁶²⁻³⁶⁴

Epinephrine is the catecholamine of choice for some, but its powerful vasoconstricting effects at higher doses necessitate thoughtful use, as any inotropic agent has the potential to increase myocardial oxygen demand and is potentially arrhythmogenic. When an insufficient response is obtained from other drugs, or if excessive tachycardia develops, epinephrine is added or substituted. The drug is initially infused at a dose of 0.01 to 0.05 µg/kg/min, which may be increased as needed.

Milrinone is also useful in patients with low cardiac output after cardiac surgery because it combines a peripheral vasodilatory action with its inotropic effect. This drug differs from catecholamines in

structure and mode of action in that it is a phosphodiesterase enzyme inhibitor in cardiac and vascular (including pulmonary) tissue, not a β -adrenergic receptor agonist. Milrinone can be used to treat RV dysfunction or pulmonary hypertension, but it may require coadministration with a vasopressor to maintain an adequate MAP. In addition, the half-life of milrinone (30–60 min) and its impact on platelet function must be considered before it is used.^{294,365} Administration can be initiated with a loading dose of 5 $\mu\text{g}/\text{kg}$ over 10 minutes (though often initiated without a bolus to minimize hypotension), followed by a maintenance dose of 0.125 to 0.5 $\mu\text{g}/\text{kg}/\text{min}$.

Additionally, 10% calcium chloride is administered in a dose of 0.1 mmol/kg, with supplemental doses, if the ionized serum calcium level is lower than 1.2 mmol/L.

Extracorporeal Life Support. Implantable VADs, whose use was first described by Cooley and colleagues in 1969,³⁶⁶ are used for support after cardiac operations, as a bridge to transplant, and as a bridge to recovery. Shorter-term, temporary cardiopulmonary support devices are referred to as ECLS, defined as a “...set of therapies that focus on oxygenation, carbon dioxide removal, cardiac support, or a combination thereof.”^{367,368}

Low cardiac output accompanying reduced ventricular function after cardiac surgery occasionally prohibits separating the patient from CPB. A thorough investigation for correctable causes of low cardiac output should be made and may include the use of Doppler ultrasound flow velocity or electromagnetic flow probes to ensure aortocoronary bypass graft patency and adequacy of flow, and the use of intraoperative TEE to evaluate the accuracy of repairs and segmental ventricular wall motion. Preload and afterload should be optimized, and appropriate pharmacologic agents should be administered. When these measures are insufficient, patients with preoperative shock or anticipated postoperative difficulty in being weaned from CPB may require a short-term ECLS device. Insertion of a long-term implantable VAD under these circumstances is inadvisable.

Temporary support of the failing circulation allows further major operative intervention to be postponed until the patient's condition improves. The temporary assist system can be used as a bridge to ventricular recovery or as a bridge to more durable mechanical circulatory support if weaning is not possible. The options, timing, and management of temporary ECLS have evolved over the past 60 years, such that we now have access to a range of devices, allowing for a more nuanced approach to the treatment of medically refractory cardiac and cardiopulmonary failure.³⁶⁹

The ECLS devices presently available include the intra-aortic balloon pump (IABP), ECMO, and percutaneous or central VADs. With any short-term ECLS device, the use should be considered in the context of the patient's age, comorbidities, neurologic function, and prospects for long-term survival and quality of life.³⁴⁴ The following discussion relates to use of various temporary ECLS devices for cardiopulmonary failure after cardiac operations.

Intra-aortic Balloon Pump. The IABP is the most widely used temporary ECLS device today. First described by Mouloupoulos and colleagues³⁷⁰ in 1962, the IABP was designed to produce diastolic augmentation of coronary and systemic blood flow. The IABP employs the principle of diastolic counterpulsation, which augments diastolic coronary perfusion pressure, reduces systolic afterload, favorably affects the myocardial oxygen supply/demand ratio, and augments cardiac output. Kantrowitz and colleagues performed the first IABP procedure in humans in 1968.³⁷¹

An IABP is used in adult patients with inadequate cardiac performance not responsive to optimized preload, afterload, and heart

rate or to moderate doses of inotropic and vasopressor support (see Appendix 4B). The rapid inflation of the IABP during diastole provides augmented diastolic coronary perfusion pressure; rapid deflation immediately presystole reduces left ventricular afterload (through both passive and active mechanisms with a Venturi-like effect) and left ventricular myocardial oxygen requirement. This action enhances coronary perfusion and, by optimizing left ventricular hemodynamics (particularly factors related to ventricular interdependence), an IABP may also augment RV function.³⁷² The absolute magnitude of support provided by an IABP depends on a multitude of factors (including balloon size and optimal positioning, aortic and systemic vascular compliance, and heart rate), but even in the best circumstances, remains relatively modest, at 0.5 to 1.0 L/min of additional cardiac output.³⁶⁹

Whenever possible, the decision to insert an IABP (if not already placed preoperatively) is made in the OR rather than postoperatively. Postoperatively, an IABP may be preferable to higher-dose catecholamines for patients with severe left ventricular dysfunction with inadequate cardiac output with preserved oxygenation or severe ventricular arrhythmias. This technique has led to the survival of some patients who would otherwise have died^{373–376}; however, more recent trials and guidelines have questioned routine IABP use and its impact on longer-term survival.^{343,377,378}

An IABP is commonly inserted in cases of MI with low cardiac output or shock. Preoperative insertion may also be helpful in patients with unstable angina, left main disease with ongoing ischemia or ischemia leading to ventricular arrhythmia. In the era of more complex arterial revascularization for ischemic heart disease (see Chapter 9), IABP support is helpful intraoperatively for pre-bypass support in patients with low ejection fraction. For patients with acute mitral regurgitation or ventricular septal rupture, insertion upon diagnosis is often lifesaving. Perhaps the most frequent indication for preoperative IABP insertion is poor perfusion from low cardiac output. Notably, better survival rates have been found when the IABP was placed preoperatively.^{344,345}

Arterial puncture is generally used for IABP insertion, despite the somewhat higher prevalence of vascular complications,³⁷⁵ because of the technical ease of balloon insertion and removal. When the patient is still on CPB (or the femoral pulse cannot be palpated because of hypotension), ultrasound guidance is advisable. If ultrasonography is not available, the traditional technique of making a small incision in only the skin over the femoral artery can be used. Through this incision, the femoral artery can often be palpated, and the arterial puncture technique applied.

Contraindications for IABP placement include moderate or worse aortic valve insufficiency, thoracic aortic aneurysmal disease, and acute or chronic aortic dissection. In addition, significant aortoiliac occlusive disease and thoracic or abdominal aortic aneurysm greatly increase the risk for vascular complications or insertion failure when the femoral route is used³⁷⁹; in these instances, circulatory support by ECMO should be considered.

The timing of the IABP begins in a 1:1 (heartbeat to balloon activation) ratio with ventricular diastole, as judged by ECG and arterial pressure pulse signals, and is largely automated with modern devices. The timing of the IABP balloon is important, as early or late deflation reduces the effectiveness of the device and can increase left ventricular afterload, thereby increasing myocardial oxygen demand.

Once the patient's hemodynamic state improves, consideration can be given to weaning the patient from the IABP, possibly as early as 6 to 12 hours after insertion, provided that infusion of inotropic and vasopressor agents has been stable or reduced.

The IABP ratio is then progressively reduced to 1:2 or 1:3, along with ongoing reassessment of the patient's physiologic response to reduced support. In most postoperative patients, the final reduction can be achieved within 12 to 48 hours.

The balloon can then be removed by using either a closed method³⁸⁰ or a preclosure device.³⁸¹ Most vascular complications occur when the balloon is inadvertently inserted through the superficial rather than the common femoral artery. After removing the balloon percutaneously using the closed method, firm pressure is applied to the groin and held for half an hour. If circulation to the leg becomes impaired or a hematoma becomes apparent, prompt exploration in the OR is indicated.

Circulation in the leg distal to the site of balloon insertion is observed systematically. If signs of ischemia appear, generally, the balloon is removed.

Extracorporeal Membrane Oxygenation. Since the first descriptions of CPB in the 1950s³⁸² and subsequent reports of successful use outside of the OR for ECMO in the 1970s,³⁸³ there has been a rapid growth in the use of this technology in patients with extreme cardiopulmonary failure.³⁴⁴ Other indications for ECMO include biventricular failure of multiple etiologies, massive pulmonary embolus or other refractory obstructive shock states, and warming the severely hypothermic patient.^{344,369}

The primary ECMO configurations are venoarterial (VA-ECMO) and venovenous (VV-ECMO), which are used for cardiopulmonary support and respiratory support, respectively. In this chapter, the focus is on the use of VA-ECMO for cardiac surgery patients with postcardiotomy shock. In current VA-ECMO systems, a centrifugal pump removes blood from the patient's venous system via an inflow cannula. The blood is oxygenated and ventilated through a temperature-regulated polymethylpentene hollow-fiber oxygenator/heat exchanger; it is filtered and returned to the patient's arterial system via an outflow cannula.³⁶⁹

Cannula placement for venous drainage and arterial inflow is limited only by vessel size and accessibility. Depending on cannula size and position (i.e., central or peripheral cannulation), flows of 5 to 10 L/min are achievable. In the postcardiotomy patient, a central cannulation strategy is an optimal first-line strategy due to the convenience of being able to switch the existing CPB circuit while leaving the aortic and central venous cannulae in place.

Several key management aspects distinguish ECMO from other ECLS therapies. The use of ECMO is associated with both bleeding and thrombotic complications.³⁸⁴ As VA-ECMO requires an oxygenator, it generally necessitates anticoagulation (except for short periods) due to the risk for oxygenator thrombosis and arterial thromboembolism, in particular when the patient is not on full support. The converse also is true: Using anticoagulation to maintain ECMO integrity increases the risk of bleeding complications.

As with the other ECLS therapies, ECMO can provide full cardiopulmonary support, but it does not completely drain all blood returning to the heart. Further, in peripherally cannulated ECMO, the retrograde nature of blood return can increase afterload and, therefore, left ventricular distention. Patients with very poor left ventricular systolic function have an increased risk for blood stasis and intracardiac thrombosis.³⁸⁵ Therefore, if the LV is distended, the patient will probably require support with inotropes to help maintain ejection, targeting a minimum pulse pressure of 10 to 15 mmHg. If pharmacologic support is insufficient, the next step is to consider IABP placement, atrial septostomy, or a surgical left ventricular

drainage vent for left ventricular decompression. More recently, the use of a transaortic VAD device (such as the Impella®, described later) in conjunction with ECMO has been reported to provide effective emptying of the LV and additional afterload reduction.³⁸⁶

When patients who are peripherally cannulated have residual or recovering heart function but insufficient lung function, deoxygenated blood can be ejected from the heart. As a consequence, oxygenated and deoxygenated blood can mix in the aorta, resulting in a "north-south" or "Harlequin" syndrome due to a hypoxic upper body (including heart and brain) and hyperoxic lower body.^{385,387} In addition to clinical findings, detecting this phenomenon requires arterial sampling from a catheter placed in the right arm (the left arm might be accurate, depending on where the interface between oxygenated and deoxygenated blood exists within the aorta). If this condition is severe and persistent and requires ongoing support, the team may need to consider adding a venovenous arterial cannula (for arterial VV-ECMO to improve venous drainage from the right heart) or converting to central cannulation.^{385,388}

When planning for weaning from VA-ECMO, the perioperative team should assess whether both biventricular performance and pulmonary function are adequate to ensure successful decannulation of the ECMO circuit. There is no one universal methodology on how best to perform this assessment^{389,390}; however, an institutional procedure designed around a multidisciplinary team should be developed.³⁴⁴

In regard to outcomes related to the use of VA-ECMO for postcardiotomy shock, high-quality evidence supporting its use remains sparse. Data from contemporary published series indicate that 40% to 60% of these patients are decannulated from ECMO, whereas only 20% to 40% survive to hospital discharge.³⁹¹⁻³⁹⁵ Given that ECMO is typically used in emergency and salvage situations as a bridge to recovery, durable device implantation, or transplantation, perhaps this outcome is not surprising; that said, the alternative would be certain death. In light of these difficult decision-making situations, it is increasingly advocated that expertise from supportive and palliative care professionals be sought early in the process of supporting any patient with ECLS.³⁹⁶ Importantly, one study has indicated that those patients who survive beyond 30 days of ECMO support have reasonable longer-term (5-year) survival and better health-related quality of life.³⁹⁵

Impella. Catheter-mounted temporary VADs were described as early as 1975, with the first in-human use reported in 1990.^{397,398} The microaxial concept evolved into today's Impella pump (Abiomed, Danvers, MA), initially made available for clinical use in 2000. The Impella system includes an external controller and a catheter-based pump that houses the electrical power and controller connections and an inlet for continuous heparin or bicarbonate flushing of the implanted motor.³⁶⁹ The pump comes in two sizes for left ventricular support (CP and 5.5) and one size for RV support (RP). Support with the larger CP device is intended for short duration (generally 4–6 days), whereas the larger 5.5 device can be used for up to 14 days.³⁹⁹

Impella devices can be inserted through a peripheral artery (either femoral or axillary, although axillary access better facilitates extubation and mobilization) or directly into the ascending aorta over a wire and positioned retrograde through the aortic valve. The RP version is inserted through the femoral vein and right heart to the pulmonary artery, unloading the right ventricle into the pulmonary artery. These devices can be inserted with fluoroscopic guidance in a cardiac catheterization laboratory, hybrid OR, or a

standard OR with portable fluoroscopy. Echocardiographic guidance is typically added for more precise positioning.

Impella devices are contraindicated for patients with severe aortic stenosis or insufficiency, left ventricular thrombus, tamponade or left ventricular rupture, conditions precluding placement (such as a mechanical aortic valve or severe obstructive peripheral vascular disease), or a contraindication for anticoagulation.³⁶⁹ Contraindications specific to the RP device include major venous or RV thrombus, inferior vena cava filters, and anatomic anomalies of the pulmonic or tricuspid valve that preclude placement.

The degree of support is titratable and can be set on the controller by adjusting the RPMs by set levels (P1–P9). The perioperative team needs to monitor the position of the device across the aortic valve; current devices accomplish this automatically by using differential pressure sensors. Also important is monitoring for potential complications, such as hemolysis, limb ischemia, aortic or mitral valve injury, stroke, arrhythmia, and vascular injury.^{400,401} Recovery of cardiac function can be assessed by scaling down the support level: The device can be removed when satisfactory hemodynamics and organ perfusion are maintained at low (P1 or P2) settings over a period of hours.³⁶⁹

Similar to the IABP, Impella CP pumps are commonly used as a means of support in centers without surgical VAD capability, either as a bridge to recovery or to allow safer transportation to a VAD center. The utility of this device continues to be investigated and refined. One RCT comparing the efficacy of a previous-generation device (Impella LP 2.5) with an IABP in nonemergency, high-risk percutaneous coronary intervention (the PROTECT II Study) was stopped early due to concerns about futility; nonetheless, the trial did reveal a trend toward superior hemodynamic support and fewer adverse events with the Impella.⁴⁰² Similarly, in an RCT investigating the management of acute MI complicated by cardiogenic shock (the EMPRESS Trial), the Impella was not superior to an IABP in terms of 30-day mortality.⁴⁰³ An ongoing RCT (SURPASS Impella 5.5 trial; registration NCT05100836) is examining the impact of the Impella 5.5 device on outcomes.

TandemHeart. The TandemHeart® (LivaNova, London, UK) is a percutaneous, centrifugal left-side VAD. It uses a 21F drainage catheter inserted through the femoral vein and right atrium and into the left atrium via a transseptal puncture. A centrifugal pump and control console unload the left heart, decreasing left ventricular end-diastolic pressure and myocardial oxygen consumption. A separate 15F to 19F arterial catheter inserted into the common femoral artery provides the inflow in a retrograde direction. In essence, this approach is similar to percutaneous ECMO, with the key difference being that the TandemHeart does not require an oxygenator (although one can be used if required).

Contraindications for TandemHeart use include right or left atrial intracardiac thrombus, severe peripheral arterial or aortoiliac atherosclerotic disease that precludes safe arterial cannula placement, or contraindication for anticoagulation.³⁶⁹ Aside from the potential access site issues of bleeding and limb ischemia, device-specific complications include left or right atrial perforation causing tamponade during insertion, cannula migration causing selective pulmonary vein intubation, tamponade from perforation of the left atrium, or shunting from backward migration into the right atrium.^{404,405} Accordingly, many institutional protocols require fixation of the device and cannulae and limitation of leg mobility, in addition to routine mechanical circulatory support care, including monitoring for distal limb ischemia or emboli, insertion-site bleeding and hematoma, and evidence of hemolysis.³⁶⁹

In patients with improved native heart function and minimal inotropic support requirements, weaning from the TandemHeart is accomplished by decreasing the pump speed and assessing hemodynamic response over time. Data on the effectiveness of this device versus others continue to evolve; one small meta-analysis failed to demonstrate the benefit of this device over an IABP.⁴⁰⁶

CentriMag and Rotaflow. The CentriMag (Abbott Laboratories, Abbott Park, IL) and Rotaflow (Getinge, Gothenburg, Sweden) devices (and a new but less-well-studied product, the LivaNova Revolution pump) are examples of intermediate duration centrifugal pump systems that can be used as bridge-to-decision mechanical circulatory support (either recovery, palliation, or conversion to a durable VAD).^{407–411} These devices were developed to overcome key issues with older-generation blood pumps, including shear stress resulting in hemolysis and thrombus, friction with heat generation, turbulent flow, and stasis.³⁸⁵ The Rotaflow pump is magnetically suspended on a sapphire bearing with no shafts or seals, whereas the CentriMag is completely magnetically levitated with no bearings, shafts, or seals. Both pumps have very small prime volumes and can be used for left ventricular, RV, or biventricular assist support. In addition, an oxygenator can be used or removed, and initial biventricular support can be simplified to univentricular support if partial cardiopulmonary recovery has occurred.

An additional potential advantage is that the use of central cannulation permits the use of shorter, large-bore cannulae that produce better hemodynamics (higher flows, better ventricular decompression). As a result, these systems are usually indicated for patients who present with postcardiotomy cardiogenic shock or who are unable to achieve adequate hemodynamic support from peripheral mechanical circulatory support systems. Specific care for the centrally cannulated mechanical circulatory support patient includes monitoring for bleeding complications, ensuring cannula and/or tubing securement, and avoiding kinking along the length of the extracorporeal tubes during mobilization. Importantly, as the cannulae are fixed to access sites, cannula migration should not be a problem.

In the ICU, moving toward extubation and active mobilization should be the goal when using these devices.³⁸⁵ Although inability to anticoagulate is a major concern, the flow characteristics and low stasis issues associated with these devices produce very few thromboembolic events, and there have been reports on managing these pumps without anticoagulation for several days to weeks after initial implantation.^{385,412} The weaning strategy is similar to that of the Impella and TandemHeart devices.

Other Considerations. The complex scientific, surgical, ethical, philosophical, and financial considerations necessarily involved in the decision to implement ECLS do not permit a simple listing of indications for these therapies. There is general agreement that survival is negatively affected by a low cardiac output (CI ≤ 1.5 L/min/m²) and elevated ventricular diastolic pressure (reflected in high left or right atrial pressure) when CPB is continued for an hour longer than usual, even with IABP support and catecholamine administration in moderate doses.⁴¹³ Trial separation from CPB is likely to fail (usually reflected in a progressive drop of systemic arterial blood pressure, elevation of atrial pressures, reduction in venous oxygen saturation, and metabolic acidosis).

Patients with continuing low cardiac output in the ICU despite IABP and catecholamine support should also be considered for temporary ECLS, even if the patient must be transferred to another institution. Use of temporary ECLS is appropriate for

patients of any age whose myocardial complications are expected to resolve within a few days, and for patients younger than 70 years of age whose myocardial complications may not resolve but who are acceptable candidates for durable VAD or cardiac transplant. In the latter group, the temporary device is used for support while it is determined whether the major organ systems are functioning and whether the patient is neurologically intact.

The care of patients on ECLS is labor intensive. Continuous bedside care, often by more than one nurse, is initially required, as is the ready availability of a cardiopulmonary perfusionist or other personnel capable of managing the extracorporeal assist system. Bleeding is frequently a nuisance and can even be a major complication, requiring frequent monitoring of ACT, which is used to monitor heparin effect; the desired ACT is approximately 160 to 200 seconds (assuming a control level of 100–120 seconds). Bleeding from the primary operative site is controlled by reversing heparin with protamine. Platelets, FFP or other blood products, and pharmacologic agents to promote normalization of the blood clotting subsystem are administered as indicated. A heparin-bonded extracorporeal circuit is adequate to prevent clotting in the short term. When bleeding from the operative site has ceased or slowed (usually within 12 hours postoperatively), heparin therapy can be restarted. Many centers now use bivalirudin for anticoagulation during ECMO support as accumulating data demonstrate its safety and possible benefit regarding patient survival.⁴¹⁴

Smooth operation of the system requires frequent infusion of blood products to maintain adequate atrial pressures. Body temperature is monitored continuously, and measures are taken to heat or cool the blood in the extracorporeal circuit or the patient's body with a heating or cooling blanket. The ECLS may be fitted with ports for hemofiltration or access for continuous venovenous renal replacement therapy to remove excess extracellular fluid if there is marked edema. Ventilatory assistance is required.

Separating the patient from temporary ventricular assistance requires judgment and patience. There is a tendency to remove systems too early to avoid complications directly related to prolonged extracorporeal circulation. Moreover, simply removing the temporary device in anticipation that the heart will sustain adequate function is usually unsuccessful. Instead, the heart's ability to support the circulation should be tested by reducing flow into the extracorporeal circuit, thereby raising atrial pressures and allowing flow through the supported ventricle. However, there is a limit to which flow in an extracorporeal circuit can be reduced without introducing danger of clotting. Flow of less than about 1 L/min should be avoided. Anticoagulation levels should be maintained, and the testing interval should be brief. Cardiac function is monitored using echocardiography and continuous measurement of arterial and pulmonary artery pressures, atrial pressures, cardiac output, and SvO_2 .

If the heart cannot sustain adequate cardiac output under conditions of low-flow bypass, the separation process should be abandoned, full flow resumed, and plans made for later attempts at separation or conversion to an implantable device. The multidisciplinary team should collaborate on the process and timing of ECLS discontinuation and plan ahead in case the patient is unable to achieve adequate cardiac output in the absence of support.³⁴⁴ This could, in some instances, include a plan not to reescalate to ECLS therapy. Thus, it is appropriate to establish a clear and transparent process and to “set the stage” with all team members, including the patient, family, and caregivers. This

process should be undertaken early, frequently, and with the appropriate documentation

Cardiac Arrhythmias

Postoperative morbidity and mortality can result from cardiac arrhythmias, which may occur either with normal myocardial function or as a complication of low cardiac output. Atrial and ventricular pacing wires, routinely placed during surgery and left for 2 to 10 days postoperatively, are of clear benefit in diagnosing and treating postoperative arrhythmias.⁴¹⁵⁻⁴¹⁸

Ventricular Electrical Instability. Ventricular electrical instability includes premature ventricular contractions (PVCs), ventricular tachycardia, and ventricular fibrillation. Whether postoperative patients with frequent PVCs and some types of ventricular tachycardia are at risk for developing ventricular fibrillation remains controversial. Despite this, it is prudent to assume that such arrhythmias soon after surgery place the patient at increased risk for sudden death. Detecting such arrhythmias requires continuous ECG monitoring during the first 24 to 72 hours postoperatively.

Unifocal isolated PVCs occurring outside the T waves fewer than 4 to 6 times per minute does not appear to increase the risk for cardiac death; however, the perioperative care team must remain vigilant for ischemia, hypothermia, abnormalities in potassium (or other electrolytes), and other postoperative complications, particularly in those patients with low left ventricular function or known history of ventricular arrhythmias.²⁹⁴ Routine use of Class IC antiarrhythmic medication (see Appendix 4C) is generally not required and indeed may cause harm.⁴¹⁹

In patients with postoperative ventricular tachycardia and an ejection fraction <40%, early electrophysiologic study may be indicated.^{294,420} Paradoxically, patients in whom such studies reveal acute suppression of PVCs or ventricular tachycardia with the administration of drugs have a good prognosis without drug treatment; these patients can probably be discharged from the hospital on no drug therapy. By contrast, patients whose ventricular arrhythmias cannot be suppressed with drugs have a relatively poor prognosis, and consideration should be given to an implantable cardioverter-defibrillator.

Often, PVCs or bursts of ventricular tachycardia can be suppressed by using temporary electrodes to establish atrial or ventricular pacing without the need for drug therapy. When the hemodynamic state is impaired by ventricular tachycardia, immediate direct current cardioversion (100 J initially in adults and, if ineffective, 200 J) is indicated. When these measures are ineffective, advice of an experienced electrophysiologist is indicated.

Atrial Fibrillation. Postoperative atrial tachyarrhythmias remain one of the most common occurrences after cardiac surgery. The perioperative team should be aware of several important scientific statements and guidelines, in particular the ACC, American Heart Association (AHA), and Heart Rhythm Society (HRS) guideline for managing AF,⁴²¹ the AHA scientific statement on AF occurring during acute hospitalization,⁴²² the European Society of Cardiology and European Association for Cardio-Thoracic Surgery guidelines for diagnosing and managing AF,⁴²³ The Society of Thoracic Surgeons 2017 clinical practice guidelines for the surgical treatment of AF,⁴²⁴ and the Society of Cardiovascular Anesthesiologists and European Association of Cardiothoracic Anaesthetists practice advisory for managing perioperative AF in patients undergoing cardiac surgery.⁴²⁵

Postoperative AF in the setting of cardiac surgery is common, affecting 20% to 60% of patients depending on the type of surgery (less prevalent after CABG, more common with isolated or concomitant valve surgery).^{422,426-428} It is most likely to occur on postoperative days 2 to 4.⁴²⁹ The traditional belief has been that new postoperative atrial fibrillation (POAF) is often self-limited and benign, although this dogma has been challenged by new evidence that patients with POAF after cardiac surgical intervention have a fivefold higher risk for permanent AF and ischemic stroke that persists to 10 years after surgical intervention.^{423,426,430,431} Further, although findings on the degree of complications are inconsistent, it is generally understood that the development of POAF after cardiac surgery is associated with longer hospitalization,⁴²⁶ greater short- and long-term morbidity (such as renal complications, infections, and bleeding),^{426-428,432,433} greater mortality,⁴³⁴ recurrent hospitalizations, and consequently greater costs of care.^{422,435,436}

Definitions. Definitions of AF are detailed in Chapter 15. Recently, the AHA issued a scientific statement emphasizing **acute atrial fibrillation**, defined as AF detected in the setting of acute care or acute illness (such as new POAF after cardiac surgery).⁴² Although there are many aspects of AF management, this section focuses on the prevention and management of new POAF. Surgical management is covered in Chapter 15.

Risk Factors. Several patient-related and perioperative triggers induce POAF in the cardiac surgery patient. These can be broadly categorized as biochemical/biomechanical (e.g., inflammation, oxidative stress, local mechanical stretch, volume shift and electrolyte imbalances, alterations in autonomic neuronal signaling); surgical (e.g., pericardial effusions, procedure type, long procedural and CPB times); and perioperative (e.g., inotropic agent use, new ischemia, bleeding, infection, pulmonary and other complications).⁴²²

Detecting patient risk remains an important topic of ongoing investigation. Various risk-prediction scoring systems that combine multiple subscores include:

- **CHA₂DS₂-VASc**, which includes congestive heart failure, hypertension, age (>75 years of age), diabetes mellitus (DM), previous stroke (or transient ischemic attack or thromboembolic event), vascular disease, age (65–74 years of age), and sex.^{438,439}
- **ATRIA** (Anticoagulation and Risk Factors in Atrial Fibrillation), which includes anemia, severe renal disease (defined as a glomerular filtration rate [GFR] of <30 mL/min), age (>75 years of age), previous hemorrhage diagnosis, and history of hypertension.^{440,441}
- **HATCH**, which includes hypertension, age, transient ischemic attack or stroke, chronic obstructive pulmonary disease, and heart failure⁴⁴²; and
- **POAF**, which includes chronic obstructive pulmonary disease (COPD), preoperative IABP use, emergency surgery, GFR of <15 mL/min or dialysis, and LV ejection fraction <30%.⁴⁴³

All of these have their advantages and disadvantages; however, the CHA₂DS₂-VASc is currently the most commonly used tool and has been shown to better predict POAF and thromboembolic complications than the other scores.^{422,444}

Prevention. Due to the high prevalence of AF and associated short-term and long-term complications, prevention is of significant interest and has been addressed in several society guidelines.^{422,423,425,445}

Beta-Blockers: The Society of Thoracic Surgeons quality metrics direct that β -adrenergic receptor blocking agents (beta-blockers) be

given within 24 hours after cardiac surgery; their use has been a Class I recommendation since the 2011 ACC/AHA/HRS focused update.^{421,423,445-448} Accordingly, beta-blockers are commonly used for POAF prophylaxis. Previous meta-analyses have shown an association between a reduction in the incidence of POAF and the use of perioperative beta-blockers. However, recent data have signaled potential harm (such as acute kidney injury [AKI]) from preoperative use,⁴⁴⁹ and a meta-analysis found that the impact of beta-blockers on POAF is improved when they are used in conjunction with the antiarrhythmic drug amiodarone.^{422,450}

Nonetheless, the use of beta-blockers presently remains the most accepted first-line agent for the prevention of POAF.^{422,423,445} The choice of beta-blocker may play a role. Small RCTs have shown that starting carvedilol at 6.25 to 12.5 mg twice daily was more efficacious in reducing POAF than perioperative metoprolol started at 25 to 50 mg twice daily.^{445,451,452} Alternatively, propranolol 10 to 20 mg may be given 3 to 4 times a day, beginning the morning after surgery, atenolol 25 to 50 mg may be given once per day, or propafenone 300 mg may be given twice daily.^{445,453}

Amiodarone: The largest trial examining amiodarone in cardiac surgery patients has been the Prophylactic Amiodarone for the Prevention of Arrhythmias that Begin Early After Revascularization, Valve Replacement, or Repair (PAPABEAR) trial. In this trial, investigators examined use of prophylactic amiodarone (10 mg/kg) versus placebo from 6 days before surgery through 6 days after surgery (13 days) to determine the effect on reducing postoperative atrial tachyarrhythmias lasting 5 minutes or longer.⁴⁵⁴ Within this study of more than 600 patients, those treated with amiodarone had a significant reduction in the occurrence of new tachyarrhythmias (16.1% vs. 29.5% in the placebo group); no significant differences were found in serious postoperative complications, in-hospital mortality, hospital readmission within 6 months of discharge, or 1-year mortality. A subsequent meta-analysis (largely informed by the PAPABEAR trial) found a 50% reduction in atrial arrhythmias and an approximate 0.6 reduction in hospital LOS with the use of amiodarone in the perioperative period.⁴⁵⁵

Several additional studies have examined the use of amiodarone alone at different doses (5 mg/kg day or lower), routes of administration (oral vs. IV), and durations in comparison with other agents (beta-blockers, calcium channel blockers, digoxin) or with other adjuvants (e.g., pacing). Despite some variability in outcome, amiodarone was safe and generally associated with less tachyarrhythmia, particularly in the most studied group (those undergoing CABG procedures).⁴⁵⁶⁻⁴⁶⁰ A recent larger-scale examination of outcomes in patients undergoing CABG at 235 hospitals found that in those using perioperative amiodarone, atrial arrhythmias were less frequent, and hospital stays were shorter.⁴⁴⁶ Nonetheless, adoption of amiodarone remains variable due to concerns about lack of outpatient monitoring for bradyarrhythmias and other side effects, in addition to feasibility and efficacy concerns.⁴²⁵

Sotalol: Sotalol is a combination agent with both beta-blocker and Class III antiarrhythmic properties. Several studies have examined the efficacy of sotalol for POAF prevention in comparison with placebo or other agents, such as beta-blockers and amiodarone (either alone or in combination).⁴⁶¹⁻⁴⁶³ Whereas sotalol may have a similar impact on POAF as other agents, because of its proarrhythmic effects, it is not commonly used as a first-line agent for preventing AF after cardiac surgery.⁴⁴⁵

Colchicine, Magnesium, and Other Agents: Initial enthusiasm about the use of colchicine for reducing POAF and hospital

LOS after cardiac surgery was spurred by results from the Colchicine for the Prevention of the Postpericardiotomy Syndrome (COPPS) study.⁴⁶⁴ However, subsequent analyses, including the Colchicine for Prevention of Postpericardiotomy Syndrome and Postoperative Atrial Fibrillation (COPPS-2) study and the Effect of Colchicine on the Incidence of Atrial Fibrillation in Open Heart Surgery Patients (END-AF) study, were unable to replicate the earlier findings. This may have been due in part to the discontinuation of colchicine because of side effects (primarily diarrhea).^{422,465,466} The Colchicine For The Prevention Of Perioperative Atrial Fibrillation In Patients Undergoing Thoracic Surgery (COP-AF) study (NCT03310125), which intends to randomize 3200 patients, was still recruiting at the time of this writing.⁴²³

Low magnesium is almost ubiquitous in postoperative patients in the ICU after cardiac surgery.⁴⁶⁷ Although magnesium administration is not a universal protocol, data compiled in several meta-analyses indicate that it is associated with less atrial and ventricular arrhythmia and minimal safety issues (low rates of bradycardia or hypotension).^{445,468-470}

Several other agents, such as renin-angiotensin system inhibitors, ACE inhibitors, nonsteroidal antiinflammatory agents and steroids, N-acetylcysteine,⁴⁷¹ statins,⁴⁷² omega-3 fatty acids (fish oil supplements), calcium-channel blockers, and digitalis have been examined for their utility in preventing POAF.^{422,423,445} At present, none of the evidence for these agents is robust enough to support a recommendation for routine use to treat POAF after cardiac surgery.

Posterior Pericardiotomy and Botulinum: The Posterior Left Pericardiotomy for the Prevention of Atrial Fibrillation After Cardiac Surgery (PALACS) study, an RCT of more than 400 patients undergoing CABG, aortic valve replacement, or aortic surgery, examined the impact of a posterior pericardiotomy to the left pleural space during surgery to drain any residual pericardial effusion in patients with POAF.⁴⁷³ With this low-cost adjunct to the procedure, the investigators observed a lower rate of POAF (17% vs. 32%) in studied patients. Although not specific to POAF, the seminal Left Atrial Appendage Occlusion During Cardiac Surgery to Prevent Stroke (LAAOS III) surgical trial randomized almost 4800 patients with known AF and a CHA₂DS₂-VASc score ≥ 2 to either intraoperative left atrial appendage occlusion or no intraoperative occlusion (control). Results indicated significant short-term and long-term reduction in ischemic stroke and systemic embolism in the intervention group.⁴⁷⁴ Multiple trials are underway to identify the role of prophylactic left atrial appendage closure in reducing the risk of POAF.⁴⁷⁵

Treatment. When AF develops postoperatively, several intervention options are appropriate, depending on the need for rate control versus conversion to sinus rhythm (see Appendix 4D). Whatever institutional protocol is implemented should involve the heart team.^{422,424}

Rate Versus Rhythm Control: In the Rate Control versus Rhythm Control for Atrial Fibrillation for Cardiac Surgery trial, 523 patients with new POAF were randomized to undergo either rate or rhythm control.⁴⁷⁸ With the primary endpoint of total hospitalization days within 60 days of randomization, neither treatment was superior; however, approximately 25% of patients in each group deviated from their assigned group.

In postoperative patients with chronic AF who have stable and effective hemodynamics and who are not in the ICU, digoxin is

usually effective for rate control and is especially indicated for patients with heart failure and LV dysfunction. In the ICU, rate control of AF with preserved ventricular function is most effectively accomplished with IV administration of beta-blockers (esmolol, metoprolol, or propranolol)⁴⁷⁹ or nondihydropyridine calcium channel antagonists (verapamil, diltiazem), although caution should be exercised when hypotension is present. IV amiodarone is also useful for rate control. It should be noted that IV digoxin or calcium-channel antagonists should not be used in patients with preexcitation syndrome, as these agents may paradoxically accelerate the ventricular response.

A recommended protocol for digoxin use in rate control is found in the Appendix 4D. When approximately two-thirds of the estimated digitalizing dose has been given without control of the ventricular rate, oral administration of propranolol should be started unless the patient has poor ventricular function or pulmonary disease. If the situation is urgent, propranolol may be given intravenously in doses of 0.5 mg every 2 minutes to a total IV dose of 4 mg in adults. Alternatively, treatment with verapamil may be initiated in a dosage of 40 mg orally 2 to 3 times daily. This drug may be infused intravenously in a dose of 0.075 to 5.0 mg/kg, but its action is very short when given in this manner.

In patients with new-onset or recurrent POAF who are hemodynamically stable and asymptomatic, a reasonable initial strategy is to target rate control at <100 bpm or rhythm control with beta-blockers or amiodarone.^{422,423} Successful conversion to sinus rhythm is usually possible and is advantageous for stabilizing postoperative hemodynamics, provided that a recent TEE has excluded left atrial appendage thrombus. Amiodarone is the most commonly used first-line drug for pharmacologic conversion of AF.⁴²¹

Ibutilide (a class III antiarrhythmic agent) is effective in pharmacologically converting new AF to sinus rhythm.^{421,480} Ibutilide is given in a 1 mg dose administered intravenously over 10 minutes; continuous ECG monitoring is required during administration. The key consideration with ibutilide is the potential for QT prolongation. Pretreating with IV magnesium (1–2 g) to reduce risk for torsades de pointes and ensuring that the serum potassium level is normalized are important elements of ibutilide use. Ibutilide should be avoided in patients with very low ejection fraction, given their high risk for ventricular arrhythmia.⁴⁸¹

Anticoagulation: Over the past several years, it has become apparent that additional scrutiny of cardiac surgical patients with POAF is warranted regarding the best treatment, the need for routine anticoagulation, and appropriate follow-up after hospital discharge.⁴²⁶ Management decisions need to balance the risk of stroke and other thromboembolic phenomena with the risk of bleeding complications.

The study protocol of the Rate Control Versus Rhythm Control for Postoperative Atrial Fibrillation trial included the use of anticoagulation (targeting an international normalized ratio [INR] of 2–3) for those patients who remained in AF for 48 hours or more and found a nonsignificant increase in bleeding complications in the rate-control group (although the overall rate was low, at approximately 3%). At present, this research team is investigating the need for anticoagulation in patients with new POAF: The Anticoagulation for New-Onset Post-Operative Atrial Fibrillation After CABG; NCT04045665 (PACES) trial was still recruiting at the time of this writing.⁴²² Recent reports suggest that the benefits of anticoagulation for POAF after cardiac surgery may not be so clear.⁴⁸²

Follow-Up. Postoperative AF recurs in up to 25% of patients within 4 to 6 weeks of hospital discharge, even when sinus rhythm is restored before discharge.^{422,426,429} Further, up to 50% of patients with atrial flutter have been found to have incidental AF. Follow-up is recommended for these patients.

The Detection of Atrial Fibrillation After Cardiac Surgery (SEARCH-AF) study sought to establish the need for follow-up in a broad range of cardiac surgery patients, including those with a CHA₂DS₂-VASc score ≥ 4 or ≥ 2 plus risk factors for POAF, no history of preoperative AF, or POAF lasting < 24 hours during hospitalization.⁴⁸³ Patients randomized to the intervention group underwent 30 days of continuous cardiac rhythm monitoring with wearable, patch-based monitors. The authors found a significant increase in the rate of POAF detection after hospital discharge in the intervention group, indicating a need for routine follow-up and further study on the benefit of anticoagulation.

Atrial Flutter. Atrial flutter can be a difficult arrhythmia to control when it occurs postoperatively; it frequently requires cardioversion.⁴²² Although not always successful, the best treatment is rapid atrial pacing achieved by using the two atrial epicardial wires placed during surgery (see Appendix 4E).

Ibutilide is also effective in converting atrial flutter ($> 80\%$ success) and is probably more effective than amiodarone.

Paroxysmal Atrial Tachycardia. Important episodes of paroxysmal atrial tachycardia or paroxysmal atrial contractions occurring after cardiac operations can also be treated with rapid atrial pacing. If these arrhythmias persist, the diagnostic and therapeutic advice of an expert electrophysiology cardiologist is needed. Specific treatment depends on the precise nature of the supraventricular tachycardia. In general, administration of catecholamines with chronotropic activity should be kept at the lowest dosage possible.

Postoperative Cardiac Arrest

The incidence of cardiac arrest after cardiac surgery has been reported as ranging from 0.7% to 8.0%.⁴⁸⁴⁻⁴⁸⁷ Importantly, unlike recovery after out-of-hospital arrest or in-hospital arrest in non-cardiac surgery settings, successful resuscitation to hospital discharge after cardiac surgery is far superior, ranging from 17% to 79%.^{241,487-489} Most likely, the reasons for this include the early identification of arrest due to the patient's being managed in a highly monitored environment and the high incidence of potentially reversible causes for the arrest. For example, ventricular fibrillation accounts for the arrhythmia in 25% to 50% of cases, and in the cardiothoracic ICU, this is immediately identified and treated. In addition, tamponade and major bleeding account for many arrests, and both conditions can be quickly relieved by prompt resuscitation and emergency re sternotomy where appropriate.^{241,489,490} This requires coordinated perioperative team activity and a high level of situational awareness. Whereas the early institution of ECMO during resuscitative efforts has been described, further study is required to determine survival benefit.^{391,491,492}

In 2007, the Clinical Guideline Committee of the European Association for Cardio-Thoracic Surgery initiated a process to create a set of clear clinical guidelines to apply specifically to resuscitation after cardiac surgery. A formal statement was published in 2009, updated in The Society of Thoracic Surgeons expert consensus document in 2017, and further supported in the most recent version of the AHA's guidelines for cardiopulmonary resuscitation and emergency cardiovascular care.^{241,489,490} The included

protocol (Fig. 4.12) is a modification of the original 2005 European Resuscitation Council advanced life support cardiac arrest algorithm and is recommended to be used in the ICU up to the 10th day postoperatively.²⁴¹

An important element of this new process is the six key roles of the ICU cardiac arrest team to facilitate successful resuscitation of the cardiac surgery patient (Fig. 4.13). Within this cardiac surgery unit cardiac arrest protocol are several notable deviations from the "standard" advanced cardiovascular life support resuscitation guidelines. Although a thorough discussion of these is beyond the scope of this chapter, they pertain to the timing of closed-chest cardiopulmonary resuscitation initiation, epinephrine and atropine administration, epicardial pacing, and chest re-opening.^{241,490} Within this cardiac surgery unit arrest protocol, the team must determine whether the arrest is being triggered by a tachyarrhythmia (e.g., ventricular tachycardia or ventricular fibrillation), a bradyarrhythmia or asystole, or pulseless electrical activity (e.g., secondary to cardiac tamponade or tension pneumothorax), as each etiology will have its resuscitation algorithm (see Fig. 4.12).

The recent AHA statement notes that "Effective education is an essential contributor to improved survival outcomes from cardiac arrest."⁴⁹³ Accordingly, it is vital for the perioperative team to engage in deliberate education and skill acquisition and retention specific to their center's clinical context so as to be prepared for these relatively uncommon but highly salvageable postoperative events.^{241,493}

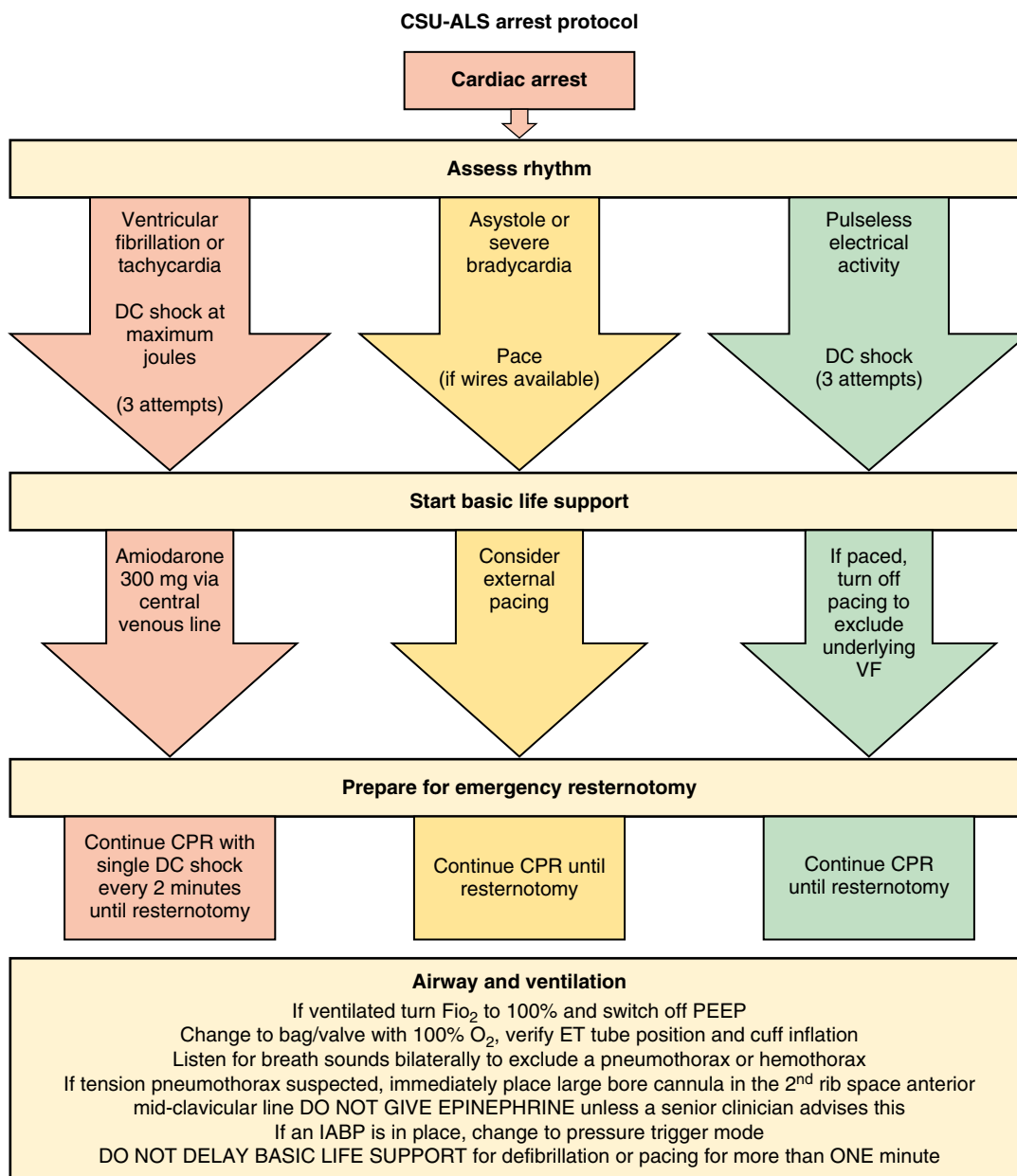
Pulmonary Subsystem

Adequate Pulmonary Function

During Intubation. The adequacy of pulmonary function early after surgery, while the patient is still intubated and receiving controlled ventilation, is judged by the response of PaO₂ and PaCO₂ to the minute volume of ventilation and ventilating gas mixture being used. The response of the arterial oxygen levels can be expressed as the alveolar-arterial oxygen difference, P(A - a)O₂, which in intact humans is normally only a few mmHg. Nearly all patients early after cardiac surgery have abnormally large alveolar-arterial oxygen differences due to intrapulmonary right-to-left shunting of 3% to 15%.⁴⁹⁴⁻⁴⁹⁶ This assumes that no right-to-left intracardiac shunting is present. Response of the carbon dioxide level can be expressed as the minute volume of ventilation required to maintain PaCO₂ at 30 to 45 mmHg, usually about 15 to 20 mL/kg in both adults and children.

When emerging from anesthesia in the OR or ICU, the patient can be converted to a spontaneous mode of mechanical ventilation, either synchronized intermittent mandatory ventilation or pressure support ventilation. The patient's respiratory rate and other hemodynamic parameters can be used to judge the adequacy of cardiac and pulmonary function. In an adult, adequacy of lung function is indicated by a patient-triggered respiratory rate of 8 to 12 breaths/min with a tidal volume of 6 to 8 mL/kg.⁴⁹⁷⁻⁵⁰⁰ and adequate oxygenation, ventilation arterial blood gas parameters, and end-expiratory pressure.

PEEP may be used routinely or as indicated, with a setting of 5 to 8 cm H₂O for adults and children older than 12 years and 4 cm H₂O for younger patients. Unless the hemodynamic state is suboptimal, using PEEP does not alter it, even in infants.⁵⁰¹ Studies suggest that PEEP is associated with larger lung volumes, fewer



• **Figure 4.12** CSU-ALS algorithm.

perfused but nonventilated alveoli during ventilation, and smaller P(A - a)O_2 after extubation.⁵⁰²

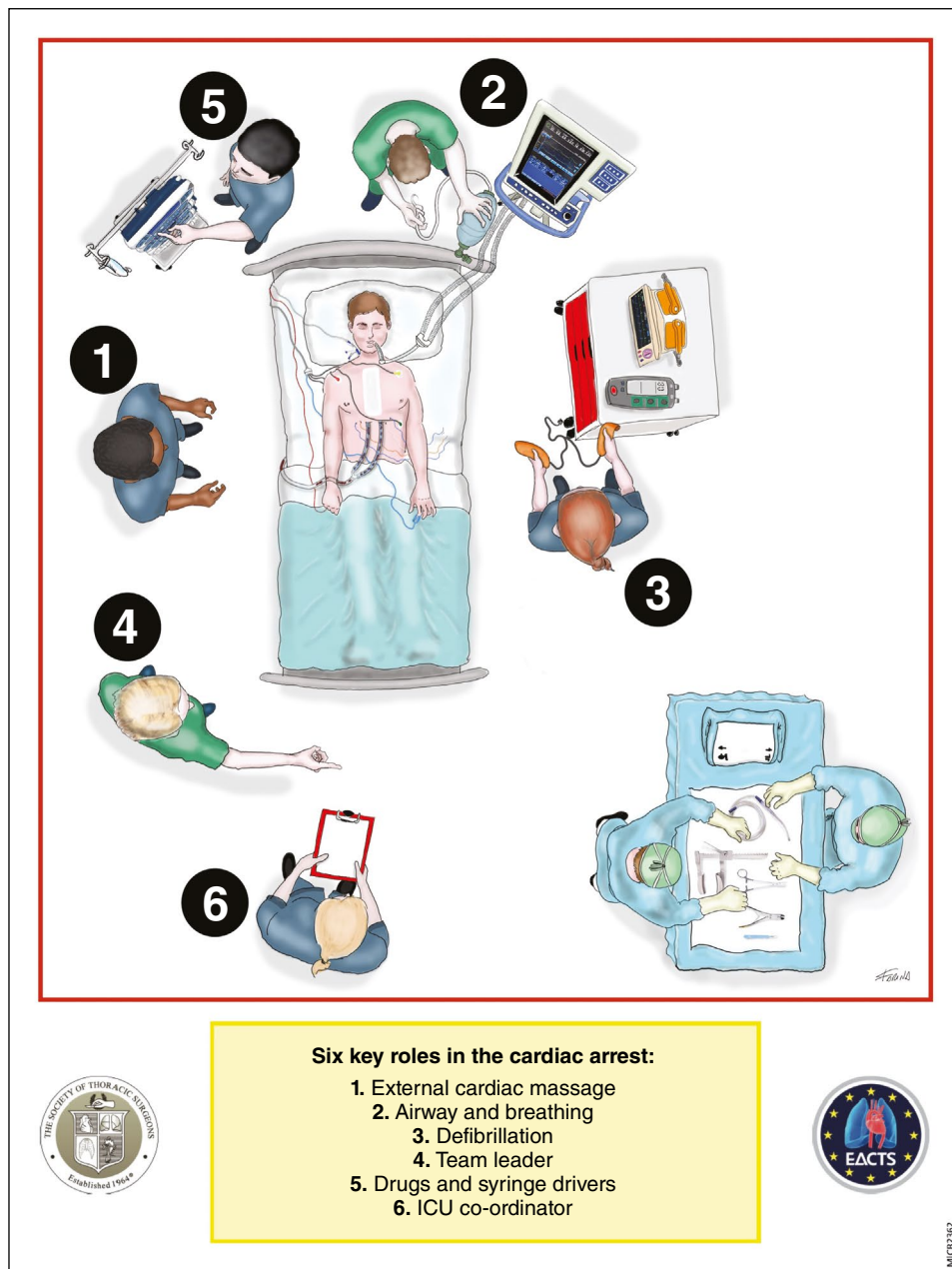
Appropriate PEEP levels need to be considered in patients with chronic obstructive lung disease (to avoid air trapping and barotrauma in those with bullous lung disease) and in infants and children who have undergone a Fontan operation or cavopulmonary anastomosis (to avoid further elevation of jugular venous pressure).

Continuous positive airway pressure (CPAP)^{503,504} may be used in infants once their cardiovascular state is stable, obviating the need for intermittent positive-pressure breathing and intermittent mandatory ventilation. Should intermittent positive-pressure breathing be required initially, the infant is transferred to intermittent mandatory ventilation and sometimes to CPAP for several hours before extubation.

Extubation. Traditionally, continued intubation after cardiac surgery has been recommended to maintain more precise control of cardiopulmonary physiology. However, early extubation, either

in the first 6 hours in the ICU (fast-track)^{230,505-508} or in the OR (ultra-fast-track)⁵⁰⁹⁻⁵¹² facilitated by lower opioid doses and neuromuscular blockade reversal, leads to shorter ICU and hospital stays and potentially lower healthcare costs.^{502,505} Therefore, a protocol beginning with anesthesia induction can be designed to expedite awakening and spontaneous breathing. This requires perioperative team readiness and a culture of safety to ensure appropriate surgical pain management and preparedness for reintubation, day or night.⁵¹³⁻⁵¹⁶

Once the patient has been extubated, useful indices of pulmonary function include Paco_2 and PaO_2 levels, hemodynamics, and visually estimated work of breathing. A Paco_2 level <45 mmHg in adults, <50 mmHg in young children, and <55 mmHg in infants generally indicates an adequate minute volume of ventilation. Higher values indicate inadequate alveolar ventilation. Of note, PaO_2 is often mildly depressed after extubation when the cardiac surgery was performed with CPB and



• **Figure 4.13** Six key roles in the cardiac arrest.

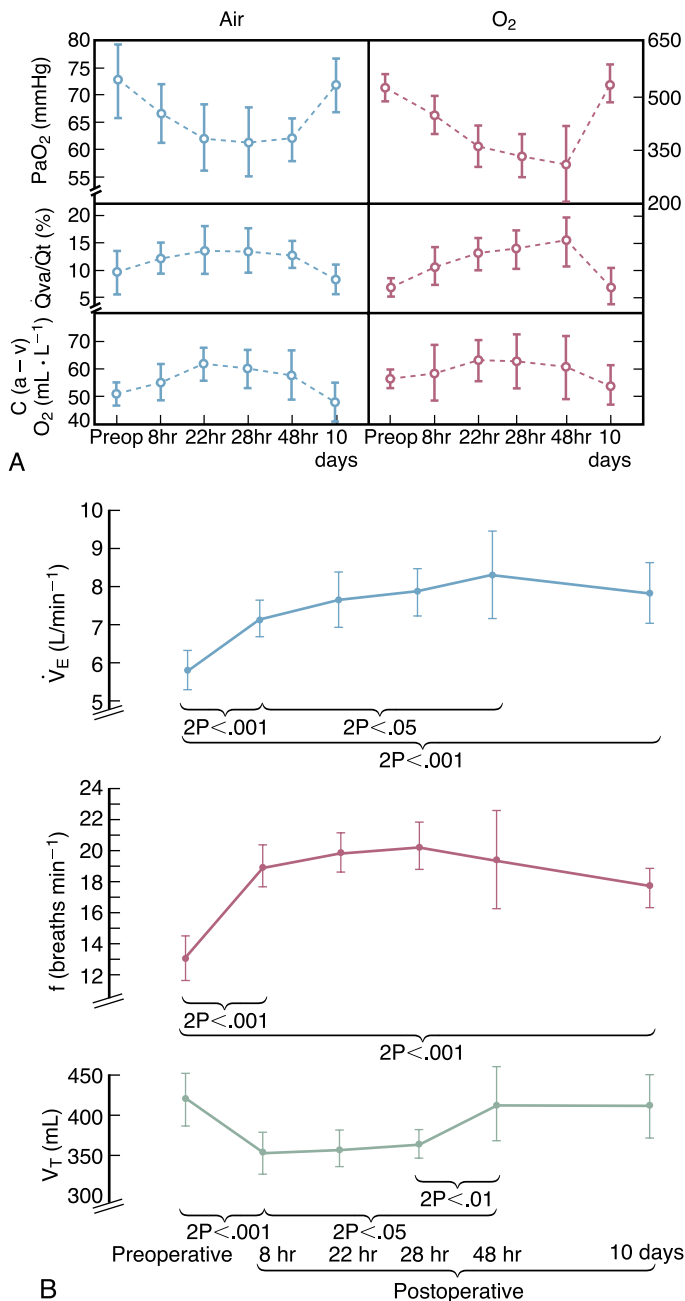
usually remains so for the first few days (Fig. 4.14A and B). This is caused by the somewhat widened alveolar-arterial oxygen difference. These measurements are valuable, but when a patient is comfortable and breathing easily and slowly, and the chest radiograph is within normal limits of a postoperative patient, it is highly probable that pulmonary function is adequate and initial recovery will be satisfactory.

Pulmonary Dysfunction After Cardiac Surgery

Causes. After cardiac surgery with CPB, the lungs are more likely to have some degree of dysfunction, albeit typically mild and transient. Pulmonary dysfunction is caused in part by the absence of pulmonary blood flow during total CPB or by its near absence during partial CPB, either of which produces very low

shear stresses in the pulmonary capillaries. This appears to accentuate neutrophil activation because neutrophils appear to be exquisitely sensitive to shear stress. Leukocytes activated by the general damaging effects of CPB incite an inflammatory response in the pulmonary vasculature.⁵¹⁷ During total CPB and lung ischemia, plasma thromboxane B₂ levels increase, which may contribute to a pulmonary vascular inflammatory response. The cytokines interleukin-6 and interleukin-8 increase with CPB and may also contribute to membrane damage and neutrophil activation in the lung.^{518,519} The alveolar-capillary barrier becomes more permeable than normal,^{520,521} and after cardiac surgery with CPB, macromolecules enter the pulmonary interstitium and ultimately the alveoli, promoting the development of pulmonary edema.

Postoperative disturbances in lung function and increases in alveolar polymorphonuclear leukocytes have been linked to an



• **Figure 4.14** A, Arterial oxygen tension (P_{aO_2}), venous admixture ($\dot{Q}_{va}/\dot{Q}_t$), and arteriovenous oxygen content difference [$C(a-v)O_2$] measured preoperatively and at intervals early postoperatively in 10 adults undergoing operation with cardiopulmonary bypass. B, Preoperative and postoperative values for minute ventilation ($\dot{V}_E$), frequency (f), and tidal volume (V_T) in the same patients. Air- and O₂-breathing results have been combined, and mean values for the group at each time are shown. (Data from Rea HH, Harris EA, Seelye ER, Whitlock RM, Withy SJ. The effects of cardiopulmonary bypass upon pulmonary gas exchange. *J Thorac Cardiovasc Surg.* 1978;75(1):104-120.)

important reduction in pulmonary surfactant activity.⁵²² In addition, multiple and not-completely-defined factors encourage the development of large areas of atelectasis, either segmental or occasionally lobar; in particular, the left lower lobe has a strong tendency to atelectasis, even in patients who are otherwise recovering normally.⁵²³

In some patients, pulmonary dysfunction is caused by direct trauma to the lungs. In addition, the high intraoperative tidal volumes (>8 mL/kg) and ventilatory driving pressure (the difference between the plateau pressure and the level of PEEP) have been associated with ventilator-induced lung injury and postoperative pulmonary complications.^{499,500,524}

A patient's inability to cough and clear secretions (due to poor inspiratory effort secondary to poor pain control) during or early after surgery also contributes to dysfunction. Left (and occasionally, right) phrenic nerve injury may occur even after carefully performed cardiac operations, increasing the tendency to pulmonary dysfunction early after surgery.⁵²⁵⁻⁵²⁸ However, left lower lobe atelectasis is considerably more common than left phrenic nerve paralysis. Markand and colleagues found the left phrenic nerve to be paralyzed in only 11% of patients who developed left lower lobe atelectasis early after open-heart operations.⁵²⁹ In most patients, the left phrenic nerve recovers within 6 months of paralysis.⁵³⁰ In patients with persistent diaphragmatic paralysis, diaphragm pacing is being investigated.^{527,531}

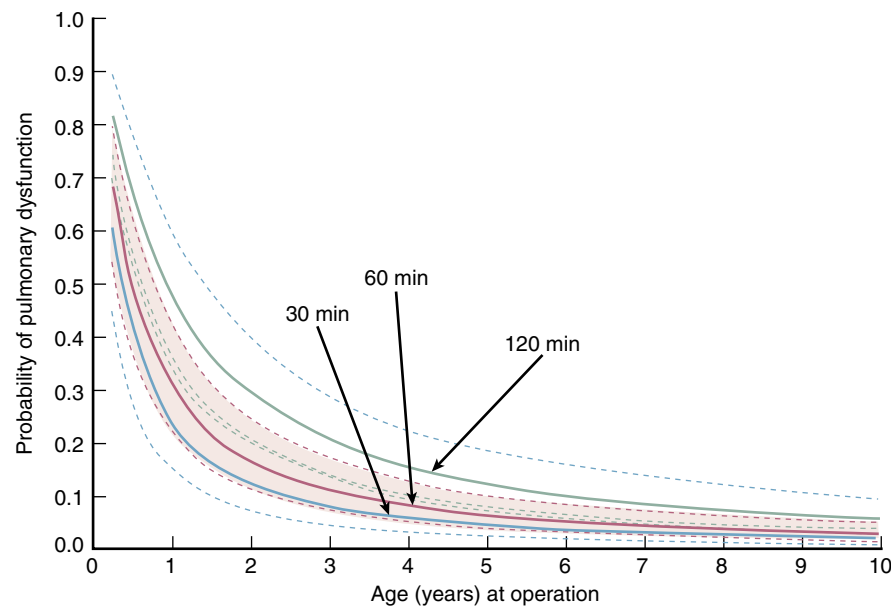
Mechanical obstruction of the lower trachea or the bronchi can produce pulmonary dysfunction that may go unrecognized unless care is taken to identify and treat it.⁵³² Localized or more extensive pulmonary edema may develop in the presence of low or normal left atrial pressure, no doubt related to changes in pulmonary venular and capillary permeability, the causes of which are only partially understood. This phenomenon seems to be more marked in older adults.⁵³³ Less commonly, frank pulmonary hemorrhage that develops as CPB is discontinued or early thereafter can cause serious bronchial obstruction and contribute in a major way to pulmonary dysfunction.⁵³⁴ This is probably a more severe result of the same factors that lead to pulmonary edema in patients with normal or low left atrial pressures.

Risk Factors

Patient-Specific Factors. Patient-specific risk factors for pulmonary dysfunction after cardiac surgery have long been recognized, but formal identification and quantification are rare. In an observational study,⁵³⁵ *young age* at operation was identified as a risk factor, particularly when the patient was younger than about 2 years old (Fig 4.15). Lell and colleagues made similar observations.⁵³⁶ Higher risk at a young age is associated with the increased tendency of the very young to develop whole-body edema after CPB. A study by Gallagher and colleagues,⁵³³ supported by broad anecdotal experience, found that *older age*, particularly older than 60 years, also has been associated with a higher prevalence of pulmonary dysfunction after cardiac surgery.

Chronic obstructive lung disease is an important risk factor for postoperative pulmonary dysfunction and increases the overall risk of the operation because it predisposes patients to increased work of breathing and air trapping.⁵³⁷ Preoperative *pulmonary arterial hypertension*, even when associated with low pulmonary arteriolar resistance, predisposes infants to pulmonary dysfunction and also to paroxysms of pulmonary arteriolar constriction and pulmonary hypertension soon after surgery. Congenital morphometric pulmonary abnormalities, such as *alveolar hypoplasia* frequently present in patients with CHD or *Down syndrome*, increase the prevalence of postoperative pulmonary dysfunction.⁵³⁸

Procedural Factors. Protective mechanical ventilation strategies using low tidal volume or high PEEP levels improve outcomes for patients who have had surgery.⁵⁰⁰ High intraoperative ventilatory driving pressure (defined earlier) and changes in the



• **Figure 4.15** Relationship between age at operation and duration of CPB (represented by solid isobars and their dashed 70% confidence limits) to probability of pulmonary dysfunction after cardiac surgery. Nomogram depicts specific solutions of a multivariable equation; value entered for C3a was $882 \text{ ng} \cdot \text{mL}^{-1}$ (see original publication for details). (From Kirklin and colleagues.⁵³⁵)

level of PEEP are associated with more postoperative pulmonary complications. However, traditional belief has been that low tidal volumes promote alveolar collapse in poorly ventilated, dependent regions of the lung⁵³⁹ with resultant lung trauma secondary to the repetitive collapse and reopening of alveolar units, leading to ventilator-induced lung injury.^{499,540} As a result, there has been some enthusiasm for the use of an “open-lung strategy.”

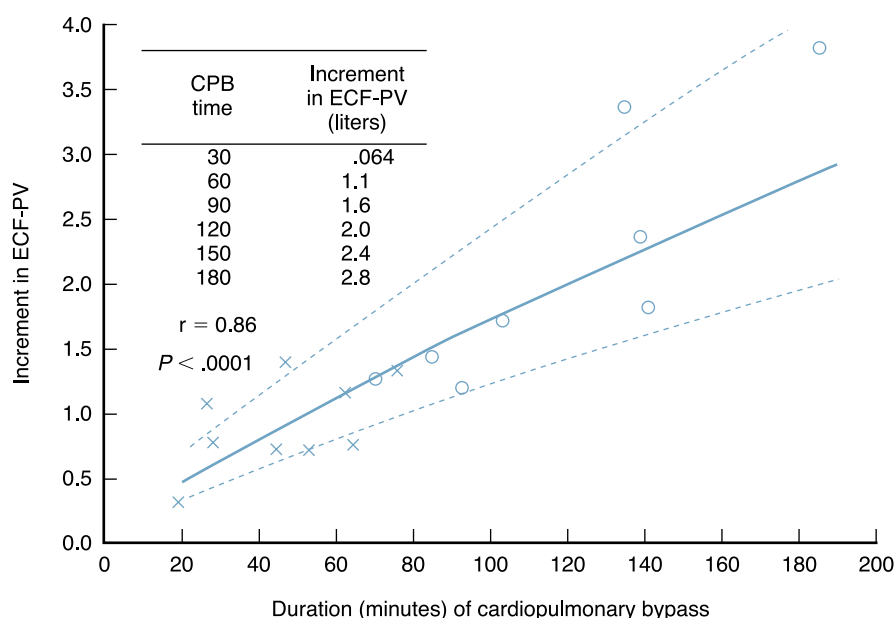
The open-lung strategy refers to ventilation management that uses mechanical ventilation during surgery along with recruitment maneuvers to prevent alveolar collapse during the procedure.^{499,524} The multinational Protective Ventilation in Cardiac Surgery (PROVECS) trial, an RCT of 493 patients, investigated whether an open-lung perioperative ventilation strategy that combined mechanical ventilation during CPB, perioperative recruitment maneuvers, and higher PEEP levels was protective against postoperative pulmonary complications, compared with usual care.⁴⁹⁹ The intervention was not found to be superior. Another trial (FLOWVENTIN HEARTSURG; German Clinical Trials Register DRKS00018956) is investigating whether a newer ventilation mode that increases control of inspiratory and expiratory airway flows will result in less postoperative pulmonary dysfunction.⁵⁴¹

The *type of oxygenator* used is associated with the amount of pulmonary dysfunction generated by the operation, with oxygenators other than membrane-type oxygenators being risk factors for pulmonary damage.⁵⁴² This is less common with contemporary oxygenators used in clinical practice. The use of *filters* in the arterial tubing may reduce pulmonary dysfunction postoperatively. *Longer-duration* CPB is also a risk factor, probably related to its direct correlation with an increase in the patient's extracellular water^{543,544} (Fig. 4.16). A *higher level of C3a* generated by complement activation during CPB is a risk factor that may be related to elevated neutrophil activation.⁵³⁵ Use of *external cardiac cooling*, particularly by ice slush rather than cold saline, increases the prevalence of left phrenic nerve paralysis and hence the tendency toward postoperative pulmonary dysfunction.⁵⁴⁵

Postoperative Factors. Postoperative events can increase the probability of pulmonary dysfunction. Elevated *left atrial pressure*, with consequent elevation of pulmonary capillary and venular pressures, aggravates the tendency toward increased lung water and pulmonary dysfunction. The longer the patient is on a ventilator, the greater the chances of ongoing pulmonary dysfunction. *Elevating the left hemidiaphragm* because of phrenic nerve paralysis predisposes patients to continuing pulmonary dysfunction, particularly small patients.^{525,526}

Course after Cardiac Surgery

Mild pulmonary dysfunction is common in normally recovering patients and will improve slowly without specific therapy other than early mobilization and breathing exercises; nonetheless, some dysfunction may still be present 10 days postoperatively.^{546,547} Occasionally, the patient who appears to be recovering normally and has been out of the ICU for 3 to 6 days may develop orthopnea and paroxysmal nocturnal dyspnea. This is particularly likely in patients who have marked left ventricular hypertrophy or poor left ventricular function preoperatively, and it can occur even though their left atrial pressure was $<15 \text{ mmHg}$ when last monitored in the ICU. The patient's chest radiograph may have been normal or may have shown evidence of a mild increase in interstitial fluid. Response to diuresis in such a patient is dramatic and typically resolves the problem. The hypothesis is that the fluid that accumulated in the interstitial spaces throughout the body during and early after CPB returns to the vascular space 24 to 72 hours after surgery. Blood volume is thereby increased at a time when the renal response is subnormal, and diuresis and control of blood volume do not follow.^{544,548} In this setting, left ventricular end-diastolic, left atrial, and pulmonary venous pressures rise, and symptoms develop, even though weight gain and other gross evidence of fluid retention may be absent.



• **Figure 4.16** Relationship between duration of CPB and increase in interstitial fluid (ECF-PV) 4 to 6 hours after operation. The x's represent patients undergoing closure of left-to-right shunts; circles represent those undergoing operation for valvular heart disease. Patients with heart failure were not included. The regression equation is: $\text{Ln}(\text{ECF} - \text{PV}) = -3.27 + 0.83 \text{ Ln}(\text{CPB time})$; $r = 0.86$; $P < 0.001$. CPB, Cardiopulmonary bypass; ECF, extracellular fluid; Ln, natural logarithm; PV, plasma volume. (Data from Cleland J, Pluth JR, Tauxe WN, Kirklin JW. Blood volume and body fluid compartment changes soon after closed and open intracardiac surgery. *J Thorac Cardiovasc Surg.* 1966;52(5):698-705.)

Although more persistent transient pulmonary dysfunction can occur after cardiac surgery, need for mechanical ventilatory support for more than 72 hours is uncommon. Occasionally, patients with an uncomplicated early recovery begin to cough up thick tracheobronchial secretions 48 to 72 hours after the operation. Seemingly paradoxically, whatever dyspnea and tachypnea may have been present often begins to lessen with these events. Presumably, protein-rich fluid that has been in the alveoli and interstitium of the lung since CPB (or soon thereafter) begins to be moved by ciliary action out of the terminal bronchioles and into the larger airways, from which it can be cleared by coughing.

Lung volumes are usually reversibly lower early after cardiac surgery, particularly vital capacity and total lung volume.⁵⁴⁹ This is probably the result of the summed effects of multiple small areas of atelectasis, occasionally left lower lobe collapse, occult pulmonary edema and pleural fluid, and reduced inspiratory effort.⁵⁴⁹ These usually revert to normal within 3 to 6 months.

More concerning causes of persistent respiratory failure include pulmonary infections (see Section III: Infections later) and acute respiratory distress syndrome (ARDS). The incidence of ARDS is low (ranging from as low as 0.6% to as high as 17%),⁵⁵⁰⁻⁵⁵² but once it occurs, the mortality rate is quite high, ranging from 40% to 80%. The occurrence of ARDS after cardiac surgery is higher in patients undergoing redo sternotomy, longer CPB times, or complex cardiac surgery procedures (reflected in higher predictive scores, such as the EuroSCORE II) and those with diabetes, multiunit blood transfusions, or lower albumin levels and impaired renal function.^{551,553}

Management and Treatment. The treatment goal for the pulmonary subsystem is the patient's earliest possible return to extubated spontaneous breathing and ambulation. These benefits include a decline in intrapleural pressure and an increase in left ventricular end-diastolic diameter (or volume), improved ventricular systolic function

(due to increased preload associated with the shifting of some blood volume toward the chest), and improved cardiac output.⁵⁵⁴

For patients with ARDS, ICU and mechanical ventilation management mirrors the principles of management in critically ill noncardiac patients. Maintaining low tidal volume ventilation is the mainstay of therapy, and vigilant fluid management, minimization of blood product transfusion, appropriate nutrition, and early physical rehabilitation may improve outcomes.^{552,555-558} In cases of refractory hypoxemia, rescue therapies such as inhaled nitric oxide or fixed-dose epoprostenol (although of unclear survival benefit),⁵⁵⁹⁻⁵⁶² neuromuscular blockage, recruitment maneuvers, and escalation to ECMO (in appropriately selected patients) may be required.^{395,557,558,563}

General Measures. Stable patients are usually extubated either in the OR or during the early hours after the operation. Otherwise, if pulmonary function is good, extubation should be delayed only under special circumstances, including the presence of cardiac assist devices and the possibility of early reoperation. Early extubation can prevent complications such as ventilator-associated infections and prolonged ventilation dependence.²⁹⁴

In general, criteria for extubation include stable and satisfactory cardiac performance, lack of important cardiac arrhythmia, and appropriate awakening with satisfactory neurologic status. There should be no anticipation of return to the OR (e.g., for bleeding) and satisfactory mechanics and ventilatory lung function, as assessed by arterial blood gas analysis and a clinical estimate of inspiratory force and volume. Increasing emphasis has been placed on the advisability of routinely extubating the normally recovering patient within 6 hours of arrival in the ICU.^{230,233,507} Using standardized protocols that leverage the interdisciplinary team can facilitate fast-track extubation after uncomplicated cardiac surgery procedures.⁵⁶⁴⁻⁵⁶⁷

Atelectasis and the associated loss of functional alveolar units may be particularly problematic in patients with pre-existing obstructive or restrictive lung disease. In such patients, the FiO_2 should be titrated to target $\text{PaO}_2 > 70$ mm and normocarbia, with a goal pH of 7.35 to 7.45.⁵⁵⁸ Standard prophylactic measures include chest physiotherapy and use of devices to encourage deep inspiration to recruit atelectatic areas.^{233,568} Several studies have found that using noninvasive ventilation and maintaining prophylactic nasal positive airway pressure at 10 cm H_2O for at least 6 hours after cardiac and aortic procedures may avoid the need for reintubation in higher-risk patients.⁵⁶⁹⁻⁵⁷² Recent findings suggest that the use of a high-flow nasal cannula (without additional dedicated positive pressure) also may maintain oxygenation and mitigate against the need for reintubation⁵⁷³⁻⁵⁷⁶; a large ongoing RCT (Reducing Reintubation Risk in High-Risk Cardiac Surgery Patients With High-Flow Nasal Cannula) is examining the utility of this approach in the postoperative cardiac surgery patient.⁵⁷⁷

Treating patients with pulmonary dysfunction who are still intubated during the early hours after surgery is largely an intensification of usual management, as specific effective therapy is not available. The FiO_2 level is appropriately adjusted upward if P(A - a)O_2 is large. Minute volume of respiration (not only measured but also judged visually by chest wall excursion) and respiratory rate are adjusted to maintain PaCO_2 at 30 to 35 mmHg. The airway is kept clear by appropriate tracheal suctioning.

When hemoconcentration develops because of plasma leakage from the intravascular space into the interstitial space of the lungs (and at times into all other organs and the pleural and peritoneal spaces), administering concentrated serum albumin can help to counteract this trend; however, the benefit of colloids after cardiac surgery remains unclear.^{578,579} The timely use of diuretics is useful because they reduce extracellular fluid volume and, in turn, extravascular lung water; nonetheless, appropriate flow to the kidneys must be preserved to avoid a prerenal acute kidney injury.³⁴⁸

All the while, efforts to reduce ventilatory support and work toward extubation of the patient must continue (see Appendix 4F). Conducting spontaneous breathing trials and decreasing levels of pressure support appear equally effective in patients who are difficult to wean from the ventilator.

Prolonged Intubation. A prolonged period of endotracheal intubation is necessary when:

- Criteria for extubation are not met
- Neurologic complications are present

- Severe dysfunction or hemodynamic instability of the cardiac subsystem is present
- Persistent chest drainage or a residual cardiac defect makes early return to the OR likely

During periods of prolonged ventilation, active decision-making is required regarding use of continuous versus intermittent sedation and neuromuscular blockade, depending on hemodynamic stability, effectiveness of ventilation, and timing of ventilator weaning. Commonly used sedation and paralytic agents are described in Tables 4.16 and 4.17.

When patients have agitated delirium after cardiac surgery, particularly during ventilator weaning, the priority is evaluating cardiac and pulmonary subsystems to ensure that agitation is not an indicator of important dysfunction.^{580,581} If these subsystems have satisfactory function, the next level of intervention targets communication between the patient and nursing or physician staff or both to ensure that adequate pain management has been achieved and nonpharmacologic approaches to allay patient anxiety are in use. When these interventions are insufficient, using pharmacologic agents at the lowest effective dosage can be considered. These may include propofol, dexmedetomidine, and, in patients with a known history of substance abuse or withdrawal, benzodiazepines (e.g., diazepam, lorazepam, midazolam) used judiciously.^{580,582} Antipsychotic medication (e.g., haloperidol) should be reserved for patients whose pain is well managed but who remain at a safety risk to themselves or the healthcare provider team.⁵⁵⁵

In those unusual circumstances when intubation in older children and adults is necessary for more than 7 to 10 days, consideration is given to tracheostomy.⁵⁸³⁻⁵⁸⁵ Some data suggest that early tracheostomy may be of benefit in patients who are likely to require prolonged mechanical ventilation after their surgery.^{586,587} Tracheostomy has few disadvantages, and it usually increases both patient comfort and ventilation effectiveness. Further, tracheostomy can be helpful in weaning the patient from the ventilator and is not associated with higher rates of sternal infection.^{583,588,589} Tracheostomy is rarely performed in neonates, infants, and young children because it is difficult to manage in this age group.

Reintubation. When the patient meets all criteria for extubation, reintubation is usually unnecessary. However, positive pressure ventilation or reintubation should be considered when:

- PaCO_2 increases to > 50 mmHg over 4 hours in the absence of appropriate respiratory compensation for metabolic alkalosis.
- Signs of decreasing cardiac output are noted.

TABLE 4.16 Commonly Used Neuromuscular Blockers for Paralysis during Ventilator Support

Drug	Action	Bolus Dose	Onset	Duration	Continuous Infusion	Comments
Rocuronium	Non-depolarizing	0.6-1.2 mg/kg	1-3 min	30-70 min	1-12 $\mu\text{g/kg/min}$	Preferred agent for patients without hepatic or renal failure. Given as intermittent bolus or continuous infusion
Vecuronium	Non-depolarizing	Initial: 0.05-0.1 mg/kg Maintenance: 0.1-0.2 mg/kg every hour	2-3 min	40-90 min	0.8-1.7 $\mu\text{g/kg/min}$	Preferred agent for patients without hepatic or renal failure. Given as intermittent bolus or continuous infusion
Cisatracurium	Non-depolarizing	0.1 mg/kg	1-2 min	25-90 min	3 $\mu\text{g/kg/min}$	Generally given as continuous infusion. Preferred for renal or hepatic failure

TABLE 4.17 Control of Anxiety and Agitation during Ventilator Support

Drug	Class	Initial Bolus	Maintenance Bolus	Continuous Infusion	Comments
Midazolam	Benzodiazepine	0.02-0.08 mg/kg up to max of 5 mg	Same as initial	0.04-0.2 mg/kg/hr · h — 1 titrate to desired sedation level	Adverse effects: hypotension, nausea, vomiting May accumulate; use with caution in patients with renal dysfunction
Fentanyl	Opioid analgesic	0.35-1.5 µg/kg	Same as initial, every 30-60 min	0.7-10 µg/kg/h	Adverse effects: bradycardia, nausea, vomiting, respiratory depression
Propofol	Short-acting lipophilic general anesthetic	1-3 mg/kg over 20-30 s		5-50 µg/kg/min	May cause hypotension and bradycardia
Dexmedetomidine	α ₂ -Adrenergic agonist; sedative	1 µg/kg infused over 10 min (often not required and may cause hypotension)		0.1-1.4 µg/kg/hr	Not recommended for more than 4 days to avoid withdrawal symptoms May cause bradycardia and hypotension
Ketamine	General anesthetic	0.3-1 mg/kg		0.5-2 mg/kg/hr	May cause hypertension, tachycardia, decompensated cardiac failure (catecholamine depleted patients), and emergence reactions

- Signs of exhaustion from spontaneous breathing are observed.
- Excessive pulmonary secretions with ineffective coughing occur.

When the situation is borderline, proper management requires careful observation by senior members of the team. A competent professional should perform reintubation.

Renal Subsystem

Adequate Renal Function

Monitoring the adequacy of renal function in the postoperative patient is necessary to avoid cardiac surgery-associated acute kidney injury (CSA-AKI), which occurs in 15% to 50% of adults undergoing cardiac surgery and is characterized by a 0.3 mg/dL or 50% increase in serum creatinine from baseline.⁵⁹⁰ The development of reversible stage 2 or 3 CSA-AKI is associated with longer lengths of stay, more readmissions, and higher costs. Most importantly, CSA-AKI increases short- and long-term risk for death: After cardiac surgery with CPB, severe renal injury (the requirement for dialysis) occurs in 1% to 2% of patients and is associated with 60% mortality.⁵⁹¹ Milder degrees of kidney failure, occurring in up to 17% of patients, are associated independently with up to a 19-fold increase in short-term mortality.^{592,593}

As a guide to the continuing evaluation of the renal subsystem, a urinary catheter is inserted in the OR to monitor urine flow during the first several hours after surgery. Serum potassium concentration is measured regularly during the first 12 to 24 postoperative hours and then as needed on the basis of the patient's clinical condition and renal function status. Serum creatinine and blood urea nitrogen levels are measured on arrival at the ICU and then each morning, typically for the first 48 hours.

Urine output has traditionally been considered adequate when urine volume exceeds 30 to 40 mL/h/m² or 0.5 to 1.0 mL/h/kg in the case of infants and small children. However, a urine volume lower than this level may not be sufficient to identify renal injury, and serum creatinine levels may not rise until well over half of

kidney function has been lost, up to 48 hours after the damage has occurred.

Cardiac Surgery-Associated Acute Kidney Injury

Definitions. Three different classification systems have been used to define CSA-AKI (Table 4.18). The Risk, Injury, Failure, Loss, and End-stage (RIFLE) kidney disease classification defines three severity grades for CSA-AKI (risk, injury, and failure) based on changes to serum creatinine and urine output and two clinical outcomes (loss, end-stage).^{594,595} The Acute Kidney Injury Network (AKIN) group sought to increase the sensitivity of the RIFLE criteria by recommending that a smaller change in serum creatinine (≥ 26.2 µmol/L) be used as the threshold to define the presence of AKI.⁵⁹⁶ In 2012, the Kidney Disease: Improving Global Outcomes (KDIGO)⁵⁹⁰ AKI working group published further criteria for defining and classifying AKI that is meant to serve as a combination of AKIN and RIFLE definitions. Specific to cardiac surgery patients are:

- Increase in serum creatinine by ≥ 26.5 µM (0.3 mg/dL) within 48 hours; or
- Increase in serum creatinine to ≥ 1.5 times baseline within 7 days.

Patient Risk Factors. Preoperative renal function impairment that is intractable to therapy considerably increases the risk for acute renal failure soon after surgery.⁵⁹⁷ Therefore, preoperative evaluation includes assessment of renal function.

The predictive value of serum creatinine and GFR on the risk for postoperative dialysis has been quantified by Mehta and colleagues⁵⁹⁸ for adult patients undergoing cardiac surgery. Although GFR is traditionally the standard measure of renal function, its formal measurement is frequently not feasible in the clinical setting. A useful approximation of creatinine clearance can be made with the Cockcroft-Gault equation^{599,600}:

$$\text{CrClearance} = \frac{(140 - \text{age} \times \text{lean body weight}[\text{kg}])}{\text{Cr}(\text{mg} \cdot \text{dL}^{-1}) \times 72}$$

TABLE 4.18 Cardiac Surgery Associate Acute Kidney Injury Classifications

Serum Creatinine Criteria		Urine Output Criteria
RIFLE		
Stage 1	Increase in serum creatinine $\geq 1.5\times$ baseline or $>25\%$ decrease in eGFR	<0.5 mL/kg/h for ≥ 6 h
Stage 2	Increase in serum creatinine $\geq 2.0\times$ baseline or $>50\%$ decrease in eGFR	<0.5 mL/kg/h for ≥ 12 h
Stage 3	Increase in serum creatinine $\geq 3.0\times$ baseline or $>75\%$ decrease in eGFR or an absolute serum creatinine ≥ 354 $\mu\text{mol/L}$ with an acute rise of at least 44 $\mu\text{mol/L}$	<0.3 mL/kg/h ≥ 24 h or anuria ≥ 12 h
AKIN		
Stage 1	Increase in serum creatinine ≥ 26.2 $\mu\text{mol/L}$ or increase to $\geq 150\%$ – 199% (1.5- to 1.9-fold) from baseline	<0.5 mL/kg/h for ≥ 6 h
Stage 2		
Stage 3	Increase in serum creatinine to 200% – 299% (>2 – 2.9 fold) from baseline Increase in serum creatinine to $\geq 300\%$ (≥ 3 -fold) from baseline or serum creatinine ≥ 354 $\mu\text{mol/L}$ with an acute rise of at least 44 $\mu\text{mol/L}$ or initiation of RRT	<0.5 mL/kg/h for ≥ 12 h <0.3 mL/kg/h ≥ 24 h or anuria ≥ 12 h

A comparison of the different definitions and classification schemes for CSA-AKI.

where Cr = creatinine. For women, multiply the result by 0.85 to account for smaller muscle mass.

Chronic heart failure (preoperative NYHA class IV) considerably increases postsurgical risk for acute renal failure, as does *congenital heart disease* after cardiac surgery in older patients.⁶⁰¹ A renal lesion is known to exist preoperatively in many such patients. This does not appear to be the case in young patients; for example, renal failure is rare in infants and children undergoing repair of tetralogy of Fallot.⁶⁰² Further, when cardiac output is importantly reduced after cardiac surgery in neonates and infants, acute renal failure rates as high as 8% to 10% have been reported,^{603,604} possibly because immature kidneys may have less ability to concentrate urine in the context of reduced renal blood flow. Compared with older patients, infants may develop more tissue hypoxia during and early after CPB, with a resulting increase in production of potassium, blood urea nitrogen, and other substances, some of which may be nephrotoxic. Uric acid levels, for example, were shown by Hencz and colleagues⁶⁰⁵ to rise to nephrotoxic levels (10 mg/dL) in some patients within 24 hours of operation, and the levels were significantly higher in patients younger than 3 years of age than in older children.

In adults, the development of CSA-AKI results from the interplay between patient susceptibility to renal injury and perioperative renal insults.⁵⁹¹ Multiple factors can affect the kidney, including hemodynamic instability, inflammation, hemolysis, and nephrotoxic agents. In the setting of cardiac surgery, it is believed that renal injury arises intraoperatively during CPB.^{606,607} Indeed, a renal tubular injury signal is measurable shortly after the initiation of CPB.

The evolution of ischemic CSA-AKI is conceptually divided into initiation, extension, maintenance, and repair phases.⁶⁰⁸⁻⁶¹⁰ The initiation phase is the period immediately after an ischemic insult initiates reversible tubular injury and GFR drops abruptly. At the tissue level, ATP depletion, reactive oxygen species generation, and inflammatory mechanism induction contribute to sublethal epithelial and endothelial damage.^{608,611,612} If the ischemic insult is brief and mild, the injury remains sublethal, and organ function recovers rapidly. Prolonged or severe ischemia followed by reperfusion, however, leads to an extension phase characterized by microvascular damage, tubule cell death, desquamation, and luminal obstruction.

Further decline in GFR ensues, and the injured endothelial and epithelial cells amplify inflammatory cascades. The maintenance

phase is characterized by a dynamic equilibrium between waning injury and evolving repair mechanisms. The balance between cell survival and death determines the severity and duration of this phase.

Karkouti and colleagues⁶⁰⁹ conducted a study of 3500 adult cardiac surgery patients from seven hospitals to identify potentially modifiable risk factors associated with CSA-AKI. Using multivariate logistic regression analysis, the investigators identified three potentially modifiable variables that were independently and strongly associated with AKI: preoperative anemia, perioperative RBC transfusions, and surgical reexploration. In addition to these potential targets, several recent clinical practice guidelines and expert consensus statements have been generated to provide an evidence-based synthesis of strategies for preventing CSA-AKI.^{233,613-615} An example of a preoperative, intraoperative, and postoperative approach provided by the Harrington Heart and Vascular Institute–University Hospitals Taskforce on Preventing CSA-AKI in the Perioperative Period identifies six components for the perioperative team to consider (see also Appendix 4G):

1. **Preoperative:** Identify patients at risk for CSA-AKI (low/moderate/high risk) and provide patient medication education.⁶¹⁶
2. **Intraoperative:** Identify AKI risk factors during OR timeout and consider medications and hemodynamic management.
3. **Postoperative:** Assess risk factors for AKI handed over to the team upon ICU arrival.⁶¹⁷
 - a. Implement goal-directed therapies for high-risk patients by using a hemodynamic algorithm in addition to blood sugar and medication management.⁶¹⁸
 - b. Implement ongoing evaluation of renal function, hemodynamic status, and volume status.
 - c. Conduct outcome measurement and monitoring.

The following sections provide a framework for this approach to patient management aimed at minimizing the risk for CSA-AKI (see Engelman and Shaw⁶¹⁹ for a protocol example).

Preoperative Risk Factors. Preoperative identification of patients at risk for CSA-AKI is a key step in mitigating it. The perioperative team performs a kidney health assessment^{233,619} that includes a full medical and medication history, determination of baseline kidney function, a history of previous episodes of AKI after illness, diagnostics, procedures or surgery, recent nephrotoxin exposure, and assessment for anemia.⁶²⁰ A complete urinalysis should be undertaken, and consultation with a nephrologist is

advisable. Further considerations include optimizing glycemic control by maintaining blood glucose at 80 to 180 mg/dL, limiting aminoglycoside antibiotics, using vancomycin judiciously, and holding ACE inhibitors and angiotensin receptor blockers (ARBs) for up to 48 hours preoperatively. In addition, consider holding all nonsteroidal antiinflammatory drugs (NSAIDs) for 1 week before surgery when possible.

The use of a dynamic predictive risk score can help the perioperative team classify patients as having either a low, moderate, or high risk.⁶¹⁶ The patient's risk for perioperative CSA-AKI will be documented within the medical record and should be discussed at the OR presurgical huddle and during handoff to the ICU team initially responsible for patient care immediately after surgery.⁶¹⁷

Intraoperative Risk Factors. Key considerations highlighted in recent guidelines include avoiding hypothermic perfusion ($>37^{\circ}\text{C}$) and using a goal-directed oxygen delivery strategy that includes appropriate hemodynamic monitoring and fluid management.⁶¹⁵

Rewarming the patient on CPB may result in differential tissue perfusion and has been proposed as a mechanism for renal dysfunction after cardiac surgery.⁶²¹ Thus, avoiding hyperthermia during this phase of the operation is recommended.^{622,623} Similarly, inadequate DO_2 has been associated with the development of CSA-AKI. A seminal single-center, prospective observational study of 1048 patients on CPB found that oxygen delivery lower than DO_2 that was indexed to body surface (DO_{2i}) of $<272 \text{ mL/min/m}^2$ was independently associated with CSA-AKI that required renal replacement therapy.⁶²⁴ This finding was further refined in subsequent analyses, indicating that a DO_{2i} of $\geq 270 \text{ mL/min/m}^2$ is required to avoid stage 1 AKI.^{331,625,626} The intraoperative team should seek to adjust arterial flow and hematocrit levels to maintain DO_{2i} above this level.^{233,613,619} At present, there is no evidence that a dopamine infusion during CPB or the use of mannitol provides any protection against CSA-AKI.⁶¹³

Postoperative Risk Factors. During the first 24 to 48 hours after a cardiac procedure, the kidney remains vulnerable to ongoing or new insults. Hemodynamic perturbations, bleeding, and administration of drugs or agents with nephrotoxic side effects can compound renal dysfunction after CPB. The ICU team should review a patient's preoperative risk score during sign-out with OR anesthesia staff and then calculate a new risk score based on a dynamic predictive scoring tool at the time of ICU admittance.⁶¹⁶ The postoperative risk score should be documented in the initial ICU history and physical and then incorporated into the genitourinary section of the patient's daily assessment and plan in the progress note. As with the intraoperative phase, ongoing care should include optimizing glycemic control, using an insulin infusion to maintain blood glucose at 120 to 180 mg/dL (see endocrine section later), limiting aminoglycoside antibiotics, and avoiding NSAIDs and IV radiocontrast agents. Consideration should also be given to withholding ACE inhibitors and ARBs in patients who are initially oliguric or classified as having moderate or high risk for CSA-AKI. The use of a urine catheter for hourly monitoring of urine output should be maintained, with the goal of the patient producing a minimum of 0.5 to 1.0 mL/h/kg.

The choice of fluid, in terms of type (balanced versus saline crystalloid or crystalloid versus albumin), remains varied and controversial.^{578,579,627-633} Regardless, excessive fluid administration is associated with venous congestion and increased rates of AKI. Fluid administration and other agents should be goal-directed to ensure adequate kidney blood flow. The perioperative team should seek to optimize intravascular blood volume by using

continuous hemodynamic hourly monitoring (either invasive or noninvasive) with administration of fluid challenges with lactated Ringer's solution for responsiveness if the patient is oliguric; when the CI is $<2.0 \text{ L/min/m}^2$ and the CVP is <8 to 10 mmHg ; if the PADP is $<14 \text{ mmHg}$ (if a pulmonary artery catheter is being used); or if stroke volume variation in a ventilated patient with normal sinus rhythm is $>13\%$ (if using an arterial pressure waveform monitor).⁶¹⁹ Diuretics should be considered if CVP is $>15 \text{ mmHg}$ or PADP is $>20 \text{ mmHg}$.

Oliguric and High-Risk Patients. All patients with oliguria or deemed at high risk for CSA-AKI should be treated with the postoperative hemodynamic algorithm known as the Kidney Disease Improving Global Outcomes (KDIGO) Bundle.⁶²³ This practice was validated by the PrevAKI RCT,⁶¹⁸ in which 1046 high-risk patients were identified through two new renal biomarkers: insulin-like growth factor-binding protein 7 (IGFBP7) and tissue inhibitor of metalloproteinases-2 (TIMP-2). These biomarkers are involved in G1 cell cycle arrest and are able to identify patients at high risk for CSA-AKI through a clinically available assay.⁶³⁴ An assay result of $[\text{TIMP-2}] \cdot [\text{IGFBP7}] >0.3$ has been associated with CSA-AKI.⁶³⁵

In the PrevAKI study, patients with a positive biomarker result were randomized to an intervention protocol that included using fluids, diuretics, and inotropes to maintain a systolic blood pressure 100 to 130 mmHg and/or a MAP of 65 to 90 mmHg, a CI $>2.2 \text{ L/min/m}^2$, urine output of at least 0.5 mL/kg/h (using lean body mass), and $\text{SvO}_2 >65\%$. Within this study protocol, the investigative team used a hemodynamic catheter-based cardiac monitoring system that combined pulse-wave contour analysis, transpulmonary continuous thermodilution, and central venous oxygen saturation monitoring to optimize volume status. Further, ACE-I, ARB medications, NSAIDs, and radiocontrast agents were avoided in high-risk patients for at least 48 hours postoperatively. In addition, an intensive insulin infusion order set was implemented for the first 12 to 24 hours to allow for optimal blood glucose management and to avoid hyperglycemia, defined as $>180 \text{ mg/dL}$ for more than 3 hours.

Patients in the intervention arm of the PrevAKI trial had lower rates of AKI versus controls (55.1% vs. 71.7%, respectively; odds ratio 0.48, 95% CI 0.29–0.80) and had significant reductions in moderate to severe AKI as well (29.7% versus 44.9%). There were no differences between the groups in the need for renal replacement therapy in those who developed CSA-AKI despite the use of the KDIGO Bundle. Similarly, the 90-day mortality, ICU LOS, and major adverse kidney events endpoints did not differ statistically between treatment groups.

Lastly, in patients with RV dysfunction, venous hypertension can have deleterious effects on renal function due to a reduced transrenal pressure gradient. It is prudent for the perioperative team to provide appropriate inotropic support to the patient with RV dysfunction or RV failure and to consider renal replacement therapy for volume management (particularly in patients with a CVP >15 or a PADP >20).

Ongoing Evaluation and Monitoring. A renal function panel should be monitored at least every 12 hours during the first 24 hours for all elective cardiac surgery cases, regardless of risk score. Serum creatinine and hourly urine output should be monitored until no further indicator of AKI persists. It is reasonable to consider reassessing AKI risk in all elective cardiac surgery patients by using a dynamic predictive scoring tool after the first 24 hours.⁶¹⁶

As outcomes related to CSA-AKI can be impactful, from both patient outcome and healthcare resource utilization viewpoints, it

is suggested that the surgical and perioperative teams monitor AKI rates as a component of ongoing quality efforts.

Summary Points

- Consider implementing a dynamic risk score for patients in the preoperative phase and communicate results to the perioperative team.
- Creatinine elevation is a lagging indicator of CSA-AKI, whereas oliguria may be an early indicator of CSA-AKI in a patient with euvolemia.
- Both hypovolemia and hypervolemia can exacerbate CSA-AKI and can be remedied by implementing goal-directed fluid therapy (the KDIGO Bundle).
- Packed RBC transfusion is independently associated with CSA-AKI development.

Neuropsychological Subsystem

Neurologic complications have been a recognized concern since the inception of cardiac surgery.⁶³⁶ Neurologic sequelae after cardiac surgery are not uncommon, and the perioperative team will often need to assess and intervene on potentially life-threatening complications rapidly. This chapter focuses on acute perioperative neuropsychological complications after cardiac surgery, the more common of which include stroke; postoperative cognitive dysfunction (including delirium, an acute confusional state characterized by fluctuating mental status, inattention, and either disorganized thinking or altered level of consciousness); mood disorders; and seizures.

Ischemic or Embolic Cerebral Events

In 1996, Roach and colleagues⁶³⁷ described rates of adverse cerebral outcomes after CABG in 2108 patients across 24 centers. They found that the rate of adverse cerebral outcomes was 6.1%, with resultant substantial increases in mortality, LOS, and resource utilization. In more recent reviews, which include patients undergoing valvular, aortic, or transcatheter aortic valve replacement (see also Huded and colleagues⁶³⁸) procedures, rates of clinically determined periprocedural stroke ranged from 1.1% to 9.7%.^{637,639-642} Importantly, mortality associated with postoperative stroke has been reported as high as 20%.⁶⁴³⁻⁶⁴⁵ In addition, in a meta-analysis, rates of clinically asymptomatic radiographic infarctions after cardiac procedures ranged from 27% to 80%, with approximately 1 in 10 patients experiencing a periprocedural stroke or transient ischemic attack.

Risk Factors. Patient and surgical risk factors for postoperative stroke include history of stroke, peripheral vascular disease, diabetes, hypertension, dialysis dependency, severe chronic lung disease, previous cardiac surgery, preoperative infection, urgent operation, CPB lasting >2 hours, need for intraoperative hemofiltration, and high transfusion requirement.^{640,644,646}

In a large retrospective review of more than 45,000 patients undergoing CABG over almost 3 decades, Tarakji and colleagues⁶⁴³ found that the risk for stroke peaks by the second postoperative day. Others have reported that the risk is reduced after that but can persist for as long as 5 years.⁶⁴⁷ Whitlock and colleagues⁶⁴⁸ classified postoperative strokes as early or late, based on etiology. Gaudino and colleagues⁶³⁹ classified stroke as intraoperative, early postoperative (first 7 days), or late (beyond 7 days).

Approximately 40% to 50% of perioperative strokes are discovered when the patient awakens from anesthesia and are therefore attributed to an intraoperative event.^{649,650} Broadly, these can be subclassified as secondary thromboembolic events (60%, arising from aortic, cardiac, or CPB sources) or hypoperfusion events (40%).⁶³⁹ The most common etiology of early postoperative stroke events is new POAF (see “Atrial Fibrillation” earlier).^{432,648} Late stroke can be attributed to the baseline atherosclerotic burden and the patient’s risk profile; persistent POAF after cardiac surgery remains a long-term risk.^{426-428,432,639,648}

Intraoperatively, several strategies may minimize the risk for cerebral hypoperfusion and embolic events: maintaining a higher mean arterial blood pressure,⁶⁵¹ using cerebral oximetry monitoring and intervening with different techniques for low cerebral saturation,⁶⁵²⁻⁶⁵⁵ using TEE and epiaortic ultrasonography to identify and avoid manipulation of aortic atheromatous disease,⁶⁵⁶ using neuraxial anesthetic adjuncts,⁶⁵⁷ maintaining a hemoglobin concentration >60 g/L (or 6 g/dL) during CPB and at least 7 g/dL (or 70 g/L) postoperatively,⁶⁵⁸ and maintaining rationalized control of blood glucose (80–180 mg/dL).⁶⁵⁶

Pharmacologically, aspirin initiated within 6 hours postoperatively continues to be the standard of care after CABG. This approach is associated with lower perioperative stroke rates if administered within 48 hours of revascularization.^{659,660} Data from the LAAOS III trial indicate that intraoperative occlusion of the left atrial appendage in patients with preexisting AF and a CHA₂DS₂-VASc score of at least 2 was associated with significant short-term and long-term reduction in ischemic stroke and systemic embolism.⁴⁷⁴

Managing Acute Stroke after Cardiac Surgery. When the perioperative team identifies a new neurologic deficit indicative of a stroke after a cardiac procedure, a hospital-specific acute stroke protocol should be activated.^{649,661} In addition to seeking neurologic consultation, the perioperative team should obtain brain and cerebrovascular imaging while maintaining adequate blood pressure and avoiding fever, hypoglycemia and hyperglycemia, and hypovolemia.⁶⁶² Imaging typically consists of a noncontrast computed tomography (CT) of the head and CT angiography of the intracranial and extracranial (carotid and vertebral) vessels. Magnetic resonance imaging (MRI) of the brain and magnetic resonance angiography may be required. Of note, concerns persist about the presence of epicardial pacing wires, although investigations have shown the safety of performing an MRI in patients with retained materials after cardiac surgery.⁶⁶³⁻⁶⁶⁵ In this circumstance, the perioperative team should consult institutional policy before proceeding.

Treatment options for hyperacute stroke include thrombolysis and thrombectomy; however, only the latter is possible in the immediate postoperative patient due to the bleeding risk from IV alteplase. The timing of mechanical thrombectomy is an important consideration, with the current indication for patients with an acute ischemic stroke and large-vessel occlusion being within the first 24 hours, on the basis of recent trials^{666,667} (but not yet updated in current guidelines). Thus, the perioperative team should examine and document last-known normal exam results to ensure that the patient has appropriate IV vascular access for the procedure. In addition, for patients with a pending mechanical thrombectomy, it is reasonable to maintain blood pressure at ≤185/110 mmHg before the procedure.⁶⁶¹

Postoperative Cognitive Dysfunction

Intelligence, problem-solving, concentration, learning, memory, error-free performance, and dexterity are components of the general

neuropsychological subsystem and are considered cognitive functions. Neurocognitive disorders are common in preoperative patients and range from mild cognitive impairment (14%–48% of older adults aged >70 years) to severe cognitive difficulties consistent with dementia (10% of these older adults).⁶⁶⁸⁻⁶⁷²

It has long been observed that cognitive difficulties affect both short-term and long-term recovery in cardiac surgery patients. A seminal paper by Newman and colleagues⁶⁷³ reported that the incidence of cognitive decline was 24% at 6 months and 42% at 5 years after cardiac surgery (and was even higher in patients who underwent TAVR).⁶⁷⁴ Several additional reports indicated that postoperative delirium places the cardiac surgery patient at risk for further cognitive decline.^{670,671,675-679}

A problem with characterizing postoperative neurocognitive changes has been the varied definitions used to report postoperative outcomes. To address this issue, in 2018, the Perioperative Cognition Nomenclature Working Group developed recommendations for standardizing the nomenclature of postoperative cognitive changes (Table 4.19).⁶⁸⁰ The term *perioperative neurocognitive disorders* is an overarching term for any cognitive impairment or change identified in the preoperative or postoperative period (up to 12 months after surgery or other procedure, assuming the cognitive decline cannot be accounted for by any other medical condition).⁶⁸⁰

Delirium has long been recognized as a complication of cardiac surgery. Postoperative delirium is estimated to occur in 20% to 25% of cardiac surgery patients.⁶⁸¹⁻⁶⁸⁴ It is more common among older adult patients and those with greater comorbid disease burden.⁶⁸⁵⁻⁶⁸⁷ Delirium is triggered by multiple potential causes,⁶⁸⁸ and in the cardiac surgery patient, it is the product of a baseline vulnerability, stress from the surgical intervention and hemodynamic perturbations, and iatrogenesis in the postoperative phase of care.⁵⁸¹ Cardiac surgery procedures are being performed on older patients with recent coronary syndromes, higher NYHA classifications, lower left ventricular ejection fractions, cardiogenic shock, and frailty (defined as an accumulation of deficits resulting in a vulnerability to stressors^{689,690})—all of which are associated with increased rates of postoperative delirium.⁶⁹¹ Postoperative

delirium is variable in duration, with most episodes lasting only a few days; however, in up to 20% of individuals, delirium may persist for weeks or months.⁶⁸⁸

The importance of delirium was previously dismissed, as it was believed to be a transient and benign entity. More recently, increasing attention has been paid to delirium's negative effects on postoperative outcomes, including long-term survival, hospital readmissions, and cognitive and functional recovery.^{670,671,678,692-697} Despite its strong association with adverse postoperative events, delirium is often underrecognized as a source of end-organ dysfunction.

Risk Factors. Patient-related factors being examined for their association with postoperative neurocognitive difficulties include silent cerebral infarction, cerebral autoregulation, alterations in blood-brain barrier function, and brain protein biomarkers such as S100Beta, glial fibrillary acidic protein, tau, neurofilament light, and neuron-specific enolase.⁶⁹⁸⁻⁷⁰² Rudolph and colleagues⁶⁸¹ identified four risk factors associated with an increased risk for postoperative delirium after cardiac surgery: previous stroke or transient ischemic attack, low Mini Mental State Examination score, abnormal serum albumin, and a Geriatric Depression Scale score indicating depression.⁷⁰³ Other researchers have further identified elevated preoperative EuroSCORE, older age (≥ 70 years), number of comorbidities, history of delirium, alcohol use, type of surgery, blood transfusion, mechanical ventilation, postoperative low cardiac output, inadequate management or overtreatment of pain, sleep deprivation, use of benzodiazepines, and infection as additional risk factors.^{683,684,704}

Management. Postoperative delirium can be difficult to manage. Thus, prevention and management should begin in the preoperative phase of care by identifying key risk factors for its occurrence in the postoperative phase.⁷⁰⁵ Risk should be documented in the patient's chart, and delirium risk should be communicated to the intraoperative team during the OR huddle and at handover to the ICU team postoperatively.^{617,706}

In the postoperative phase, a systematic delirium screening tool that assesses the full spectrum of delirium presentation (hypoactive, hyperactive, or mixed) should be incorporated as part of routine monitoring in the ICU at least once per nursing shift.⁷⁰⁷

TABLE 4.19 Perioperative Neurocognitive Disorders

TIME PERIOD		TERM AND DEFINITION		COMMENTS
Preoperative		Mild NCD	Major NCD	As in community
Perioperative Cognitive Disorders				
Emergence from anesthesia	Emergence excitation or delirium			
<i>From:</i> Immediately postoperative	Delirium (postoperative**) OR	<i>Delayed</i>	<i>Delayed</i>	The time for expected resolution is based on perioperative conditions (e.g., complications/infection/prolonged hospitalization)
<i>Until:</i> Expected recovery (between 30 days and 3 months)*	<i>Delayed Neurocognitive Recovery</i>	<i>Neurocognitive Recovery#</i>	<i>Neurocognitive Recovery#</i>	
<i>From:</i> Expected recovery (30 days to 3 months)		Mild NCD postoperative	Major NCD postoperative	"Postoperative" is an indicator of the temporal association with the anesthesia/surgery event
<i>Until:</i> 12 months				
Beyond 12 months		Mild NCD	Major NCD	As in community if a <i>new</i> diagnosis after this time

* The time point from at least 30 days when the effects of anaesthesia and surgery should have resolved.

**The postoperative specifier will be applied where delirium is persistent beyond hospital discharge.

#Also applies to assessments prior to medical readiness for discharge, although it should be noted any decline prior to readiness for discharge is unlikely to represent a delay in recovery.

Adapted with permission from Evered L, Silbert B, Knopman DS, et al. Recommendations for the Nomenclature of Cognitive Change Associated with Anaesthesia and Surgery-2018. *Anaesthesiology*. 2018;129(5):872-879

NCD, neurocognitive disorder.

When an active screening program in the ICU is lacking, delirium goes undiagnosed in more than 70% of cases.^{708,709} A systematic methodology for delirium screening in the cardiothoracic ICU will use either a confusion assessment method^{710,711} or an intensive care delirium screening checklist.⁷¹² Delirium screening should be undertaken in tandem with arousal assessment. The Riker Sedation Agitation Scale⁷¹³ and the Richmond Agitation-Sedation Scale⁷¹⁴ are some of the more widely used tools for determining levels of sedation and agitation.

Once delirium is detected, it is important to determine the underlying cause (e.g., pain, hypoxemia, low cardiac output, sepsis) and initiate appropriate treatment.^{580,688,705,715} Nonpharmacologic strategies are the first-line component of management and should include the multicomponent Society of Critical Care Medicine ICU “ABCDEF Liberation Bundle” in the postoperative cardiac surgery ICU (see: <https://www.sccm.org/ICULiberation/Home>).^{233,555} A multicenter randomized trial examining a multicomponent family-support intervention found that the intervention was associated with higher ratings of the quality of communication and a shorter ICU LOS.⁷¹⁶ Although these data were not specific to the cardiac surgery patient, it is reasonable for the perioperative team to consider incorporating this approach within their delirium management protocols.

Pharmacologic strategies using prophylactic medicines such as clonidine or antipsychotics have been found to reduce delirium rates, as shown in several RCTs.⁷¹⁷⁻⁷²² Current recommendations indicate that the use of antipsychotic medications should be limited to patients who have particularly distressing symptoms or who pose safety risks to themselves or the healthcare team.⁵⁸⁰ An exception may be dexmedetomidine, a selective α_2 -adrenergic agonist⁷²³ that may facilitate lighter sedation, shorten the duration of mechanical ventilation, reduce the risk for delirium, and provide analgesia in critically ill patient populations.^{721,724-726} However, dexmedetomidine is also associated with bradycardia and lowered blood pressure. In a recent rapid practice guideline, the writing committee stated that the use of dexmedetomidine could be considered “...if the desirable effects including a reduction in delirium are valued over the undesirable effects, including an increase in hypotension and bradycardia...”⁵⁸²

Importantly, McPherson and colleagues⁶⁸² reported that the use of chemical restraints (such as benzodiazepines) or physical restraints and restraining devices predisposed patients to a greater risk for delirium, pointing to potential areas for quality improvement in the cardiac surgical ICU.

Mood Dysfunction

The patient’s preoperative mood characteristics play a major role in postoperative mood state. Contemporary studies, however, estimate the rates of preoperative depression to be 10% to 30% in patients generally and 15% to 47% in cardiac surgery patients, and it is associated with new postoperative delirium and longer-term mood and cognitive issues.

Risk Factors. The absence of daily access to an empathetic individual who is not part of the surgical team, both before and after surgery, is an important risk factor for increased anxiety and depression postoperatively. Postoperative depression after cardiac surgery is an independent risk factor for subsequent cardiac events⁷²⁷⁻⁷²⁹ and is associated with increased risk for hospital readmission⁷³⁰ and death.^{727,728,731} Postoperative depression is probably influenced by a complex interaction between genetic and psychosocial predisposing factors, by neuroendocrine activation, and by the release of proinflammatory factors.⁷³²⁻⁷³⁵

Management and Treatment. The most important methods for preventing mood dysfunction include daily access to family or caregivers and physician attention to the patient as a sensitive and threatened human being.⁷³⁶⁻⁷³⁸ When the mood state becomes severely abnormal, psychiatric assistance is required. Lastly, appropriate communication to the patient’s primary caregiver at the time of hospital discharge should include follow-up of both cardiac rehabilitation and mental health needs.⁷³⁹

Other Neurologic Issues

Visual Defects. Visual-field defects are sometimes perceived by the patient and confirmed by testing; these usually regress or disappear spontaneously.⁷⁴⁰ Difficulty focusing, such as is required for reading, is sometimes experienced in the early postoperative period (up to 30 days). These issues can be classified as mild-delay neurocognitive recovery⁶⁸⁰ and often resolve without specific treatment.

Seizures. Postoperative seizures are relatively uncommon after cardiac surgery but can have important consequences for the patient.^{741,742} Incidence rates of clinically evident seizures range from 0.5% to 8.0%. Postoperative seizures usually appear within the first 2 days after surgery and are typically short (2–3 minutes), but they can deteriorate to status epilepticus.⁷⁴¹ When seizures do occur, recurrence rates of 40% to 66% have been reported.⁷⁴²

Goldstone and colleagues,⁷⁴³ in an analysis of approximately 2600 patients undergoing cardiac surgery, determined that 71% of observed seizures were of the generalized tonic-clonic type, 26% were simple/complex partial seizures, and 3% were status epilepticus. The lowest seizure rates were for isolated CABG, and higher rates were observed for combined valve/CABG and aortic procedures. Postoperative seizure was associated with a longer ICU and hospital LOS and nearly fivefold higher operative mortality (29% vs. 6% in patients with no seizure). Reported causes include ischemic strokes (the most common etiology in this cohort), DHCA, aortic calcification or atheroma, older age (>75 years old), previous strokes, critical preoperative state, cerebral air embolism, medication toxicity or withdrawal, substance (alcohol) withdrawal, and other operative drugs such as larger doses of tranexamic acid.⁷⁴³⁻⁷⁴⁸

Immediate management should focus on the “ABCs” of acute resuscitation: airway, breathing (with an increase in FiO_2), and circulation. If the seizure has not self-terminated, IV benzodiazepine is the appropriate first-line agent. If this is unsuccessful, one of several antiseizure medications (e.g., valproic acid, levetiracetam, phenytoin) can be given. If seizures do not terminate with one antiseizure medication load, the perioperative team should next infuse propofol or midazolam and plan to secure the airway with endotracheal intubation.

Once the patient is stabilized, appropriate neurologic imaging (CT) should be undertaken to rule out acute ischemic stroke or hemorrhage. An electroencephalogram may be required to rule out nonclonic seizures in patients with remaining somnolence or delirium. The patient’s medical history should be queried for a previous history of seizures or other conditions that may predispose the patient to seizure activity (e.g., substance withdrawal). Decisions about the need for longer-term antiepileptics should be made in conjunction with a neurology consultation.⁷⁴¹

Summary Points

- Both clinical and subclinical neurologic injuries, such as ischemic infarctions, are common after cardiac surgery.

- Early diagnosis and timely management can improve surgical outcomes.
- Immediate neurologic consultation (such as with a brain attack team) is critical for patients with acute focal neurologic deficits.
- Further research on how these subclinical ischemic infarctions affect long-term outcomes is necessary.

Gastrointestinal Subsystem

Nutrition

It has long been understood that a patient's nutritional status before cardiac surgery is an important driver of clinical outcomes.⁷⁴⁹ Well-nourished patients have better outcomes and lower postoperative mortality after cardiac surgery.^{750,751} That said, malnutrition, defined as an unintentional imbalance in nutritional intake (not necessarily decreased intake), is evident in 20% of patients before cardiac surgery.^{752,753} As the overall population ages and patients undergoing cardiac surgery are generally more frail, identifying and reversing malnutrition have become increasingly important for the perioperative team.^{754,755}

The typical perioperative fasting, inflammatory, and regenerative processes (such as wound healing, immune function, and acute-phase reactants) induce an exponential increase in nutritional demand that results in a demand-supply imbalance (i.e., a state of metabolic stress).⁷⁵⁶ Traditional practices such as fasting before surgery (typically for 6–12 hours) and delaying or inadequately initiating postoperative nutrition (typically for 2 days because of complications like swallowing difficulties or postoperative nausea and vomiting) further compound issues with malnutrition.⁷⁵⁷⁻⁷⁵⁹ In addition, the metabolic and immune response to injury induces insulin resistance, further aggravating a calorie-deficient state.^{760,761} Ultimately, this perioperative demand-supply imbalance produces an extensive insult to a reserve-deficient physiologic system (as in a typical frail patient), disproportionately diminishing the patient's health status and hindering recovery to baseline functional capacity.^{221,762-764}

To address this vulnerability, an expert consensus panel on enhancing recovery after surgery has recommended the following management strategies.²³³

1. **At baseline:** The perioperative team should determine the patient's baseline frailty, nutritional status (using a dedicated screening tool and a serum albumin), and glycemic control (by measuring hemoglobin A1C).^{752,754,762,765-767} **Rationale:** To establish baseline risk and identify possible targets for therapeutic intervention and optimization.
2. **Preoperatively:** Clear liquids should be continued until 2 to 4 hours before surgery, and solid foods until 6 hours before surgery; carbohydrate loading (24 g of a complex carbohydrate beverage) should be initiated 2 hours before surgery. **Rationale:** Preoperative intake of a complex carbohydrate load is associated with reduced insulin resistance and tissue glycosylation, improved postoperative glucose control, and enhanced return of gastrointestinal function; it does not appear to be associated with increased risk for aspiration.^{231,768-770}
3. **Preoperatively:** When possible, oral nutritional supplementation should be used in malnourished patients for 7 to 10 days before cardiac surgery. **Rationale:** Evidence from the noncardiac surgery literature suggests that supplementation with whey protein provides essential branched-chain amino acids, notably leucine, and is superior to other protein sources for

stimulating muscle protein synthesis and attenuating muscle loss in older adults.⁷⁷¹⁻⁷⁷⁶

In critically ill patients for whom oral intake cannot be initiated in a timely fashion, particularly those with an anticipated long ICU LOS, early enteral nutrition or combined enteral and parenteral nutrition has been recommended.^{777,778} Nonetheless, recent data from both critical care and cardiac surgery patients suggest that the perioperative team should exercise caution when using early feeds with mineral supplementation, given the lack of proven benefit,⁷⁷⁹ risk for bowel ischemia (particularly in patients taking multiple vasopressors), and greater risk for renal dysfunction.^{780,781} Decisions about the timing, route, and dosage of early postoperative feeds in a critically ill patient who has undergone cardiac surgery should be made on an individual basis, in conjunction with the interdisciplinary team.

Postoperative Nausea and Vomiting

Postoperative nausea and vomiting (PONV) is one of the most frequently reported complications after anesthesia and surgery. Despite advances in anesthesia practice and antiemetic therapies, the overall incidence of PONV has remained relatively unchanged over the past 4 decades.⁷⁸² Whereas the incidence of PONV is 20% to 30% in the noncardiac surgery population, PONV rates in the cardiac surgery population are reportedly as high as 70%.^{759,782,783}

Current guidelines recommend using a preoperative risk score to identify a patient's propensity for PONV⁷⁸² and considering prophylaxis with ondansetron, betamethasone plus droperidol, or perhaps aromatherapy in high-risk patients.⁷⁸⁴⁻⁷⁸⁶ Using a nasogastric tube after cardiac surgery has not been found to consistently reduce PONV.⁷⁸⁷ A large Cochrane review reported an association between postoperative chewing gum use and quicker return of gastrointestinal function; however, this approach has not been studied extensively in cardiac surgery patients specifically.⁷⁸⁸ Therefore, extrapolating current guidelines to the cardiac surgery patient should be undertaken with consideration of any potential proarrhythmic side effects of the agents being administered.⁷⁵⁹

Gastrointestinal Complications

Acute dysfunction of the gastrointestinal subsystem is generally uncommon but has been reported in 0.3% to 4.5% of patients who undergo cardiac surgery.⁷⁸⁹⁻⁷⁹³ Significant abdominal complications, including deep sternal wound infections, renal failure requiring dialysis, and multisystem organ failure, can be devastating. As a result, these complications are associated with high morbidity and mortality rates that can approach 50%.⁷⁹⁴ Key in management is early identification and rapid confirmatory diagnosis by the perioperative team, followed by definitive therapy to improve outcomes in these patients.^{789,795,796}

The most prevalent abdominal complications include ileus (most common), gastrointestinal bleeding and ulcerations (next most common), mesenteric ischemia and perforated viscus (most lethal), cholecystitis, pancreatitis, and hepatic dysfunction or failure.⁷⁹⁵ The risk factors associated with gastrointestinal complications include: age >65 years, advanced congestive heart failure (NYHA classification III/IV), endocarditis, COPD, peripheral vascular disease, low serum albumin, complex surgery (reflected in longer CPB and cross-clamp times), postoperative low cardiac output syndrome, hemodynamic instability on admission, prolonged need for inotropic or vasopressor support, use of an IABP,

end-stage renal disease requiring hemodialysis, and acute hepatic dysfunction.^{789,790,796-799}

Ileus. Postoperative ileus is a temporary lack of normal intestinal muscle contraction that occurs when gas in the gastrointestinal tract produces abdominal distention, generally within 24 to 48 hours of surgery. Bowel sounds can be heard initially. Although not a component of normal recovery, this complication is usually benign and subsides without specific treatment. However, when severe, postoperative ileus can be associated with significant morbidity, including prolonged hospital LOS and mortality.^{794,800}

Key mechanisms of postoperative ileus include fluid overload, use of opioids for analgesia, neurohormonal dysfunction, and gastrointestinal stretching and inflammation.⁸⁰¹ If untreated, a severe colonic pseudo-obstruction can lead to cecal overdistension and possible perforation, which usually carries a high risk of death.⁷⁹⁴ The critical cecal diameter at which perforation is more likely ranges from 9 to 12 cm.⁸⁰²

Management of postoperative ileus begins with prevention. Enhanced recovery guidelines include limiting preoperative fasting, using neuraxial and multimodal analgesia, avoiding fluid overload, removing urinary catheters early after surgery to promote mobilization, and using coffee and chewing gum to promote the return of gastrointestinal function and to facilitate recovery.

Should postoperative ileus develop, conservative measures (which include maintaining preventive strategies) and the use of neostigmine remain the mainstay of treatment and are sufficient in most cases. Colonic decompression or surgical intervention is not usually required, unless the ileus is severe or the bowel has perforated.⁷⁹⁴ A newer agent, alvimopan, is a peripherally acting μ -opioid receptor antagonist recommended for postoperative ileus in the enhanced recovery guidelines for noncardiac surgery. However, its utility and safety in the cardiac surgery setting have not been well studied to date.

Gastrointestinal Bleeding. Gastrointestinal bleeding after open cardiac operations usually comes from the upper part of the gastrointestinal tract (where the pathology is typically hemorrhagic gastritis or duodenitis) and less commonly from the lower tract. Patients who are mechanically ventilated or at high risk for gastric or duodenal stress ulcers and not on enteral nutrition should receive proton pump inhibitor prophylaxis (in preference to H₂ histamine blockers) to minimize the risk for gastrointestinal bleeding.^{803,804}

Postoperative gastrointestinal bleeding is more common after valve procedures than after CABG.⁷⁹⁷ A possibly related factor is a preoperative history of AF and anticoagulation with warfarin. Further, combined anticoagulation/antiplatelet therapy, intermittent episodes of hypoperfusion, and use of vasoactive medications (which can cause splanchnic vasoconstriction) have been associated with gastrointestinal bleeding after cardiac surgery.⁷⁸⁹ Additional risk factors include age ≥ 70 years, ejection fraction $< 35\%$, congestive heart failure, cerebrovascular disease, chronic kidney disease, and gastrointestinal disease. Although these conditions can be managed by the perioperative team with effective resuscitation^{805,806} and endoscopic therapies and with CT angiogram-directed coil embolization, mortality remains high, ranging from 9% to 38%.^{780,789,790,796,797}

Bleeding from the lower gastrointestinal tract, although less common than upper gastrointestinal bleeding, can be serious, with a reported mortality rate of 17%.⁷⁹² The most common sources of bleeding include diverticula and vascular malformations, colitis, neoplasia, and anorectal bleeding.⁸⁰⁷ In addition,

10% of lower gastrointestinal bleeding has traditionally been presumed to have an upper gastrointestinal source,⁷⁹² and hemodynamically unstable patients with hematochezia may first require upper endoscopy.⁸⁰⁸ Current American College of Gastroenterology guidelines suggest that, in hemodynamically stable patients, appropriate bowel preparation followed by colonoscopy should be the initial diagnostic procedure and should be performed within 24 hours. Radiographic interventions (tagged RBC scintigraphy, CT angiography, angiography) should be considered in high-risk patients with ongoing bleeding who do not respond adequately to resuscitation and who are unlikely to tolerate bowel preparation and colonoscopy. Importantly, although the American College of Gastroenterology generally recommends against the use of NSAIDs, patients with high-risk cardiovascular disease and lower gastrointestinal bleeding are advised to continue any ongoing aspirin therapy (secondary prophylaxis).⁸⁰⁸ The decision to maintain or discontinue dual antiplatelet therapy should be based on individual patient assessment by the interdisciplinary team.⁷⁹²

Mesenteric Ischemia and Bowel Perforation. Intestinal necrosis develops uncommonly but insidiously, particularly in sedated patients after cardiac operations. The estimated incidence rate ranges from 0.06% to 0.20%. Initially, intestinal necrosis may not be readily apparent until gastrointestinal metabolic acidosis (lactate) develops, signs of sepsis or septic shock appear, or the inevitable perforation in the necrotic area causes pneumoperitoneum. Diagnostic delay can occur as a result of this latency, such that mortality associated with postoperative ischemic bowel is high, ranging from 46% to 75%.^{790,799,809} Risk factors include the presence of peripheral vascular disease, hemodynamic instability or vasoplegia requiring high-dose vasopressors, and low cardiac output.

The perioperative team must have a high index of suspicion for bowel ischemia, particularly in patients who develop an intolerance to enteral nutrition, clinical signs of shock (with or without an elevated lactate), and abdominal pain or distention. Initial management consists of IV fluid administration, antimicrobial therapy, and correction of hypoperfusion (if possible). Rapid investigation includes an abdominal radiograph, a CT scan with contrast, and/or endoscopy.⁸⁰⁰

In all instances, early diagnosis and surgical intervention remain the only effective means of reducing mortality from bowel ischemia after cardiac surgery.^{809,810} Bowel perforation secondary to a duodenal ulcer additionally requires an initial infusion of proton pump inhibitors and assessment for *Helicobacter pylori*. Nonocclusive mesenteric ischemia induced by postoperative low cardiac output can be difficult to diagnose, as the typical CT scan findings (portal venous gas, pneumatosis intestinalis, "string of sausages," abnormal contrast-induced bowel wall enhancement) are poorly sensitive. Thus, endoscopy or surgical exploration may be required for diagnosis.⁸¹⁰

Acute Cholecystitis. Two large retrospective series found that the rate of acute cholecystitis after cardiac surgery ranges from 0.03% to 0.08%, with acalculous cholecystitis being more common than calculous cholecystitis.^{811,812} When cholecystitis develops after cardiac surgery, it produces the same clinical syndrome it causes in other patients: severe pain, leukocytosis, and right upper quadrant tenderness. The precise mechanism underlying acalculous cholecystitis is not well understood, but hypoperfusion of the gallbladder, bile stasis, and inflammatory mediators have been proposed. Liver function tests often display the usual abnormalities seen with cholestasis. Workup consists of ultrasonography or CT, depending on physician preference.⁸⁰⁰

Rady and colleagues⁸¹² found that presentation for acute cholecystitis can be quite delayed (a median 26 days after the cardiac surgery procedure). Perhaps because of this delay, mortality among patients with this complication is high, ranging from 15% to 42%.^{811,812} Patients with acute cholecystitis but without overt peritonitis should initially be treated conservatively with appropriate antibiotics. However, a lack of significant improvement within 48 hours or a worsening clinical picture should lead to placement of a percutaneous cholecystostomy drainage tube and/or surgical intervention.⁸¹¹

Acute Pancreatitis. Pancreatitis early following cardiac surgery was first described in 1976.⁸¹³ Some evidence of pancreatic cellular injury⁸¹⁴ is present in about 25% of patients after open cardiac surgery, typically presenting with hyperamylasemia, increased serum lipase, or both.^{794,815} In most cases, these increased levels gradually decline, with few, if any, clinical sequelae. Occasionally (in <5% of patients with elevated amylase), severe pancreatitis develops, along with pancreatic abscess or hemorrhagic pancreatitis, and this condition is associated with high rates of mortality.⁸¹⁵

Current guidelines recommend that, in addition to establishing a diagnosis on the basis of clinical symptoms and laboratory testing, additional contrast-enhanced CT and/or MRI of the pancreas should be obtained in patients for whom the diagnosis is unclear or who do not improve clinically.^{816,817} Patients with acute pancreatitis and concurrent acute cholangitis should be evaluated as to whether endoscopic retrograde cholangiopancreatography is needed. Resuscitation with appropriate fluid administration should be done within the first 12 to 24 hours if clinically indicated. Routine use of prophylactic antibiotics is not recommended, even in patients with severe acute pancreatitis or sterile necrosis. Conversely, in patients with infected necrosis, appropriate antibiotics may be useful in delaying intervention.⁸¹⁷

In milder presentations, oral feeding can be started immediately if the patient is extubated and has no nausea and vomiting. In more severe presentations, enteral nutrition is recommended over parenteral feeding to prevent infectious complications; in fact, parenteral nutrition should be avoided.^{816,817} In stable patients with infected necrosis, any surgical, radiologic, or endoscopic drainage should be delayed for 4 weeks, when possible, to allow a wall to develop around the necrosis.⁸¹⁷

Liver Dysfunction. Descriptions of hyperbilirubinemia (serum bilirubin concentration of 3–4 mg/dL) after cardiac surgery date back to the early years of the field.⁸¹⁸ Hyperbilirubinemia develops in 10% to 40% of patients after cardiac surgery.^{819,820} The most common preoperative risk factor is venous congestion due to heart failure. Postoperative jaundice is largely secondary to an increase in conjugated bilirubin and has been associated with prolonged CPB and cross-clamp times, the use of IABP, the administration of inotropes, and the number of blood and plasma transfusions.^{820,821} More recent reports have indicated that hyperbilirubinemia, although often benign, is associated with adverse patient outcomes such as prolonged ICU stay, new-onset infection, low-output syndrome, greater need for invasive ventilation and renal replacement therapy, and higher mortality rates (ranging from 4%–17%).^{819,821}

Frank hepatic necrosis may develop when cardiac output is acutely and severely reduced. In patients with fulminant liver failure (<0.4%), mortality can exceed 90%.^{819,822} A recent large meta-analysis examining the severity of preexisting liver dysfunction (identified by using the Model for End-stage Liver Disease score preoperatively and/or the Child-Turcotte-Pugh score after cardiac surgery⁸²³) found that liver dysfunction in patients

undergoing cardiac surgery was associated with a three- to fivefold higher risk for short- and long-term mortality. In addition, an increase in other perioperative adverse events, such as neurologic complications, prolonged ventilation, CSA-AKI, sepsis, and bleeding, was observed.

Farag and colleagues⁸²¹ suggested that the timing and level of peak bilirubin concentration may help differentiate patients with an underlying liver function pathology due to systemic hypoperfusion. Specifically, delayed and steep hyperbilirubinemia (median 3.5 postoperative days with an optimal bilirubin cutoff value of 5.35 mg/dL) was linked to hypoperfusion and poor outcomes.

Managing a patient with hyperbilirubinemia and liver dysfunction requires the perioperative team to ensure adequate cardiac flow, optimize volume and other organ status, and supply adequate nutritional support.

Watery Diarrhea and *Clostridium Difficile* Infection. As a complication of cardiac surgery, watery diarrhea may accompany or follow abdominal distention or appear de novo toward the end of the first week. This can be secondary to electrolyte replacement, enteral feeding, and medications. The perioperative team should be alerted if the diarrhea becomes more frequent, explosive, or if it is accompanied by mucus and/or blood, particularly if associated with leukocytosis. Discontinuing oral antibiotics should be considered when diarrhea develops. Diarrhea can also be caused by ischemia of the bowel secondary to a long period of low cardiac output, and this ischemia may lead to small or large bowel infarction (see earlier), or it may be a form of ulcerative colitis or proctitis. Diagnosis can be made by using contrast CT or endoscopy. If the bowel ischemia and ulceration progress, laparotomy and bowel resection are required—but even with such treatment, prognosis is poor.

A common, important, and potentially serious cause of watery diarrhea after cardiac surgery is *Clostridium difficile*, which colonizes the intestinal tract after the normal gut flora have been altered by broad-spectrum antibiotics. The prevalence of this complication ranges from 0.7% to 5.4% (although the higher rate may be due to colonization rather than true infection).^{789,824} Risk factors for development of *C difficile*-associated diarrhea include advanced age, greater cumulative antibiotic days, longer ICU stay, blood product transfusion, and longer time on a ventilator. *C difficile* is associated with increased mortality when severe (i.e., antibiotic-associated pseudomembranous colitis).⁸²⁴

Recent guidelines have classified *C difficile* infections as mild to moderate (mild abdominal pain and cramps that may progress to altered electrolyte balances and dehydration) and severe (mild-moderate symptoms with fever, leukocytosis, and hypoalbuminemia).⁸²⁵ Progression to fulminant *C difficile* infection (often heralded by the absence of diarrhea) is associated with higher mortality rates due to the development of toxic megacolon, bowel perforation, septic shock, and eventual multiorgan dysfunction. Lastly, *C difficile* infections may recur in 10% to 13% of cases and in as many as 60% of patients after one or two recurrences.⁸²⁵

The current guideline-recommended management bundle for *C difficile* infection includes⁸²⁵:

- Infection control with antibiotic stewardship, discontinuation of unnecessary proton pump inhibitors, hand hygiene, isolation measures, and environmental disinfection.
- Contact (enteric) precautions with *C difficile* carriers.
- Testing of diarrheal stools from at-risk patients with clinically significant diarrhea (≥ 3 loose stools in 24 hours) and no obvious alternative explanation. In patients with ileus, polymerase chain reaction testing of perirectal swabs is an alternative approach.

- CT imaging for patients with clinical manifestations of severe or complicated *C difficile* colitis. Ultrasonography or flexible sigmoidoscopy may be useful in critically ill patients suspected to have pseudomembranous colitis and who cannot be transported to the CT scanner.
- Avoidance of empirical therapy for *C difficile* infection unless there is a strong suspicion of *C difficile* infection. Oral metronidazole should be limited to treating an initial episode of mild to moderate *C difficile* infection, with oral vancomycin reserved for patients who do not respond to metronidazole. Repeated or prolonged courses of metronidazole should be avoided due to risk of cumulative and potentially irreversible neurotoxicity. Either oral vancomycin or fidaxomicin is recommended for treatment of all patients with severe *C difficile* infections.
- Early surgical consultation for all patients with *C difficile* infection who progress to sepsis or septic shock.

Information on the use of probiotics is limited, other than that they could be useful as an adjunct to antibiotics for recurrent *C difficile* infection; however, this requires additional study in cardiac surgery patients. Consultation with infection control or an infectious disease specialist, particularly in cases of recurrent *C difficile* infection, should be undertaken per institutional protocols.

Summary Points

- Well-nourished patients have better outcomes and lower postoperative mortality after cardiac surgery. Developing a perioperative strategy to identify nutritional deficiencies, manage PONV, and establish nutrition postoperatively are important considerations for the perioperative team.
- Mild gastrointestinal complications after cardiac surgery are relatively common. However, significant complications such as gastrointestinal bleeding or mesenteric ischemia, although less common, are associated with a significant risk of death and require prompt identification and aggressive management.

Endocrine Subsystem

Glycemic Control

Hyperglycemia (in both the OR and ICU) is associated with adverse outcomes after cardiac surgery. The prevalence of DM among cardiac surgery patients varies from 7% to 40%.⁸²⁶⁻⁸²⁸ Thus, screening for suboptimal glycemic control or DM is recommended for all patients undergoing cardiac surgery.^{233,829} Glycated hemoglobin (HbA1C) is a useful clinical measure for quantifying long-term glucometabolic state. Clinical practice guidelines recommend a target HbA1C <7% in patients with DM; however, approximately 25% of patients undergoing cardiac surgery have suboptimal glycemic control (HbA1C >7%), and 10% of these patients have undiagnosed DM.^{828,830} In patients undergoing CABG or valvular procedures, higher HbA1C is a strong predictor of mortality and morbidity, hospital LOS, and long-term survival, irrespective of previous DM status,^{828,831-836} and a perioperative HbA1C level >7% is associated with increased risk for postoperative complications (acute renal failure, cerebrovascular accident, postoperative MI, sternal wound infection) in patients undergoing CABG.^{767,826,830} A serum glucose level >200 mg/dL is associated with an increased risk for death and adverse events, regardless of whether or not DM is present.⁸²⁶

Due in part to seminal work by Van den Berghe,⁸³⁷ increased attention is being given to appropriate glycemic management in critically ill patients. In a large population undergoing cardiac surgery, the investigators studied the utility of continuous insulin infusion for tightly regulating blood glucose levels at 80–110 mg/dL. The study found a significant reduction in overall in-hospital mortality, mechanical ventilation time, number of blood transfusions, bloodstream infections, acute renal failure requiring dialysis, and other morbidities. Further work by Lazar and colleagues⁸³⁸ further showed that tight glycemic control (using a combination of glucose, insulin, and potassium) in patients with DM undergoing cardiac surgery was associated with improved survival, fewer ischemic events, and fewer wound complications. Nonetheless, more recent clinical trials of tight glycemic control have not replicated these benefits (or have shown an increase in mortality), and concerns have been raised about the potential downside of hypoglycemia.⁸³⁹⁻⁸⁴¹

More recently, a tailored glycemic control strategy was developed that takes into account a patient's baseline HbA1C. This process, referred to as "time in band," seeks to reduce relative hypoglycemia for a given patient based on the adequacy of baseline glycemic control.⁸⁴² In other words, those with suboptimal glycemic control (HbA1C >8%) are permitted to maintain a higher perioperative serum glucose level. Relative hypoglycemia has been associated with higher rates of mortality in the critical care population⁸⁴²; however, this has not been extensively studied in postoperative cardiac surgery patients.

As a result, current recommendations pertaining to perioperative glycemic control include the following^{843,844}:

Preoperatively

- Perform routine HbA1C screening for risk stratification.
- Seek to achieve a blood glucose level of 120 to 180 mg/dL (6.7–10.0 mmol/L) while the patient awaits surgery; consider using short-acting subcutaneous insulin, with blood glucose checks every 4 hours.

Intraoperatively

- In non-DM patients, consider using a small (5 IU) IV bolus of insulin if the blood glucose level is >180 mg/dL (>10.0 mmol/L), as well as hourly checks.
- In non-DM patients, consider using a continuous IV insulin infusion to maintain blood glucose of 150 to 180 mg/dL (8.3–10.0 mmol/L) during surgery if blood glucose is persistently >180 mg/dL (>10.0 mmol/L).
- In patients with DM, start a continuous IV insulin infusion at the beginning of surgery and continue throughout to maintain a blood glucose level >150 (>8.3 mmol/L) and <180 mg/dL (<10.0 mmol/L).

In the ICU

- In patients with or without DM, use a continuous insulin infusion if the blood glucose level is >180 mg/dL (>10.0 mmol/L) for a target of 150 to 180 mg/dL (8.3–10.0 mmol/L) during the ICU stay; this should be accompanied by hourly blood glucose monitoring, which can be deescalated to every 4 hours if the patient is stable.

Post-ICU

- Consider initiating a combination of short-acting and long-acting subcutaneous insulin at 50% of the total previous 24-hour insulin dose (in the ICU) and then titrate by checking

the blood glucose level every 4 hours to a target of 150 to 180 mg/dL (8.3–10.0 mmol/L).

- Consider restarting oral anti-diabetes medications at 50% of the preoperative dose when the patient is on oral feeding.

At Hospital Discharge

- In patients with preexisting or newly identified DM, consult with an endocrinologist or DM specialist before the patient is discharged.
- For all patients with DM and those who have started a new glycemic control regimen, provide inpatient education regarding glucose monitoring and medication administration and communicate appropriately with the patient's primary care provider.

Sodium-glucose co-transporter-2 (SGLT2) inhibitors and short-acting glucagon-like peptide 1 receptor agonists are newer classes of medications that are being used with increased frequency due to their glycemic control, improved outcomes in patients undergoing CABG, and their impact on heart failure and renal dysfunction.⁸⁴⁵⁻⁸⁴⁸ Although a detailed review of these agents is beyond the scope of this chapter,⁸⁴⁹⁻⁸⁵¹ the perioperative team should be aware of several key issues.

- Neither of these medication classes is associated with hypoglycemia, but they can potentiate hypoglycemia when combined with a sulfonylurea or insulin.^{849,851}
- Some SGLT2 inhibitors are associated with genital infection and, less commonly, euglycemic diabetic ketoacidosis when precipitated by stress. Therefore, SGLT2 inhibitors should be held before surgery (per local institutional guidelines) and avoided in patients with type 1 DM.
- Because SGLT2 inhibitors have been associated with hypovolemia-related side effects (osmotic diuresis), they should be held in patients with a systolic blood pressure <100 mmHg.
- Generally, SGLT2 inhibitors are not recommended if the estimated GFR is <45 to 60 mL/min/1.73 m² due to a decrease in glycemic efficacy.
- Reinitiating these agents in the postoperative period remains an area for further study.⁸⁴⁹⁻⁸⁵¹

Summary Points

- Estimates suggest that 40% of patients undergoing CABG have DM, and >10% have undiagnosed DM at the time of surgery and increased morbidity and mortality compared with non-DM patients. Glycemic control in the perioperative period reduces morbidity and mortality.
- Glycemic control with a continuous insulin infusion and a blood glucose goal of 140 to 180 mg/dL confers the most benefit. The perioperative team should be aware of current guidelines on the optimal preoperative, intraoperative, and postoperative glycemic management of patients with and without DM.
- Both SGLT2 inhibitors and glucagon-like peptide 1 receptor agonists should be held before surgery.

Adrenal Insufficiency

Acute adrenal insufficiency after cardiac surgery was first described by Alford and colleagues⁸⁵² in 1979. In that review of 4364 patients, 5 (0.1%) had symptoms consistent with acute adrenal insufficiency. Although the etiology was not well

understood, in some of the reported cases, hypotension was preceded by hypertension, lending to the notion that the subsequent adrenal insufficiency was secondary to a hemorrhagic infarction of the adrenal gland.⁸⁵² It was thought that this might have been precipitated by the use of CPB with anticoagulant therapy, an added risk. Symptoms generally appear between postoperative days 4 and 10 and consist of abdominal and flank pain, abdominal distention, altered mental status, fever, and occasionally shock.

Assessing for adrenal insufficiency should include ruling out other potential causes of shock and using biochemical analysis to confirm the diagnosis. A cosyntropin stimulation test can provide credence to the diagnosis if the serum cortisol level remains low 60 minutes after administration of a 250-μg dose.^{390,852-854} Confirmatory evidence (e.g., a CT or MRI scan) of acute adrenal insufficiency should be obtained.^{855,856} Initial treatment includes prompt glucocorticoid and mineralocorticoid replacement, use of vasopressors, and fluid resuscitation. Subsequent lifetime oral treatment is indicated.⁸⁵⁶

Thyroid Dysfunction

A reduction in plasma free triiodothyronine (T₃), but not thyroxine (T₄), can occur early after cardiac surgery and remain low for at least 24 hours.⁸⁵⁷ Classic work by Timek and colleagues⁸⁵⁸ found that exogenous T₃ may improve the amplitude of myocardial fiber shortening by accelerating Ca²⁺ transients in an in vivo preparation of catecholamine stimulation. There is some clinical evidence that exogenous T₃ (0.6–0.8 μg/kg) may improve cardiac output immediately after CABG in patients with depressed ejection fraction.^{859,860} RCTs examining the utility of T₃ administration have found an association with an increase in cardiac output but no consistent improvement in postoperative outcomes.^{861,862} Thus, routine T₃ administration is not current practice.

Hematologic Subsystem

In cardiac surgical patients, managing hematologic function is a critical component of perioperative care. As a result, a team-based approach to PBM is essential to contemporary practice. This ranges from managing anticoagulation preoperatively to monitoring coagulation intraoperatively to managing bleeding postoperatively. Specific considerations for postoperative bleeding are as follows.

Postoperative Bleeding

Although the reasons for bleeding after CPB and heparin reversal are multifactorial, a surgical (mechanical) cause should always be considered. Procedures that confer a high risk for excessive bleeding include reoperation, complex procedures with long CPB times, and aortic surgery. Multiple factors contribute to coagulopathy after CPB and cardiac surgery, including preoperative anticoagulation use, reoperations, time and duration of CPB, and pharmacologic strategies, including antifibrinolytic agents.

The reported rates of surgical reexploration for bleeding range from 2% to 8%,⁸⁶³ depending on institutional criteria and case mix. Moulton and colleagues⁸⁶⁴ identified patient older age, preoperative renal insufficiency, non-CABG procedures, and prolonged CPB time as independent risk factors for reexploration,

which was associated with increased operative mortality and morbidity.

Inflammatory responses are associated with CPB for various reasons—for example, nonendothelial surface interactions that cause contact activation and proinflammatory responses (including fibrinolysis and platelet and neutrophil activation).⁵³⁵ Platelet abnormalities, both quantitative and qualitative, universally develop during and after CPB,^{865,866} as does coagulation upregulation as a result of dilution, activation, and consumption of critical hemostatic factors. During CPB, hypothermia, mechanical injury, and consumption of platelets impair platelet function and are important contributors to coagulation defects. Despite adequate rewarming after CPB-related hypothermia, hemostatic dysfunction can persist.

Transfusion of hemostatic factors, factor concentrates, and platelets is the mainstay therapy for managing bleeding, as is the use of bleeding-management algorithms, usually with viscoelastic testing and the use of intraoperative hemostatic checklists.^{863,867-869} Transfusing allogeneic blood products should be limited or avoided whenever possible for several reasons, for example, higher risk for death, infection, and other adverse events, although patient-related risk factors could also cause these outcomes.^{868,870,871} Heparin rebound can occur after initial reversal, in which case additional protamine can be administered, usually as an infusion of ~25 mg/h for several hours.

Protocols and institutional practices vary, with different transfusion thresholds. Unfortunately, most patient blood-management practices focus on RBCs. In critically ill, actively bleeding, or in those with unstable hemodynamics, higher thresholds may be warranted. Comparative studies examining RBC/hemoglobin thresholds did not report the physiologic implications of different hemoglobin concentrations.^{244,866}

In addition to traditionally used allogeneic blood products, several coagulation factor concentrates and procoagulants have been added to the perioperative teams' armamentarium, including fibrinogen concentrates, factor 8 inhibiting bypass activity, recombinant activated factor VIIa, fibrinogen concentrates, and prothrombin complex concentrates. These evolving therapies are effective in managing coagulopathic bleeding from cardiac surgery and may reduce the need for surgical reexploration.⁸⁷²⁻⁸⁷⁹

When excessive bleeding continues after a reoperation in which no specific bleeding point is identified, prompt hematologic consultation and specific treatment are indicated. It is tempting to regress to the indiscriminate, costly, and sometimes hazardous use of FFP and platelet concentrates and additional doses of protamine. Despite the common use of FFP, no data support its efficacy, and FFP can only correct INR to ~1.5. Optimally, specific management protocols include (1) repeated measurements of ACT and administration of additional protamine only when indicated; (2) consideration and possibly measurement of excessive fibrinolysis, along with use of tranexamic acid, although this is best used prophylactically; and (3) consideration of possible platelet dysfunction, depletion, or both, and administration of platelet concentrates. Platelet transfusions continue to be empirical, as we cannot monitor platelet function, and most algorithms use platelet numbers for transfusions. When warfarin reversal is urgently needed, prothrombin complex concentrates should be administered according to current guidelines. Fibrinogen is an important target in bleeding patients, in whom fibrinogen levels <200 mg/dL should be treated with cryoprecipitate or fibrinogen concentrates, depending on their availability.

Chest Tubes

Chest drainage tubes must be properly placed in the OR for them to function well in the early postoperative period and keep the pericardial and pleural spaces free of blood. When the patient is being transported to the ICU, chest tubes are connected to an underwater seal. Once the patient is in the unit, the tubes drain into an initially empty container in which a negative pressure of 10 to 20 cm H₂O is continuously maintained. Routine "stripping" of the chest tubes by nursing personnel is unnecessary and potentially dangerous, and the technique should be used only for clearing clots.^{233,880,881} Provisions are made for continuous precise measurement of drainage from the chest tubes; this can be incorporated into an automated care system.

When chest drainage is excessive, prompt reoperation for bleeding is indicated. The surgical and perioperative team should develop specific institutional quantitative guidelines for communication on threshold to discuss concerns about excessive chest tube output, with or without hemodynamic instability. Although reoperation for bleeding is not risk-free, relatively early reoperation for bleeding usually stops excessive bleeding, even when no specific bleeding points are found. This approach may avoid pericardial tamponade, minimize the administration of allogeneic blood, and ease the management of all subsystems.

Summary Points

- The surgical and perioperative teams should conjointly develop an approach to PBM for postoperative bleeding that includes a mechanism for effective communication and an algorithmic approach to blood product and procoagulant agent use.

Special Considerations after Cardiac Surgery

Acid-Base Balance

As indicated earlier, some *metabolic acidosis* (base deficit ≥ 2 mEq/L) may be present during the early hours after cardiac surgery, even when the patient is convalescing normally. It is left untreated if arterial pH is ≥ 7.4 and Paco_2 is ≥ 30 mmHg. When Paco_2 is <30 mmHg, the base deficit is treated before adjusting Paco_2 appropriately upward. When pH is <7.4, the base deficit is treated. However, if convalescence is otherwise normal, treatment is delayed for 4 to 8 hours in adults and for 2 to 4 hours in infants, by which time the base deficit may have cleared spontaneously. In infants, particularly, it is best to avoid an additional sodium load whenever possible.

A mild *metabolic alkalosis* may be present 24 hours after operation in normally convalescing patients, probably related in part to the citrate load contained in the anticoagulant solution of banked blood. Metabolic alkalosis is self-correcting under these circumstances and is not treated.

When large amounts of homologous blood containing sodium citrate have been transfused, important metabolic alkalosis can occur. Metabolism of the citrate leaves bicarbonate as the only anion available to balance sodium ions. This situation may be exacerbated after 2 to 4 days by hydrogen ion migration from the cells because potassium ions replace hydrogen ions, and serum potassium falls, sometimes precipitously, in association with potassium loss in urine. Relatively large amounts of potassium chloride must be given under such circumstances (up to 200 mEq in 24 hours), but the dose should be titrated against 4-hourly serum

potassium measurements. The alkalosis is associated with a fall in serum chloride and a compensatory rise in PaCO_2 . Provided renal function is adequate, the situation will correct itself. If it continues to worsen, the carbonic anhydrase inhibitor acetazolamide is administered, usually intravenously at a dose of 5 mg/kg up to 250 mg in adults once or twice a day as needed until the alkalosis is corrected.

Cardiac Tamponade

In an echocardiographic study, Weitzman and colleagues⁸⁸² found that 103 of 122 consecutive patients (84%; CL 80%–88%) had pericardial effusions during recovery from cardiac surgery; Ikäheimo and colleagues found a similar prevalence.⁸⁸³ Large pericardial effusions 4 to 10 days postoperatively develop in 30% of patients and are more common when early postoperative bleeding is excessive⁸⁸⁴ or after heart transplant. Effusions reach their maximum size in most patients around postoperative day 10, generally regressing spontaneously thereafter. Many smaller pericardial effusions are asymptomatic and require no treatment.

Detection and Treatment

Cardiac tamponade develops in only about 1% of patients with pericardial fluid.^{882,883} Recognizing and treating acute tamponade early after cardiac surgery can be challenging (see also “Acute Cardiac Tamponade” and “Treating Low Cardiac Output” in Section I).⁸⁸⁵ A chest radiograph usually shows enlargement of the cardiac silhouette before symptoms appear; nonetheless, tamponade may not become apparent until as late as 4 months after operation.⁸⁸⁶

Cardiac tamponade occasionally develops in the ICU, in which case prompt reoperation is required (see “Acute Cardiac Tamponade” in Section I). Delayed cardiac tamponade may develop several days to several weeks after the patient leaves the OR. Although delayed tamponade is uncommon, it is more common among patients being treated with anticoagulants and possibly when smaller incisions are used. If unrecognized, delayed cardiac tamponade can cause serious morbidity and mortality.^{887–890} Clinical signs and symptoms may be subtle, but one or more of the following are usually present: progressive and unexplained weakness and lethargy; progressive dyspnea on exertion or orthopnea; unexplained hepatic congestion or hepatomegaly, ascites, or peripheral edema; elevated jugular venous pressure; pulsus paradoxus; widening of the cardiac silhouette on chest radiography; and unexplained pre-renal dysfunction.⁸⁸⁵

Treatment consists of pericardial decompression by either pericardiocentesis, preferred whenever possible, or surgical drainage. In one review of 21,416 cardiac surgical patients, 327 (1.5%) had postoperative pericardial effusion, which was evacuated by echocardiography-guided pericardiocentesis ($n = 169$, 52%) or surgical drainage ($n = 75$, 23%); the remaining patients were managed conservatively.⁸⁹¹ With surgical drainage, the subxyphoid portion of the incision is reopened, and the pericardial space is entered at the diaphragm. If a week or more has elapsed from surgery, great care is required to gently dissect along the diaphragmatic surface until the fluid collection site is entered. Undue haste can result in accidental entry into the right ventricle, which can be friable and soft. Mortality has occurred under such circumstances.

Summary Points

- The signs and symptoms of delayed tamponade can be subtle.
- The perioperative team should have a high index of suspicion when there is an unexplained failure to progress and widened mediastinum on chest x-ray, particularly in patients on anticoagulation.

Infection

Infection and wound complications, although uncommon after cardiac surgery, are associated with high rates of morbidity and mortality. The many people involved and the length and complexity of the operations make bacterial contamination more likely in cardiac surgery than it is in shorter and simpler procedures.⁸⁹² Control of the surgical and ICU environments and awareness among personnel of the possibility of contamination significantly reduce the prevalence of infection.⁸⁹³ Meticulous aseptic technique by the surgical, anesthetic, and perfusionist teams is an essential part of prophylaxis against infection in cardiac operations.

Prevention

Infection within or around the heart, particularly when prostheses have been used, is a major threat to the patient's life. Therefore, preventative measures should be employed, such as appropriate nutritional support in patients who are malnourished, body weight control in obese patients, smoking cessation, perioperative glycemic control, and pulmonary physiotherapy (particularly in smokers and patients with COPD).^{233,894} Prophylactic antibiotic therapy is recommended for all cardiac operations,^{895–897} as is *Staphylococcus aureus* decolonization with intranasal mupirocin ointment and chlorhexidine bath, with appropriate skin preparation in the OR.^{894,898,899} Hair removal (if necessary) can be done by using an appropriate clipper immediately before the operation; shaving is strongly discouraged, as it has been shown to increase the risk for microinjuries and bacterial growth on the patient's skin.⁸⁹⁴

The Joint Commission (www.jointcommission.org), initially founded in 1951, has set forth a number of national quality core measures, including the Surgical Care Improvement Project, which presents specific recommendations for perioperative antibiotic prophylaxis for patients undergoing cardiac surgery.⁹⁰⁰ Antibiotic levels in the blood must be adequate immediately before and during the operation and while any intravascular or endotracheal device is in place. The perioperative team should engage in tracking compliance with their chosen surgical site infection prophylaxis approach.

Typically, skin flora consists of *Staphylococcus* and *Streptococcus* species and will be covered by a first-generation or second-generation cephalosporin. The current Surgical Care Improvement Project recommends the use of cefazolin 2 gm (3 gm for patients weighing >120 kg) for this purpose, to be given 1 hour before the skin incision.⁹⁰⁰ If a patient has a known beta-lactam allergy or requires additional gram-negative coverage with a fluoroquinolone, such as vancomycin, this should be administered 2 hours before the skin incision due to the infusion time.^{895,896} In patients with methicillin-resistant *S aureus*, vancomycin is given preferentially; however, data from a meta-analysis suggest that both cefazolin and vancomycin should be given in this context to ensure adequate coverage of both methicillin-sensitive and methicillin-resistant *S aureus*.^{896,901} Intraoperative repeat doses (every 3–4 hours) may be required for longer

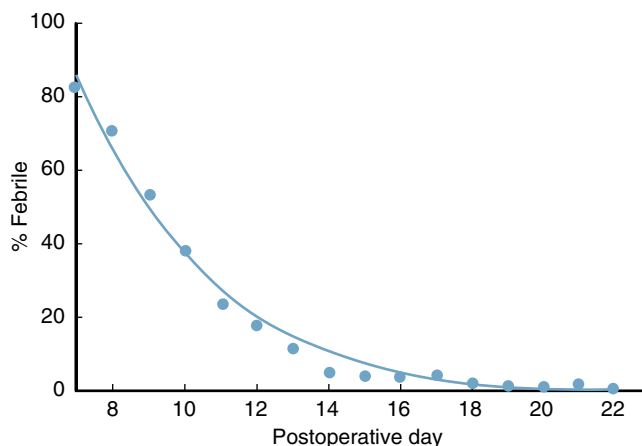
operations. Current guidelines recommend that antibiotic prophylaxis be continued for up to 48 hours. Coverage beyond 48 hours is not recommended, even in patients with chest tubes, lines, or other drains, as it is not associated with benefits and may increase the risk for *C difficile* infection and the development of antimicrobial-resistant pathogens.^{897,902} Evidence supporting the efficacy of additional topical antibiotics,⁹⁰³ implantable gentamicin-collagen sponges,^{904,905} or irrigation^{906,907} is inconsistent at best, and high-quality RCTs will be required to establish benefit from these approaches.

The perioperative team should seek to minimize the risk of infection by removing intravascular lines, endotracheal devices, and urinary catheters as early as possible postoperatively. Infection can also be minimized through simplified postoperative care, early transfer out of the ICU, early ambulation and oral alimentation, and early hospital discharge. Negative-pressure wound therapy has been suggested as a modality for reducing surgical-site infections in patients at higher risk for postoperative wound complications.^{324,908-910} A large RCT examining surgical-site infection rates with mono- or dual-antibiotic prophylaxis in combination with conventional wound dressing versus negative-pressure wound therapy is currently recruiting (Prevention of Infections in Cardiac Surgery [PICS] Prevena® Study; NCT: 03402945).

Body Temperature

Variation in body temperature is common and can persist for 2 weeks or longer after surgery (Fig. 4.17).⁹¹¹⁻⁹¹³ During this period, routine orders for repeated blood, sputum, and urine cultures and other special studies, when there is no indication other than fever, typically have low yield and increase healthcare expenditures.⁹¹⁴ Using antibiotics empirically in this context should be avoided, given the risks mentioned earlier for *C difficile* infection and antimicrobial-resistant and opportunistic pathogens.

By contrast, severe and increasing hyperthermia during the first 6 to 36 hours after surgery is a potentially lethal complication that should prompt the team to investigate and treat for malignant hyperthermia, neuroleptic malignant syndrome, or serotonin



• **Figure 4.17** Proportion of patients febrile (temperature > 37.8°C) after sixth postoperative day, starting with the group febrile on the sixth day. Note continuation of fever in some patients to nearly 3 weeks after operation. (From Livelli FD, Johnson RA, McEnany MT, et al. Unexplained in-hospital fever following cardiac surgery. Natural history, relationship to postpericardiotomy syndrome, and a prospective study of therapy with indomethacin versus placebo. *Circulation*. 1978;57(5):968-975.)

syndrome (particularly in cases with recent methylene blue or selective serotonin reuptake inhibitor use).⁹¹⁵

Surgical Site and Sternal Wound Infection

Sternotomy closure after cardiac surgery is most commonly done by using wire cerclage, which reopposes the cut edges of the sternum with a wire or band wrapped around or through the two portions of bone; the wire or band is tightened to pull the two parts together. Alternatively, sternal plating (i.e., rigid sternal fixation) has been used on the premise that it reduces side-to-side movement (which might not be achieved with wire cerclage) and has been associated with less postoperative pain, quicker mobility progression, improved quality of life,^{916,917} and fewer sternal complications (i.e., mediastinitis and nonunion), thereby offsetting the additional cost of the rigid fixation plates within 6 months of surgery.⁹¹⁸⁻⁹²⁰ Surgeons may opt for this closure method in patients at higher risk for sternal complications, such as those with a high body mass index, previous chest wall radiation, severe COPD, or steroid use.²³³

The sternal wound and graft-harvesting locations are the most common sites of infection. Healthcare-associated infections, such as central line-associated bloodstream infections, catheter-associated urinary tract infections, and ventilator-associated infections, are the most common source of sepsis and provide key metrics for evaluating perioperative team performance.⁹²¹ Quality-improvement initiatives for these events are well described and should be regularly reviewed and audited by the perioperative team as part of ongoing performance improvement initiatives.⁹²²

Sternal Wound Complications. Sternal wound complications can be classified as superficial or deep and as early or late.⁸⁹⁴ Superficial sternal wound infections are limited to the pectoralis fascia and above (i.e., the skin and subcutaneous tissue), whereas deep sternal wound infections involve tissues below the fascia. Early deep sternal wound infections are more common than late infections; late infections are often a combination of superficial and deep infection.⁹²³

Progression to mediastinitis, defined as an infection occurring within 1 year after surgery, is one of the most serious postoperative infections in cardiac surgery patients. The US Centers for Disease Control and Prevention characterize mediastinitis as having at least one of the following^{923,924}: positive culture from mediastinal tissue or fluid showing bacterial organisms; evidence of mediastinitis on gross anatomic or histopathologic examination; signs or symptoms of fever (>38°C), chest pain, or sternal instability with purulent drainage from the mediastinal area; and/or mediastinal widening on imaging. A deep sternal wound infection with mediastinitis may include sternal osteomyelitis and further retrosternal, posterior mediastinal, and/or pleural space involvement.^{894,923,925}

The extent of the infection should be further characterized with CT imaging of the chest and mediastinum. Although CT evidence of retrosternal fluid (some edema or blood in the anterior mediastinum is expected in the early postoperative setting) and sternal dehiscence is not unequivocally diagnostic of infection, air pockets or sternal ostial destruction usually indicates mediastinitis.

Previous reports of mediastinitis and sternal dehiscence after bilateral internal thoracic artery-to-CABG have ranged from 0.8% to 8.0%, with mortality as high as 70%.⁹²⁶⁻⁹²⁸ Risk factors for mediastinitis include^{923,925,929,930}:

- **Preoperatively:** DM, obesity, advanced age, COPD, smoking, frailty, heart failure, female sex, renal dysfunction or failure,

prolonged preoperative hospital stay, urgent or emergency surgery.

- **Intraoperatively:** use of bilateral internal mammary artery grafts (particularly in patients who are frail, malnourished, or have poor glycemic control), prolonged or complex surgery, redo operations, method of sternal closure.
- **Postoperatively:** prolonged respiratory failure, prolonged ICU LOS.

Expert consensus guidelines developed to prevent sternal wound infection have reduced estimated deep sternal wound infection rates to 0.5% to 5.0%. When sternal wound infection does occur, the associated healthcare costs, morbidity, and mortality can be very high.^{923,929-931}

Surgical Prevention and Treatment. In addition to the prophylaxis and prevention strategies described earlier, current surgical recommendations include use of a pedicled internal mammary artery graft and low-dose electrocautery (or harmonic scalpel) during harvest, meticulous hemostasis and chest closure, minimal use of bone wax, perioperative glycemia control, use of an external chest support vest, negative-pressure wound therapy (particularly in patients with a delayed sternal closure), and appropriate postoperative sternal precautions.^{324,923,929,930,932,933}

Sternal precautions are important for recovery after sternotomy; however, in a departure from traditional methods, nowadays, the early use of patient-centered mobilization protocols is recommended. The use of less-restrictive sternal precautions based on kinesiology principles is a safe alternative that encourages movement and promotes more rapid return to normal functioning.⁹³⁴⁻⁹³⁶ One example of this process is “Keep Your Move in the Tube,” wherein patients imagine themselves in a truncal tube, keeping their upper arms close to their bodies to allow for load-bearing activity.⁹³⁷ This produces a safe body position that allows patients to undertake previously contraindicated movements with fewer load and time restrictions. Gach and colleagues⁹³⁸ found that this approach was associated with quicker functional independence for patients and higher rates of discharge to home (versus to inpatient rehabilitation or a skilled nursing facility). Perioperative teams are encouraged to examine their practices and to use a team-based early-mobilization approach to facilitate recovery and potentially reduce the risk of surgical site infection.²³³

When sternotomy-related superficial sternal wound infection, deep sternal wound infection, or mediastinitis occurs, the standard of care involves pain management, antibiotic therapy tailored to cultures and microbial sensitivities, and surgical debridement.^{894,923,930} The extensiveness of the surgical treatment is guided by clinical examination and CT imaging of the chest. For superficial infections, surgical incision and drainage at the surgical site can be employed (for a detailed review of surgical treatment, see van Wingerden and colleagues⁹³⁹). Often, deep infections can be managed by debridement and daily dressing changes followed by primary closure if the tissue is healthy and free of inflammation. If the infection is extensive involving the sternum to the extent that primary wound closure is not possible, sternal wound flap closure with pectoralis, latissimus dorsi, rectus abdominis flaps, or, when needed, omental transfer can be undertaken.⁹⁴⁰⁻⁹⁴⁷

In this situation, negative-pressure wound therapy can be used in the interval to permit drainage for local sepsis control, sternal stabilization, earlier extubation, mobilization or ICU rehabilitation, and wound healing and granulation.^{338,948-952} In general, complications related to negative-pressure wound therapy are minor, although a few case reports have described erosion into underlying cardiac and vascular structures.^{953,954} Current guidelines

recommend using a barrier dressing between the gauze or foam and the heart to minimize the delivery of excessive negative pressure to the exposed, potentially friable myocardium, great vessels, or bypass conduits.^{923,930}

Summary Points

- An institutional approach to antimicrobial prophylaxis should incorporate current society recommendations, and local compliance should be audited.
- The perioperative team should seek to minimize the risk of infection by removing intravascular lines, endotracheal devices, and urinary catheters as early as possible postoperatively.
- The perioperative team should seek to use less-restrictive sternal precautions based on kinesiology principles as a safe alternative that encourages movement and promotes more rapid return to normal functioning.

Chylothorax

Chylothorax, the accumulation of lymphatic fluid (chyle) in the pleural space, is uncommon (estimates of 0.3%–1.5% in cardiothoracic procedures) but morbid and potentially life-threatening after cardiac surgery.^{955,956} Chyle contains a high level of triglycerides in the form of chylomicrons, which gives the lymphatic fluid a milky appearance and can lead to electrolyte and nutritional deficiencies.⁹⁵⁵ In addition, chyle is rich in immunoglobulins and fat-soluble vitamins absorbed from the intestines. Chyle flow increases with dietary intake of fat, particularly long-chain triglycerides, which are absorbed into the lymphatic system from the intestines. The potential for relative immunodeficiency to develop during prolonged chylothorax relates to the high content of lymphocytes (primarily T lymphocytes) in chyle. Chyle accumulation in the pleural space may originate from large lymphatic vessels that have been divided or from generalized lymphatic leakage, presumably from elevated right atrial pressure.

Specifically, chylothorax is thought to result from the cutting of large tributaries of the thoracic duct (a 2–3-mm thin-walled structure that begins at the cisterna chyli at the second lumbar vertebra and enters the chest through the aortic hiatus) or, less frequently, from injury to the thoracic duct itself.^{955,957} This has been reported to occur after repair of coarctation of the aorta, after placement of Blalock-Taussig or (less frequently) polytetrafluoroethylene interposition shunts, after repair of PDA, and less commonly after CABG procedures.^{956,958,959}

More complex and difficult to manage is chylothorax (and on occasion, pericardial accumulations of chyle with tamponade) that develops after operations through a median sternotomy, such as atrial switch procedures for transposition of the great arteries, SVC/right pulmonary artery anastomoses, and the Fontan procedure. In these circumstances, chylothorax is thought to result from the combination of small lymph channel transections (probably in the thymus gland) and SVC pressure elevation that follows these procedures.

Chylothorax may not be immediately apparent after injury if the patient has not resumed a normal diet, and effluent may appear to be serous. A lymphangiogram demonstrating extravasation is the “gold standard” test⁹⁶⁰; however, analysis of the pleural fluid revealing chylomicrons and a triglyceride concentration >110 mg/dL is generally diagnostic of chylothorax.^{955,961} When identified, treatment for chylothorax includes initial conservative measures (nutritional support and drainage) and surgery (thoracic duct ligation).

Initial management priorities consist of using medium-chain triglyceride formulas to ensure adequate nutrition and reducing enteral long-chain triglyceride intake to reduce chyle production. In pediatric patients, defatted human milk has been suggested.⁹⁶² Parenteral nutrition and bowel rest may be required for patients with high-output chylothorax pleural drainage and/or as a supplemental resource for patients with malnutrition resistant to oral intake alone. The use of octreotide (100 µg every 8 hours), a synthetic polypeptide that mimics natural somatostatin by inhibiting the release of multiple intestinal enzymes, may be a useful adjunct, even for moderate to large effusions; most reports have shown benefit within 2 to 3 days of initiating treatment.^{956,961,963-968} The perioperative team should seek to leverage the knowledge of a dietician and a pharmacist to ensure that optimal nutritional and medication adjustments are undertaken.⁹⁶¹

In addition to nutritional support, chest tube drainage and suction of the chylothorax effusion are required. If the drainage persists for 7 days or so, doxycycline or povidone-iodine pleurodesis can be instilled through the chest tube, plus appropriate pain management can be undertaken.^{969,970} Surgery may be required for persistent and high-output drainage (1000 mL/day over 5 days) despite conservative treatment.^{955,957,960,971-974} A thorascopic or percutaneous procedure may be the first choice, with open surgery for thoracic duct ligation reserved for those in whom less-invasive options would not be feasible or effective.^{956,960,972,975-978}

Summary Points

- Chylothorax, although uncommon, can lead to significant life-threatening electrolyte, nutritional, and immunologic deficiencies.
- Initial treatment includes nutritional support with a medium-chain triglyceride oral diet, use of octreotide, and pleural chest tube drainage.
- A dietician and pharmacist should be consulted as part of the care plan.
- Percutaneous, thorascopic, or open surgical treatment may be required in patients with persistent high-output chylothorax despite conservative measures.

Interdisciplinary Approaches to Care

The “heart team” model has become common in the contemporary cardiac surgery arena. This concept is based on patient-centric care that brings together various cardiac specialists (traditionally, interventional cardiologists, cardiac surgeons, imaging specialists, and the patient’s general cardiologist) to evaluate optimal treatment strategies for valvular and revascularization procedures. Implementing a heart team now receives a Class I or IIa recommendation in several recent guidelines and has been incorporated into several clinical trial designs.^{659,979-981} The interdisciplinary team’s integral influence on the outcomes of patients undergoing cardiac procedures is increasingly appreciated.

Through contemporary quality improvement networks, several key insights into the causes of death after cardiac procedures have been elucidated. Phase-of-care mortality analysis focuses on identifying and addressing root causes of death at the level of single seminal events (specifically, avoidable adverse events) that occur within a specific phase of care and that are found to trigger patient death.⁹⁸²⁻⁹⁸⁴ Failure-to-rescue metrics can be used to identify institutional factors responsible for poor perioperative outcomes.²²⁵ ERPs have the potential to standardize evidence-based perioperative care, reduce practitioner variability, and improve quality metrics and the patient experience.

Phase-of-Care Mortality Analysis

Phase-of-care mortality analysis was developed as an analytic quality improvement tool on the premise that a patient’s perioperative course is an interconnected sequence of clinical events, decisions, interventions, and treatment responses.⁹⁸² The analysis process examines five perioperative phases: preoperative, operative, in the ICU, on the ward, and during discharge planning. Careful review and analysis of these phases allow each patient’s death to be categorized as avoidable or unavoidable, with an aim toward better identifying areas of care that may be amenable to mortality-reducing interventions. In seminal work by Shannon and colleagues,²²⁴ up to 60% of all perioperative deaths can be attributed to seminal events in the preoperative or postoperative ICU phases; similarly, other studies have found that 50% to 80% of these deaths may be attributable to avoidable adverse events during the perioperative period outside of the OR.^{484,982} Importantly, a significant number of preventable deaths occur in patients whose predicted risk for perioperative death was relatively low.^{982,985}

The most common preoperative seminal events were related to the patient’s risk-factor profile (e.g., heart failure, cardiogenic shock, myocardial viability), surgeon judgment regarding timing or the risk/benefit ratio of the surgery, and noncardiac risk factors related to patient selection (e.g., renal failure, chronic obstructive pulmonary disease, cerebrovascular disease, peripheral vascular disease). The most common postoperative ICU seminal events were catastrophic events (unexpected fatal events) and perioperative team deficiencies in identifying and treating acute decompensation. Importantly, phase-of-care analysis has found that more than 40% of reviewed mortalities were “avoidable” (defined as having a >50% likelihood that the patient would have survived in the absence of the seminal event). For the perioperative team, this can provide an opportunity to target both specific death-triggering events and problematic phases of care.⁹⁸⁶⁻⁹⁸⁹ Several groups have made significant improvements to perioperative outcomes by addressing these avoidable terminal events.⁹⁹⁰

Failure to Rescue

Most deaths after cardiac surgery are not triggered by a catastrophic event resulting in sudden death but rather are secondary to the cascade mentioned earlier of clinical events, decisions, and interventions before the terminal event.⁹⁸² Each death can, therefore, be thought of as two separate failures: failure to prevent the seminal adverse event and failure to rescue patients from the complications of these events.⁹⁹¹

Initial insight into failure to rescue was derived from a landmark study by Silber and colleagues⁹⁹¹ in 1992. Defining *failure to rescue* as a death during hospitalization among patients who had a complication, the investigative team studied the relationship between adverse events and failure to rescue in nearly 6000 noncardiac surgery patients across 531 US hospitals.⁹⁹¹ The key finding from this work was that adverse event rates were primarily associated with patient risk, whereas failure-to-rescue rates were primarily associated with hospital characteristics. Validation of this work in the cardiac surgery patient population^{484,990,992} has fostered advocacy for risk-adjusted failure to rescue (defined as deaths arising from one or more complications that include prolonged ventilation, stroke, reoperation, and renal failure) as a complementary metric to existing preoperative risk in the patient undergoing cardiac surgery.²²⁵ Including failure-to-rescue metrics

aims specifically to assess the institutional effectiveness of the postoperative care team, allowing hospitals to target quality improvement efforts.²²⁵

Enhanced Recovery Protocols

To address the needs of an increasingly aging and frail population, a group of general surgeons in 1997 began developing a formalized, interdisciplinary process for ensuring patient readiness for surgery and promoting quicker recovery after the procedure.⁹⁹³ Specific components included optimizing perioperative nutritional status, promoting analgesia without opioids, and early postoperative feeding and mobilization.⁹⁹⁴⁻⁹⁹⁷ The use of such ERPs has been associated with significantly fewer perioperative complications, shorter hospital LOS, and lower costs as compared with conventional perioperative management.⁹⁹⁸⁻¹⁰⁰⁰

Each of the surgical subspecialties has generated consensus documents providing ERP recommendations for their patients (see also www.erassociety.org).¹⁰⁰¹⁻¹⁰⁰⁴ Even though standardized protocols to facilitate fast-track extubation after cardiac surgery have been commonplace for several decades,⁵⁶⁴⁻⁵⁶⁷ cardiac-specific perioperative ERPs are a more recent development.^{231,232,1005} Cardiac surgery patients are especially vulnerable, given their increasingly advanced age, greater frailty and heightened comorbidity levels, and the likelihood of postoperative ICU admission^{221,222,765,766,1006,1007}; nonetheless, these patients have been found to benefit from ERP protocols.

Cardiac-specific ERPs should incorporate the interdisciplinary team's expertise to apply evidence-based measures that address preoperative vulnerabilities and CPB-related sequelae and incorporate best practices of care in the ICU and postoperative unit—all with the goals of optimizing the patient experience, improving outcomes, and using resources efficiently.^{334,1008} To this end, a guidance document developed specifically for cardiac surgery perioperative teams recommends 22 potential interventions that aggregate marginal gains and synergies provided by bundling preoperative, intraoperative, and postoperative elements.^{233,1009} The level and quality of the evidence supporting many of the ERP elements pertaining to the cardiac surgery patient are still under investigation,¹⁰⁰⁹⁻¹⁰¹¹ and an updated version of proposed recommendations is underway. It is reasonable for the interdisciplinary perioperative team to consider developing local ERPs so that elements specific to its particular setting can be incorporated.

Enhanced recovery is more than simply the implementation of a protocol.¹⁰⁰⁹ Importantly, this process requires a shift in culture, crucial interdisciplinary team building, education, planning, and processes needed to develop and sustain a successful cardiac ERP program based on best practices scaled to the clinical context of the local institution or healthcare system.¹⁰¹² In addition, patient value can be enhanced by involving a patient-engagement panel to provide additional insights for the perioperative team about ERP elements they seek to implement.^{738,1013-1015} Lastly, ERPs should include accountability measures so that key elements can be tracked, benchmarked, and correlated with standard and patient-centered outcomes.¹⁰¹⁶

Evidence-Based Perioperative Management and Team Training

The perioperative care of cardiac surgery patients is interdisciplinary by design. As patients move through the five phases of care,

they travel through multiple parts of the hospital and are cared for by several different teams of physicians, nurses, and allied health-care professionals.¹⁰¹⁷ Coordinating care across multiple disciplines and areas is of paramount importance. As discussed earlier, the use of ERPs is one method for improving the quality of care for surgical patients.¹⁰¹⁸ Although further study in patients undergoing cardiac surgery is needed, noncardiac surgical programs have seen reductions in complication rates and mortality when ERPs are used.^{1019,1020}

One area of rescue that has garnered significant interest over the past two decades is cardiac arrest after cardiac surgery, which occurs in 0.7% to 8.0% of patients⁴⁸⁴ but has a high degree of reversibility, with approximately half of patients surviving to hospital discharge.^{488,489} A key aspect in minimizing failure-to-rescue rates after cardiac arrest is its rapid recognition and protocol-driven management of hypotensive emergencies in the postoperative ICU. To this end, a cardiac surgery-specific decision-making and team-based training protocol was developed by Dunning and colleagues¹⁰²¹⁻¹⁰²³ and endorsed by both The Society of Thoracic Surgeons and the European Association of Cardio-Thoracic Surgery (see “Postoperative Cardiac Arrest” in Section I). Maccaroni and colleagues¹⁰²⁴ found a decrease in re sternotomy rates and an increase in postarrest survival after implementing the protocol. A similar study also found a significant decrease (from 53% to 13%) in failure-to-rescue rates after cardiac arrest.¹⁰²⁵ However, both studies are small single-center studies; larger studies are required to conclusively show an association between protocol implementation and a reduction in failure-to-rescue rates.¹⁰²⁶

In summary, a combination of patient and hospital factors contribute to perioperative deaths. Recently developed protocols and recommendations target many of these seminal, potentially death-triggering events to improve patient outcomes after cardiac surgery. However, the literature on the efficacy of these quality improvement initiatives remains sparse. Larger future studies with multi-institutional data sharing will be key to facilitate ongoing knowledge translation in this field.

Summary Points

- The cardiac team should consider incorporating the perioperative interdisciplinary team in the “heart team” model of care.
- Modern quality improvement methodology, including phase-of-care mortality analysis and failure-to-rescue analysis, has the potential to significantly improve outcomes after cardiac surgery.
- Preliminary studies indicate that using ERPs to optimize the preoperative, intraoperative, and postoperative phases has improved patient-centered outcomes in patients undergoing cardiac surgery.
- Developing ERPs necessarily requires interdisciplinary collaboration (including patient input) that incorporates evidence-based best practices nested in a local culture of patient-centered care.
- The perioperative team should include essential elements for data reporting in an ERP, to inform quality improvement efforts and facilitate accountability.
- Larger future studies with multi-institutional data sharing are needed to facilitate ongoing quality improvement and knowledge translation in this field.

SECTION IV: SPECIAL CONSIDERATIONS IN THE POSTOPERATIVE CARE OF NEONATES AND INFANTS

General Comments

Neonates, infants, and some small young children present postoperative care problems that differ in some ways from those of adults. Principles of subsystems and overall care remain the same, but the internal milieu of even normal neonates and infants is different in certain respects from adults. The cardiac operation has generally been more urgently indicated, and therefore, these young patients often have more preoperative derangements and require more extensive operations. Cardiac reserve of neonates and infants is often less than that of adults after cardiac surgery because of the nature and seriousness of their preoperative cardiac condition. Their small airways are more apt to be suddenly obstructed by secretions, and they are subject to special problems such as episodes of pulmonary arterial hypertensive crises. Serum potassium level rises much more rapidly during oliguria than it does in older children and adults. These considerations do not necessarily indicate that young age per se is a risk factor for death after cardiac surgery; they do indicate that special attention must be given to them. As in Section III of this chapter, we provide the general aspects of postoperative care of neonates and infants here, with details presented in the discussions of the various subsystems.

The endotracheal tube, a urinary catheter, intracardiac recording devices, epicardial myocardial wires, and intravenous access to large central veins are usually in place when the patient leaves the OR, just as in adults. Intravenous access in the caval or internal jugular veins is generally avoided in patients having cavopulmonary connections because of the risk of thrombosis. In the ICU, the patient is placed on a ventilator specifically designed for small patients. Often, low-dose inotropic support, (e.g., dopamine 2.5 to $5 \mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$), is used, even in patients convalescing normally, and this is generally continued until the morning of postoperative day 1.

Because measuring devices and infusion of medications require some fluid, and because it is desirable to limit the total fluid intake in neonates and infants just as it is in adults, all continuously infused medications are mixed in sufficient concentration that the infusion rate does not exceed 2 to $3 \text{ mL} \cdot \text{h}^{-1}$, the minimal rate at which reasonably accurate infusions can be given. No “calculated fluids” are planned for the day of operation, and this can be reconsidered early on the morning of postoperative day 1. Calculations are made as to the amount of fluid administered not only by the intravenous route but also in flushing pressure-measuring devices, and at 7 AM of postoperative day 1, consideration is given to beginning administration of “calculated fluids” or continuing without them.

Once the patient is settled on the ventilator, the status of anesthetic and sedating medications is assessed. The goal after complex operations in general is to keep the patient heavily sedated and paralyzed at least until the morning of postoperative day 1. For paralysis, appropriate doses of pancuronium (or a comparable agent) are administered intravenously. This is insufficient to overcome many possibly deleterious reflexes. Therefore, fentanyl is also given as a continuous infusion of about $10 \mu\text{g} \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$, and at times at the increased dose of 15 to $25 \mu\text{g} \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$, not only for sedation but also for control of sympathetic activity.^{1027,1028} Particularly when the endotracheal tube is suctioned, the patient must be under the influence of fentanyl.¹⁰²⁹ Ventilation is arranged to keep arterial Paco_2 at about 30 mmHg , arterial Pao_2 above 120

mmHg when a biventricular repair has been accomplished, and if possible, pH at 7.45 or higher. This minimizes the tendency for pulmonary vascular resistance to rise paroxysmally.

We recommend irrigating the endotracheal tube with saline and suctioning every 2 hours, although some who do not use fentanyl favor no suctioning. Whatever the policy, it must be recognized that this is a potentially hazardous procedure, and it should be undertaken only by skilled nurses with adequate backup, should an emergency develop during or within a few minutes after suctioning.

Early on the morning of postoperative day 1, consideration is given to extubating the patient. If all subsystems appear to be functioning satisfactorily, the protocol leading to extubation is begun. Then, if the patient is awake and active, extubation is accomplished. After extubation, the baby is maintained in a humidified oxygen-enriched environment for at least another 24 hours in the ICU, although care is taken to avoid excessively high Pao_2 .

If all subsystems are not functioning well on the morning after operation, there should be no hesitancy to continue control of ventilation as well as paralysis and sedation for another 24 hours. However, about 70% of normally convalescing neonates and small infants can be extubated the morning after operation and the remainder usually on the morning of postoperative day 2.

Following extubation and the passage of a few hours, oral feedings can be started. For the feedings, the patient is held at least in a semi-upright position, observing the usual precautions necessary in feeding small babies. Neonates and young infants have little nutritional reserve. When oral intake is not possible on postoperative day 1 or 2, nasogastric tube feeding is begun. If this is not promptly effective, intravenous alimentation is considered.

To avoid bleeding after their removal, intracardiac catheters should *not* be removed before postoperative day 2. It is prudent to leave all chest tubes/Blake drains in place until after intracardiac lines have been removed.

When convalescence is not normal for any reason, management is altered appropriately, generally as described in the earlier parts of this chapter. A special situation is presented early postoperatively by neonates and young infants when the operation leaves pulmonary blood flow coming solely from a surgically created systemic–pulmonary artery shunt (see Special Features of Postoperative Care in Chapters 34 and 52). In these intubated patients, the magnitude of pulmonary blood flow is largely determined by the relation of systemic to pulmonary vascular resistance.¹⁰³⁰ When pulmonary resistance is low, pulmonary blood flow tends to be excessive and systemic blood flow small. Should pulmonary vascular resistance gradually or suddenly rise, pulmonary blood flow declines, sometimes rapidly, and hypoxia develops. In this setting, pulmonary vascular resistance is largely determined by arterial Paco_2 and arterial pH. Manipulation of Paco_2 is the best method for regulating pulmonary vascular resistance. When pulmonary blood flow appears to be too high and systemic blood flow too low, arterial Paco_2 is deliberately elevated modestly (to 40 to 45 mmHg) to increase pulmonary vascular resistance, decrease pulmonary blood flow, and increase systemic blood flow. When pulmonary blood flow seems too low and hypoxia too prominent, pulmonary arteriolar resistance is lowered by lowering the arterial Paco_2 appropriately, usually to 25 to 35 mmHg . Alterations in arterial Paco_2 are conveniently accomplished by altering tidal volume and respiratory rate. The adjustments required are usually sufficiently small that they produce no change in other important variables.

Pulmonary Hypertensive Crises

Pulmonary hypertensive crisis is a phrase used to describe a serious syndrome of hyperacute rise in pulmonary arterial pressure, usually accompanied by bronchospasm, often followed within seconds or accompanied by profound reduction in cardiac output and fall in Sao_2 ; these sequelae may be irreversible. The syndrome occurs most commonly, but not exclusively, among neonates and infants who are intubated after an operation for repair of a congenital cardiac defect associated with pulmonary arterial hypertension. Pulmonary artery pressure is usually essentially normal or mildly elevated after repair, until an episode of pulmonary hypertensive crisis. The crisis may appear spontaneously, but usually occurs during or shortly after endotracheal tube suctioning, particularly if the endotracheal tube is too far down the trachea and makes contact with the carina. Multiple crises may occur in a single patient. The prevalence of pulmonary hypertensive crises is greatest about 18 hours after operation, but they can occur before or after that time.¹⁰²⁸ During an acute hypertensive crisis, atypical cardiac tamponade can result from acute RV dilation.^{1031,1032}

Incremental Risk Factors

Although recognized for many years, its precise cause at the time it occurs is unknown.¹⁰³³⁻¹⁰³⁵ Congenital cardiac defects accompanied by large pulmonary blood flow and relatively low pulmonary vascular resistance are strong *patient risk factors* for their development. Chief among them are truncus arteriosus and AV septal defect. The histology of the pulmonary arteries and arterioles is normal or mildly abnormal, with all changes considered clearly reversible.¹⁰²⁸ The syndrome is not seen in patients with fixed pulmonary hypertension or severe pulmonary vascular disease. *Procedural risk factors* have not been clearly identified, although absence of pulmonary blood flow during CPB, accumulation of neutrophils in the lung, and damage to pulmonary endothelial cells are known to occur and provide a favorable setting for development of pulmonary hypertensive crises. Release of arachidonic metabolites from some cells in the lung (probably pulmonary endothelial cells),¹⁰³⁶ failure of pulmonary endothelial cells to inactivate bradykinin,¹⁰³⁷ and release by injured pulmonary endothelial cells of the vasoconstrictive platelet-activating factor¹⁰³⁸ are probably all encouraged by events during CPB, perhaps particularly if these cells were preconditioned before operation by a high pulmonary blood flow. *Postoperative risk factors* nearly certainly include acute hypoxia, which is known to produce acute hypoxic vasoconstriction, probably using some of the mechanisms just described.^{1039,1040} Catecholamine administration appears to predispose patients to these crises.

Prevention and Treatment

Because pulmonary hypertensive crisis may be fatal despite intense therapy, its prevention is critical. An important preventive measure in neonates and infants with high-risk conditions is maintenance of paralysis and sedation (with pancuronium [or a similar agent] and fentanyl) for at least the first 20 to 24 postoperative hours, and for at least another 24 hours or longer if the patient remains intubated. Fentanyl has been shown to be a safe agent in neonates and adults, with little adverse effect on cardiac function or systemic and pulmonary circulations.¹⁰²⁹ A paralytic agent is given along with fentanyl to prevent rigidity. Fentanyl (in a dose of $25 \mu\text{g} \cdot \text{kg}^{-1}$), along with pancuronium, should be given

before suctioning the endotracheal tube in any susceptible neonate or infant who is not already well-paralyzed and sedated because it appears to minimize the possibility of a pulmonary hypertensive crisis. In addition to sedation, aggressive treatment of acidosis, including hyperventilation to $\text{Pco}_2 = 30$ to 34 mmHg , decreases prevalence and severity of pulmonary hypertensive crises. When the monitored pulmonary artery pressure becomes paroxysmally elevated early postoperatively, efforts are made to control and reduce it. A high FiO_2 is maintained because this is as useful an agent as exists for reducing pulmonary vascular resistance.

There are encouraging results from use of inhaled nitric oxide to reduce pulmonary artery pressure in cardiac transplant recipients,¹⁰⁴¹ patients on LV assist devices,¹⁰⁴² and generally in infants and children prone to pulmonary hypertensive crises.¹⁰⁴³ Journis and colleagues¹⁰⁴⁴ found that 15 of 17 children having critical pulmonary hypertension following operation for congenital heart disease responded to inhaled nitric oxide (20 ppm) after failure of conventional medical treatment. Pulmonary arterial pressure decreased without a change in systemic arterial pressure but with concomitant increase of Sao_2 and SVO_2 . Other investigators have also shown an increase in cardiac index and improvement of indexes of RV function after nitric oxide treatment.¹⁰⁴⁵ Intrapulmonary shunt fraction may decrease as well.¹⁰⁴⁶ The response in postoperative patients with valvar heart disease may be less satisfactory.¹⁰⁴⁷

Junctional Ectopic Tachycardia (JET)

JET may be a particularly difficult problem in children, with a prevalence of 5% to 8%.¹⁰⁴⁸ It is associated with operations to close VSDs, perioperative transient AV block, repair of AV septal defect,¹⁰⁴⁸ and young age. The appearance of JET in the postoperative period may be associated with residual lesions, which should be ruled out. In 70 of 71 patients with postoperative JET, Walsh and colleagues obtained control of the arrhythmia and hemodynamic stability using a combination of sedation, hypothermia to 34°C , and procainamide.¹⁰⁴⁹ Amiodarone may also be effective.¹⁰⁵⁰

Optimal Recovery following Infant Pediatric Cardiac Surgery

Recently, there has been an increased focus on optimization of the postoperative evolution of neonates and infants following cardiac surgery. The American Association for Thoracic Surgery has published an expert consensus document on that topic.¹⁰⁵¹ In the perioperative period, several important elements need to be assessed, including the quality of repair, blood product management, pain management, and respiratory management, among others. The heterogeneity of cardiac malformations, surgical repairs, patients' age and weight, preoperative status, and presence or absence of syndromes makes the standardization of postoperative care challenging. In the end, each patient's care should be individualized within a structured protocol-driven platform.

References

Complete references in this chapter can be accessed in the online version.

Protocol for Reducing Arterial Blood Pressure and Afterload

Sodium nitroprusside: Sodium nitroprusside is continuously administered via IV as required. It acts directly on arterial (and to a lesser extent, venous) smooth muscle and thus decreases systemic vascular resistance, PVR, and systemic venous tone. Its onset and duration of action are immediate. The dose is 1 to $5 \mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ (doses larger than this are *not* used), regulated in most cases to maintain a mean arterial blood pressure 10% above the normal value for the patient's age. In patients with thick left ventricular walls or coronary artery disease, concern about coronary perfusion pressure makes a mean arterial blood pressure 20% above normal desirable (or 150 mmHg in adults). Twenty-five to 50 mg of sodium nitroprusside are dissolved in 100 to 250 mL of 5% glucose in water; however, this may vary from institution to institution. The drug may be administered with a servo-pump using a closed-loop system⁶⁶⁶ under computer control^{668,669} or with a slow-infusion pump. When nitroprusside is being administered, methods should be available for measuring blood thiocyanate levels, because values of $>10 \text{ g} \cdot \text{dL}^{-1}$ are potentially toxic.^{670,671} Toxicity is manifested by signs of intracellular suppression of oxygen consumption (elevation of mixed venous oxygen levels, narrowing of arterial/venous oxygen difference, and metabolic acidosis) and by anorexia, muscular spasms, disorientation, and convulsions. Toxicity is due largely to the formation of cyanide, the major metabolic product of sodium nitroprusside.^{671,672} Toxicity is treated by stopping the infusion and administration of the hydroxocobalamin (in a cyanide antidote kit) or less commonly an IV infusion over 15 minutes of $150 \mu\text{g} \cdot \text{kg}^{-1}$ of a 25% solution of sodium thiosulfate ($10 \mu\text{g} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$).

Nitroglycerin: Nitroglycerin decreases both venous tone and coronary resistance. It is therefore particularly useful when

myocardial ischemia is present. Infusion rates of 5 to 200 $\mu\text{g}/\text{min}$ titrated to desired clinical response is recommended. Because nitroglycerin is absorbed into polyvinyl tubing used for IV infusion, the concentration reaching the patient will be less than planned until the tubing becomes saturated. Nitroglycerin is not as effective as nitroprusside in lowering arterial pressure.

Clevidipine: Clevidipine is a dihydropyridine L-type calcium-channel blocker that acts on arterial smooth muscle. It is indicated for blood pressure reduction when oral therapy is not feasible or not desirable. The half-life of clevidipine is approximately 1 minute, and the agent is metabolized by esterase in blood to an inactive metabolite and not affected by hepatic or renal function. The initial IV infusion is usually commenced at 1 to 2 mg/h. The dose can be bidirectionally titrated by $\leq 50\%$ every 90 seconds to 10 minutes. When approaching the desired blood pressure goal, the frequency of dose adjustment can be lengthened to every 5 to 10 minutes to avoid blood pressure target overshoot. A 1 to 2 mg/h increase will generally produce an additional 2 to 4 mmHg decrease in systolic pressure. The maintenance dose is usually 4 to 6 mg/h, with a maximum dose of 16 mg/h up to 72 hours.

Adverse effects of clevidipine include risks for hypotension and reflex tachycardia; headache, nausea, and vomiting also have been reported. Key contraindications to use include allergy to soybeans, soy products, eggs, or egg products and a defective lipid metabolism, given that the vehicle in which the agent is stored contains phospholipids, which can support microbial growth; a good aseptic technique is needed. Lastly, in light of the potential lipid load, a maximum of 1000 mL/24 hours is recommended. The agent contains 2 kcal/mL from fat, which should be counted as a caloric source.

Appendix Table 4A-1 Insulin Infusion Adjustment (Cleveland Clinic)*

Blood Glucose	If Blood Glucose <i>Decreases</i> ≥ 30 mg · dL ⁻¹ Since Last Level	If Blood Glucose Is <i>Stable</i> (change < 30 mg · dL ⁻¹) Since Last Level	If Blood Glucose <i>Increases</i> ≥ 30 mg · dL ⁻¹ Since Last Level
≤60	Stop insulin infusion See Hypoglycemia protocol	Stop insulin infusion See Hypoglycemia protocol	—
61-70	Stop insulin infusion See Hypoglycemia protocol	Stop insulin infusion See Hypoglycemia protocol	—
71-85	Stop insulin infusion See Hypoglycemia protocol	Decrease rate by 50%	
86-100	Decrease rate by 50%	Decrease rate by 50%	—
101-115	Decrease rate by 50%	Continue current rate	—
116-150	Decrease rate by 50%	Increase rate by 25%	Increase rate by 25%
151-200	Decrease rate by 25%	Increase rate by 25%	Bolus 2 units/increase rate by 25%
201-250	Continue current rate	Bolus 2 units/ Increase rate by 25%	Bolus 4 units/ Increase rate by 25%
251-300	Continue current rate	Bolus 4 units/ Increase rate by 50%	Bolus 6 units/ Increase rate by 50%
301-350	Continue current rate	Bolus 6 units/ Increase rate by 50%	Bolus 8 units/ Increase rate by 50%
351-400	Continue current rate	Bolus 8 units/ Increase rate by 50%	Bolus 10 units/ Increase rate by 50%
>400	Notify staff anesthesiologist ^a	Notify staff anesthesiologist ^a	Notify staff anesthesiologist ^a

Note: If insulin rate is ≥30 units · h⁻¹, notify staff anesthesiologist.
[†]Do not adjust insulin rate every hour; make adjustments every 2 hours.
^aSevere hyperglycemia is treated per anesthesiologist's discretion.

Standard Infusion Rates of Inotropic and Vasopressor Agents

Dopamine	1–20 µg/kg/min
Dobutamine	1–20 µg/kg/min
Epinephrine	0.01–0.2 µg/kg/min
Norepinephrine	0.01–1 µg/kg/min
Milrinone	0.25–0.75 µg/kg/min
Vasopressin	0.01–0.06 units/min
Angiotensin II	10–40 ng/kg/min

Protocol for Managing Aspects of Ventricular Electrical Instability

Early Interventions for Ventricular Electrical Instability

1. Give lidocaine as an IV bolus injection (the dose is 1 mg/kg for adults and children, although in adults the usual dose is 50–100 mg) if the arrhythmia is a PVC or ventricular tachycardia with a good hemodynamic state. If there is ventricular tachycardia and reduction of cardiac output, use *immediate synchronized DC cardioversion* (100 J and then 200 J).
2. The perioperative team should seek to maintain the appropriate levels of electrolytes with replacement based on each specific institution.
3. If the ventricular rate is <80 to 90 beats/min, initiate pacing. If basic rhythm is sinus or AV junctional, use atrial pacing.

When cardiac rhythm is other than sinus or AV junctional, or atrial pacing fails to result in 1:1 AV conduction, use ventricular pacing (with the ventricular wire electrode attached to negative pole of pacer). In the presence of second- or third-degree AV block, consider AV sequential pacing.

Intervention after Control of the Urgent Situation

If the arrhythmia recurs promptly or is not controlled by these simple measures, begin continuous IV lidocaine infusion of 1 to 4 mg/min titrated based on serum lidocaine concentration levels and patient clinical response.

Protocols for Acute Management of Postoperative Atrial Fibrillation

The treatment of postoperative atrial fibrillation (POAF) can be approached with either a rate-control or a rhythm-control strategy. Gillinov and colleagues⁴⁷⁸ found no significant superiority for either approach in clinical outcomes such as hospitalization length of stay, complication rates, and rates of persistent atrial fibrillation at 60 days.

In hemodynamically stable adult patients, the most commonly used first-line agents are beta-blockers for rate control and amiodarone for rhythm control. Other agents such as verapamil, diltiazem, and digoxin are less commonly used as first-line agents but can be considered if the patient is intolerant or has a contraindication to the usual first-line agents.

Rate Control Methods

- Metoprolol 5 mg IV every 3 to 5 minutes (total 15 mg) and/or initiate and titrate oral beta-blockade administration.
- Diltiazem 0.25 mg/kg IV bolus $\times$ 1, followed by diltiazem infusion 5 to 15 mg/h titrated to heart rate <100 and with medication hold parameters for a heart rate <60 or a systolic blood pressure <100 .
- Digoxin 0.5 mg IV $\times$ 1, followed by 0.25 mg IV every 8 hours $\times$ 2 doses. In the older adult (>65 years of age), lower body weight or in the presence of compromised renal function, the therapeutic window is narrow, and toxicity may occur. Further, if co-administered with amiodarone, digoxin concentrations will increase so lower dosing will be required and monitored.

Rhythm Control Methods

- Amiodarone 150 mg IV loading dose over 20 to 30 minutes, followed by 1 mg/min for 6 hours, then 0.5 mg/min for 18 hours. Transition to amiodarone 200 mg PO BID $\times$ 5 day, then 200 mg PO daily. Hold agent if the patient becomes bradycardic (i.e., heart rate <60) or has a prolonged QTc >450 . Target a total amiodarone load of 10 g.

- In the hemodynamically unstable patient, managing the patient's airway, breathing, and circulation should be prioritized with plans to undertake timely direct DC cardioversion.

Anticoagulation

The prevalence of stroke attributable to postoperative AF is unknown. Although guidelines recommend that anticoagulation be initiated for POAF, the optimal duration of paroxysmal AF that triggers this therapy is a subject of controversy. The justification to initiate anticoagulation if AF persists longer than 48 hours or is recurrent should be balanced between the patient's risk for thromboembolic complications versus the risk for postoperative bleeding. Further, the role of novel oral anticoagulants (e.g., apixaban) versus warfarin remains controversial at the time of this writing. If anticoagulation is being considered:

- **Warfarin with a target INR of 2–3:** It is important to consider the interaction between warfarin bioavailability in patients concurrently being managed with amiodarone and the use of UFH infusion (with titration to a goal activated PTT 2 to 3 times normal or anti-Xa level per the institution's nomogram) as a bridge in appropriate higher-risk patients.
- **Apixaban 5 mg BID for patients with nonvalvular atrial fibrillation:** If the patient meets two of three mitigating criteria (age ≥ 80 , weight ≤ 60 kg, creatinine ≥ 1.5 g/dL), the dose may need to be reduced to 2.5 mg.
- Alternative: rivaroxaban 20 mg daily, dose adjusted based on renal function (starting when creatinine clearance (CrCL) <50 mL/min).
- If elective cardioversion is planned, IV heparin is recommended and is prescribed in therapeutic doses.
- If the patient is on dual antiplatelet therapy, strongly consider holding one medication if triple-therapy bleeding risk exceeds benefit.

Protocol for Rapid Atrial Pacing with Atrial Wires

The technique of rapid atrial pacing is applied to *atrial flutter*, defined as a general atrial rate of 250 to 350 beats/min with a constant beat-to-beat cycle length that *can be* interrupted by rapid atrial pacing. This is rarely possible in *atrial flutter-fibrillation*, a more rapid type of atrial flutter with a rate >350 beats/min, and it is not possible in AF.

ECG limb leads are placed on the patient for monitoring, and the two atrial wires are connected to the rapid atrial pacer. Because the atrial pacing threshold is usually high during atrial flutter, output is set at 10 to 20 mA. Bipolar atrial pacing is used because the stimulus artifact then rarely distorts the ECG tracing, and the atrial complex in the ECG is clearly seen so that atrial capture can be verified.

A relatively slow pacing rate is used first, because occasionally, immediate one-to-one conduction occurs with rapid ventricular tachycardia, which is to be avoided. After this, several possible maneuvers are available for controlling atrial flutter and its rapid ventricular response, as described by Waldo and MacLean.⁴¹⁷

Ramp Technique

Atrial pacing is initiated at a rate 10 beats/min faster than the atrial flutter rate. The rate of atrial pacing is then gradually increased. When the typical negative atrial complex in lead II changes to a positive atrial complex, indicating capture by pacing, atrial pacing is either abruptly stopped or gradually slowed until the ventricular rate is considered satisfactory.

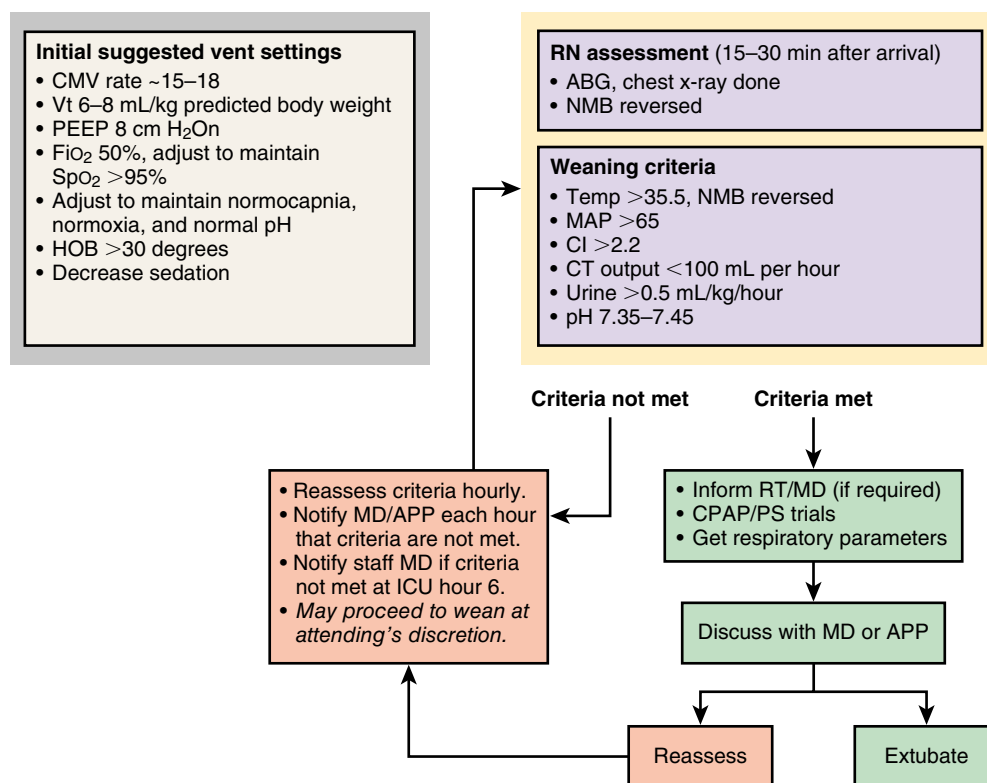
Constant Rate Technique

Pacing is initiated at a rate 10 beats/min faster than the spontaneous atrial flutter rate. After pacing at this rate for about 30 seconds, pacing is either abruptly stopped or the pacing rate quickly slowed until the ventricular rate is considered satisfactory. If these maneuvers are unsuccessful in interrupting the atrial flutter, they are repeated, with the initial atrial pacing rate increased in increments of 10 beats/min.

Protocol for Initial Mechanical Ventilation Management and Early Extubation

This section is an amalgamation of protocols and checklists from the cardiac surgery section and cardiac surgery ICU from

St. Boniface Hospital (Winnipeg, Manitoba, Canada) and University Hospitals (Cleveland, Ohio, US).



Checklist for Extubation

Respiratory

- pH 7.35–7.45
- pCO₂ 35–45 mmHg
- Po₂ ≥80 mmHg or SpO₂ ≥95%
- Hemoglobin ≥7.5
- FiO₂ ≤0.5
- Ve <10 L/min
- Able to cough spontaneously

- Minimal respiratory secretions

Hemodynamics

- HR 50–110 (may be paced)
- MAP 60–110 mmHg (may be on stable dose of inotrope/vasopressor)
- Chest tube losses <100 mL/h
- Hgb >7.5

General

- Temperature $>35.5^{\circ}\text{C}$
- Pain control adequate
- Able to lift head off pillow for 5 sec
- Conscious and responding to commands
- Able to cough

Special Considerations

Was patient difficult to intubate?

Yes

☐

No

☐

Does patient currently have significant facial or oropharyngeal edema?

Yes

☐

No

☐

**If Yes appears for any of the earlier special considerations, a patient may be weaned by the protocol, but a physician order must be obtained to extubate, and a physician must be present for the extubation.

Was a physician presence needed?

Yes

☐

No

☐

Time extubated: _____ Form completed by:

Protocol for Acute Kidney Injury Risk Assessment and Prevention

This protocol is adapted with permission from the Harrington Heart and Vascular Institute, University Hospitals (Cleveland, Ohio, US) AKI Prevention protocol.

Step 1: Preoperative Evaluation and Education

Patients at risk for cardiac surgery-associated acute kidney injury AKI can be identified preoperatively by using a dynamic predictive scoring tool (such as that adapted from Jiang and colleagues⁶¹⁶ shown later). Patients undergoing elective cardiac surgery should be assigned to a low, moderate, or high-risk category based on a scoring tool used in perioperative medicine. The patient's risk for perioperative AKI should be documented in the patient's chart as part of the preoperative history and physical (H&P). Patients should continue to receive preoperative cardiac surgery education with emphasis on holding all NSAIDs at least 1 week before surgery. Angiotensin-converting enzyme and angiotensin receptor blocker agents (ACE/ARBs) should be discontinued at least 48 hours preoperatively.

Variables	Points
Male	1
Age: < 40 y	0
Age: 41-60 y	1
Age: 61-80 y	2
Age: > 81 y	3
Chronic kidney disease	1
New York Heart Association score >2	1
Previous cardiac surgery	1

Step 2: Intraoperative Identification of Risk Factors and Patient Care Considerations

The patient's AKI risk score will be discussed during the intraoperative pre-incision huddle and added to any intraoperative checklist. Intraoperative hemodynamic monitoring and fluid management will continue per cardiac anesthesia discretion, keeping the patient's risk status in mind. High-risk patients may receive a pulmonary artery catheter or other minimally invasive cardiac output-monitoring device. Anesthesia should ensure preoperative methicillin-resistant *Staphylococcus aureus* (MRSA) swab

results are reviewed; intraoperative vancomycin should be avoided if possible.

Step 3: Postoperative Assessment of AKI Risk Factor

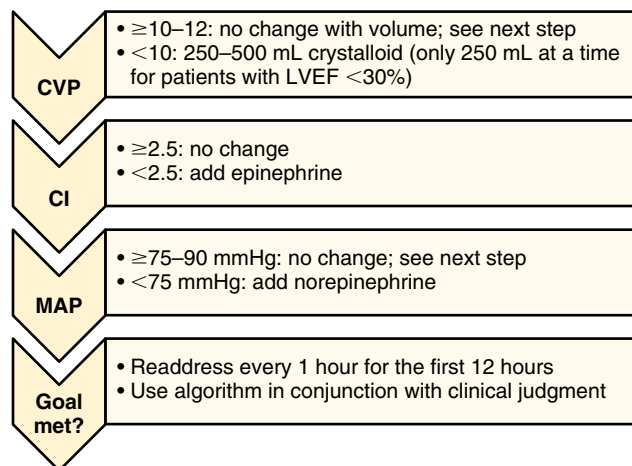
The receiving team in the ICU should review the patient's preoperative risk score during the OR-to-ICU handover. The ICU provider will then calculate a new risk score based on the dynamic predictive scoring tool at the time of ICU admittance, as shown here. The patient's postoperative risk score should be documented within the initial ICU H&P and incorporated into the genitourinary section of the patient's daily assessment and plan in the progress note. If vancomycin was administered in the OR, the resident or advanced practice provider should review MRSA swab status and send swab if necessary. Only order vancomycin postoperatively if the patient is MRSA positive or if swab results are still pending.

Variables	Points
Male	1
Age: <40 y	0
Age: 41-60 y	1
Age: 61-80 y	2
Age: >81 y	3
Chronic kidney disease	1
New York Heart Association score >2	1
Previous cardiac surgery	1
Valve and coronary artery bypass grafting	1
Erythrocyte transfusion (per unit)	1/unit

Step 4: Implementation of Goal-Directed Therapies for High-Risk Patients

All patients at high risk for AKI should be treated according to a postoperative hemodynamic algorithm like the one later. Overall goals for high-risk patients should include a mean arterial pressure (MAP) of 75 to 90 mmHg, a CVP of 10 to 12 mmHg, and a cardiac index (CI) >2.5 L/min/m². A postoperative ICU cardiac surgery intensive insulin infusion order set should be used for all high-risk patients during the first 12 to 24 hours to allow for

optimal blood glucose management and to avoid hyperglycemia, defined as >180 mg/dL for more than 3 hours. No ACE/ARB medications or NSAIDs should be given to high-risk patients for at least 48 hours postoperatively. Avoid radiocontrast agents for at least 48 hours postoperatively in high-risk patients. No hydroxyethyl starch, gelatin, or chloride-rich solutions; first-choice crystalloid fluid should be lactated Ringer's or Plasmalyte. Use normal saline if patient has, or is at risk for developing, clinically significant hyperkalemia.



Step 5: Ongoing ICU Evaluation and Monitoring

The renal function panel should be monitored at least every 12 hours during the first 24 hours for all elective cardiac surgery cases, regardless of risk score. For all elective cardiac surgery patients, AKI risk should be reassessed at the 24-hour mark by using the dynamic predictive scoring tool, as shown here.

Variables	Points
Male	1
Age: < 40 y	0

Variables	Points
Age: 41–60 y	1
Age: 61–80 y	2
Age: > 81 y	3
Chronic kidney disease	1
Cardiopulmonary bypass application	1
Previous cardiac surgery	1
Valve and coronary artery bypass grafting	1
Erythrocyte transfusion (per unit)	1/unit
Low cardiac output syndrome	1

Step 6: Outcome Measurement and Monitoring

Outcomes will be reported and monitored as a component of ongoing quality efforts, including participation in STS registries. Use of the dynamic predictive scoring tool will be tracked through chart reviews of H&Ps and daily progress notes.

Risk Stratification of CSA-AKI Development at Indicated Time Points

Sum Points	Preoperative	ICU Admittance	24 Hours after ICU Admittance
0–1 pts	Low	Low	Low
2–3 pts	Medium	Low	Low
4 pts	High	Low	Low
5–9 pts	High	Medium	Medium
> 10 pts	High	High	High

Example of an ERAS Implementation Checklist

Adapted from Salenger et al.¹⁰¹²

ENHANCED RECOVERY AFTER SURGERY	
IMPLEMENTATION CHECKLIST	
A. Creating a case for change	
1. Gap analysis of a service line a. Follow patient through the surgical journey b. Process/value stream mapping c. Teamwork: determine communication gaps between providers in all phases of care	☐
2. Create vision of future pathways a. How will the interdisciplinary team work in delivering pathways? b. Look at the evidence and decide which elements to incorporate c. Obtain patient feedback on their experience of surgical case d. Identify clinical stakeholders necessary to make the change e. Recognize what is going well and what needs to change	☐
3. Collect baseline data of service line: a. Process measures b. Existing protocols c. Outcome measures d. Existing dashboards/STS outcomes	
B. Communicate the case for change	
1. Done by executive sponsor and clinical lead (champion) 2. Communicate all information collected during Step #1 3. Communicate to all internal clinical areas 4. Communication to be consistent 5. Address any and all questions/pushback 6. Clinical Change: All clinical change is vetted through the ERAS Committee with opportunity for input and approval from key stakeholders	☐
C. Building the ERAS team: Include name and contact information	
1. Surgeon Champion(s)	☐
2. Anesthesiologist Champion(s)	☐
3. Nurse Champion(s)/ERAS Coordinator	☐
4. Hospital Administration	☐

5. CRNA(s)	☐
6. Service Line Navigator	☐
7. Pharmacist	☐
8. Allied Staff: Physical Therapist, Occupational Therapist, Respiratory Therapist, CAN, Exercise Tech, Nutritionist	☐
9. Administrative support staff	☐
10. Case Manager	☐
11. Quality Improvement	☐
12. Dimensions	☐
13. Capital Equipment Committee member	☐
14. Other	☐
D. Clarify ERAS Committee Roles, Responsibilities, and Goals	
1. Goals Established	☐
2. ERAS Committee Meeting Schedule Established **who owns the meeting invite and agenda development?	☐
3. Committee Chair(s)	☐
4. Protocol Development Team established **set deadline for protocol completion	☐
5. Patient Education Materials Team established **set deadline for patient education materials	☐
6. Nursing Education Team established	☐
7. Order Set Development Team established	☐
8. ERAS Data Collection Team established	☐
9. Capital Needs Assessment: Complete and submit EARLY , should be minimal	☐
E. Education Planning: Plan to begin when protocol nears finalization	
1. All care settings (mandatory: in-person education sessions) *include APP, Residents, RN, CNA, PT, OT, RT, Exercise Tech, Rehab, Pharmacist, Case Manager • Office • Preop Clinic • Preop holding • PACU • OR • ICU • Floor	☐
2. Anesthesiologist/CRNA education	☐
F. Patient Education Materials	
1. Begin early in process and finalize after program finalization	☐
2. Considerations a. Hospital process to approve patient materials b. Patient input during development c. Different learning styles d. Ability to modify materials in future	☐

G. Questions to Consider 1. All patients or a subset? 2. Who will provide patient education and where? 3. Will message always be the same?	☐ ☐ ☐
H. Create ERAS dashboard/database 1. Determine who will collect and report the data 2. Process Metrics 3. Outcomes Dashboard	☐ ☐ ☐
I. Other Completion Items 1. Protocol Checklists in ORs (in binders or on computer desktop) 2. Protocols/Checklists in central location or EHR	☐ ☐
J. Implementation Day 1. Have patient sign marketing release (possible story on hospital intranet) 2. Make sure all MD orders are complete and correct 3. Designated person in care areas, following patient, available for questions/help as needed 4. Celebrate	☐ ☐ ☐ ☐
K. Future Planning 1. Review outcome and process data – Key! 2. Report data to ERAS Committee and disseminate to all care areas 3. If low compliance, address with nursing leadership AND ask front line staff challenges 4. Review PDSA 5. Plan for expansion of program/further innovation	☐ ☐ ☐ ☐ ☐

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